CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 demo varianti

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Koordinata (holat);

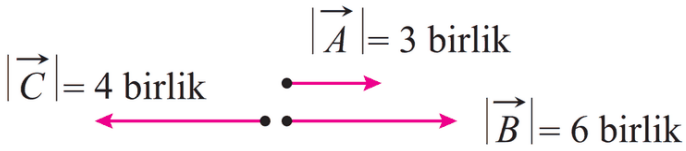
II. Ko'chish;

III. Bosib o'tilgan yo'l

kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) Faqat I **B)** Faqat II **C)** I va II **D)** I va III

2. B Rasmda bir to'g'ri chiziq bo'ylab joylashgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



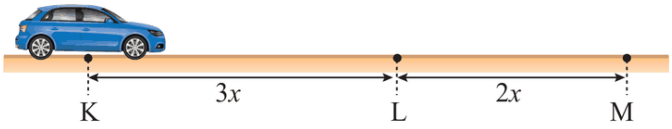
A) 5 **B)** 7 **C)** 9 **D)** 10

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

I ustun	II ustun
1. Jismni harakatga keltiruvchi kuch jism bilan bevosita kontaktda bo'lishi shart.	A) Noto'g'ri, chunki kontakt kuchi jismni tortishi ham mumkin.
2. Kontakt kuchlari har doim itaruvchi kuch sifatida ta'sir qiladi.	B) To'g'ri, elektr va magnit kuchlar tortish yoki itarish tarzida ta'sir qiladi.
3. Maydon kuchlari tortish yoki itarish tarzida ta'sir qiladi.	C) Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi.
	D) To'g'ri, chunki har qanday kuch faqat bevosita kontakt orqali uzatiladi.

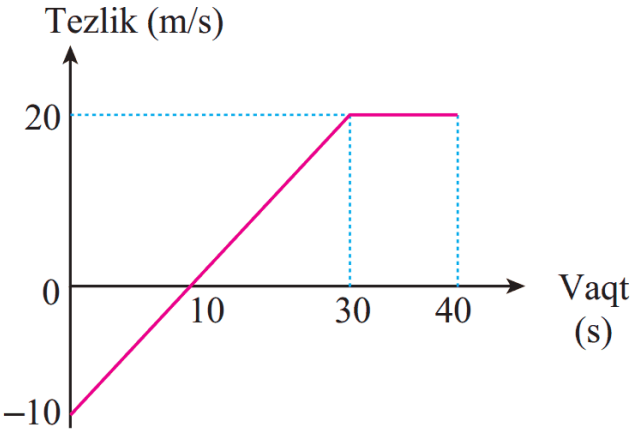
A) 1–A, 2–C, 3–D **B)** 1–C, 2–A, 3–B **C)** 1–D, 2–B, 3–A **D)** 1–B, 2–D, 3–C

4. Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K nuqtadan L nuqtagacha tekis tezlanuvchan, L nuqtadan M nuqtagacha esa L nuqtadagi o'zgarmas tezligi bilan harakat qiladi. Avtomobilning KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball



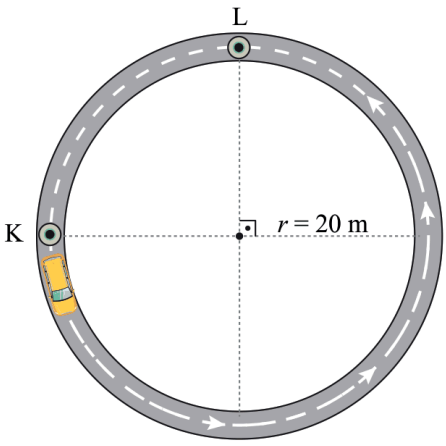
A) 1 **B)** 2 **C)** 3 **D)** 4

5. To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 350 **B)** 375 **C)** 400 **D)** 450

6. Q Radiusi 20 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan yo'lga chiqib, rasmda ko'rsatilgan yo'nalish bo'ylab harakatlanadi va 2 s dan keyin L nuqtaga yetib keladi. Avtomobilning K va L nuqtalar orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? Hisoblashda $\pi = 3$ deb oling. 2 ball

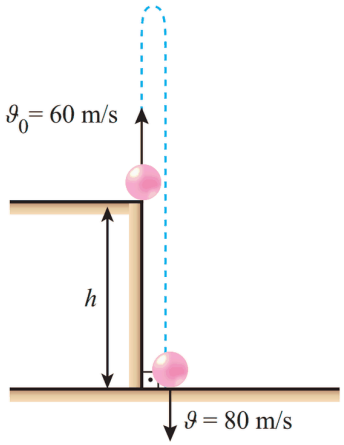


A) $\vartheta_{o'rt} = 45$ m/s; $|\vec{\vartheta}_{o'rt}| = 10\sqrt{2}$ m/s
B) $\vartheta_{o'rt} = 30$ m/s; $|\vec{\vartheta}_{o'rt}| = 30$ m/s
C) $\vartheta_{o'rt} = 10$ m/s; $|\vec{\vartheta}_{o'rt}| = 10\sqrt{2}$ m/s
D) $\vartheta_{o'rt} = 10$ m/s; $|\vec{\vartheta}_{o'rt}| = 10$ m/s

7. Q Jism $s = 5t - 0{,}25t^2$ qonuniyat bilan harakatlanadi. Uning dastlabki 4 s ichida o'tgan yo'lini toping (m). 2 ball

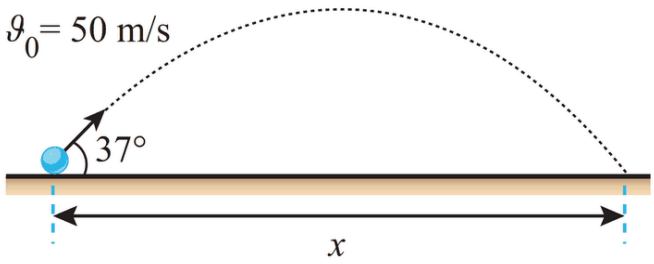
A) 16 B) 10 C) 12 D) 5

8. Q Jism h balandlikdan vertikal yuqoriga 60 m/s boshlang'ich tezlik bilan otildi. Jism yerga 80 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball



A) 150 B) 140 C) 130 D) 120

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 37° burchak ostida 50 m/s boshlang'ich tezlik bilan otildi. Jismning gorizontal yo'nalishda bosib o'tgan masofasi x necha metr? ($\sin 37^\circ = 0{,}6$; $\cos 37^\circ = 0{,}8$; $g = 10$ m/s²) 2 ball



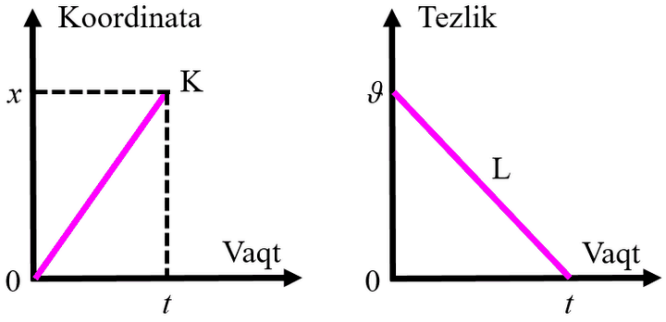
A) 120 **B)** 160 **C)** 180 **D)** 240

10. To'g'ri chiziqli yo'lda harakatlanayotgan K avtomobilning koordinata–vaqt va L avtomobilning tezlik–vaqt grafiklari rasmda ko'rsatilgan. Grafiklarga ko'ra quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. K avtomobildagi yo'lovchilarning harakat holati o'zgarmaydi.

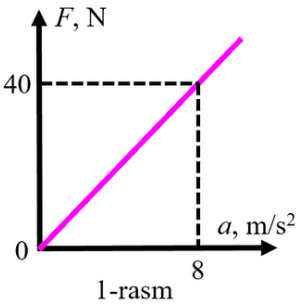
II. L avtomobildagi yo'lovchilar o'z harakat holatini saqlashi uchun oldinga tomon qalqiydi (siljiydi).

III. K avtomobilga ta'sir etuvchi natijaviy kuch nolga teng, L avtomobil esa noldan farqli natijaviy kuch ta'sirida harakatlanmoqda.

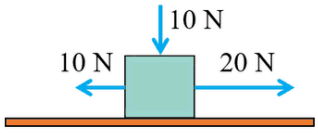


A) Faqat I **B)** I va II **C)** I va III **D)** I, II va III

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi 1-rasmda ko'rsatilgan. Jismga 2-rasmda ko'rsatilgan kuchlar ta'sir ettirilganda uning tezlanishi necha m/s^2 bo'ladi? 2 ball



1-rasm



2-rasm

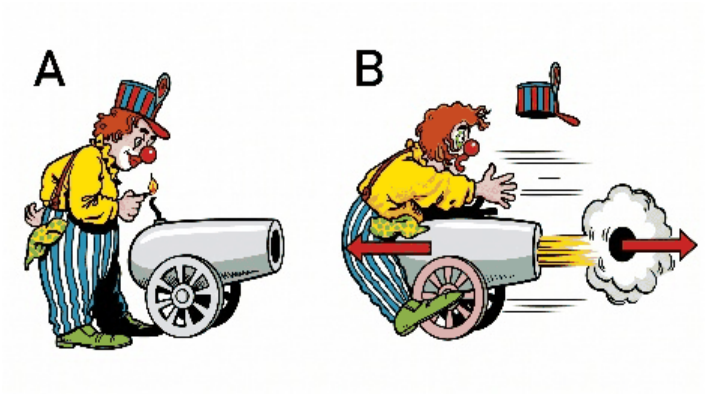
A) 1 **B)** 2 **C)** 4 **D)** 6

12. Fizikani bilmaydigan masxaraboz to'pdan o'q uzish paytida qayerda turish kerakligini hisobga olmagani uchun II rasmda tasvirlangan holatga tushib qoladi. Bunga ko'ra quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Har qanday ta'sirga miqdor jihatidan teng va qarama-qarshi yo'nalgan aks ta'sir mavjud.

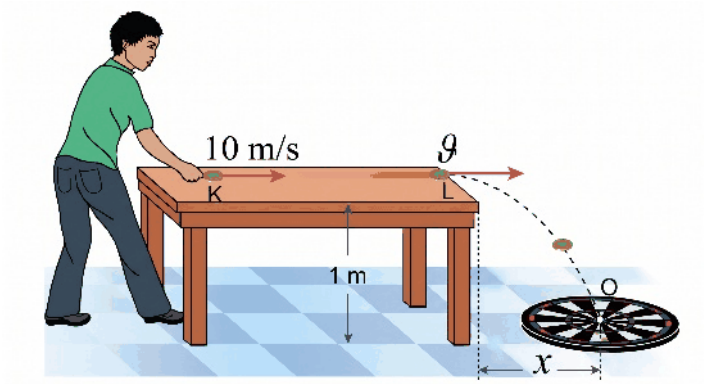
II. Ta'sir va aks ta'sir kuchlar miqdor jihatidan teng va qarama-qarshi yo'nalganligi sababli snaryad to'piga ta'sir qiluvchi natijaviy kuch nolga teng.

III. Ta'sir kuchi snaryad to'piga, aks ta'sir kuchi esa to'p aravachasiga ta'sir qiladi.



A) I, II va III **B)** Faqat II **C)** I va II **D)** I va III

13. Q Salim metall tangani stolning K nuqtasidan 10 m/s boshlang'ich tezlik bilan sirpantirib yubordi. Tanga 2,5 m/s² tezlanish moduli bilan sekinlashib, 2 s dan so'ng stolning L nuqtasidan gorizontal yo'nalishda uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofada joylashgan O nuqtaga tushdi. Stolning balandligi 1 m bo'lsa, x masofa necha metrga teng? $g = 10$ m/s². Havoning qarshiligi hisobga olinmasin. 2 ball



A) 1 **B)** 2 **C)** $2\sqrt{2}$ **D)** $\sqrt{5}$

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharchaning massasi, aravachaning tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Barcha tajribalarda ip uzunligi bir xil, $g = 10$ m/s². Sharcha aravachaga nisbatan tinch holatda bo'lgan. Burchakni o'lchash xatoligi ±0,3°; massa va tezlanishning o'lchash xatoliklari hisobga olinmaydi. 3 ball

Tajriba	Sharcha massasi, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,1	2	11,3°
2	0,2	4	21,8°
3	0,2	2	11,4°

O'quvchi 1- va 2-tajribalarni solishtirib: "Sharchaning massasi oshsa, ipning og'ish burchagi ham ortadi", degan xulosaga keldi. Jadval asosida qaysi xulosa eng asosli?

A) Xulosa asosli: 1 va 2 da massa ortishi bilan burchak ham ortgan.

B) Xulosa asosli: 1 va 3 dagi burchaklar farqi massa ta'sirini ko'rsatadi.

C) Xulosa asoslanmagan: 1 va 2 da massa bilan birga tezlanish ham o'zgargan.

D) Xulosa asoslanmagan: 2 va 3 da massalar tengligi burchaklar tengligini talab qiladi.

15. M Balandligi 40 m bo'lgan qoya tepasidan tosh gorizontga nisbatan 30° burchak ostida yuqoriga, 20 m/s boshlang'ich tezlik bilan otildi. Videoyozuvdan toshning yer sathidan balandligi aniqlandi. Vaqtni o'lchash xatoligi hisobga olinmaydi, balandlikni o'lchash xatoligi esa ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	40	45	42	25	0

Havo qarshiligi hisobga olinmaydigan model qabul qilingan. $g = 10$ m/s²; $\sin 30^\circ = 0{,}5$.

Ikki o'quvchi quyidagicha fikr bildirdi:

• Aziz: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan".

• Malika: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak".

Qaysi baho ilmiy jihatdan eng asosli?

A) Azizning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

B) Malikaning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

C) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar berilgan xatolik doirasida modelga mos.

D) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 demo varianti

1 B C	2 B A	3 B B	4 Q C	5 Q A	6 Q A	7 Q A
8 Q B	9 Q D	10 Q D	11 Q B	12 Q D	13 Q D	14 M C
15 M B						

1. C) Koordinata (sanoq boshiga nisbatan holat) va ko'chish — yo'nalishga ega kattaliklar. Bosib o'tilgan yo'l — skalyar, faqat son qiymatga ega.

2. A) $\vec{A}$ va $\vec{B}$ o'ngga, $\vec{C}$ chapga yo'nalgan: $3+6-4=5$ birlik.

3. B) 1 — noto'g'ri: maydon kuchlari kontaktsiz ta'sir qiladi (C). 2 — noto'g'ri: masalan, ip tortish kuchi ham kontakt kuchi (A). 3 — to'g'ri: elektr va magnit kuchlar tortadi ham, itaradi ham (B).

4. C) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1=\frac{3x}{\vartheta/2}=\frac{6x}{\vartheta}$. LM da $t_2=\frac{2x}{\vartheta}$. $t_1:t_2=3$.

5. A) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–10 s: $-\frac{10\cdot 10}{2}=-50$; 10–30 s: $\frac{20\cdot 20}{2}=200$; 30–40 s: $20\cdot 10=200$. Jami 350 m.

6. A) Ko'rsatilgan yo'nalishda K dan L gacha aylananing $\frac{3}{4}$ qismi: $s=\frac{3}{4}\cdot 2\pi R=90$ m, $\vartheta_{o'rt}=\frac{90}{2}=45$ m/s. Ko'chish — vatar: $R\sqrt{2}=20\sqrt{2}$ m, $|\vec{\vartheta}_{o'rt}|=10\sqrt{2}$ m/s.

7. A) $\vartheta=5-0{,}5t$, jism 10 s da to'xtaydi, demak 4 s ichida yo'nalish o'zgarmaydi. $s(4)=20-4=16$ m.

8. B) Energiya saqlanishidan: $\vartheta^2=\vartheta_0^2+2gh$. $6400=3600+20h\to h=140$ m.

9. D) $\vartheta_x=40$ m/s, $\vartheta_y=30$ m/s. Uchish vaqti $t=\frac{2\cdot 30}{10}=6$ s. $x=40\cdot 6=240$ m.

10. D) K: koordinata vaqtga chiziqli bog'liq — tekis harakat, natijaviy kuch nol (I, III). L: tezlik kamaymoqda — tormozlanish, yo'lovchilar inersiya bo'yicha oldinga siljiydi (II), natijaviy kuch noldan farqli. Uchala fikr to'g'ri.

11. B) 1-rasmdan: $m=\frac{F}{a}=\frac{40}{8}=5$ kg. 2-rasmda vertikal kuch harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $20-10=10$ N. $a=\frac{10}{5}=2$ m/s².

12. D) Nyutonning III qonuni: ta'sir va aks ta'sir teng va qarama-qarshi (I to'g'ri). Lekin ular turli jismlarga qo'yilgan — biri snaryadga, ikkinchisi to'pga (III to'g'ri), shuning uchun bir-birini muvozanatlamaydi (II noto'g'ri).

13. D) L nuqtadagi tezlik: $10-2{,}5\cdot 2=5$ m/s. Tushish vaqti: $t=\sqrt{\frac{2h}{g}}=\sqrt{0{,}2}$ s. $x=5\sqrt{0{,}2}=\sqrt{5}$ m.

14. C) 1- va 2-tajribada ikki kattalik — massa ham, tezlanish ham — o'zgargan, shuning uchun burchak o'zgarishini massaga bog'lab bo'lmaydi. 1- va 3-tajribada faqat massa o'zgargan va burchak xatolik doirasida bir xil ($\operatorname{tg}\alpha=a/g$).

15. B) Model: $h=40+10t-5t^2$: 40; 45; 40; 25; 0. Faqat $t=2$ s dagi qiymat (42 m) xatolik doirasidan chiqadi, qolganlari mos. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Malika).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 1-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

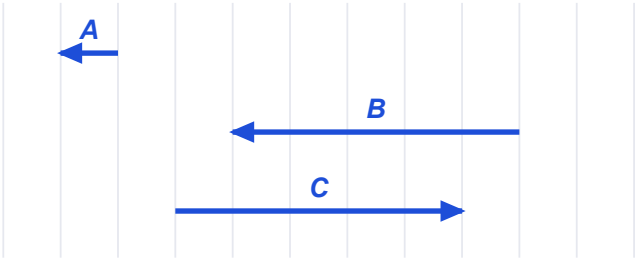
Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Jism massasi;
II. Ko'chish;
III. Kuch;
kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) I, II va III **B)** Faqat I **C)** II va III **D)** I va III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



1 katak = 1 birlik

A) 11 **B)** 1 **C)** 3 **D)** 5

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

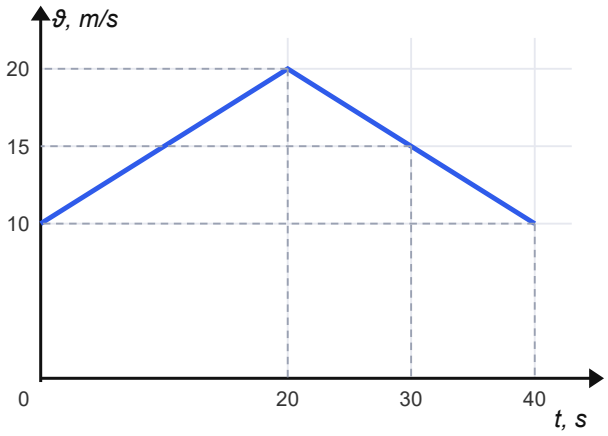
I ustun	II ustun
1. Ishqalanish kuchi kontakt kuchiga misol bo'ladi.	A) To'g'ri, chunki har qanday kuch faqat bevosita kontakt orqali uzatiladi.
2. Kuch jismning tezligini ham, shaklini ham o'zgartirishi mumkin.	B) To'g'ri, kuch tezlanish beradi yoki deformatsiya hosil qiladi.
3. Maydon kuchlari tortish yoki itarish tarzida ta'sir qiladi.	C) To'g'ri, elektr va magnit kuchlar tortish yoki itarish tarzida ta'sir qiladi.
	D) To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi.

A) 1–C, 2–D, 3–B **B)** 1–C, 2–A, 3–D **C)** 1–D, 2–B, 3–C **D)** 1–B, 2–D, 3–C

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 120 m) tekis tezlanuvchan, L dan M gacha (LM = 80 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 1,5 **B)** 6 **C)** 3 **D)** 4

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 625 **B)** 650 **C)** 550 **D)** 600

6. Q Radiusi 60 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 3 s da aylananing $\frac{3}{4}$ qismini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 90$ m/s; $|\vec{\vartheta}_{o'rt}| = 20\sqrt{2}$ m/s

B) $\vartheta_{o'rt} = 90$ m/s; $|\vec{\vartheta}_{o'rt}| = 40$ m/s

C) $\vartheta_{o'rt} = 20\sqrt{2}$ m/s; $|\vec{\vartheta}_{o'rt}| = 20\sqrt{2}$ m/s

D) $\vartheta_{o'rt} = 90$ m/s; $|\vec{\vartheta}_{o'rt}| = 90$ m/s

7. Q Jism $s = 5t - 0{,}25t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 4 s ichida o'tgan yo'lini toping (m). 2 ball

A) 18 **B)** 20 **C)** 24 **D)** 16

8. Q Jism h balandlikdan vertikal pastga 10 m/s boshlang'ich tezlik bilan otildi. Jism yerga 30 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 45 **B)** 50 **C)** 20 **D)** 40

9. Q Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 53° burchak ostida 25 m/s boshlang'ich tezlik bilan otildi. Jismning gorizontal yo'nalishda bosib o'tgan masofasi x necha metr? ($\sin 53^\circ = 0{,}8$; $\cos 53^\circ = 0{,}6$; $g = 10$ m/s²) 2 ball

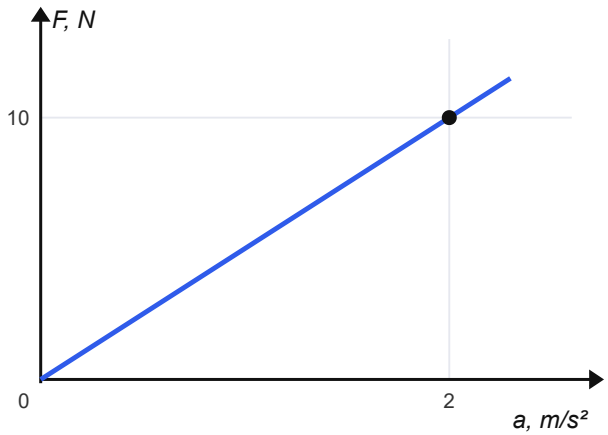
A) 60 **B)** 120 **C)** 80 **D)** 30

10. Q To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 20 + 12t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 5 s davomida 24 m/s dan 16 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. K avtomobildagi yo'lovchilarning harakat holati o'zgarmaydi.
II. K ga ta'sir etuvchi natijaviy kuch nolga teng, L ga esa noldan farqli.
III. K avtomobilning tezlanishi 12 m/s².

A) Faqat I **B)** II va III **C)** Faqat II **D)** I va II

11. Q Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 2$ m/s², $F = 10$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 40 N, chapga 20 N va vertikal yuqoriga 5 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 4 **B)** 12 **C)** 5 **D)** 8

12. Q Ikki konkichi muz ustida bir-birini itardi. A ning massasi B nikidan ikki marta katta. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. A kuchliroq bo'lgani uchun B ga kattaroq kuch bilan ta'sir qiladi.
II. Ular bir-biriga teng kuch bilan ta'sir qiladi.
III. B ning tezlanishi A nikidan ikki marta katta.

A) II va III **B)** Faqat I **C)** Faqat II **D)** I va II

13. Q Kamola metall tangani stolning K nuqtasidan 10 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 2 m/s² tezlanish moduli bilan sekinlashib, 1 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 1,8 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 4,8 **B)** 6 **C)** 28,8 **D)** 9,6

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,2	1	5,7°
2	0,4	2	11,4°
3	0,4	1	5,8°

O'quvchi 2- va 3-tajribalarni solishtirib: "Aravacha tezlanishi oshsa, ipning og'ish burchagi ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asoslanmagan: burchak farqi o'lchash xatoligi doirasida.

B) Xulosa asoslanmagan: 2 va 3 da massa ham o'zgargan.

C) Xulosa asosli: 1 va 2 da massa o'zgarmagan.

D) Xulosa asosli: 2 va 3 da faqat tezlanish o'zgargan, burchak farqi xatolikdan katta.

15. M Balandligi 45 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	45	48	45	30	5

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Aziz: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

• Nilufar: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

Qaysi baho ilmiy jihatdan eng asosli?

A) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

B) Azizning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

C) Nilufarning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

D) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 1-variant

1 B C	2 B B	3 B C	4 Q C	5 Q D	6 Q A	7 Q D
8 Q D	9 Q A	10 Q D	11 Q A	12 Q A	13 Q A	14 M D
15 M B						

1. C) Jism massasi — skalyar kattalik. Ko'chish — vektor kattalik. Kuch — vektor kattalik. Javob: II va III.

2. B) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $-1-5+5=-1$. Modul: 1 birlik.

3. C) 1 — To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi. (D). 2 — To'g'ri, kuch tezlanish beradi yoki deformatsiya hosil qiladi. (B). 3 — To'g'ri, elektr va magnit kuchlar tortish yoki itarish tarzida ta'sir qiladi. (C).

4. C) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1=\frac{120}{\vartheta/2}=\frac{240}{\vartheta}$. LM da $t_2=\frac{80}{\vartheta}$. $t_1:t_2=3$.

5. D) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–20 s: 300; 20–30 s: 175; 30–40 s: 125. Ko'chish: 600 m.

6. A) Yo'l: $s=\frac{3}{4}\cdot 2\pi R=270$ m, $\vartheta_{o'rt}=\frac{270}{3}=90$ m/s. Ko'chish — vatar: $R\sqrt{2}=60\sqrt{2}$ m, $|\vec{\vartheta}_{o'rt}|=20\sqrt{2}$ m/s.

7. D) $\vartheta=5-0{,}5t$, jism 10 s da to'xtaydi, demak 4 s ichida yo'nalish o'zgarmaydi. $s(4)=16$ m.

8. D) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2=\vartheta_0^2+2gh$. $900=100+20h\to h=40$ m.

9. A) $\vartheta_x=15$ m/s, $\vartheta_y=20$ m/s. Uchish vaqti $t=\frac{2\cdot 20}{10}=4$ s. $x=15\cdot 4=60$ m.

10. D) I: K tekis harakatlanadi — to'g'ri. II: K da tezlanish yo'q, L da bor — to'g'ri. III: 12 m/s — tezlik, tezlanish nolga teng — noto'g'ri. Javob: I va II.

11. A) Grafikdan: $m=\frac{F}{a}=\frac{10}{2}=5$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $40-20=20$ N. $a=\frac{20}{5}=4$ m/s².

12. A) I: kuchlar har doim teng — noto'g'ri. II: III qonun — to'g'ri. III: massa ikki marta kichik — to'g'ri. Javob: II va III.

13. A) L dagi tezlik: $10-2\cdot 1=8$ m/s. Tushish vaqti: $t=\sqrt{\frac{2h}{g}}=0{,}6$ s. $x=8\cdot 0{,}6=4{,}8$ m.

14. D) 2 va 3 da massa bir xil (0,4 kg), faqat tezlanish farq qiladi; burchaklar farqi ±0,3° dan ancha katta ($\operatorname{tg}\alpha=a/g$).

15. B) Model: $h=45+10t-5t^2$: 45; 50; 45; 30; 5. Faqat $t=1$ s dagi qiymat (48 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Aziz).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 2-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

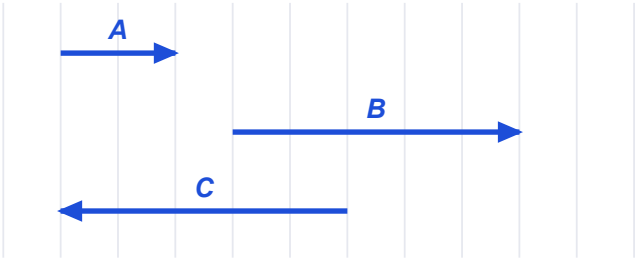
Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Bosib o'tilgan yo'l;
II. Koordinata (holat);
III. Kuch;
kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) Faqat II **B)** II va III **C)** I, II va III **D)** I va III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



1 katak = 1 birlik

A) 2 **B)** 12 **C)** 4 **D)** 6

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

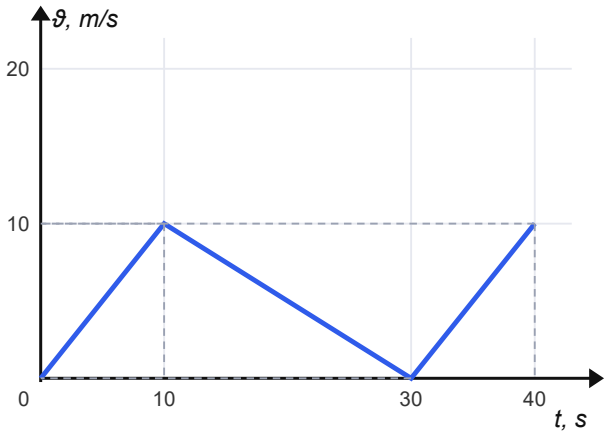
I ustun	II ustun
1. Jismni harakatga keltiruvchi kuch jism bilan bevosita kontaktda bo'lishi shart.	A) Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin.
2. Og'irlik kuchi faqat jism yerga tegib turganda ta'sir qiladi.	B) Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi.
3. Kontakt kuchlari har doim itaruvchi kuch sifatida ta'sir qiladi.	C) Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi.
	D) To'g'ri, chunki maydon kuchlari faqat itarish tarzida ta'sir qiladi.

A) 1–B, 2–C, 3–D **B)** 1–D, 2–C, 3–A **C)** 1–D, 2–B, 3–A **D)** 1–B, 2–C, 3–A

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 80 m) tekis tezlanuvchan, L dan M gacha (LM = 40 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 5 **B)** 4 **C)** 8 **D)** 2

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 150 **B)** 225 **C)** 250 **D)** 200

6. Q Radiusi 60 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 1 s da aylananing yarmini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 120$ m/s; $|\vec{\vartheta}_{o'rt}| = 120$ m/s

B) $\vartheta_{o'rt} = 180$ m/s; $|\vec{\vartheta}_{o'rt}| = 60\sqrt{2}$ m/s

C) $\vartheta_{o'rt} = 180$ m/s; $|\vec{\vartheta}_{o'rt}| = 120$ m/s

D) $\vartheta_{o'rt} = 180$ m/s; $|\vec{\vartheta}_{o'rt}| = 180$ m/s

7. Q Jism $s = 4t - 2t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 5 s ichida o'tgan yo'lini toping (m). 2 ball

A) 20 **B)** 30 **C)** 70 **D)** 34

8. Jism h balandlikdan gorizontal 10 m/s boshlang'ich tezlik bilan otildi. Jism yerga 50 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 120 **B)** 125 **C)** 130 **D)** 80

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 37° burchak ostida 25 m/s boshlang'ich tezlik bilan otildi. Jismning ko'tarilishining maksimal balandligi H necha metr? ($\sin 37^\circ = 0{,}6$; $\cos 37^\circ = 0{,}8$; $g = 10$ m/s²) 2 ball

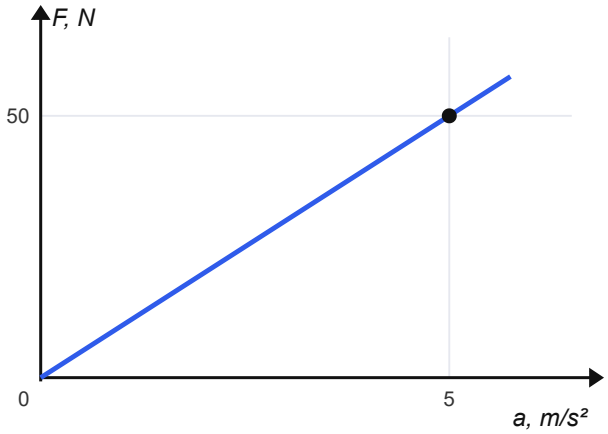
A) 22,5 **B)** 11,25 **C)** 31,25 **D)** 20

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 12t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 2 s davomida 24 m/s dan 4 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. K avtomobilning tezlanishi 12 m/s².
II. K avtomobildagi yo'lovchilarning harakat holati o'zgarmaydi.
III. K ga ta'sir etuvchi natijaviy kuch nolga teng, L ga esa noldan farqli.

A) Faqat I **B)** I va III **C)** Faqat II **D)** II va III

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 5$ m/s², $F = 50$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 40 N, chapga 5 N va vertikal yuqoriga 15 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 4,5 **B)** 5 **C)** 4 **D)** 3,5

12. Q To'pdan o'q uzilganda to'p orqaga tepiladi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Har qanday ta'sirga miqdor jihatidan teng va qarama-qarshi yo'nalgan aks ta'sir mavjud.

II. Ta'sir kuchi snaryadga, aks ta'sir kuchi esa to'pga qo'yilgan.

III. Ta'sir va aks ta'sir kuchlari teng va qarama-qarshi bo'lgani uchun snaryadga ta'sir qiluvchi natijaviy kuch nolga teng.

A) Faqat II **B)** I va III **C)** I va II **D)** Faqat III

13. Q Salim metall tangani stolning K nuqtasidan 6 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 2,5 m/s² tezlanish moduli bilan sekinlashib, 1 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 0,45 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 3,15 **B)** 1,8 **C)** 1,05 **D)** 2,1

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,2	1	5,6°
2	0,3	2	11,3°
3	0,3	1	5,7°

O'quvchi 2- va 3-tajribalarni solishtirib: "Aravacha tezlanishi oshsa, ipning og'ish burchagi ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asosli: 2 va 3 da faqat tezlanish o'zgargan, burchak farqi xatolikdan katta.

B) Xulosa asosli: 1 va 2 da massa o'zgarmagan.

C) Xulosa asoslanmagan: burchak farqi o'lchash xatoligi doirasida.

D) Xulosa asoslanmagan: 2 va 3 da massa ham o'zgargan.

15. M Balandligi 45 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	45	50	42	30	5

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Aziz: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

• Nilufar: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

Qaysi baho ilmiy jihatdan eng asosli?

A) Nilufarning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

B) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

C) Azizning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

D) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 2-variant

1 B	B	2 B	A	3 B	D	4 Q	B	5 Q	D	6 Q	C	7 Q	D
8 Q	A	9 Q	B	10 Q	D	11 Q	D	12 Q	C	13 Q	C	14 M	A
15 M	A												

1. B) Bosib o'tilgan yo'l — skalyar kattalik. Koordinata (holat) — sanoq boshiga nisbatan yo'nalishga ega. Kuch — vektor kattalik. Javob: II va III.

2. A) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $2 + 5 - 5 = 2$. Modul: 2 birlik.

3. D) 1 — Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi. (B). 2 — Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi. (C). 3 — Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin. (A).

4. B) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1 = \frac{80}{\vartheta/2} = \frac{160}{\vartheta}$. LM da $t_2 = \frac{40}{\vartheta}$. $t_1 : t_2 = 4$.

5. D) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–10 s: 50; 10–30 s: 100; 30–40 s: 50. Ko'chish: 200 m.

6. C) Yo'l: $s = \frac{1}{2} \cdot 2\pi R = 180$ m, $\vartheta_{o'rt} = \frac{180}{1} = 180$ m/s. Ko'chish — vatar: $2R = 120$ m, $|\vec{\vartheta}_{o'rt}| = 120$ m/s.

7. D) $\vartheta = 4 - 4t$, jism $t = 1$ s da to'xtab, orqaga harakatlanadi. Yo'l: $2 + 32 = 34$ m (koordinata esa -30 m).

8. A) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2 = \vartheta_0^2 + 2gh$. $2500 = 100 + 20h \to h = 120$ m.

9. B) $\vartheta_x = 20$ m/s, $\vartheta_y = 15$ m/s. Uchish vaqti $t = \frac{2 \cdot 15}{10} = 3$ s. $H = \frac{\vartheta_y^2}{2g} = 11{,}25$ m.

10. D) I: 12 m/s — tezlik, tezlanish nolga teng — noto'g'ri. II: K tekis harakatlanadi — to'g'ri. III: K da tezlanish yo'q, L da bor — to'g'ri. Javob: II va III.

11. D) Grafikdan: $m = \frac{F}{a} = \frac{50}{5} = 10$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $40 - 5 = 35$ N. $a = \frac{35}{10} = 3{,}5$ m/s².

12. C) I: Nyutonning III qonuni — to'g'ri. II: kuchlar turli jismlarga qo'yilgan — to'g'ri. III: kuchlar turli jismlarga qo'yilgan — noto'g'ri. Javob: I va II.

13. C) L dagi tezlik: $6 - 2{,}5 \cdot 1 = 3{,}5$ m/s. Tushish vaqti: $t = \sqrt{\frac{2h}{g}} = 0{,}3$ s. $x = 3{,}5 \cdot 0{,}3 = 1{,}05$ m.

14. A) 2 va 3 da massa bir xil (0,3 kg), faqat tezlanish farq qiladi; burchaklar farqi ±0,3° dan ancha katta ($\mathrm{tg}\,\alpha = a/g$).

15. A) Model: $h = 45 + 10t - 5t^2$: 45; 50; 45; 30; 5. Faqat $t = 2$ s dagi qiymat (42 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Nilufar).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 3-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Tezlik moduli (spidometr ko'rsatkichi);

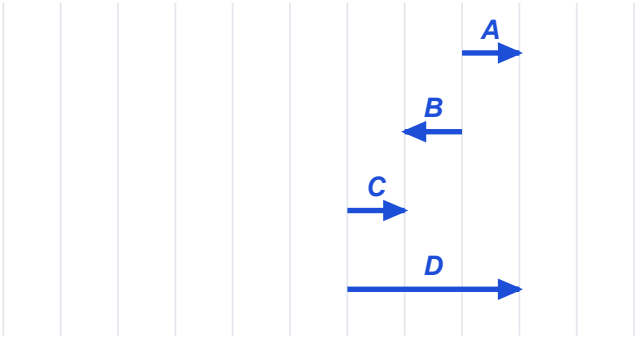
II. Harakat vaqti;

III. Tezlanish;

kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) Faqat I **B)** I, II va III **C)** I va II **D)** Faqat III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



A) 4 **B)** 2 **C)** 6 **D)** 8

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

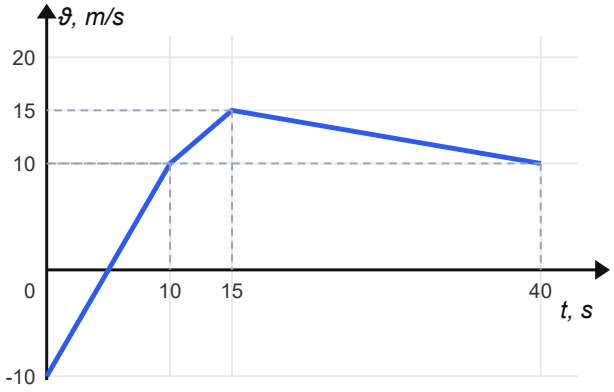
I ustun	II ustun
1. Kontakt kuchlari har doim itaruvchi kuch sifatida ta'sir qiladi.	A) Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi.
2. Og'irlik kuchi faqat jism yerga tegib turganda ta'sir qiladi.	B) Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin.
3. Ishqalanish kuchi kontakt kuchiga misol bo'ladi.	C) To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi.
	D) To'g'ri, chunki har qanday kuch faqat bevosita kontakt orqali uzatiladi.

A) 1–B, 2–A, 3–C **B)** 1–A, 2–C, 3–B **C)** 1–D, 2–A, 3–B **D)** 1–A, 2–B, 3–D

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 100 m) tekis tezlanuvchan, L dan M gacha (LM = 40 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 6 **B)** 2,5 **C)** 10 **D)** 5

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 375 **B)** 425 **C)** 325 **D)** 400

6. Q Radiusi 60 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 1 s da aylananing $\frac{3}{4}$ qismini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 60\sqrt{2}$ m/s; $|\vec{\vartheta}_{o'rt}| = 60\sqrt{2}$ m/s

B) $\vartheta_{o'rt} = 270$ m/s; $|\vec{\vartheta}_{o'rt}| = 120$ m/s

C) $\vartheta_{o'rt} = 270$ m/s; $|\vec{\vartheta}_{o'rt}| = 60\sqrt{2}$ m/s

D) $\vartheta_{o'rt} = 270$ m/s; $|\vec{\vartheta}_{o'rt}| = 270$ m/s

7. Q Jism $s = 4t - 0{,}5t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 8 s ichida o'tgan yo'lini toping (m). 2 ball

A) 18 **B)** 16 **C)** 32 **D)** 64

8. Jism h balandlikdan vertikal yuqoriga 10 m/s boshlang'ich tezlik bilan otildi. Jism yerga 40 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 75 **B)** 45 **C)** 80 **D)** 85

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 37° burchak ostida 30 m/s boshlang'ich tezlik bilan otildi. Jismning uchish vaqti necha sekund? ($\sin 37^\circ = 0{,}6$; $\cos 37^\circ = 0{,}8$; $g = 10$ m/s²) 2 ball

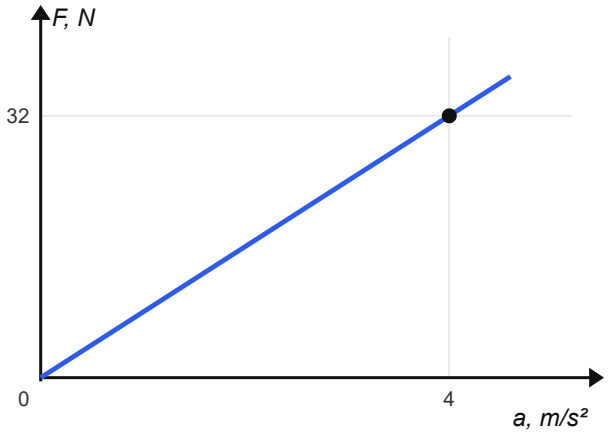
A) 4,8 **B)** 3,6 **C)** 1,8 **D)** 4,6

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 50 + 15t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 2 s davomida 24 m/s dan 6 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. L avtomobil tezlanishining moduli 9 m/s².
II. K ga ta'sir etuvchi natijaviy kuch nolga teng, L ga esa noldan farqli.
III. K avtomobildagi yo'lovchilarning harakat holati o'zgarmaydi.

A) Faqat III **B)** I, II va III **C)** Faqat II **D)** I va III

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 4$ m/s², $F = 32$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 30 N, chapga 20 N va vertikal yuqoriga 10 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 3,75 **B)** 2,5 **C)** 1,25 **D)** 6,25

12. Q Puflangan havo shari og'zi ochib qo'yib yuborilganda uchib ketdi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Shar va havoga ta'sir qiluvchi kuchlar miqdoran teng.

II. Shar havoni bir tomonga itaradi, havo sharni qarama-qarshi tomonga itaradi.

III. Ta'sir va aks ta'sir bir-birini muvozanatlagani uchun shar harakatlanmasligi kerak edi.

A) Faqat I **B)** I, II va III **C)** I va II **D)** Faqat III

13. Q Salim metall tangani stolning K nuqtasidan 8 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 2,5 m/s² tezlanish moduli bilan sekinlashib, 2 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 0,2 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 1,2 **B)** 1,6 **C)** 0,6 **D)** 1,1

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,3	3	16,8°
2	0,5	4	21,7°
3	0,5	3	16,7°

O'quvchi 2- va 3-tajribalarni solishtirib: "Aravacha tezlanishi oshsa, ipning og'ish burchagi ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asoslanmagan: 2 va 3 da massa ham o'zgargan.

B) Xulosa asosli: 2 va 3 da faqat tezlanish o'zgargan, burchak farqi xatolikdan katta.

C) Xulosa asoslanmagan: burchak farqi o'lchash xatoligi doirasida.

D) Xulosa asosli: 1 va 2 da massa o'zgarmagan.

15. M Balandligi 60 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	60	65	63	45	20

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Nilufar: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

• Malika: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

Qaysi baho ilmiy jihatdan eng asosli?

A) Nilufarning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

B) Malikaning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

C) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

D) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 3-variant

1 B D	2 B A	3 B A	4 Q D	5 Q A	6 Q C	7 Q B
8 Q A	9 Q B	10 Q B	11 Q C	12 Q C	13 Q C	14 M B
15 M B						

1. D) Tezlik moduli (spidometr ko'rsatkichi) – skalyar – faqat son qiymat. Harakat vaqti – skalyar kattalik. Tezlanish – vektor kattalik. Javob: Faqat III.

2. A) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $1 - 1 + 1 + 3 = 4$. Modul: 4 birlik.

3. A) 1 – Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin. (B). 2 – Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi. (A). 3 – To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi. (C).

4. D) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1 = \frac{100}{\vartheta/2} = \frac{200}{\vartheta}$. LM da $t_2 = \frac{40}{\vartheta}$. $t_1 : t_2 = 5$.

5. A) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–10 s: 0; 10–15 s: $62{,}5$; 15–40 s: $312{,}5$. Ko'chish: 375 m. Yo'l esa yuzalar modullari yig'indisi: 425 m.

6. C) Yo'l: $s = \frac{3}{4} \cdot 2\pi R = 270$ m, $\vartheta_{o'rt} = \frac{270}{1} = 270$ m/s. Ko'chish – vatar: $R\sqrt{2} = 60\sqrt{2}$ m, $|\vec{\vartheta}_{o'rt}| = 60\sqrt{2}$ m/s.

7. B) $\vartheta = 4 - t$, jism $t = 4$ s da to'xtab, orqaga harakatlanadi. Yo'l: $8 + 8 = 16$ m (koordinata esa 0 m).

8. A) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2 = \vartheta_0^2 + 2gh$. $1600 = 100 + 20h \to h = 75$ m.

9. B) $\vartheta_x = 24$ m/s, $\vartheta_y = 18$ m/s. Uchish vaqti $t = \frac{2 \cdot 18}{10} = 3{,}6$ s. Uchish vaqti $t = 3{,}6$ s.

10. B) I: $a = \frac{18}{2} = 9$ m/s² – to'g'ri. II: K da tezlanish yo'q, L da bor – to'g'ri. III: K tekis harakatlanadi – to'g'ri. Javob: I, II va III.

11. C) Grafikdan: $m = \frac{F}{a} = \frac{32}{4} = 8$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $30 - 20 = 10$ N. $a = \frac{10}{8} = 1{,}25$ m/s².

12. C) I: III qonun – to'g'ri. II: reaktiv harakat – to'g'ri. III: kuchlar turli jismlarga qo'yilgan – noto'g'ri. Javob: I va II.

13. C) L dagi tezlik: $8 - 2{,}5 \cdot 2 = 3$ m/s. Tushish vaqti: $t = \sqrt{\frac{2h}{g}} = 0{,}2$ s. $x = 3 \cdot 0{,}2 = 0{,}6$ m.

14. B) 2 va 3 da massa bir xil (0,5 kg), faqat tezlanish farq qiladi; burchaklar farqi ±0,3° dan ancha katta ($\operatorname{tg}\alpha = a/g$).

15. B) Model: $h = 60 + 10t - 5t^2$: 60; 65; 60; 45; 20. Faqat $t = 2$ s dagi qiymat (63 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi – avval o'lchovni tekshirish kerak (Malika).

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 4-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Koordinata (holat);

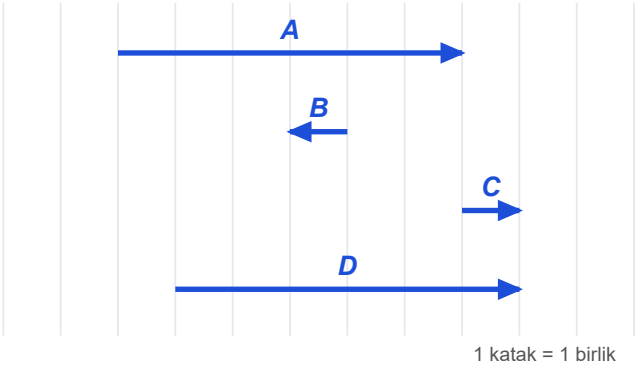
II. Ko'chish;

III. Tezlik;

kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) Faqat II **B)** I va III **C)** I, II va III **D)** Faqat I

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



A) 14 **B)** 16 **C)** 10 **D)** 12

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

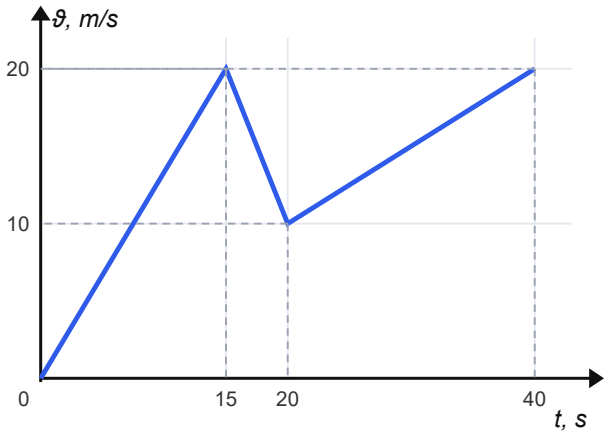
I ustun	II ustun
1. Kuch jismning tezligini ham, shaklini ham o'zgartirishi mumkin.	A) To'g'ri, kuch tezlanish beradi yoki deformatsiya hosil qiladi.
2. Jismga hech qanday kuch ta'sir qilmasa, u albatta tinch turadi.	B) Noto'g'ri, u to'g'ri chiziqli tekis harakatini davom ettirishi ham mumkin.
3. Magnit temir mixni masofadan turib tortadi.	C) To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi.
	D) To'g'ri, chunki har qanday kuch faqat bevosita kontakt orqali uzatiladi.

A) 1–B, 2–C, 3–A **B)** 1–B, 2–D, 3–A **C)** 1–C, 2–D, 3–A **D)** 1–A, 2–B, 3–C

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 120 m) tekis tezlanuvchan, L dan M gacha (LM = 160 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 1,5 **B)** 3 **C)** 0,75 **D)** 2,5

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 475 **B)** 550 **C)** 575 **D)** 525

6. Q Radiusi 40 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 5 s da aylananing yarmini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 24$ m/s; $|\vec{\vartheta}_{o'rt}| = 8\sqrt{2}$ m/s **B)** $\vartheta_{o'rt} = 24$ m/s; $|\vec{\vartheta}_{o'rt}| = 16$ m/s

C) $\vartheta_{o'rt} = 16$ m/s; $|\vec{\vartheta}_{o'rt}| = 16$ m/s **D)** $\vartheta_{o'rt} = 24$ m/s; $|\vec{\vartheta}_{o'rt}| = 24$ m/s

7. Q Jism $s = 3t - t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 4 s ichida o'tgan yo'lini toping (m). 2 ball

A) 28 **B)** 8,5 **C)** 12 **D)** 4

8. Q Jism h balandlikdan gorizontal 40 m/s boshlang'ich tezlik bilan otildi. Jism yerga 90 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 335 **B)** 125 **C)** 405 **D)** 325

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 37° burchak ostida 50 m/s boshlang'ich tezlik bilan otildi. Jismning gorizontal yo'nalishda bosib o'tgan masofasi x necha metr? ($\sin 37^\circ = 0{,}6$; $\cos 37^\circ = 0{,}8$; $g = 10$ m/s²) 2 ball

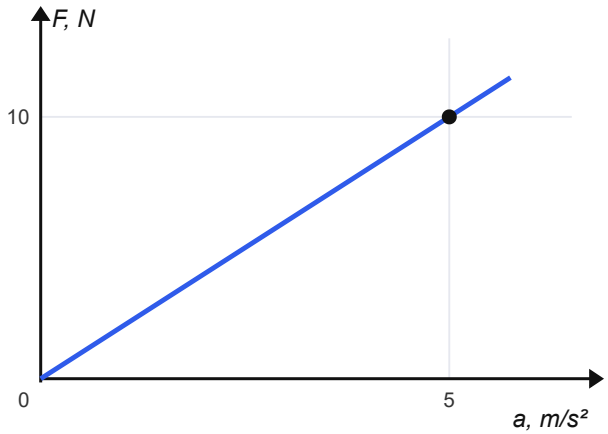
A) 480 **B)** 120 **C)** 180 **D)** 240

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 50 + 20t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 4 s davomida 12 m/s dan 16 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. K avtomobildagi yo'lovchilarning harakat holati o'zgarmaydi.
II. K avtomobilning tezlanishi 20 m/s².
III. Ikkala avtomobilga ta'sir etuvchi natijaviy kuch nolga teng.

A) Faqat I **B)** Faqat III **C)** I va III **D)** I va II

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 5$ m/s², $F = 10$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 30 N, chapga 10 N va vertikal yuqoriga 15 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 20 **B)** 15 **C)** 10 **D)** 17,5

12. Kitob stol ustida tinch yotibdi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Og'irlik kuchi va stolning tayanch reaksiya kuchi ta'sir-aks ta'sir juftligi hisoblanadi.
II. Kitob stolga, stol kitobga teng kuch bilan ta'sir qiladi.
III. Kitobga ta'sir qiluvchi natijaviy kuch nolga teng.

A) Faqat I **B)** II va III **C)** I, II va III **D)** Faqat III

13. Q Dilshod shayba stolning K nuqtasidan 8 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 1 m/s² tezlanish moduli bilan sekinlashib, 2 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 0,8 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 9,6 **B)** 3,2 **C)** 2,4 **D)** 4,8

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,1	2	11,2°
2	0,2	4	21,7°
3	0,2	2	11,4°

O'quvchi 2- va 3-tajribalarni solishtirib: "Aravacha tezlanishi oshsa, ipning og'ish burchagi ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asosli: 2 va 3 da faqat tezlanish o'zgargan, burchak farqi xatolikdan katta.

B) Xulosa asoslanmagan: burchak farqi o'lchash xatoligi doirasida.

C) Xulosa asoslanmagan: 2 va 3 da massa ham o'zgargan.

D) Xulosa asosli: 1 va 2 da massa o'zgarmagan.

15. M Balandligi 60 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 10 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	60	60	52	30	0

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Nilufar: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

• Bobur: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

Qaysi baho ilmiy jihatdan eng asosli?

A) Nilufarning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

B) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

C) Boburning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

D) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 4-variant

1 B C	2 B D	3 B D	4 Q A	5 Q D	6 Q B	7 Q B
8 Q D	9 Q D	10 Q A	11 Q C	12 Q B	13 Q C	14 M A
15 M A						

1. C) Koordinata (holat) — sanoq boshiga nisbatan yo'nalishga ega. Ko'chish — vektor kattalik. Tezlik — vektor kattalik. Javob: I, II va III.

2. D) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $6 - 1 + 1 + 6 = 12$. Modul: 12 birlik.

3. D) 1 — To'g'ri, kuch tezlanish beradi yoki deformatsiya hosil qiladi. (A). 2 — Noto'g'ri, u to'g'ri chiziqli tekis harakatini davom ettirishi ham mumkin. (B). 3 — To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi. (C).

4. A) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1 = \frac{120}{\vartheta/2} = \frac{240}{\vartheta}$. LM da $t_2 = \frac{160}{\vartheta}$. $t_1 : t_2 = 1{,}5$.

5. D) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–15 s: 150; 15–20 s: 75; 20–40 s: 300. Ko'chish: 525 m.

6. B) Yo'l: $s = \frac{1}{2} \cdot 2\pi R = 120$ m, $\vartheta_{o'rt} = \frac{120}{5} = 24$ m/s. Ko'chish – vatar: $2R = 80$ m, $|\vec{\vartheta}_{o'rt}| = 16$ m/s.

7. B) $\vartheta = 3 - 2t$, jism $t = 1{,}5$ s da to'xtab, orqaga harakatlanadi. Yo'l: $2{,}25 + 6{,}25 = 8{,}5$ m (koordinata esa -4 m).

8. D) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2 = \vartheta_0^2 + 2gh$. $8100 = 1600 + 20h \to h = 325$ m.

9. D) $\vartheta_x = 40$ m/s, $\vartheta_y = 30$ m/s. Uchish vaqti $t = \frac{2 \cdot 30}{10} = 6$ s. $x = 40 \cdot 6 = 240$ m.

10. A) I: K tekis harakatlanadi — to'g'ri. II: 20 m/s — tezlik, tezlanish nolga teng — noto'g'ri. III: L ning tezligi o'zgarmoqda — noto'g'ri. Javob: Faqat I.

11. C) Grafikdan: $m = \frac{F}{a} = \frac{10}{5} = 2$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $30 - 10 = 20$ N. $a = \frac{20}{2} = 10$ m/s².

12. B) I: ular bitta jismga qo'yilgan — noto'g'ri. II: III qonun — to'g'ri. III: kitob tinch — to'g'ri. Javob: II va III.

13. C) L dagi tezlik: $8 - 1 \cdot 2 = 6$ m/s. Tushish vaqti: $t = \sqrt{\frac{2h}{g}} = 0{,}4$ s. $x = 6 \cdot 0{,}4 = 2{,}4$ m.

14. A) 2 va 3 da massa bir xil (0,2 kg), faqat tezlanish farq qiladi; burchaklar farqi ±0,3° dan ancha katta ($\operatorname{tg}\alpha = a/g$).

15. A) Model: $h = 60 + 5t - 5t^2$: 60; 60; 50; 30; 0. Faqat $t = 2$ s dagi qiymat (52 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Nilufar).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 5-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

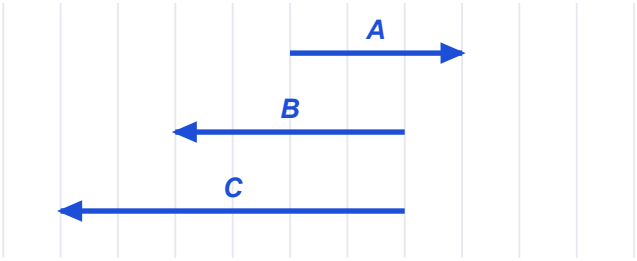
Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Kuch;
II. Koordinata (holat);
III. Harakat vaqti;
kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) I va III **B)** Faqat I **C)** I va II **D)** II va III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



1 katak = 1 birlik

A) 7 **B)** 5 **C)** 13 **D)** 9

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

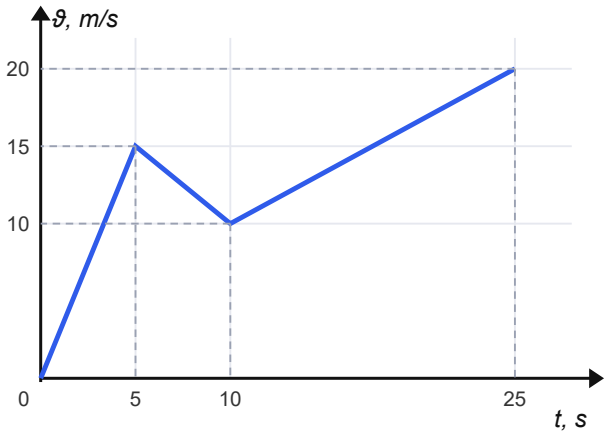
I ustun	II ustun
1. Og'irlik kuchi faqat jism yerga tegib turganda ta'sir qiladi.	A) To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi.
2. Ishqalanish kuchi kontakt kuchiga misol bo'ladi.	B) Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi.
3. Kontakt kuchlari har doim itaruvchi kuch sifatida ta'sir qiladi.	C) Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin.
	D) To'g'ri, chunki maydon kuchlari faqat itarish tarzida ta'sir qiladi.

A) 1–B, 2–C, 3–A **B)** 1–B, 2–A, 3–C **C)** 1–D, 2–A, 3–C **D)** 1–D, 2–C, 3–B

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 180 m) tekis tezlanuvchan, L dan M gacha (LM = 80 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 2,25 **B)** 4,5 **C)** 9 **D)** 5,5

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 375 **B)** 275 **C)** 325 **D)** 350

6. Q Radiusi 40 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 1 s da aylananing yarmini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 80$ m/s; $|\vec{\vartheta}_{o'rt}| = 80$ m/s

B) $\vartheta_{o'rt} = 120$ m/s; $|\vec{\vartheta}_{o'rt}| = 120$ m/s

C) $\vartheta_{o'rt} = 120$ m/s; $|\vec{\vartheta}_{o'rt}| = 40\sqrt{2}$ m/s

D) $\vartheta_{o'rt} = 120$ m/s; $|\vec{\vartheta}_{o'rt}| = 80$ m/s

7. Q Jism $s = 5t - 0{,}5t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 10 s ichida o'tgan yo'lini toping (m). 2 ball

A) 50 **B)** 27 **C)** 25 **D)** 100

8. Jism h balandlikdan vertikal yuqoriga 30 m/s boshlang'ich tezlik bilan otildi. Jism yerga 40 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 80 **B)** 35 **C)** 5 **D)** 45

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 53° burchak ostida 50 m/s boshlang'ich tezlik bilan otildi. Jismning gorizontal yo'nalishda bosib o'tgan masofasi x necha metr? ($\sin 53^\circ = 0{,}8$; $\cos 53^\circ = 0{,}6$; $g = 10$ m/s²) 2 ball

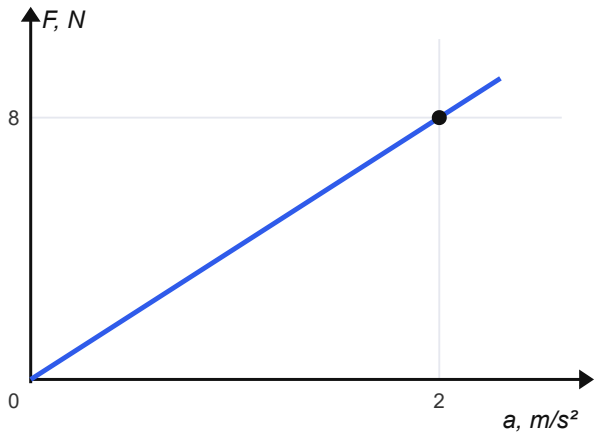
A) 120 **B)** 480 **C)** 240 **D)** 320

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 50 + 20t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 4 s davomida 12 m/s dan 16 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. L avtomobildagi yo'lovchilar orqaga tomon siljiydi.
II. L avtomobil tezlanishining moduli 2 m/s².
III. K ga ta'sir etuvchi natijaviy kuch nolga teng, L ga esa noldan farqli.

A) Faqat II **B)** I va III **C)** Faqat III **D)** II va III

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 2$ m/s², $F = 8$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 10 N, chapga 5 N va vertikal yuqoriga 10 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 2,5 **B)** 2,25 **C)** 1,25 **D)** 3,75

12. Q Qayiqdan qirg'oqqa sakragan bola qayiqni orqaga itaradi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Bolaga ta'sir qiluvchi kuch qayiqqa ta'sir qiluvchidan katta.
II. Qayiq massasi kichik bo'lsa, u kattaroq tezlanish oladi.
III. Bola qayiqqa, qayiq bolaga teng kuch bilan ta'sir qiladi.

A) I, II va III **B)** Faqat II **C)** I va II **D)** II va III

13. Q Kamola shayba stolning K nuqtasidan 8 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 1 m/s² tezlanish moduli bilan sekinlashib, 2 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 0,2 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 1,6 **B)** 1,2 **C)** 1,7 **D)** 2,4

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,1	1	5,6°
2	0,3	3	16,7°
3	0,3	1	5,6°

O'quvchi 1- va 2-tajribalarni solishtirib: "Sharcha massasi oshsa, ipning og'ish burchagi ham ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asosli: 1 va 2 da massa ortishi bilan burchak ham ortgan.
B) Xulosa asoslanmagan: 1 va 2 da massa bilan birga tezlanish ham o'zgargan.
C) Xulosa asoslanmagan: 2 va 3 da massalar tengligi burchaklar tengligini talab qiladi.
D) Xulosa asosli: 1 va 3 dagi burchaklar farqi massa ta'sirini ko'rsatadi.

15. M Balandligi 60 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	60	65	63	45	20

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Bobur: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

• Malika: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

Qaysi baho ilmiy jihatdan eng asosli?

A) Malikaning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

B) Boburning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

C) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

D) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 5-variant

1 B C	2 B A	3 B B	4 Q B	5 Q C	6 Q D	7 Q C
8 Q B	9 Q C	10 Q B	11 Q C	12 Q D	13 Q B	14 M B
15 M B						

1. C) Kuch — vektor kattalik. Koordinata (holat) — sanoq boshiga nisbatan yo'nalishga ega. Harakat vaqti — skalyar kattalik. Javob: I va II.

2. A) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $3-4-6=-7$. Modul: 7 birlik.

3. B) 1 — Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi. (B). 2 — To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi. (A). 3 — Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin. (C).

4. B) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1=\frac{180}{\vartheta/2}=\frac{360}{\vartheta}$. LM da $t_2=\frac{80}{\vartheta}$. $t_1:t_2=4{,}5$.

5. C) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–5 s: $37{,}5$; 5–10 s: $62{,}5$; 10–25 s: 225. Ko'chish: 325 m.

6. D) Yo'l: $s=\frac{1}{2}\cdot 2\pi R=120$ m, $\vartheta_{o'rt}=\frac{120}{1}=120$ m/s. Ko'chish – vatar: $2R=80$ m, $|\vec{\vartheta}_{o'rt}|=80$ m/s.

7. C) $\vartheta=5-t$, jism $t=5$ s da to'xtab, orqaga harakatlanadi. Yo'l: $12{,}5+12{,}5=25$ m (koordinata esa 0 m).

8. B) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2=\vartheta_0^2+2gh$. $1600=900+20h\to h=35$ m.

9. C) $\vartheta_x=30$ m/s, $\vartheta_y=40$ m/s. Uchish vaqti $t=\frac{2\cdot 40}{10}=8$ s. $x=30\cdot 8=240$ m.

10. B) I: L tezlashmoqda — inersiya bo'yicha orqaga siljiydi — to'g'ri. II: $a=1$ m/s² — noto'g'ri. III: K da tezlanish yo'q, L da bor — to'g'ri. Javob: I va III.

11. C) Grafikdan: $m=\frac{F}{a}=\frac{8}{2}=4$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $10-5=5$ N. $a=\frac{5}{4}=1{,}25$ m/s².

12. D) I: kuchlar teng — noto'g'ri. II: $a=F/m$ — to'g'ri. III: III qonun — to'g'ri. Javob: II va III.

13. B) L dagi tezlik: $8-1\cdot 2=6$ m/s. Tushish vaqti: $t=\sqrt{\frac{2h}{g}}=0{,}2$ s. $x=6\cdot 0{,}2=1{,}2$ m.

14. B) 1- va 2-tajribada massa ham, tezlanish ham o'zgargan — burchak o'zgarishini massaga bog'lab bo'lmaydi. 1 va 3 da faqat massa o'zgargan va burchak xatolik doirasida bir xil ($\operatorname{tg}\alpha=a/g$).

15. B) Model: $h=60+10t-5t^2$: 60; 65; 60; 45; 20. Faqat $t=2$ s dagi qiymat (63 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Bobur).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 6-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Tezlanish;

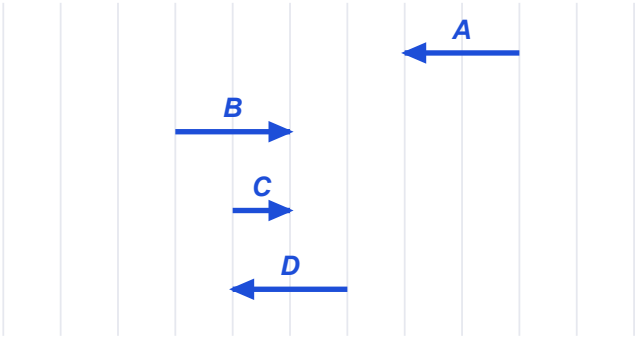
II. Jism massasi;

III. Kuch;

kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) II va III **B)** I va III **C)** I, II va III **D)** Faqat III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



A) 1 **B)** 3 **C)** 7 **D)** 5

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

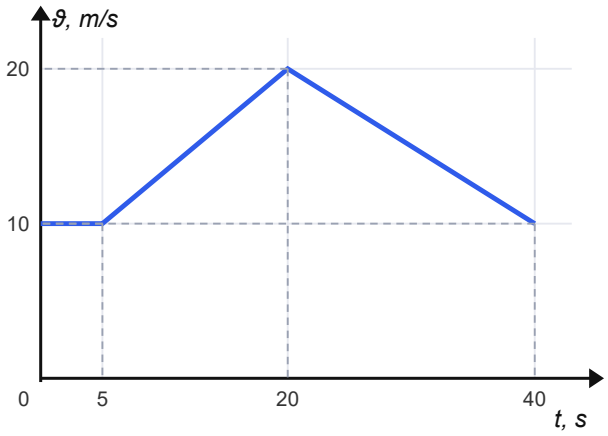
I ustun	II ustun
1. Kontakt kuchlari har doim itaruvchi kuch sifatida ta'sir qiladi.	A) To'g'ri, chunki maydon kuchlari faqat itarish tarzida ta'sir qiladi.
2. Magnit temir mixni masofadan turib tortadi.	B) To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi.
3. Jismga hech qanday kuch ta'sir qilmasa, u albatta tinch turadi.	C) Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin.
	D) Noto'g'ri, u to'g'ri chiziqli tekis harakatini davom ettirishi ham mumkin.

A) 1–C, 2–B, 3–D **B)** 1–B, 2–C, 3–A **C)** 1–D, 2–C, 3–A **D)** 1–C, 2–A, 3–D

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 40 m) tekis tezlanuvchan, L dan M gacha (LM = 160 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 1,5 **B)** 0,5 **C)** 1 **D)** 0,25

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 575 **B)** 625 **C)** 600 **D)** 525

6. Q Radiusi 20 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 1 s da aylananing chorak qismini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 30$ m/s; $|\vec{\vartheta}_{o'rt}| = 20\sqrt{2}$ m/s

B) $\vartheta_{o'rt} = 30$ m/s; $|\vec{\vartheta}_{o'rt}| = 40$ m/s

C) $\vartheta_{o'rt} = 30$ m/s; $|\vec{\vartheta}_{o'rt}| = 30$ m/s

D) $\vartheta_{o'rt} = 20\sqrt{2}$ m/s; $|\vec{\vartheta}_{o'rt}| = 20\sqrt{2}$ m/s

7. Q Jism $s = 3t - 0{,}5t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 10 s ichida o'tgan yo'lini toping (m). 2 ball

A) 80 **B)** 30 **C)** 29 **D)** 20

8. Jism h balandlikdan gorizontal 20 m/s boshlang'ich tezlik bilan otildi. Jism yerga 50 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 125 **B)** 105 **C)** 115 **D)** 45

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 37° burchak ostida 20 m/s boshlang'ich tezlik bilan otildi. Jismning gorizontal yo'nalishda bosib o'tgan masofasi x necha metr? ($\sin 37^\circ = 0{,}6$; $\cos 37^\circ = 0{,}8$; $g = 10$ m/s²) 2 ball

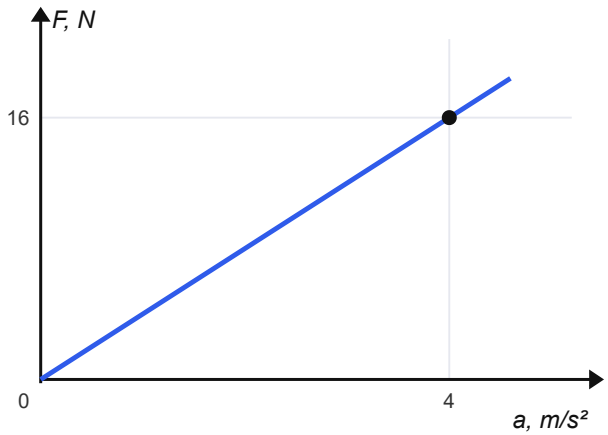
A) 76,8 **B)** 19,2 **C)** 38,4 **D)** 28,8

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 50 + 12t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 5 s davomida 20 m/s dan 4 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. L avtomobil tezlanishining moduli 6,4 m/s².
II. K avtomobilning tezlanishi 12 m/s².
III. L avtomobildagi yo'lovchilar oldinga tomon siljiydi.

A) I va III **B)** I va II **C)** Faqat III **D)** Faqat II

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 4$ m/s², $F = 16$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 50 N, chapga 20 N va vertikal yuqoriga 5 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 12,5 **B)** 7,5 **C)** 17,5 **D)** 8,75

12. Q Bolg'a mixga urilganda mix taxtaga kiradi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Bolg'a mixga mix bolg'aga qaraganda kattaroq kuch bilan ta'sir qiladi.
II. Bolg'aning to'xtashi mixning unga ta'siri bilan bog'liq.
III. Mix ham bolg'aga teng kuch bilan ta'sir qiladi.

A) I va III **B)** Faqat I **C)** Faqat III **D)** II va III

13. Q Jasur shayba stolning K nuqtasidan 6 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 0,5 m/s² tezlanish moduli bilan sekinlashib, 1 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 0,8 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 2,2 **B)** 2,4 **C)** 8,8 **D)** 4,4

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,3	2	11,2°
2	0,5	3	16,8°
3	0,5	2	11,3°

O'quvchi 1- va 2-tajribalarni solishtirib: "Sharcha massasi oshsa, ipning og'ish burchagi ham ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asosli: 1 va 2 da massa ortishi bilan burchak ham ortgan.

B) Xulosa asoslanmagan: 2 va 3 da massalar tengligi burchaklar tengligini talab qiladi.

C) Xulosa asoslanmagan: 1 va 2 da massa bilan birga tezlanish ham o'zgargan.

D) Xulosa asosli: 1 va 3 dagi burchaklar farqi massa ta'sirini ko'rsatadi.

15. M Balandligi 60 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 10 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	60	60	50	32	0

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Zarina: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

• Malika: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

Qaysi baho ilmiy jihatdan eng asosli?

A) Malikaning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

B) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

C) Zarinaning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

D) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 6-variant

1 B	B	2 B	A	3 B	A	4 Q	B	5 Q	A	6 Q	A	7 Q	C
8 Q	B	9 Q	C	10 Q	C	11 Q	B	12 Q	D	13 Q	A	14 M	C
15 M	A												

1. B) Tezlanish — vektor kattalik. Jism massasi — skalyar kattalik. Kuch — vektor kattalik. Javob: I va III.

2. A) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $-2+2+1-2=-1$. Modul: 1 birlik.

3. A) 1 — Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin. (C). 2 — To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi. (B). 3 — Noto'g'ri, u to'g'ri chiziqli tekis harakatini davom ettirishi ham mumkin. (D).

4. B) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1=\frac{40}{\vartheta/2}=\frac{80}{\vartheta}$. LM da $t_2=\frac{160}{\vartheta}$. $t_1:t_2=0{,}5$.

5. A) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–5 s: 50; 5–20 s: 225; 20–40 s: 300. Ko'chish: 575 m.

6. A) Yo'l: $s=\frac{1}{4}\cdot 2\pi R=30$ m, $\vartheta_{o'rt}=\frac{30}{1}=30$ m/s. Ko'chish – vatar: $R\sqrt{2}=20\sqrt{2}$ m, $|\vec{\vartheta}_{o'rt}|=20\sqrt{2}$ m/s.

7. C) $\vartheta=3-t$, jism $t=3$ s da to'xtab, orqaga harakatlanadi. Yo'l: $4{,}5+24{,}5=29$ m (koordinata esa -20 m).

8. B) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2=\vartheta_0^2+2gh$. $2500=400+20h\to h=105$ m.

9. C) $\vartheta_x=16$ m/s, $\vartheta_y=12$ m/s. Uchish vaqti $t=\frac{2\cdot 12}{10}=2{,}4$ s. $x=16\cdot 2{,}4=38{,}4$ m.

10. C) I: $a=3{,}2$ m/s² — noto'g'ri. II: 12 m/s — tezlik, tezlanish nolga teng — noto'g'ri. III: L tormozlanmoqda — inersiya bo'yicha oldinga siljiydi — to'g'ri. Javob: Faqat III.

11. B) Grafikdan: $m=\frac{F}{a}=\frac{16}{4}=4$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $50-20=30$ N. $a=\frac{30}{4}=7{,}5$ m/s².

12. D) I: kuchlar teng — noto'g'ri. II: aks ta'sir kuchi — to'g'ri. III: III qonun — to'g'ri. Javob: II va III.

13. A) L dagi tezlik: $6-0{,}5\cdot 1=5{,}5$ m/s. Tushish vaqti: $t=\sqrt{\frac{2h}{g}}=0{,}4$ s. $x=5{,}5\cdot 0{,}4=2{,}2$ m.

14. C) 1- va 2-tajribada massa ham, tezlanish ham o'zgargan — burchak o'zgarishini massaga bog'lab bo'lmaydi. 1 va 3 da faqat massa o'zgargan va burchak xatolik doirasida bir xil ($\operatorname{tg}\alpha=a/g$).

15. A) Model: $h=60+5t-5t^2$: 60; 60; 50; 30; 0. Faqat $t=3$ s dagi qiymat (32 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Malika).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 7-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

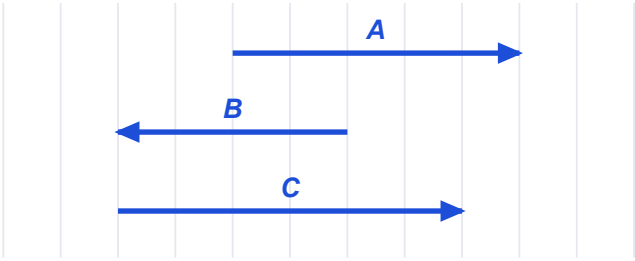
Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Koordinata (holat);
II. Tezlanish;
III. Jism massasi;
kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) II va III **B)** Faqat III **C)** I va II **D)** Faqat II

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



1 katak = 1 birlik

A) 9 **B)** 5 **C)** 7 **D)** 15

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

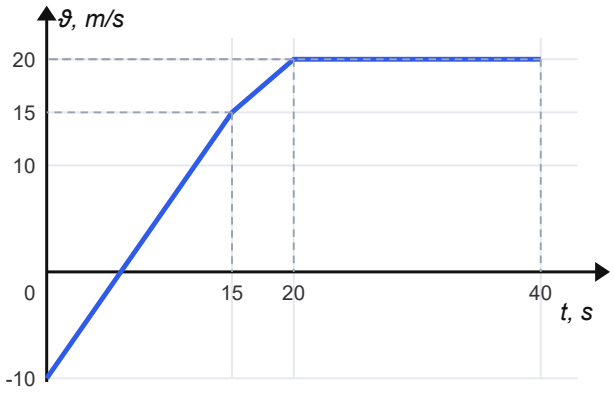
I ustun	II ustun
1. Kuch jismning tezligini ham, shaklini ham o'zgartirishi mumkin.	A) To'g'ri, elektr va magnit kuchlar tortish yoki itarish tarzida ta'sir qiladi.
2. Maydon kuchlari tortish yoki itarish tarzida ta'sir qiladi.	B) To'g'ri, kuch tezlanish beradi yoki deformatsiya hosil qiladi.
3. Jismni harakatga keltiruvchi kuch jism bilan bevosita kontaktda bo'lishi shart.	C) To'g'ri, chunki har qanday kuch faqat bevosita kontakt orqali uzatiladi.
	D) Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi.

A) 1–B, 2–D, 3–C **B)** 1–D, 2–B, 3–C **C)** 1–C, 2–B, 3–A **D)** 1–B, 2–A, 3–D

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 150 m) tekis tezlanuvchan, L dan M gacha (LM = 120 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 2,5 **B)** 5 **C)** 1,25 **D)** 3,5

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning bosib o'tgan yo'li necha metrga teng? 2 ball



A) 585 **B)** 535 **C)** 635 **D)** 525

6. Q Radiusi 20 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 1 s da aylananing yarmini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 60$ m/s; $|\vec{\vartheta}_{o'rt}| = 20\sqrt{2}$ m/s **B)** $\vartheta_{o'rt} = 60$ m/s; $|\vec{\vartheta}_{o'rt}| = 40$ m/s

C) $\vartheta_{o'rt} = 40$ m/s; $|\vec{\vartheta}_{o'rt}| = 40$ m/s **D)** $\vartheta_{o'rt} = 60$ m/s; $|\vec{\vartheta}_{o'rt}| = 60$ m/s

7. Q Jism $s = 3t - 2t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 6 s ichida o'tgan yo'lini toping (m). 2 ball

A) 56,25 **B)** 90 **C)** 54 **D)** 18

8. Q Jism h balandlikdan vertikal yuqoriga 20 m/s boshlang'ich tezlik bilan otildi. Jism yerga 40 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 70 **B)** 60 **C)** 20 **D)** 80

9. Q Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 53° burchak ostida 10 m/s boshlang'ich tezlik bilan otildi. Jismning gorizontal yo'nalishda bosib o'tgan masofasi x necha metr? ($\sin 53^\circ = 0{,}8$; $\cos 53^\circ = 0{,}6$; $g = 10$ m/s²) 2 ball

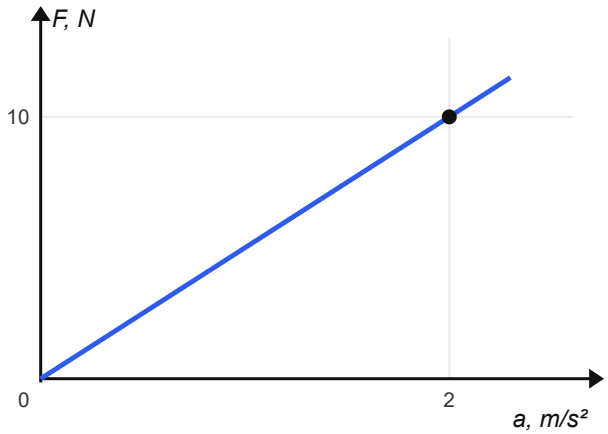
A) 19,2 **B)** 9,6 **C)** 12,8 **D)** 4,8

10. Q To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 10 + 20t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 4 s davomida 24 m/s dan 4 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. L avtomobil tezlanishining moduli 10 m/s².
II. K avtomobilning tezlanishi 20 m/s².
III. K ga ta'sir etuvchi natijaviy kuch nolga teng, L ga esa noldan farqli.

A) Faqat III **B)** Faqat I **C)** I va III **D)** II va III

11. Q Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 2$ m/s², $F = 10$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 50 N, chapga 5 N va vertikal yuqoriga 5 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 10 **B)** 11 **C)** 9 **D)** 18

12. Q To'pdan o'q uzilganda to'p orqaga tepiladi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Ta'sir kuchi snaryadga, aks ta'sir kuchi esa to'pga qo'yilgan.
II. Ta'sir va aks ta'sir kuchlari teng va qarama-qarshi bo'lgani uchun snaryadga ta'sir qiluvchi natijaviy kuch nolga teng.
III. Har qanday ta'sirga miqdor jihatidan teng va qarama-qarshi yo'nalgan aks ta'sir mavjud.

A) I, II va III **B)** Faqat III **C)** I va III **D)** I va II

13. Q Jasur kichik kubikni stolning K nuqtasidan 4 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 2 m/s² tezlanish moduli bilan sekinlashib, 1 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 0,45 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 0,6 **B)** 1,8 **C)** 1,2 **D)** 1,1

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,1	3	16,8°
2	0,2	5	26,7°
3	0,2	3	16,6°

O'quvchi 2- va 3-tajribalarni solishtirib: "Aravacha tezlanishi oshsa, ipning og'ish burchagi ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asoslanmagan: burchak farqi o'lchash xatoligi doirasida.

B) Xulosa asoslanmagan: 2 va 3 da massa ham o'zgargan.

C) Xulosa asosli: 1 va 2 da massa o'zgarmagan.

D) Xulosa asosli: 2 va 3 da faqat tezlanish o'zgargan, burchak farqi xatolikdan katta.

15. M 3 ball

Balandligi 45 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m.

Vaqt, s	0	1	2	3	4
Balandlik, m	45	50	47	30	5

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Bobur: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

• Sardor: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

Qaysi baho ilmiy jihatdan eng asosli?

A) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

B) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

C) Sardorning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

D) Boburning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 7-variant

1 B C	2 B C	3 B D	4 Q A	5 Q A	6 Q B	7 Q A
8 Q B	9 Q B	10 Q A	11 Q C	12 Q C	13 Q A	14 M D
15 M C						

1. C) Koordinata (holat) — sanoq boshiga nisbatan yo'nalishga ega. Tezlanish — vektor kattalik. Jism massasi — skalyar kattalik. Javob: I va II.

2. C) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $5 - 4 + 6 = 7$. Modul: 7 birlik.

3. D) 1 — To'g'ri, kuch tezlanish beradi yoki deformatsiya hosil qiladi. (B). 2 — To'g'ri, elektr va magnit kuchlar tortish yoki itarish tarzida ta'sir qiladi. (A). 3 — Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi. (D).

4. A) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1 = \frac{150}{\vartheta/2} = \frac{300}{\vartheta}$. LM da $t_2 = \frac{120}{\vartheta}$. $t_1 : t_2 = 2{,}5$.

5. A) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–15 s: $37{,}5$; 15–20 s: $87{,}5$; 20–40 s: 400. Ko'chish: 525 m. Yo'l esa yuzalar modullari yig'indisi: 585 m.

6. B) Yo'l: $s = \frac{1}{2} \cdot 2\pi R = 60$ m, $\vartheta_{o'rt} = \frac{60}{1} = 60$ m/s. Ko'chish — vatar: $2R = 40$ m, $|\vec{\vartheta}_{o'rt}| = 40$ m/s.

7. A) $\vartheta = 3 - 4t$, jism $t = 0{,}75$ s da to'xtab, orqaga harakatlanadi. Yo'l: $1{,}125 + 55{,}125 = 56{,}25$ m (koordinata esa -54 m).

8. B) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2 = \vartheta_0^2 + 2gh$. $1600 = 400 + 20h \to h = 60$ m.

9. B) $\vartheta_x = 6$ m/s, $\vartheta_y = 8$ m/s. Uchish vaqti $t = \frac{2 \cdot 8}{10} = 1{,}6$ s. $x = 6 \cdot 1{,}6 = 9{,}6$ m.

10. A) I: $a = 5$ m/s² — noto'g'ri. II: 20 m/s — tezlik, tezlanish nolga teng — noto'g'ri. III: K da tezlanish yo'q, L da bor — to'g'ri. Javob: Faqat III.

11. C) Grafikdan: $m = \frac{F}{a} = \frac{10}{2} = 5$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $50 - 5 = 45$ N. $a = \frac{45}{5} = 9$ m/s².

12. C) I: kuchlar turli jismlarga qo'yilgan — to'g'ri. II: kuchlar turli jismlarga qo'yilgan — noto'g'ri. III: Nyutonning III qonuni — to'g'ri. Javob: I va III.

13. A) L dagi tezlik: $4 - 2 \cdot 1 = 2$ m/s. Tushish vaqti: $t = \sqrt{\frac{2h}{g}} = 0{,}3$ s. $x = 2 \cdot 0{,}3 = 0{,}6$ m.

14. D) 2 va 3 da massa bir xil (0,2 kg), faqat tezlanish farq qiladi; burchaklar farqi ±0,3° dan ancha katta ($\operatorname{tg}\alpha = a/g$).

15. C) Model: $h = 45 + 10t - 5t^2$: 45; 50; 45; 30; 5. Faqat $t = 2$ s dagi qiymat (47 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Sardor).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 8-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Ko'chish;

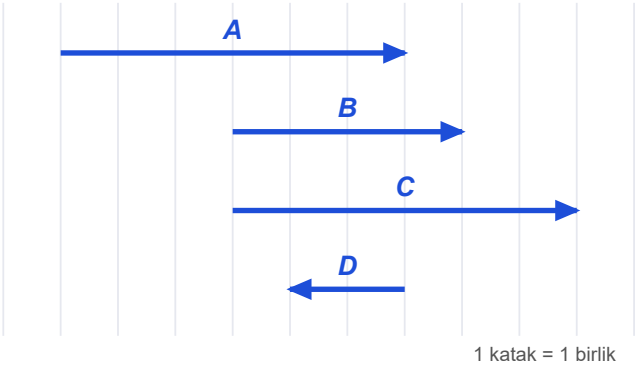
II. Jism massasi;

III. Harakat vaqti;

kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) II va III **B)** I, II va III **C)** Faqat I **D)** Faqat III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



A) 14 **B)** 18 **C)** 16 **D)** 12

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

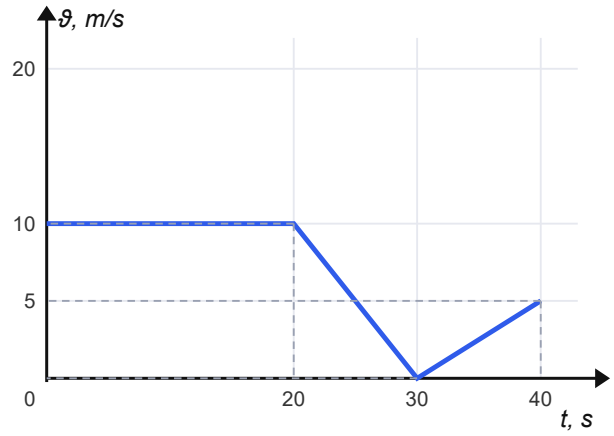
I ustun	II ustun
1. Ishqalanish kuchi kontakt kuchiga misol bo'ladi.	A) Noto'g'ri, chunki kuch faqat jism harakatlanayotganda ta'sir qiladi.
2. Magnit temir mixni masofadan turib tortadi.	B) Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin.
3. Kontakt kuchlari har doim itaruvchi kuch sifatida ta'sir qiladi.	C) To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi.
	D) To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi.

A) 1–C, 2–A, 3–B **B)** 1–C, 2–B, 3–A **C)** 1–C, 2–D, 3–B **D)** 1–A, 2–C, 3–B

4. Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 120 m) tekis tezlanuvchan, L dan M gacha (LM = 40 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 3 **B)** 7 **C)** 12 **D)** 6

5. To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 325 **B)** 225 **C)** 300 **D)** 275

6. Radiusi 20 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 5 s da aylananing yarmini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 12$ m/s; $|\vec{\vartheta}_{o'rt}| = 4\sqrt{2}$ m/s **B)** $\vartheta_{o'rt} = 12$ m/s; $|\vec{\vartheta}_{o'rt}| = 12$ m/s

C) $\vartheta_{o'rt} = 8$ m/s; $|\vec{\vartheta}_{o'rt}| = 8$ m/s **D)** $\vartheta_{o'rt} = 12$ m/s; $|\vec{\vartheta}_{o'rt}| = 8$ m/s

7. Jism $s = 5t - 0{,}5t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 2 s ichida o'tgan yo'lini toping (m). 2 ball

A) 10 **B)** 12 **C)** 8 **D)** 16

8. Jism h balandlikdan gorizontal 20 m/s boshlang'ich tezlik bilan otildi. Jism yerga 90 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 245 **B)** 405 **C)** 395 **D)** 385

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 37° burchak ostida 30 m/s boshlang'ich tezlik bilan otildi. Jismning ko'tarilishining maksimal balandligi H necha metr? ($\sin 37^\circ = 0{,}6$; $\cos 37^\circ = 0{,}8$; $g = 10$ m/s²) 2 ball

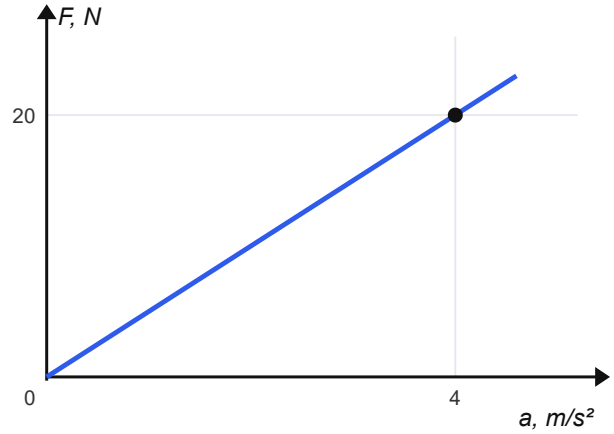
A) 45 **B)** 16,2 **C)** 32,4 **D)** 28,8

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 20t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 5 s davomida 24 m/s dan 6 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. L avtomobil tezlanishining moduli 3,6 m/s².
II. K avtomobildagi yo'lovchilarning harakat holati o'zgarmaydi.
III. L avtomobildagi yo'lovchilar oldinga tomon siljiydi.

A) I, II va III **B)** Faqat III **C)** Faqat II **D)** Faqat I

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 4$ m/s², $F = 20$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 10 N, chapga 5 N va vertikal yuqoriga 15 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 4 **B)** 1 **C)** 3 **D)** 2

12. Suzuvchi hovuz devoridan oyog'i bilan itarilib, oldinga suzib ketdi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Devor suzuvchiga teng va qarama-qarshi kuch bilan ta'sir qiladi.
II. Devor qo'zg'almagani uchun unga hech qanday kuch ta'sir qilmagan.
III. Suzuvchini oldinga harakatlantirgan kuch devor tomonidan unga qo'yilgan.

A) II va III **B)** Faqat III **C)** I va III **D)** Faqat II

13. Q Kamola kichik kubikni stolning K nuqtasidan 10 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 1 m/s² tezlanish moduli bilan sekinlashib, 1 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 1,25 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 4,5 **B)** 22,5 **C)** 9 **D)** 5

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,2	3	16,6°
2	0,3	6	30,9°
3	0,3	3	16,8°

O'quvchi 1- va 2-tajribalarni solishtirib: "Sharcha massasi oshsa, ipning og'ish burchagi ham ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asoslanmagan: 1 va 2 da massa bilan birga tezlanish ham o'zgargan.

B) Xulosa asosli: 1 va 3 dagi burchaklar farqi massa ta'sirini ko'rsatadi.

C) Xulosa asoslanmagan: 2 va 3 da massalar tengligi burchaklar tengligini talab qiladi.

D) Xulosa asosli: 1 va 2 da massa ortishi bilan burchak ham ortgan.

15. M Balandligi 45 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	45	48	45	30	5

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Aziz: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

• Zarina: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

Qaysi baho ilmiy jihatdan eng asosli?

A) Azizning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

B) Zarinaning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

C) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

D) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 8-variant

1 B C	2 B A	3 B C	4 Q D	5 Q D	6 Q D	7 Q C
8 Q D	9 Q B	10 Q A	11 Q B	12 Q C	13 Q A	14 M A
15 M A						

1. C) Ko'chish — vektor kattalik. Jism massasi — skalyar kattalik. Harakat vaqti — skalyar kattalik. Javob: Faqat I.

2. A) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $6+4+6-2=14$. Modul: 14 birlik.

3. C) 1 — To'g'ri, ishqalanish sirtlar bir-biriga tegib turganda yuzaga keladi. (C). 2 — To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi. (D). 3 — Noto'g'ri, chunki kontakt kuchi (masalan, ipning taranglik kuchi) jismni tortishi ham mumkin. (B).

4. D) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1=\frac{120}{\vartheta/2}=\frac{240}{\vartheta}$. LM da $t_2=\frac{40}{\vartheta}$. $t_1:t_2=6$.

5. D) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–20 s: 200; 20–30 s: 50; 30–40 s: 25. Ko'chish: 275 m.

6. D) Yo'l: $s=\frac{1}{2}\cdot 2\pi R=60$ m, $\vartheta_{o'rt}=\frac{60}{5}=12$ m/s. Ko'chish — vatar: $2R=40$ m, $|\vec{\vartheta}_{o'rt}|=8$ m/s.

7. C) $\vartheta=5-t$, jism 5 s da to'xtaydi, demak 2 s ichida yo'nalish o'zgarmaydi. $s(2)=8$ m.

8. D) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2=\vartheta_0^2+2gh$. $8100=400+20h\to h=385$ m.

9. B) $\vartheta_x=24$ m/s, $\vartheta_y=18$ m/s. Uchish vaqti $t=\frac{2\cdot 18}{10}=3{,}6$ s. $H=\frac{\vartheta_y^2}{2g}=16{,}2$ m.

10. A) I: $a=\frac{18}{5}=3{,}6$ m/s² — to'g'ri. II: K tekis harakatlanadi — to'g'ri. III: L tormozlanmoqda — inersiya bo'yicha oldinga siljiydi — to'g'ri. Javob: I, II va III.

11. B) Grafikdan: $m=\frac{F}{a}=\frac{20}{4}=5$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $10-5=5$ N. $a=\frac{5}{5}=1$ m/s².

12. C) I: III qonun — to'g'ri. II: devorga ham kuch ta'sir qiladi — noto'g'ri. III: aks ta'sir kuchi — to'g'ri. Javob: I va III.

13. A) L dagi tezlik: $10-1\cdot 1=9$ m/s. Tushish vaqti: $t=\sqrt{\frac{2h}{g}}=0{,}5$ s. $x=9\cdot 0{,}5=4{,}5$ m.

14. A) 1- va 2-tajribada massa ham, tezlanish ham o'zgargan — burchak o'zgarishini massaga bog'lab bo'lmaydi. 1 va 3 da faqat massa o'zgargan va burchak xatolik doirasida bir xil ($\operatorname{tg}\alpha=a/g$).

15. A) Model: $h=45+10t-5t^2$: 45; 50; 45; 30; 5. Faqat $t=1$ s dagi qiymat (48 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Aziz).

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 9-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

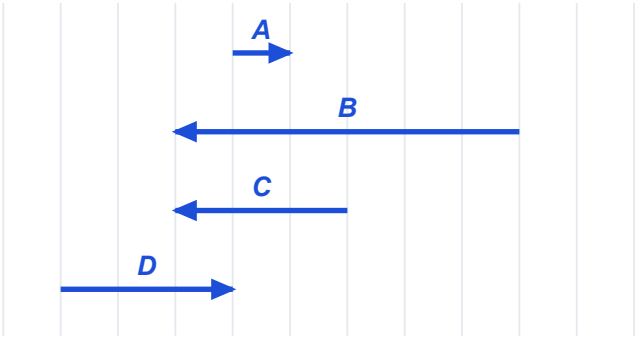
Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Bosib o'tilgan yo'l;
II. Ko'chish;
III. Jism massasi;
kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) I va II **B)** Faqat II **C)** I va III **D)** II va III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



A) 7 **B)** 3 **C)** 5 **D)** 13

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

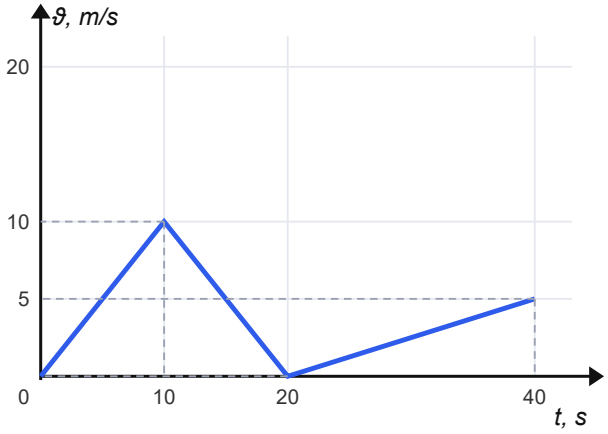
I ustun	II ustun
1. Jismga hech qanday kuch ta'sir qilmasa, u albatta tinch turadi.	A) Noto'g'ri, u to'g'ri chiziqli tekis harakatini davom ettirishi ham mumkin.
2. Og'irlik kuchi faqat jism yerga tegib turganda ta'sir qiladi.	B) Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi.
3. Jismni harakatga keltiruvchi kuch jism bilan bevosita kontaktda bo'lishi shart.	C) To'g'ri, chunki maydon kuchlari faqat itarish tarzida ta'sir qiladi.
	D) Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi.

A) 1–C, 2–A, 3–D **B)** 1–A, 2–D, 3–B **C)** 1–B, 2–D, 3–C **D)** 1–A, 2–B, 3–D

4. Q Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 100 m) tekis tezlanuvchan, L dan M gacha (LM = 100 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 4 **B)** 3 **C)** 2 **D)** 1

5. Q To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 175 **B)** 200 **C)** 150 **D)** 100

6. Q Radiusi 30 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 5 s da aylananing $\frac{3}{4}$ qismini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 27$ m/s; $|\vec{\vartheta}_{o'rt}| = 27$ m/s

B) $\vartheta_{o'rt} = 27$ m/s; $|\vec{\vartheta}_{o'rt}| = 12$ m/s

C) $\vartheta_{o'rt} = 6\sqrt{2}$ m/s; $|\vec{\vartheta}_{o'rt}| = 6\sqrt{2}$ m/s

D) $\vartheta_{o'rt} = 27$ m/s; $|\vec{\vartheta}_{o'rt}| = 6\sqrt{2}$ m/s

7. Q Jism $s = 8t - 0{,}1t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 10 s ichida o'tgan yo'lini toping (m). 2 ball

A) 72 **B)** 90 **C)** 80 **D)** 70

8. Jism h balandlikdan gorizontal 10 m/s boshlang'ich tezlik bilan otildi. Jism yerga 80 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 315 **B)** 320 **C)** 245 **D)** 325

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 37° burchak ostida 20 m/s boshlang'ich tezlik bilan otildi. Jismning ko'tarilishining maksimal balandligi H necha metr? ($\sin 37^\circ = 0{,}6$; $\cos 37^\circ = 0{,}8$; $g = 10$ m/s²) 2 ball

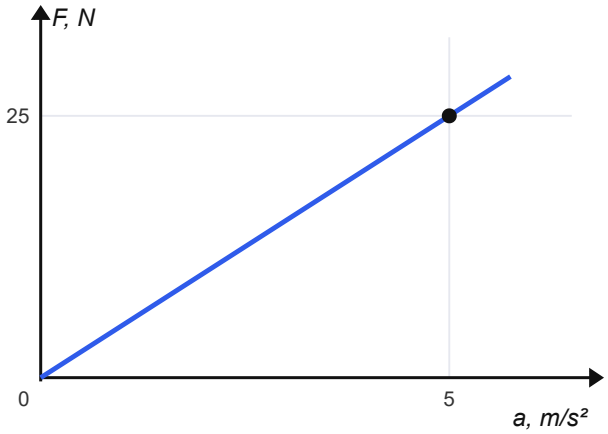
A) 7,2 **B)** 14,4 **C)** 12,8 **D)** 20

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 50 + 15t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 5 s davomida 10 m/s dan 16 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. L avtomobildagi yo'lovchilar orqaga tomon siljiydi.
II. K ga ta'sir etuvchi natijaviy kuch nolga teng, L ga esa noldan farqli.
III. L avtomobil tezlanishining moduli 1,2 m/s².

A) I va II **B)** I va III **C)** II va III **D)** I, II va III

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 5$ m/s², $F = 25$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 50 N, chapga 15 N va vertikal yuqoriga 10 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 7 **B)** 10 **C)** 13 **D)** 9

12. Q Ikki konkichi muz ustida bir-birini itardi. A ning massasi B nikidan ikki marta katta. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. B ning tezlanishi A nikidan ikki marta katta.
II. Ular bir-biriga teng kuch bilan ta'sir qiladi.
III. A kuchliroq bo'lgani uchun B ga kattaroq kuch bilan ta'sir qiladi.

A) Faqat III **B)** II va III **C)** Faqat II **D)** I va II

13. Q Nodira kichik kubikni stolning K nuqtasidan 5 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 2 m/s² tezlanish moduli bilan sekinlashib, 2 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 0,2 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 1 **B)** 0,2 **C)** 0,4 **D)** 0,7

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,2	2	11,4°
2	0,4	5	26,6°
3	0,4	2	11,3°

O'quvchi 1- va 2-tajribalarni solishtirib: "Sharcha massasi oshsa, ipning og'ish burchagi ham ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asoslanmagan: 1 va 2 da massa bilan birga tezlanish ham o'zgargan.
B) Xulosa asoslanmagan: 2 va 3 da massalar tengligi burchaklar tengligini talab qiladi.
C) Xulosa asosli: 1 va 3 dagi burchaklar farqi massa ta'sirini ko'rsatadi.
D) Xulosa asosli: 1 va 2 da massa ortishi bilan burchak ham ortgan.

15. M Balandligi 60 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	60	65	60	48	20

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Aziz: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

• Malika: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

Qaysi baho ilmiy jihatdan eng asosli?

A) Azizning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

B) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

C) Malikaning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

D) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 9-variant

1 B	B	2 B	C	3 B	B	4 Q C	5 Q C	6 Q D	7 Q D
8 Q	A	9 Q	A	10 Q	D	11 Q A	12 Q D	13 Q B	14 M A
15 M	C								

1. B) Bosib o'tilgan yo'l — skalyar kattalik. Ko'chish — vektor kattalik. Jism massasi — skalyar kattalik. Javob: Faqat II.

2. C) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $1 - 6 - 3 + 3 = -5$. Modul: 5 birlik.

3. B) 1 — Noto'g'ri, u to'g'ri chiziqli tekis harakatini davom ettirishi ham mumkin. (A). 2 — Noto'g'ri, og'irlik kuchi maydon kuchi bo'lib, uchayotgan jismga ham ta'sir qiladi. (D). 3 — Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi. (B).

4. C) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1 = \frac{100}{\vartheta/2} = \frac{200}{\vartheta}$. LM da $t_2 = \frac{100}{\vartheta}$. $t_1 : t_2 = 2$.

5. C) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–10 s: 50; 10–20 s: 50; 20–40 s: 50. Ko'chish: 150 m.

6. D) Yo'l: $s = \frac{3}{4} \cdot 2\pi R = 135$ m, $\vartheta_{o'rt} = \frac{135}{5} = 27$ m/s. Ko'chish – vatar: $R\sqrt{2} = 30\sqrt{2}$ m, $|\vec{\vartheta}_{o'rt}| = 6\sqrt{2}$ m/s.

7. D) $\vartheta = 8 - 0{,}2t$, jism 40 s da to'xtaydi, demak 10 s ichida yo'nalish o'zgarmaydi. $s(10) = 70$ m.

8. A) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2 = \vartheta_0^2 + 2gh$. $6400 = 100 + 20h \to h = 315$ m.

9. A) $\vartheta_x = 16$ m/s, $\vartheta_y = 12$ m/s. Uchish vaqti $t = \frac{2 \cdot 12}{10} = 2{,}4$ s. $H = \frac{\vartheta_y^2}{2g} = 7{,}2$ m.

10. D) I: L tezlashmoqda — inersiya bo'yicha orqaga siljiydi — to'g'ri. II: K da tezlanish yo'q, L da bor — to'g'ri. III: $a = \frac{6}{5} = 1{,}2$ m/s² — to'g'ri. Javob: I, II va III.

11. A) Grafikdan: $m = \frac{F}{a} = \frac{25}{5} = 5$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $50 - 15 = 35$ N. $a = \frac{35}{5} = 7$ m/s².

12. D) I: massa ikki marta kichik — to'g'ri. II: III qonun — to'g'ri. III: kuchlar har doim teng — noto'g'ri. Javob: I va II.

13. B) L dagi tezlik: $5 - 2 \cdot 2 = 1$ m/s. Tushish vaqti: $t = \sqrt{\frac{2h}{g}} = 0{,}2$ s. $x = 1 \cdot 0{,}2 = 0{,}2$ m.

14. A) 1- va 2-tajribada massa ham, tezlanish ham o'zgargan — burchak o'zgarishini massaga bog'lab bo'lmaydi. 1 va 3 da faqat massa o'zgargan va burchak xatolik doirasida bir xil ($\operatorname{tg}\alpha = a/g$).

15. C) Model: $h = 60 + 10t - 5t^2$: 60; 65; 60; 45; 20. Faqat $t = 3$ s dagi qiymat (48 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Malika).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · FIZIKA · CHSB-1

ChSB-1 tayyorgarlik: 10-variant

15 ta savol · 40 daqiqa · 29 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B To'g'ri chiziq bo'ylab harakatlanayotgan jismning harakatini tavsiflashda: 1 ball

I. Jism massasi;

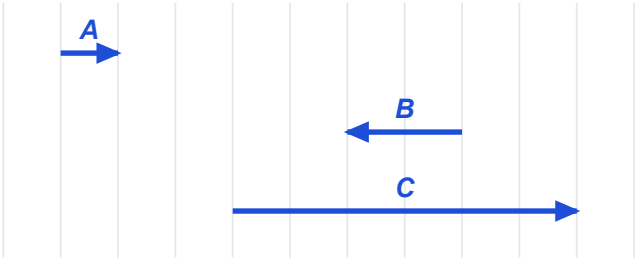
II. Tezlanish;

III. Tezlik;

kattaliklaridan qaysilarini ifodalash uchun yo'nalish ko'rsatilishi kerak?

A) Faqat II **B)** I va II **C)** I, II va III **D)** II va III

2. B Rasmda bir to'g'ri chiziq bo'ylab yo'nalgan vektorlar tasvirlangan. Ularning teng ta'sir etuvchi vektori modulini aniqlang. 1 ball



1 katak = 1 birlik

A) 3 **B)** 7 **C)** 5 **D)** 9

3. B Kuchlar haqidagi fikrlarni ularga mos izohlar bilan moslashtiring. 1 ball

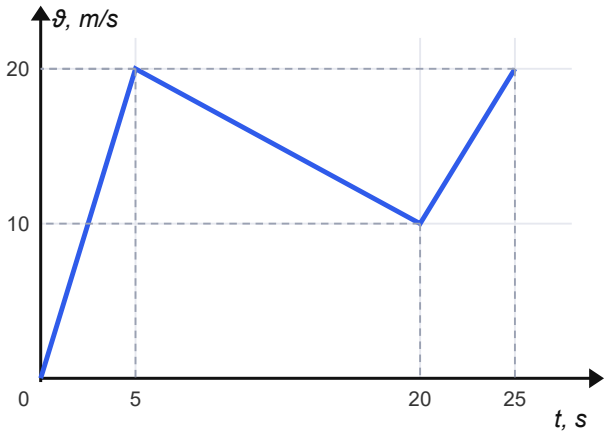
I ustun	II ustun
1. Magnit temir mixni masofadan turib tortadi.	A) To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi.
2. Jismni harakatga keltiruvchi kuch jism bilan bevosita kontaktda bo'lishi shart.	B) To'g'ri, elektr va magnit kuchlar tortish yoki itarish tarzida ta'sir qiladi.
3. Maydon kuchlari tortish yoki itarish tarzida ta'sir qiladi.	C) To'g'ri, chunki har qanday kuch faqat bevosita kontakt orqali uzatiladi.
	D) Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi.

A) 1–B, 2–D, 3–C **B)** 1–A, 2–D, 3–B **C)** 1–B, 2–C, 3–A **D)** 1–A, 2–C, 3–D

4. Avtomobil to'g'ri chiziqli yo'lning K nuqtasida tinch holatda turibdi. U K dan L gacha (KL = 120 m) tekis tezlanuvchan, L dan M gacha (LM = 120 m) L nuqtadagi tezligi bilan tekis harakat qiladi. KL masofani bosib o'tish vaqti LM masofani bosib o'tish vaqtidan necha marta katta? 2 ball

A) 4 **B)** 3 **C)** 2 **D)** 1

5. To'g'ri chiziqli yo'lda harakatlanayotgan avtomobilning tezlik–vaqt grafigi rasmda ko'rsatilgan. Grafikka ko'ra avtomobilning ko'chishi necha metrga teng? 2 ball



A) 375 **B)** 350 **C)** 400 **D)** 300

6. Radiusi 30 m bo'lgan aylana shaklidagi yo'lda harakatlanayotgan avtomobil K nuqtadan chiqib, 3 s da aylananing yarmini bosib o'tdi va L nuqtaga yetdi. Avtomobilning K va L orasidagi o'rtacha tezlik moduli $\vartheta_{o'rt}$ hamda o'rtacha tezlik vektorining moduli $|\vec{\vartheta}_{o'rt}|$ necha m/s? $\pi = 3$ deb oling. 2 ball

A) $\vartheta_{o'rt} = 30$ m/s; $|\vec{\vartheta}_{o'rt}| = 30$ m/s
B) $\vartheta_{o'rt} = 30$ m/s; $|\vec{\vartheta}_{o'rt}| = 10\sqrt{2}$ m/s
C) $\vartheta_{o'rt} = 30$ m/s; $|\vec{\vartheta}_{o'rt}| = 20$ m/s
D) $\vartheta_{o'rt} = 20$ m/s; $|\vec{\vartheta}_{o'rt}| = 20$ m/s

7. Jism $s = 8t - 2t^2$ qonuniyat bilan harakatlanadi (SI birliklarida). Uning dastlabki 8 s ichida o'tgan yo'lini toping (m). 2 ball

A) 80 **B)** 82 **C)** 192 **D)** 64

8. Jism h balandlikdan gorizontal 10 m/s boshlang'ich tezlik bilan otildi. Jism yerga 60 m/s tezlik bilan urildi. Havo qarshiligi hisobga olinmasa, h balandlik necha metrga teng? ($g = 10$ m/s²) 2 ball

A) 125 **B)** 175 **C)** 180 **D)** 185

9. Havo qarshiligi hisobga olinmaydigan sharoitda jism gorizontga 53° burchak ostida 20 m/s boshlang'ich tezlik bilan otildi. Jismning gorizontal yo'nalishda bosib o'tgan masofasi x necha metr? ($\sin 53^\circ = 0{,}8$; $\cos 53^\circ = 0{,}6$; $g = 10$ m/s²) 2 ball

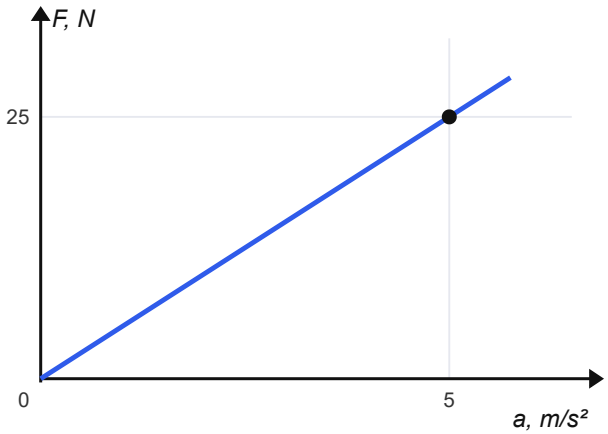
A) 51,2 **B)** 38,4 **C)** 76,8 **D)** 19,2

10. To'g'ri chiziqli yo'lda K avtomobilning koordinatasi $x = 10 + 15t$ (m) qonun bo'yicha o'zgaradi, L avtomobilning tezligi esa 4 s davomida 10 m/s dan 28 m/s gacha bir tekis o'zgardi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. K avtomobilning tezlanishi 15 m/s².
II. L avtomobildagi yo'lovchilar oldinga tomon siljiydi.
III. K ga ta'sir etuvchi natijaviy kuch nolga teng, L ga esa noldan farqli.

A) Faqat I **B)** I va III **C)** I, II va III **D)** Faqat III

11. Ishqalanishsiz gorizontal tekislikda turgan jismga ta'sir etuvchi gorizontal natijaviy kuchning tezlanishga bog'liqlik grafigi rasmda ko'rsatilgan (grafik $a = 5$ m/s², $F = 25$ N nuqtadan o'tadi). Endi jismga gorizontal o'ngga 30 N, chapga 5 N va vertikal yuqoriga 15 N kuch ta'sir ettirildi (jism tekislikdan uzilmaydi). Uning tezlanishi necha m/s² bo'ladi? 2 ball



A) 8 **B)** 7 **C)** 6 **D)** 5

12. Bolg'a mixga urilganda mix taxtaga kiradi. Quyidagi fikrlardan qaysilari to'g'ri? 2 ball

I. Mix ham bolg'aga teng kuch bilan ta'sir qiladi.
II. Bolg'a mixga mix bolg'aga qaraganda kattaroq kuch bilan ta'sir qiladi.
III. Bolg'aning to'xtashi mixning unga ta'siri bilan bog'liq.

A) I va II **B)** II va III **C)** I, II va III **D)** I va III

13. Q Nodira metall tangani stolning K nuqtasidan 12 m/s boshlang'ich tezlik bilan sirpantirib yubordi. U 3 m/s² tezlanish moduli bilan sekinlashib, 1 s dan so'ng stolning L chetidan gorizontal uchib chiqdi va stol chetidan gorizontal yo'nalishda x masofadagi O nuqtaga tushdi. Stolning balandligi 1,25 m bo'lsa, x necha metr? $g = 10$ m/s². Havo qarshiligini hisobga olmang. 2 ball

A) 9 **B)** 6 **C)** 4,5 **D)** 22,5

14. M Gorizontal yo'lda tezlanayotgan aravachaga ip orqali sharcha osildi. Uchta tajribada sharcha massasi, aravacha tezlanishi va ipning vertikaldan og'ish burchagi qayd etildi. Ip uzunligi bir xil, $g = 10$ m/s², sharcha aravachaga nisbatan tinch. Burchakni o'lchash xatoligi ±0,3°. 3 ball

Tajriba	Massa, kg	Tezlanish, m/s²	Og'ish burchagi
1	0,1	1	5,7°
2	0,3	4	21,8°
3	0,3	1	5,6°

O'quvchi 1- va 2-tajribalarni solishtirib: "Sharcha massasi oshsa, ipning og'ish burchagi ham ortadi", degan xulosaga keldi. Jadval asosida qaysi baho eng asosli?

A) Xulosa asoslanmagan: 1 va 2 da massa bilan birga tezlanish ham o'zgargan.

B) Xulosa asoslanmagan: 2 va 3 da massalar tengligi burchaklar tengligini talab qiladi.

C) Xulosa asosli: 1 va 3 dagi burchaklar farqi massa ta'sirini ko'rsatadi.

D) Xulosa asosli: 1 va 2 da massa ortishi bilan burchak ham ortgan.

15. M Balandligi 40 m bo'lgan qoya tepasidan tosh gorizontga 30° burchak ostida yuqoriga 20 m/s tezlik bilan otildi. Videoyozuvdan toshning balandligi aniqlandi; balandlikni o'lchash xatoligi ±0,2 m. 3 ball

Vaqt, s	0	1	2	3	4
Balandlik, m	40	45	40	28	0

Havo qarshiligi hisobga olinmaydigan model qabul qilingan ($g = 10$ m/s²; $\sin 30^\circ = 0{,}5$). Ikki o'quvchi fikr bildirdi:

• Nilufar: "Avval modelga mos kelmagan nuqtadagi o'lchovni qayta tekshirish kerak."

• Sardor: "Bitta qiymat modelga mos kelmadi. Demak, butun harakat davomida havo qarshiligi katta bo'lgan."

Qaysi baho ilmiy jihatdan eng asosli?

A) Nilufarning fikri asosli: bitta nomos qiymat uning sababini aniqlash uchun yetarli emas.

B) Sardorning fikri asosli: bitta nomos qiymat havo qarshiligi katta bo'lganini isbotlaydi.

C) Har ikkala fikr asosli: qayta o'lchashdan oldin havo qarshiligi sabab bo'lganini tasdiqlash mumkin.

D) Har ikkala fikr asossiz: jadvaldagi barcha qiymatlar xatolik doirasida modelga mos.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 10-variant

1 B D	2 B C	3 B B	4 Q C	5 Q B	6 Q C	7 Q A
8 Q B	9 Q B	10 Q D	11 Q D	12 Q D	13 Q C	14 M A
15 M A						

1. D) Jism massasi — skalyar kattalik. Tezlanish — vektor kattalik. Tezlik — vektor kattalik. Javob: II va III.

2. C) O'ngga yo'nalganlarni musbat, chapga yo'nalganlarni manfiy olamiz: $1 - 2 + 6 = 5$. Modul: 5 birlik.

3. B) 1 — To'g'ri, magnit kuchi maydon kuchi bo'lib, kontakt talab qilmaydi. (A). 2 — Noto'g'ri, chunki maydon kuchlari jismlarga kontaktsiz ta'sir qiladi. (D). 3 — To'g'ri, elektr va magnit kuchlar tortish yoki itarish tarzida ta'sir qiladi. (B).

4. C) KL da o'rtacha tezlik $\frac{\vartheta}{2}$: $t_1 = \frac{120}{\vartheta/2} = \frac{240}{\vartheta}$. LM da $t_2 = \frac{120}{\vartheta}$. $t_1 : t_2 = 2$.

5. B) Ko'chish grafik ostidagi yuzalarning algebraik yig'indisi: 0–5 s: 50; 5–20 s: 225; 20–25 s: 75. Ko'chish: 350 m.

6. C) Yo'l: $s = \frac{1}{2} \cdot 2\pi R = 90$ m, $\vartheta_{o'rt} = \frac{90}{3} = 30$ m/s. Ko'chish – vatar: $2R = 60$ m, $|\vec{\vartheta}_{o'rt}| = 20$ m/s.

7. A) $\vartheta = 8 - 4t$, jism $t = 2$ s da to'xtab, orqaga harakatlanadi. Yo'l: $8 + 72 = 80$ m (koordinata esa -64 m).

8. B) Energiya saqlanishidan (yo'nalishga bog'liq emas): $\vartheta^2 = \vartheta_0^2 + 2gh$. $3600 = 100 + 20h \to h = 175$ m.

9. B) $\vartheta_x = 12$ m/s, $\vartheta_y = 16$ m/s. Uchish vaqti $t = \frac{2 \cdot 16}{10} = 3{,}2$ s. $x = 12 \cdot 3{,}2 = 38{,}4$ m.

10. D) I: 15 m/s — tezlik, tezlanish nolga teng — noto'g'ri. II: L tezlashmoqda — yo'lovchilar orqaga siljiydi — noto'g'ri. III: K da tezlanish yo'q, L da bor — to'g'ri. Javob: Faqat III.

11. D) Grafikdan: $m = \frac{F}{a} = \frac{25}{5} = 5$ kg. Vertikal kuch gorizontal harakatga ta'sir qilmaydi, gorizontal natijaviy kuch $30 - 5 = 25$ N. $a = \frac{25}{5} = 5$ m/s².

12. D) I: III qonun — to'g'ri. II: kuchlar teng — noto'g'ri. III: aks ta'sir kuchi — to'g'ri. Javob: I va III.

13. C) L dagi tezlik: $12 - 3 \cdot 1 = 9$ m/s. Tushish vaqti: $t = \sqrt{\frac{2h}{g}} = 0{,}5$ s. $x = 9 \cdot 0{,}5 = 4{,}5$ m.

14. A) 1- va 2-tajribada massa ham, tezlanish ham o'zgargan — burchak o'zgarishini massaga bog'lab bo'lmaydi. 1 va 3 da faqat massa o'zgargan va burchak xatolik doirasida bir xil ($\operatorname{tg}\alpha = a/g$).

15. A) Model: $h = 40 + 10t - 5t^2$: 40; 45; 40; 25; 0. Faqat $t = 3$ s dagi qiymat (28 m) xatolik doirasidan chiqadi. Bitta nomos qiymat havo qarshiligini isbotlamaydi — avval o'lchovni tekshirish kerak (Nilufar).

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 1-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

If you return a product and the shop gives your money back, you get a _____.

A) discount **B)** bargain **C)** refund **D)** receipt

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who easily feels hurt by what others say?

A) generous **B)** confident **C)** reliable **D)** sensitive

3. B **Choose the best answer.** 2 ball

Our striker kicked the ball past the goalkeeper and _____.

A) won a goal **B)** made the referee **C)** scored a goal **D)** beat a goal

4. Q **Choose the best answer.** 2,8 ball

Which natural event happens when the ground shakes suddenly?

A) tornado **B)** drought **C)** flood **D)** earthquake

5. Q **Choose the correct sentence.** 2,8 ball

We didn't have _____ time, so we took a taxi.

A) much **B)** a few **C)** many **D)** several

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) She gave me some useful advice. **B)** She gave me some useful advices.

C) She gave me many useful advice. **D)** She gave me an useful advice.

7. Q **Which sentence is grammatically correct?** 2,8 ball

A) She enjoys read detective stories. **B)** She enjoys reads detective stories.

C) She enjoys to read detective stories. **D)** She enjoys reading detective stories.

8. Q **Choose the best answer.** 2,8 ball

Ali has bought his tickets. He _____ to Istanbul on Friday.

A) is flying **B)** flew **C)** has flown **D)** flies

9. Q **Choose the best answer.** 2,8 ball

My sister _____ to the gym three times a week, but this week she _____ because she's ill.

A) is going / doesn't go **B)** goes / isn't going

C) go / isn't going **D)** goes / doesn't go

10. Q **Choose the best answer.** 2,8 ball

Why _____ you _____ at me? I haven't done anything wrong.

A) are / shouting **B)** do / shouting **C)** do / shout **D)** are / shout

11. Q **Choose the best answer.** 2,8 ball

When I was a child, I _____ afraid of the dark, but now I'm not.

A) used to be **B)** use to was **C)** am used to be **D)** was using to be

12. Q **Choose the best answer.** 2,8 ball

When the fire alarm rang, the teacher _____ the door, _____ the students outside and _____ the fire service.

A) looked / followed / answered **B)** said / went / returned

C) opened / arrived / waited **D)** opened / led / called

13. Q **Choose the best answer.** 2,8 ball

I _____ dinner when my friend _____ me.

A) cook / called **B)** was cooking / was calling

C) cooked / was calling **D)** was cooking / called

14. M **Read the text and choose the correct answer to the question.** 3 ball

Phone Problem

Aziz's phone screen broke. The first repair shop wanted $ 80 and three days. A second shop fixed it for $ 50 in one hour and gave a six-month guarantee.

Why was the second shop a better choice?

A) It was cheaper, faster and gave a guarantee.

B) It was closer to Aziz's school.

C) It gave Aziz a free case only.

D) It sold new phones.

15. M **Read the text and answer question.** 3 ball

School Garden

Five students created a garden at school. Ali bought seeds, Sara planned the beds, Tom watered the plants, and Lola and Mia made signs. Because everyone did their part, the garden was ready in two weeks.

Which statement best explains why the project succeeded?

A) Everyone had a task and did it.

B) Ali did all the work alone.

C) The teacher planted everything.

D) The garden was very small.

Javoblar kaliti — ChSB-1 tayyorgarlik: 1-variant

1 B C	2 B D	3 B C	4 Q D	5 Q A	6 Q A	7 Q D
8 Q A	9 Q B	10 Q A	11 Q A	12 Q D	13 Q D	14 M A
15 M A						

1. C) ✅ To'g'ri javob: refund
Pul qaytarilishi — **refund**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **receipt** — chek
• **discount** — chegirma
• **bargain** — arzon xarid

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. D) ✅ To'g'ri javob: sensitive
Tez xafa bo‘ladigan — **sensitive**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **reliable** — ishonchli
• **confident** — o'ziga ishongan
• **generous** — saxiy

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. C) ✅ To'g'ri javob: scored a goal
Gol urdi — **scored a goal**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **beat a goal** — “beat a goal” — gol "score" qilinadi
• **won a goal** — “won a goal” — gol yutilmaydi, kiritiladi
• **made the referee** — "hakamni qildi" — ma'nosiz

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: earthquake
Yer silkinishi — **earthquake**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **flood** — toshqin
• **drought** — qurg'oqchilik
• **tornado** — quyun

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. A) ✅ To'g'ri javob: much
time — sanalmaydi, inkor: **much**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **many** — “many” — time sanalmaydi
• **a few** — “a few” — sanaladigan
• **several** — “several” — sanaladigan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. A) ✅ To'g'ri javob: She gave me some useful advice.
advice — sanalmaydi: **some advice**.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **She gave me some useful advices.** — “advices” — advice -s olmaydi
• **She gave me an useful advice.** — “an useful advice” — a/an ishlatilmaydi (ustiga "useful" — undosh tovush, a bo'lardi)
• **She gave me many useful advice.** — “many” — sanalmaydigan ot bilan emas

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. D) ✅ To'g'ri javob: She enjoys reading detective stories.
enjoy + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **She enjoys to read detective stories.** — bu fe'ldan keyin “to + fe'l” emas, “-ing” shakl keladi
• **She enjoys read detective stories.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **She enjoys reads detective stories.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. A) ✅ To'g'ri javob: is flying
Kelishilgan reja — **is flying**.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **flies** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **flew** — Past Simple — harakat kelajakda, o'tmishda emas
• **has flown** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. B) ✅ To'g'ri javob: goes / isn't going
Odat — **goes**; bu hafta — **isn't going**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is going / doesn't go** — “is going / doesn't go” — zamonlar almashgan
• **goes / doesn't go** — “goes / doesn't go” — this week (vaqtinchalik) → isn't going
• **go / isn't going** — “go” — my sister (she) → goes

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. A) ✅ To'g'ri javob: are / shouting
Hozir — **are ... shouting**.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / shout** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / shout** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / shouting** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. A) ✅ To'g'ri javob: used to be
O‘tmish holati — **used to be**.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **am used to be** — “am used to be” — "ko'nikkanman" tuzilmasi, ma'no va shakl xato
• **use to was** — “use to was” — used to + asosiy shakl (be)
• **was using to be** — “was using to be” — mavjud bo'lmagan tuzilma

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. D) ✅ To'g'ri javob: opened / led / called
Ketma-ket harakatlar: **opened, led, called**.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **opened / arrived / waited** — “arrived the students” — arrive to'ldiruvchi olmaydi; “waited the fire service” — wait for
• **said / went / returned** — “said the door” — ma'nosiz
• **looked / followed / answered** — “looked the door … followed the students” — o'qituvchi yetaklaydi, ergashmaydi

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: “arrived the group” (arrive to'ldiruvchi olmaydi), “followed a goal”, “looked the menu” (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. D) ✅ To'g'ri javob: was cooking / called
Davomli + to‘satdan: **was cooking / called**.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **cooked / was calling** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **was cooking / was calling** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **cook / called** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. A) ✅ To'g'ri javob: It was cheaper, faster and gave a guarantee.
Arzonroq, tezroq va kafolat bilan.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was … a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **It sold new phones.** — yangi telefonlar sotishi — matnda yo'q
• **It was closer to Aziz's school.** — maktabga yaqinligi — matnda yo'q
• **It gave Aziz a free case only.** — "faqat g'ilof" — matnda kafolat, arzon va tez

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. A) ✅ To'g'ri javob: Everyone had a task and did it.
Har kimning vazifasi bor edi va bajarildi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Ali did all the work alone.** — Ali hammasini yolg'iz qilgan — teskari
• **The garden was very small.** — bog' juda kichik — sabab emas
• **The teacher planted everything.** — o'qituvchi ekkan — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 2-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

The piece of paper that shows what you paid for is a _____.

A) refund **B)** discount **C)** promotion **D)** receipt

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who worries a lot about what might happen?

A) relaxed **B)** brave **C)** anxious **D)** cheerful

3. B **Choose the best answer.** 2 ball

The referee showed him a red card because he _____.

A) won the rules **B)** scored the rules **C)** beat the rules **D)** broke the rules

4. Q **Choose the best answer.** 2,8 ball

Which natural event happens when a river overflows and covers the land with water?

A) avalanche **B)** volcano **C)** drought **D)** flood

5. Q **Choose the correct sentence.** 2,8 ball

There were _____ people at the concert that we couldn't find a seat.

A) so much **B)** so many **C)** a little **D)** much

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) We need some new furniture for the living room.

B) We need some new furnitures for the living room.

C) We need many new furniture for the living room.

D) We need a new furniture for the living room.

7. **Which sentence is grammatically correct?** 2,8 ball

A) They finished paint the room at midnight.

B) They finished painted the room at midnight.

C) They finished to paint the room at midnight.

D) They finished painting the room at midnight.

8. **Choose the best answer.** 2,8 ball

We _____ a party next Saturday. Would you like to come?

A) have had **B)** are having **C)** had **D)** have

9. **Choose the best answer.** 2,8 ball

Dad usually _____ the news at 8, but tonight he _____ a documentary.

A) watch / is watching **B)** is watching / watches

C) watches / watches **D)** watches / is watching

10. **Choose the best answer.** 2,8 ball

Why _____ she _____ the music so loud? The baby is asleep.

A) is / play **B)** does / play **C)** is / playing **D)** does / playing

11. **Choose the best answer.** 2,8 ball

My dad _____ football every weekend, but he stopped after his injury.

A) use to played **B)** plays **C)** is playing **D)** used to play

12. **Choose the best answer.** 2,8 ball

When we arrived at the hotel, we _____ in, _____ our bags upstairs and _____ to the beach.

A) said / brought / stayed **B)** looked / carried / returned to

C) checked / took / went **D)** checked / arrived / left

13. **Choose the best answer.** 2,8 ball

We _____ along the beach when we _____ a dolphin.

A) were walking / were seeing **B)** walk / saw

C) walked / were seeing **D)** were walking / saw

14. M **Read the text and choose the correct answer to the question.** 3 ball

Train or Plane?

Lola needed to travel 600 km. The plane ticket was $ 120 and she had to be at the airport two hours early. The fast train cost $ 40 and took four hours from the city centre.

Why did Lola choose the train?

A) The plane was cancelled.

B) She was afraid of trains.

C) It was much cheaper and left from the city centre.

D) It was faster than the plane.

15. M **Read the text and answer question.** 3 ball

Class Newspaper

The class made a newspaper. Some students wrote articles, others took photos, and two students checked spelling. They shared ideas in weekly meetings and finished before the deadline.

Which statement best explains why the group succeeded?

A) One student wrote all the articles.

B) The newspaper had no photos.

C) The deadline was cancelled.

D) They divided the work and communicated regularly.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 2-variant

1 B D	2 B C	3 B D	4 Q D	5 Q B	6 Q A	7 Q D
8 Q B	9 Q D	10 Q C	11 Q D	12 Q C	13 Q D	14 M C
15 M D						

1. D) ✅ **To'g'ri javob: receipt**
Chek — **receipt**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **refund** — pul qaytarish
• **discount** — chegirma
• **promotion** — aksiya

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. C) ✅ **To'g'ri javob: anxious**
Ko‘p xavotirlanadigan — **anxious**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **relaxed** — xotirjam — teskari
• **brave** — jasur
• **cheerful** — quvnoq

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. D) ✅ To'g'ri javob: broke the rules
Qoidani buzdi — **broke the rules**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **won the rules** — “won the rules” — ma'nosiz
• **scored the rules** — “scored the rules” — ma'nosiz
• **beat the rules** — “beat the rules” — ma'nosiz; qoida buziladi (break)

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: flood
Daryo toshishi — **flood**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **drought** — qurg'oqchilik — teskari, suvsizlik
• **avalanche** — qor ko'chkisi
• **volcano** — vulqon

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. B) ✅ To'g'ri javob: so many
people — sanaladi: **so many**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **so much** — “so much” — people sanaladigan → so many
• **a little** — “a little” — sanalmaydigan va ma'no teskari
• **much** — “much” — sanalmaydigan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. A) ✅ To'g'ri javob: We need some new furniture for the living room.
furniture — sanalmaydi.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **We need some new furnitures for the living room.** — “furnitures” — -s olmaydi
• **We need a new furniture for the living room.** — “a new furniture” — a/an olmaydi
• **We need many new furniture for the living room.** — “many ... furniture” — sanalmaydi

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. D) ✅ To'g'ri javob: They finished painting the room at midnight.
finish + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **They finished to paint the room at midnight.** — bu fe'ldan keyin “to + fe'l” emas, “-ing” shakl keladi
• **They finished paint the room at midnight.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **They finished painted the room at midnight.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. B) ✅ To'g'ri javob: are having
Reja — **are having**.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **have** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **had** — Past Simple — harakat kelajakda, o'tmishda emas
• **have had** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. D) ✅ To'g'ri javob: watches / is watching
Odat — **watches**; bugun — **is watching**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is watching / watches** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **watch / is watching** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **watches / watches** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. C) ✅ To'g'ri javob: is / playing
is ... playing.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **does / play** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **is / play** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **does / playing** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. D) ✅ To'g'ri javob: used to play
used to play.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **is playing** — “is playing” — hozirgi zamon; u to'xtagan
• **plays** — “plays” — hozirgi zamon
• **use to played** — “use to played” — used to + asosiy shakl

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. C) ✅ To'g'ri javob: checked / took / went
checked in, took, went.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **checked / arrived / left** — “arrived our bags” — ma'nosiz; “left to the beach” — left the hotel / went to
• **said / brought / stayed** — “said in” — check in kerak
• **looked / carried / returned to** — “looked in ... returned to to the beach” — ma'no va tuzilma xato

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: “arrived the group” (arrive to'ldiruvchi olmaydi), “followed a goal”, “looked the menu” (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. D) ✅ To'g'ri javob: were walking / saw
were walking / saw.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **walked / were seeing** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **were walking / were seeing** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **walk / saw** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. C) ✅ To'g'ri javob: It was much cheaper and left from the city centre.
Ancha arzon va shahar markazidan jo‘naydi.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **It was faster than the plane.** — poyezd samolyotdan tezroq — teskari, 4 soat
• **She was afraid of trains.** — poyezddan qo'rqqan — matnda yo'q
• **The plane was cancelled.** — samolyot bekor qilingan — matnda yo'q

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. D) ✅ To'g'ri javob: They divided the work and communicated regularly.
Ishni bo‘lishib, muntazam muloqot qilishdi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **One student wrote all the articles.** — bitta o'quvchi hamma maqolani yozgan — teskari
• **The newspaper had no photos.** — gazetada rasm yo'q — teskari, suratga olishgan
• **The deadline was cancelled.** — muddat bekor qilingan — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 3-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

A product you buy for much less than its real value is a _____.

A) bargain **B)** receipt **C)** queue **D)** refund

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who wants what other people have?

A) patient **B)** modest **C)** generous **D)** jealous

3. B **Choose the best answer.** 2 ball

Neither team won; the match ended in a _____.

A) draw **B)** goal **C)** trophy **D)** referee

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a long time without rain?

A) hurricane **B)** flood **C)** tsunami **D)** drought

5. Q **Choose the correct sentence.** 2,8 ball

Could you give me _____ advice about my project?

A) several **B)** a few **C)** some **D)** many

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) How much luggages do you have? **B)** How many luggage do you have?

C) How many luggages do you have? **D)** How much luggage do you have?

7. Q **Which sentence is grammatically correct?** 2,8 ball

A) I can't stand wait in long queues. **B)** I can't stand to waited in long queues.

C) I can't stand waiting in long queues. **D)** I can't stand waited in long queues.

8. Q **Choose the best answer.** 2,8 ball

I've already made an appointment. I _____ the dentist tomorrow at 10.

A) see **B)** have seen **C)** am seeing **D)** saw

9. Q **Choose the best answer.** 2,8 ball

We _____ Spanish at school, but this term we _____ French too.

A) study / are studying **B)** are studying / study

C) studies / are studying **D)** study / study

10. Q **Choose the best answer.** 2,8 ball

What _____ you _____ in my bag? That's private!

A) are / looking for **B)** are / look for

C) do / look for **D)** do / looking for

11. Q **Choose the best answer.** 2,8 ball

There _____ a cinema in our street, but now it's a supermarket.

A) was used be **B)** used to be **C)** is **D)** uses to be

12. Q **Choose the best answer.** 2,8 ball

After the accident, the driver _____ out of the car, _____ the damage and _____ the police.

A) went / saw / arrived **B)** got / checked / phoned

C) came / followed / answered **D)** got / looked / talked

13. Q **Choose the best answer.** 2,8 ball

She _____ the bus when she _____ her old teacher.

A) waited for / was meeting **B)** was waiting for / met

C) waits for / met **D)** was waiting for / was meeting

14. M **Read the text and choose the correct answer to the question.** 3 ball

Online Shopping

Ben ordered shoes online. When they arrived, they were too small. Because the shop offered free returns, he sent them back and got the right size at no extra cost.

Why was the shop's policy helpful?

A) The shoes were very cheap. **B)** Ben got two pairs of shoes.

C) Ben didn't have to pay at all. **D)** Ben could exchange the shoes for free.

15. M **Read the text and answer question.** 3 ball

Charity Run

A school organised a charity run. Teachers planned the route, parents gave water, and students found sponsors. They raised twice as much money as last year.

What was the main reason for their success?

A) The run was very short.

B) Last year was cancelled.

C) Teachers, parents and students worked together.

D) Only teachers took part.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 3-variant

1 B A	2 B D	3 B A	4 Q D	5 Q C	6 Q D	7 Q C
8 Q C	9 Q A	10 Q A	11 Q B	12 Q B	13 Q B	14 M D
15 M C						

1. A) ✅ **To'g'ri javob: bargain**
Arzon va sifatli xarid — **bargain**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **refund** — pul qaytarish
• **receipt** — chek
• **queue** — navbat

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. D) ✅ **To'g'ri javob: jealous**
Boshqalarnikini xohlaydigan — **jealous**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **generous** — saxiy — teskari
• **modest** — kamtar
• **patient** — sabrli

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. A) ✅ To'g'ri javob: draw
Durang — **draw**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **goal** — gol
• **referee** — hakam
• **trophy** — kubok — durang emas

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: drought
Uzoq yomg‘irsizlik — **drought**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **flood** — toshqin — teskari
• **tsunami** — tsunami — to'lqin
• **hurricane** — dovul

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. C) ✅ **To'g'ri javob: some**
advice — sanalmaydi: **some**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **many** — "many" — advice sanalmaydi
• **a few** — "a few" — sanaladigan
• **several** — "several" — sanaladigan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. D) ✅ **To'g'ri javob: How much luggage do you have?**
luggage — sanalmaydi: **how much luggage**.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **How many luggages do you have?** — "How many luggages" — luggage sanalmaydi
• **How many luggage do you have?** — "How many luggage" — many emas, much
• **How much luggages do you have?** — "luggages" — -s olmaydi

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. C) ✅ To'g'ri javob: I can't stand waiting in long queues.
can't stand + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **I can't stand to waited in long queues.** — bu fe'ldan keyin “to + fe'l” emas, “-ing” shakl keladi
• **I can't stand wait in long queues.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **I can't stand waited in long queues.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. C) ✅ To'g'ri javob: am seeing
Kelishilgan — **am seeing**.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **see** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **saw** — Past Simple — harakat kelajakda, o'tmishda emas
• **have seen** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. A) ✅ To'g'ri javob: study / are studying
Doimiy — **study**; bu chorak — **are studying**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **are studying / study** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **studies / are studying** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **study / study** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. A) ✅ To'g'ri javob: are / looking for
are ... looking for.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / look for** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / look for** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / looking for** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. B) ✅ To'g'ri javob: used to be
used to be.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **uses to be** — “uses to be” — mavjud bo'lmagan shakl
• **is** — “is” — hozir supermarket
• **was used be** — “was used be” — xato tuzilma

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. B) ✅ To'g'ri javob: got / checked / phoned
got out, checked, phoned.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **got / looked / talked** — "looked the damage" — look AT; "talked the police" — talk TO
• **went / saw / arrived** — "went out ... saw ... arrived the police" — arrive to'ldiruvchi olmaydi
• **came / followed / answered** — "followed the damage ... answered the police" — ma'nosiz

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: "arrived the group" (arrive to'ldiruvchi olmaydi), "followed a goal", "looked the menu" (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. B) ✅ To'g'ri javob: was waiting for / met
was waiting / met.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **waited for / was meeting** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **was waiting for / was meeting** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **waits for / met** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. D) ✅ To'g'ri javob: Ben could exchange the shoes for free.
Bepul almashtirish imkoniyati.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Ben got two pairs of shoes.** — ikki juft poyabzal — matnda yo'q
• **The shoes were very cheap.** — poyabzal juda arzon — matnda yo'q
• **Ben didn't have to pay at all.** — umuman to'lamagan — u to'lagan

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. C) ✅ To'g'ri javob: Teachers, parents and students worked together.
Hamma birgalikda ishladi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **The run was very short.** — yugurish juda qisqa — matnda yo'q
• **Only teachers took part.** — faqat o'qituvchilar qatnashgan — teskari
• **Last year was cancelled.** — o'tgan yil bekor qilingan — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 4-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

The place in a supermarket where you pay for your shopping is the _____.

A) shelf **B)** changing room **C)** checkout **D)** warehouse

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who doesn't talk about their own success?

A) rude **B)** modest **C)** arrogant **D)** bossy

3. B **Choose the best answer.** 2 ball

After winning the final, the captain lifted the _____.

A) referee **B)** draw **C)** trophy **D)** stadium

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a large amount of snow falling down a mountain?

A) flood **B)** avalanche **C)** drought **D)** earthquake

5. Q **Choose the correct sentence.** 2,8 ball

I have _____ friends in this city, so I'm never lonely.

A) much **B)** a little **C)** any **D)** a lot of

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) There are too many traffics in the city centre.

B) There are too much traffics in the city centre.

C) There is too many traffic in the city centre.

D) There is too much traffic in the city centre.

7. **Which sentence is grammatically correct?** 2,8 ball

A) He avoided answering the difficult question.

B) He avoided to answer the difficult question.

C) He avoided answered the difficult question.

D) He avoided answer the difficult question.

8. **Choose the best answer.** 2,8 ball

My parents _____ to a new flat next month; they've signed the contract.

A) move **B)** have moved **C)** moved **D)** are moving

9. **Choose the best answer.** 2,8 ball

Ali _____ breakfast at home, but today he _____ at a café with friends.

A) have / is eating **B)** has / is eating **C)** is having / eats **D)** has / eats

10. **Choose the best answer.** 2,8 ball

Why _____ they _____ so fast? There is a speed limit here.

A) are / driving **B)** do / driving **C)** do / drive **D)** are / drive

11. **Choose the best answer.** 2,8 ball

Ten years ago, people _____ letters, but now they send messages.

A) write **B)** use to wrote **C)** are writing **D)** used to write

12. **Choose the best answer.** 2,8 ball

In the morning, Lola _____ up, _____ a shower and _____ her bag for school.

A) got / did / made **B)** stood / washed / put

C) woke / took / carried to **D)** woke / had / packed

13. **Choose the best answer.** 2,8 ball

They _____ football when it _____ to rain.

A) were playing / was starting **B)** were playing / started

C) played / was starting **D)** play / started

14. M **Read the text and choose the correct answer to the question.** 3 ball

A Gift for Mum

Kate wanted to buy flowers for her mum. The flower shop near her house was expensive, so she bought a small plant at the market. It was cheaper, and her mum could keep it for years.

Why was the plant a good choice?

A) It cost less and lasted longer.
B) It was more expensive than flowers.
C) The flower shop was closed.
D) Mum didn't like flowers.

15. M **Read the text and answer question.** 3 ball

Robot Competition

Four friends built a robot. Each one was responsible for one part: design, programming, testing or presentation. When problems appeared, they solved them together. They won second place.

Why did the team do well?

A) Clear roles and teamwork
B) Expensive materials
C) A very simple robot
D) Help from professional engineers

Javoblar kaliti — ChSB-1 tayyorgarlik: 4-variant

1 B C	2 B B	3 B C	4 Q B	5 Q D	6 Q D	7 Q A
8 Q D	9 Q B	10 Q A	11 Q D	12 Q D	13 Q B	14 M A
15 M A						

1. C) ✅ **To'g'ri javob: checkout**
To‘lov joyi — **checkout**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **changing room** — kiyim kiyib ko'rish xonasi
• **warehouse** — ombor
• **shelf** — javon

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. B) ✅ **To'g'ri javob: modest**
O‘z yutug‘ini maqtamaydigan — **modest**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **arrogant** — takabbur — teskari
• **bossy** — buyruq beruvchi
• **rude** — qo'pol

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. C) ✅ To'g'ri javob: trophy
Kubok — **trophy**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **referee** — hakam — ko'tarilmaydi
• **draw** — durang
• **stadium** — stadion

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. B) ✅ To'g'ri javob: avalanche
Qor ko‘chkisi — **avalanche**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **earthquake** — zilzila
• **drought** — qurg'oqchilik
• **flood** — toshqin

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. D) ✅ To'g'ri javob: a lot of
friends — ko‘p: **a lot of**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **much** — “much” — friends sanaladigan
• **a little** — “a little” — sanalmaydigan
• **any** — “any” — tasdiq gapda odatda ishlatilmaydi

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. D) ✅ To'g'ri javob: There is too much traffic in the city centre.
traffic — sanalmaydi.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **There are too many traffics in the city centre.** — “are ... traffics” — sanalmaydi, birlik
• **There is too many traffic in the city centre.** — “too many traffic” — much kerak
• **There are too much traffics in the city centre.** — “are too much traffics” — ikki xato

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. A) ✅ To'g'ri javob: He avoided answering the difficult question.
avoid + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **He avoided to answer the difficult question.** — bu fe'ldan keyin “to + fe'l” emas, “-ing” shakl keladi
• **He avoided answer the difficult question.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **He avoided answered the difficult question.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. D) ✅ To'g'ri javob: are moving
are moving.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **move** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **moved** — Past Simple — harakat kelajakda, o'tmishda emas
• **have moved** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. B) ✅ To'g'ri javob: has / is eating
Odat — **has**; bugun — **is eating**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is having / eats** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **have / is eating** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **has / eats** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. A) ✅ To'g'ri javob: are / driving
are ... driving.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / drive** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / drive** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / driving** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. D) ✅ To'g'ri javob: used to write
used to write.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **write** — “write” — hozirgi zamon
• **are writing** — “are writing” — hozirgi zamon
• **use to wrote** — “use to wrote” — used to + asosiy shakl

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. D) ✅ To'g'ri javob: woke / had / packed
woke up, had a shower, packed.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **woke / took / carried to** — "took a shower ... carried to her bag" — to ortiqcha
• **got / did / made** — "got up ... did a shower ... made her bag" — do a shower, make a bag — noto'g'ri birikma
• **stood / washed / put** — "stood up ... washed a shower" — ma'nosiz

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: "arrived the group" (arrive to'ldiruvchi olmaydi), "followed a goal", "looked the menu" (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. B) ✅ To'g'ri javob: were playing / started
were playing / started.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **played / was starting** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **were playing / was starting** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **play / started** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. A) ✅ To'g'ri javob: It cost less and lasted longer.
Arzonroq va uzoq yashaydi.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **It was more expensive than flowers.** — gullardan qimmat — teskari
• **Mum didn't like flowers.** — onasi gulni yoqtirmagan — matnda yo'q
• **The flower shop was closed.** — do'kon yopiq — matnda qimmat edi

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. A) ✅ To'g'ri javob: Clear roles and teamwork
Aniq rollar va hamkorlik.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Expensive materials** — qimmat materiallar — matnda yo'q
• **A very simple robot** — juda oddiy robot — matnda yo'q
• **Help from professional engineers** — professional muhandislar yordami — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 5-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

A shop's special offer, like “buy one, get one free”, is a _____.

A) queue **B)** receipt **C)** refund **D)** promotion

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who is calm and not worried?

A) relaxed **B)** moody **C)** anxious **D)** nervous

3. B **Choose the best answer.** 2 ball

Every morning the athletes _____ in the gym to stay fit.

A) beat **B)** score **C)** train **D)** draw

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a huge wave caused by an earthquake under the sea?

A) tsunami **B)** wildfire **C)** drought **D)** tornado

5. Q **Choose the correct sentence.** 2,8 ball

There's only _____ milk left. We need to buy more.

A) many **B)** a few **C)** a little **D)** several

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) I bought a loaf of bread and some cheese.

B) I bought two breads and a cheese.

C) I bought many bread and much cheeses.

D) I bought a bread and some cheeses.

7. Q **Which sentence is grammatically correct?** 2,8 ball

A) We decided to go to the cinema. **B)** We decided going to the cinema.

C) We decided went to the cinema. **D)** We decided go to the cinema.

8. **Choose the best answer.** 2,8 ball

Lola _____ her grandparents this weekend; she has already packed her bag.

A) is visiting **B)** has visited **C)** visited **D)** visits

9. **Choose the best answer.** 2,8 ball

Mum _____ in a hospital, but this month she _____ in a school clinic.

A) is working / works **B)** work / is working

C) works / works **D)** works / is working

10. **Choose the best answer.** 2,8 ball

Why _____ you _____ your phone in class? Put it away.

A) are / using **B)** are / use **C)** do / using **D)** do / use

11. **Choose the best answer.** 2,8 ball

I _____ sweets every day, but now I eat fruit instead.

A) used to eat **B)** am eating **C)** eat **D)** use to ate

12. **Choose the best answer.** 2,8 ball

At the restaurant, we _____ the menu, _____ our food and _____ for the bill.

A) looked / ate / paid **B)** read / ordered / asked

C) saw / chose / gave **D)** opened / cooked / waited

13. **Choose the best answer.** 2,8 ball

Ali _____ his homework when the lights _____ out.

A) did / were going **B)** was doing / were going

C) does / went **D)** was doing / went

14. M **Read the text and choose the correct answer to the question.** 3 ball

Gym Membership

Ali compared two gyms. Gym A cost $ 30 a month but was 10 km away. Gym B cost $ 35 but was next to his school. He chose Gym B and went every day after lessons.

Why did Ali choose Gym B?

A) It had no other members.

B) Its location made it easy to go regularly.

C) It was the cheapest gym.

D) Gym A was closed.

15. M **Read the text and answer question.** 3 ball

Cleaning Day

Residents of a street decided to clean their area. Some collected rubbish, some painted benches, and children planted flowers. By evening, the street looked new.

What is the main idea?

A) Working together can quickly improve a place.

B) Cleaning is the city's job only.

C) Children shouldn't plant flowers.

D) Benches don't need paint.

Javoblar kaliti — ChSB-1 tayyorgarlik: 5-variant

1 B D	2 B A	3 B C	4 Q A	5 Q C	6 Q A	7 Q A
8 Q A	9 Q D	10 Q A	11 Q A	12 Q B	13 Q D	14 M B
15 M A						

1. D) ✅ **To'g'ri javob: promotion**
Aksiya — **promotion**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **receipt** — chek
• **refund** — pul qaytarish
• **queue** — navbat

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. A) ✅ **To'g'ri javob: relaxed**
Xotirjam — **relaxed**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **anxious** — xavotirli — teskari
• **nervous** — asabiy — teskari
• **moody** — kayfiyati o'zgaruvchan

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. C) ✅ To'g'ri javob: train
Mashq qilishadi — **train**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **score** — gol kiritish — mashq emas
• **beat** — yengmoq — raqib kerak
• **draw** — durang qilmoq

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. A) ✅ To'g'ri javob: tsunami
Ulkan to‘lqin — **tsunami**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **tornado** — quyun
• **drought** — qurg'oqchilik
• **wildfire** — o'rmon yong'ini

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. C) ✅ **To'g'ri javob: a little**
milk — sanalmaydi, oz: **a little**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **a few** — “a few” — milk sanalmaydi
• **many** — “many” — sanaladigan
• **several** — “several” — sanaladigan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. A) ✅ **To'g'ri javob: I bought a loaf of bread and some cheese.**
bread — sanalmaydi: **a loaf of bread**.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **I bought a bread and some cheeses.** — “a bread” — bread sanalmaydi (a loaf of bread)
• **I bought two breads and a cheese.** — “two breads and a cheese” — sanalmaydi
• **I bought many bread and much cheeses.** — “many bread, much cheeses” — xato birikmalar

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. A) ✅ To'g'ri javob: We decided to go to the cinema.
decide + to.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **We decided going to the cinema.** — "decided going" — decide dan keyin to + fe'l keladi
• **We decided go to the cinema.** — "decided go" — to tushib qolgan
• **We decided went to the cinema.** — "decided went" — to + asosiy shakl kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. A) ✅ To'g'ri javob: is visiting
is visiting.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **visits** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **visited** — Past Simple — harakat kelajakda, o'tmishda emas
• **has visited** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. D) ✅ To'g'ri javob: works / is working
Doimiy — **works**; bu oy — **is working**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is working / works** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **work / is working** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **works / works** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. A) ✅ To'g'ri javob: are / using
are ... using.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / use** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / use** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / using** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. A) ✅ To'g'ri javob: used to eat
used to eat.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **eat** — “eat” — hozir meva yeydi
• **am eating** — “am eating” — hozirgi zamon
• **use to ate** — “use to ate” — used to + asosiy shakl

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. B) ✅ To'g'ri javob: read / ordered / asked
read, ordered, asked for.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **looked / ate / paid** — “looked the menu” — look AT; “paid for the bill” — avval so'raladi (ask for)
• **saw / chose / gave** — “saw ... gave for the bill” — ma'nosiz
• **opened / cooked / waited** — “cooked our food” — restoranda mijoz pishirmaydi

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: “arrived the group” (arrive to'ldiruvchi olmaydi), “followed a goal”, “looked the menu” (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. D) ✅ To'g'ri javob: was doing / went
was doing / went out.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **did / were going** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **was doing / were going** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **does / went** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. B) ✅ To'g'ri javob: Its location made it easy to go regularly.
Joylashuvi muntazam borishni osonlashtirdi.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **It was the cheapest gym.** — eng arzon — teskari, $35 > $30
• **It had no other members.** — a'zolari yo'q — matnda yo'q
• **Gym A was closed.** — Gym A yopiq — matnda yo'q

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($45 < $60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. A) ✅ To'g'ri javob: Working together can quickly improve a place.
Birgalikda ishlash joyni tez obod qiladi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Children shouldn't plant flowers.** — bolalar gul ekmasligi kerak — teskari
• **Cleaning is the city's job only.** — tozalash faqat shahar ishi — teskari
• **Benches don't need paint.** — o'rindiqlarga bo'yoq kerak emas — teskari

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 6-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

If you pay for something later in small amounts, you buy it in _____.

A) bargains **B)** instalments **C)** receipts **D)** refunds

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who always works hard?

A) careless **B)** hard-working **C)** selfish **D)** lazy

3. B **Choose the best answer.** 2 ball

Our team _____ the match 2–1 and went to the final.

A) beat **B)** drew **C)** won **D)** scored

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a large fire that spreads quickly in a forest?

A) flood **B)** hailstorm **C)** avalanche **D)** wildfire

5. Q **Choose the correct sentence.** 2,8 ball

Only _____ students passed the difficult exam.

A) a few **B)** a little **C)** any **D)** much

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) A news was very surprising. **B)** The newses were very surprising.

C) The news were very surprising. **D)** The news was very surprising.

7. Q **Which sentence is grammatically correct?** 2,8 ball

A) My dad gave up smoked last year. **B)** My dad gave up to smoke last year.

C) My dad gave up smoking last year. **D)** My dad gave up smoke last year.

8. Q **Choose the best answer.** 2,8 ball

The team _____ in the final on Sunday. Everything is arranged.

A) has played **B)** plays **C)** is playing **D)** played

9. Q **Choose the best answer.** 2,8 ball

I _____ to school by bus, but this week I _____ because the buses are on strike.

A) am going / walk
B) goes / am walking
C) go / am walking
D) go / walk

10. Q **Choose the best answer.** 2,8 ball

Why _____ he _____ the door? Someone is trying to sleep.

A) does / bang **B)** is / banging **C)** is / bang **D)** does / banging

11. Q **Choose the best answer.** 2,8 ball

My grandparents _____ in a small village before they moved to the city.

A) use to lived **B)** used to live **C)** live **D)** are living

12. Q **Choose the best answer.** 2,8 ball

When the plane landed, the passengers _____ their seat belts, _____ their bags and _____ the plane.

A) unfastened / took / left
B) fastened / brought / arrived
C) removed / followed / entered
D) opened / carried / went

13. Q **Choose the best answer.** 2,8 ball

I _____ when the alarm clock _____.

A) was sleeping / rang
B) was sleeping / was ringing
C) slept / was ringing
D) sleep / rang

14. M **Read the text and choose the correct answer to the question.** 3 ball

Second-hand Books

Mia needed five books for school. New books cost $ 75. She found the same books in good condition at a second-hand shop for $ 25 and gave the saved money to charity.

What did Mia do with the money she saved?

A) She kept it for a trip.
B) She bought more books.
C) She gave it to charity.
D) She bought a phone.

15. M **Read the text and answer question.** 3 ball

School Play

The drama club prepared a play. Some students acted, others made costumes and sets, and one student managed the lights. They practised together every week, and the show was a big success.

Which statement best explains the play's success?

A) Only one student acted.

B) The play was very short.

C) The audience was very small.

D) Each member had a role and they practised together.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 6-variant

1 B	2 B	3 C	4 D	5 A	6 D	7 C
8 C	9 C	10 B	11 B	12 A	13 A	14 C
15 D						

1. B) ✅ To'g'ri javob: instalments
Bo‘lib to‘lash — **instalments**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **receipts** — cheklar
• **refunds** — pul qaytarishlar
• **bargains** — arzon xaridlar

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. B) ✅ To'g'ri javob: hard-working
Mehnatkash — **hard-working**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **lazy** — dangasa — teskari
• **careless** — e'tiborsiz
• **selfish** — xudbin

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. C) ✅ To'g'ri javob: won
O‘yinni yutdi — **won the match**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **beat** — “beat the match” — beat + raqib (beat the team); o'yin esa yutiladi (won)
• **scored** — “scored the match” — gol kiritiladi
• **drew** — “drew” — 2–1 durang emas

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: wildfire
O‘rmon yong‘ini — **wildfire**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **flood** — toshqin
• **avalanche** — qor ko'chkisi
• **hailstorm** — do'lli bo'ron

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. A) ✅ **To'g'ri javob: a few**
students — sanaladi, oz: **a few**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **a little** — “a little” — students sanaladigan
• **much** — “much” — sanalmaydigan
• **any** — “any” — tasdiq gapda "only any" ishlatilmaydi

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. D) ✅ **To'g'ri javob: The news was very surprising.**
news — birlik: **was**.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **The news were very surprising.** — “were” — news birlik: was
• **The newses were very surprising.** — “newses” — mavjud emas
• **A news was very surprising.** — “A news” — a olmaydi (a piece of news)

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. C) ✅ To'g'ri javob: My dad gave up smoking last year.
give up + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **My dad gave up to smoke last year.** — bu fe'ldan keyin "to + fe'l" emas, "-ing" shakl keladi
• **My dad gave up smoke last year.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **My dad gave up smoked last year.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. C) ✅ To'g'ri javob: is playing
is playing.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **plays** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **played** — Past Simple — harakat kelajakda, o'tmishda emas
• **has played** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. C) ✅ To'g'ri javob: go / am walking
Odat — **go**; bu hafta — **am walking**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **am going / walk** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **goes / am walking** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **go / walk** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. B) ✅ To'g'ri javob: is / banging
is ... banging.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **does / bang** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **is / bang** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **does / banging** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. B) ✅ To'g'ri javob: used to live
used to live.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **live** — “live” — ular ko'chib ketgan
• **are living** — “are living” — hozirgi zamon
• **use to lived** — “use to lived” — used to + asosiy shakl

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. A) ✅ To'g'ri javob: unfastened / took / left
unfastened, took, left.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **fastened / brought / arrived** — “fastened” — qo'nganda kamar yechiladi; “arrived the plane” — xato
• **opened / carried / went** — “opened their seat belts” — noto'g'ri birikma; “went the plane” — xato
• **removed / followed / entered** — “followed their bags ... entered the plane” — ma'no teskari

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: “arrived the group” (arrive to'ldiruvchi olmaydi), “followed a goal”, “looked the menu” (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. A) ✅ To'g'ri javob: was sleeping / rang
was sleeping / rang.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **slept / was ringing** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **was sleeping / was ringing** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **sleep / rang** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. C) ✅ To'g'ri javob: She gave it to charity.
Tejalgan pulni xayriyaga berdi.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **She bought more books.** — ko'proq kitob — matnda xayriya
• **She bought a phone.** — telefon — matnda yo'q
• **She kept it for a trip.** — sayohat uchun saqlagan — matnda yo'q

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. D) ✅ To'g'ri javob: Each member had a role and they practised together.
Har kimning roli bor va birga mashq qilishdi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Only one student acted.** — faqat bitta o'quvchi o'ynagan — teskari
• **The play was very short.** — spektakl qisqa — matnda yo'q
• **The audience was very small.** — tomoshabinlar kam — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 7-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

A person who buys things in a shop is a _____.

A) supplier **B)** cashier **C)** manager **D)** customer

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who doesn't want to change their opinion?

A) sensitive **B)** flexible **C)** stubborn **D)** cheerful

3. B **Choose the best answer.** 2 ball

The person who controls a football match and makes sure players follow the rules is the _____.

A) coach **B)** fan **C)** goalkeeper **D)** referee

4. Q **Choose the best answer.** 2,8 ball

Which natural event happens when a mountain sends out hot lava and ash?

A) avalanche **B)** flood

C) drought **D)** volcanic eruption

5. Q **Choose the correct sentence.** 2,8 ball

How _____ luggage are you taking on the trip?

A) much **B)** few **C)** many **D)** several

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) We had a lot of homework yesterday. **B)** We had a homework yesterday.

C) We had many homeworks yesterday. **D)** We had a lot of homeworks yesterday.

7. Q **Which sentence is grammatically correct?** 2,8 ball

A) Would you mind closing the window? **B)** Would you mind closed the window?

C) Would you mind to close the window? **D)** Would you mind close the window?

8. Q **Choose the best answer.** 2,8 ball

I can't meet you tonight because I _____ dinner with my cousins.

A) have had **B)** am having **C)** had **D)** have

9. Q **Choose the best answer.** 2,8 ball

Kate _____ fiction, but at the moment she _____ a history book.

A) read / is reading **B)** is reading / reads

C) reads / reads **D)** reads / is reading

10. Q **Choose the best answer.** 2,8 ball

What _____ the children _____ in the kitchen? It's such a mess!

A) are / doing **B)** do / do **C)** do / doing **D)** are / do

11. Q **Choose the best answer.** 2,8 ball

Lola _____ long hair, but she cut it short last year.

A) used to have **B)** has **C)** use to had **D)** is having

12. Q **Choose the best answer.** 2,8 ball

During the match, our striker _____ the ball, _____ two defenders and _____ a goal.

A) took / passed / scored **B)** got / followed / shot

C) took / beat / won **D)** kicked / ran / made

13. Q **Choose the best answer.** 2,8 ball

Mum _____ the car when she _____ a strange noise.

A) drives / heard **B)** drove / was hearing

C) was driving / heard **D)** was driving / was hearing

14. M **Read the text and choose the correct answer to the question.** 3 ball

The Sale

Tom saw a jacket on sale for 50% off. Before buying, he checked it carefully and noticed a broken zip. He asked for another jacket and got one without problems.

Why was checking the jacket a good idea?

A) He found a cheaper shop. **B)** He got the jacket for free.

C) He avoided buying a damaged product. **D)** The sale finished early.

15. M **Read the text and answer question.** 3 ball

Science Fair

Two students worked on a project. Ali did all the research while Vali only watched. Their project was incomplete. Another pair shared the work equally and won first prize.

What can we learn from the text?

A) Prizes are given randomly.

B) Research is not important.

C) Watching is the best way to learn.

D) Sharing work fairly leads to better results.

Javoblar kaliti — ChSB-1 tayyorgarlik: 7-variant

1 B D	2 B C	3 B D	4 Q D	5 Q A	6 Q A	7 Q A
8 Q B	9 Q D	10 Q A	11 Q A	12 Q A	13 Q C	14 M C
15 M D						

1. D) ✅ **To'g'ri javob: customer**
Xaridor — **customer**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **cashier** — kassir — pul oluvchi xodim
• **manager** — menejer
• **supplier** — yetkazib beruvchi

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. C) ✅ **To'g'ri javob: stubborn**
Fikridan qaytmaydigan — **stubborn** (o‘jar).

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **flexible** — moslashuvchan — teskari
• **sensitive** — ta'sirchan
• **cheerful** — quvnoq

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. D) ✅ To'g'ri javob: referee
Hakam — **referee**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **goalkeeper** — darvozabon
• **coach** — murabbiy — o'yinni hakam boshqaradi
• **fan** — muxlis

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: volcanic eruption
Lava va kul — **volcanic eruption**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **avalanche** — qor ko'chkisi
• **drought** — qurg'oqchilik
• **flood** — toshqin

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. A) ✅ To'g'ri javob: much
luggage — sanalmaydi: **how much**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **many** — "many" — luggage sanalmaydi
• **few** — "few" — How few — noodatiy, sanaladigan
• **several** — "several" — sanaladigan va so'roqda How bilan kelmaydi

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. A) ✅ To'g'ri javob: We had a lot of homework yesterday.
homework — sanalmaydi.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **We had a lot of homeworks yesterday.** — "homeworks" — sanalmaydi
• **We had many homeworks yesterday.** — "many homeworks" — ikki xato
• **We had a homework yesterday.** — "a homework" — a olmaydi

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. A) ✅ To'g'ri javob: Would you mind closing the window?
mind + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Would you mind to close the window?** — bu fe'ldan keyin "to + fe'l" emas, "-ing" shakl keladi
• **Would you mind close the window?** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **Would you mind closed the window?** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. B) ✅ To'g'ri javob: am having
Reja — **am having**.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **have** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **had** — Past Simple — harakat kelajakda, o'tmishda emas
• **have had** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. D) ✅ To'g'ri javob: reads / is reading
Odat — **reads**; hozir — **is reading**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is reading / reads** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **read / is reading** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **reads / reads** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. A) ✅ To'g'ri javob: are / doing
are ... doing.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / do** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / do** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / doing** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. A) ✅ To'g'ri javob: used to have
used to have.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **has** — “has” — sochini kesgan
• **is having** — “is having” — hozirgi zamon
• **use to had** — “use to had” — used to + asosiy shakl

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. A) ✅ To'g'ri javob: took / passed / scored
took, passed, scored.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **took / beat / won** — “beat two defenders” — mumkin, lekin “won a goal” — gol "score" qilinadi
• **kicked / ran / made** — “made a goal” — score a goal
• **got / followed / shot** — “followed two defenders ... shot a goal” — ma'no buziladi

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: “arrived the group” (arrive to'ldiruvchi olmaydi), “followed a goal”, “looked the menu” (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. C) ✅ To'g'ri javob: was driving / heard
was driving / heard.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **drove / was hearing** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **was driving / was hearing** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **drives / heard** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. C) ✅ To'g'ri javob: He avoided buying a damaged product.
Nuqsonli mahsulot sotib olmadi.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **He got the jacket for free.** — bepul olgan — matnda yo'q
• **He found a cheaper shop.** — arzonroq do'kon — matnda yo'q
• **The sale finished early.** — chegirma erta tugagan — matnda yo'q

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. D) ✅ To'g'ri javob: Sharing work fairly leads to better results.
Ishni teng bo‘lish yaxshi natija beradi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Research is not important.** — tadqiqot muhim emas — matnda yo'q
• **Watching is the best way to learn.** — kuzatish eng yaxshi usul — teskari
• **Prizes are given randomly.** — sovrinlar tasodifiy — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 8-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

The person who takes your money at the checkout is the _____.

A) customer **B)** cashier **C)** refund **D)** supplier

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who likes meeting people and talking a lot?

A) shy **B)** quiet **C)** outgoing **D)** anxious

3. B **Choose the best answer.** 2 ball

Our basketball team _____ their rivals by ten points.

A) won **B)** drew **C)** scored **D)** beat

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a very strong tropical storm with violent winds?

A) hurricane **B)** avalanche **C)** earthquake **D)** drought

5. Q **Choose the correct sentence.** 2,8 ball

How _____ countries have you visited?

A) little **B)** much **C)** many **D)** a little

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) There isn't many money in my wallet.
B) There aren't much money in my wallet.
C) There isn't much money in my wallet.
D) There aren't many moneys in my wallet.

7. Q **Which sentence is grammatically correct?** 2,8 ball

A) They promised help us with the project.
B) They promised helped us with the project.
C) They promised helping us with the project.
D) They promised to help us with the project.

8. Q **Choose the best answer.** 2,8 ball

Our class _____ the science museum next Tuesday; the bus is booked.

A) visits **B)** is visiting **C)** has visited **D)** visited

9. Q **Choose the best answer.** 2,8 ball

The train _____ at 7:15 every day, but today it _____ late.

A) leave / is running **B)** leaves / is running

C) leaves / runs **D)** is leaving / runs

10. Q **Choose the best answer.** 2,8 ball

Why _____ you _____ so quickly? I can't understand you.

A) are / speaking **B)** do / speaking **C)** do / speak **D)** are / speak

11. Q **Choose the best answer.** 2,8 ball

We _____ to the seaside every summer when I was young.

A) are going **B)** use to went **C)** used to go **D)** go

12. Q **Choose the best answer.** 2,8 ball

When Ali got home, he _____ his shoes off, _____ some juice and _____ on the sofa.

A) took / poured / slept to **B)** put / ate / lay

C) made / had / went **D)** took / drank / sat

13. Q **Choose the best answer.** 2,8 ball

The students _____ a test when the fire alarm _____.

A) were writing / was going off **B)** were writing / went off

C) write / went off **D)** wrote / was going off

14. M **Read the text and choose the correct answer to the question.** 3 ball

Taxi App

Sara had to get to the airport at 4 a.m. Instead of waiting in the street, she booked a taxi through an app the night before. The car arrived on time and the price was fixed.

Why was booking in advance useful?

A) She got a reliable car at a known price. **B)** The taxi was free.

C) The driver was her friend. **D)** She didn't need to go to the airport.

15. M **Read the text and answer question.** 3 ball

Moving House

The Karimov family moved to a new flat. Dad carried furniture, Mum packed the kitchen, and the children packed their own rooms. They finished in one day instead of three.

Why did the family finish so quickly?

A) A company did all the work.

B) Everyone helped with a different job.

C) They had very few things.

D) The new flat was next door.

Javoblar kaliti — ChSB-1 tayyorgarlik: 8-variant

1 B — B	2 B — C	3 B — D	4 Q — A	5 Q — C	6 Q — C	7 Q — D
8 Q — B	9 Q — B	10 Q — A	11 Q — C	12 Q — D	13 Q — B	14 M — A
15 M — B						

1. B) ✅ To'g'ri javob: cashier
Kassir — **cashier**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **customer** — xaridor — pul to'lovchi
• **supplier** — yetkazib beruvchi
• **refund** — pul qaytarish — odam emas

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. C) ✅ To'g'ri javob: outgoing
Kirishimli — **outgoing**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **shy** — uyatchan — teskari
• **quiet** — jim — teskari
• **anxious** — xavotirli

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. D) ✅ To'g'ri javob: beat
Raqibni yengdi — **beat**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **won** — “won their rivals” — win + o'yin; raqib yengiladi (beat)
• **scored** — “scored their rivals” — ma'nosiz
• **drew** — “drew” — 10 ochko farq bilan durang bo'lmaydi

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. A) ✅ To'g'ri javob: hurricane
Tropik bo‘ron — **hurricane**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **drought** — qurg'oqchilik
• **earthquake** — zilzila
• **avalanche** — qor ko'chkisi

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. C) ✅ To'g'ri javob: many
countries — sanaladi: **how many**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **much** — "much" — countries sanaladigan
• **little** — "little" — sanalmaydigan
• **a little** — "a little" — sanalmaydigan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. C) ✅ To'g'ri javob: There isn't much money in my wallet.
money — sanalmaydi: **much money**.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **There aren't many moneys in my wallet.** — "aren't many moneys" — money sanalmaydi
• **There isn't many money in my wallet.** — "isn't many money" — much kerak
• **There aren't much money in my wallet.** — "aren't much money" — is kerak

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. D) ✅ To'g'ri javob: They promised to help us with the project.
promise + to.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **They promised helping us with the project.** — "promised helping" — promise dan keyin to + fe'l
• **They promised help us with the project.** — "promised help" — to tushib qolgan
• **They promised helped us with the project.** — "promised helped" — to + asosiy shakl kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. B) ✅ To'g'ri javob: is visiting
is visiting.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **visits** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **visited** — Past Simple — harakat kelajakda, o'tmishda emas
• **has visited** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. B) ✅ To'g'ri javob: leaves / is running
Jadval — **leaves**; bugun — **is running late**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is leaving / runs** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **leave / is running** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **leaves / runs** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. A) ✅ To'g'ri javob: are / speaking
are ... speaking.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / speak** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / speak** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / speaking** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. C) ✅ To'g'ri javob: used to go
used to go.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **go** — “go” — gap bolalik haqida
• **are going** — “are going” — hozirgi zamon
• **use to went** — “use to went” — used to + asosiy shakl

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. D) ✅ To'g'ri javob: took / drank / sat
took off, drank, sat.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **put / ate / lay** — “put his shoes off” — take off; “lay” — ma'no boshqa
• **made / had / went** — “made his shoes off” — ma'nosiz
• **took / poured / slept to** — “slept to on the sofa” — to ortiqcha

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: “arrived the group” (arrive to'ldiruvchi olmaydi), “followed a goal”, “looked the menu” (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. B) ✅ To'g'ri javob: were writing / went off
were writing / went off.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **wrote / was going off** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **were writing / was going off** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **write / went off** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. A) ✅ To'g'ri javob: She got a reliable car at a known price.
Ishonchli mashina, oldindan ma’lum narx.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **The taxi was free.** — taksi bepul — matnda narx belgilangan (fixed)
• **She didn't need to go to the airport.** — aeroportga borishi shart emas — teskari
• **The driver was her friend.** — haydovchi do'sti — matnda yo'q

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. B) ✅ To'g'ri javob: Everyone helped with a different job.
Har kim boshqa ish bilan yordam berdi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **They had very few things.** — narsalari kam — matnda yo'q
• **A company did all the work.** — kompaniya qilgan — teskari
• **The new flat was next door.** — yangi uy qo'shnida — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 9-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

When a shop sells products at lower prices for a short time, there is a _____.

A) sale **B)** refund **C)** receipt **D)** queue

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who thinks of new and original ideas?

A) reliable **B)** jealous **C)** creative **D)** lazy

3. B **Choose the best answer.** 2 ball

The person who trains a team and plans its tactics is the _____.

A) fan **B)** trophy **C)** referee **D)** coach

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a mass of earth and rocks sliding down a hill?

A) drought **B)** hurricane **C)** tsunami **D)** landslide

5. Q **Choose the correct sentence.** 2,8 ball

We need _____ information before we can decide.

A) more **B)** many **C)** several **D)** a few

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) He has made a lot of progresses in English.

B) He has made many progress in English.

C) He has made a progress in English.

D) He has made a lot of progress in English.

7. **Which sentence is grammatically correct?** 2,8 ball

A) I suggest to take the train instead of the bus.

B) I suggest taking the train instead of the bus.

C) I suggest take the train instead of the bus.

D) I suggest took the train instead of the bus.

8. **Choose the best answer.** 2,8 ball

Tom _____ his driving test tomorrow morning.

A) takes **B)** has taken **C)** is taking **D)** took

9. **Choose the best answer.** 2,8 ball

My friends _____ football on Sundays, but this Sunday they _____ a match on TV.

A) are playing / watch **B)** play / are watching

C) play / watch **D)** plays / are watching

10. **Choose the best answer.** 2,8 ball

Why _____ your brother _____ my laptop without asking?

A) does / use **B)** is / use **C)** is / using **D)** does / using

11. **Choose the best answer.** 2,8 ball

This town _____ very quiet, but now it's full of tourists.

A) use to was **B)** is being **C)** used to be **D)** is

12. **Choose the best answer.** 2,8 ball

At the shop, Mia _____ a dress, _____ it on and _____ it.

A) chose / tried / bought **B)** looked / dressed / sold

C) saw / put / took **D)** chose / wore / paid

13. **Choose the best answer.** 2,8 ball

Tom _____ his bike when he _____ off and hurt his arm.

A) was riding / fell **B)** was riding / was falling

C) rides / fell **D)** rode / was falling

14. M **Read the text and choose the correct answer to the question.** 3 ball

Comparing Laptops

Nodir wanted a laptop for school. The first model had a better camera, but the second had a longer battery life and was lighter. He chose the second because he carries it every day.

Why did Nodir choose the second laptop?

A) It was the most expensive.

B) His friend had the same one.

C) It had a better camera.

D) It was better for carrying and using all day.

15. M **Read the text and answer question.** 3 ball

Debate Team

Our debate team lost every round last year because one student prepared all the arguments. This year, everyone researched one topic and practised together. The team reached the final.

What was the main reason for the improvement?

A) The team shared the preparation.

B) The topics were easier.

C) They had a new teacher only.

D) The judges were friendlier.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 9-variant

1 B A	2 B C	3 B D	4 Q D	5 Q A	6 Q D	7 Q B
8 Q C	9 Q B	10 Q C	11 Q C	12 Q A	13 Q A	14 M D
15 M A						

1. A) ✅ **To'g'ri javob: sale**
Chegirmalar davri — **sale**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **receipt** — chek
• **queue** — navbat
• **refund** — pul qaytarish

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. C) ✅ **To'g'ri javob: creative**
Yangi g‘oyali — **creative**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **lazy** — dangasa
• **reliable** — ishonchli
• **jealous** — hasadgo'y

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. D) ✅ To'g'ri javob: coach
Murabbiy — **coach**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **referee** — hakam — jamoani o'qitmaydi
• **fan** — muxlis
• **trophy** — kubok — odam emas

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: landslide
Yer ko‘chkisi — **landslide**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **tsunami** — tsunami
• **drought** — qurg'oqchilik
• **hurricane** — dovul — tuproq ko'chishi emas

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. A) ✅ **To'g'ri javob: more**
information — sanalmaydi: **more**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **many** — "many" — information sanalmaydi
• **a few** — "a few" — sanaladigan
• **several** — "several" — sanaladigan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. D) ✅ **To'g'ri javob: He has made a lot of progress in English.**
progress — sanalmaydi.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **He has made a lot of progresses in English.** — "progresses" — sanalmaydi
• **He has made many progress in English.** — "many progress" — much/a lot of
• **He has made a progress in English.** — "a progress" — a olmaydi

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. B) ✅ To'g'ri javob: I suggest taking the train instead of the bus.
suggest + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **I suggest to take the train instead of the bus.** — bu fe'ldan keyin “to + fe'l” emas, “-ing” shakl keladi
• **I suggest take the train instead of the bus.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **I suggest took the train instead of the bus.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. C) ✅ To'g'ri javob: is taking
is taking.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **takes** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **took** — Past Simple — harakat kelajakda, o'tmishda emas
• **has taken** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. B) ✅ To'g'ri javob: play / are watching
Odat — **play**; bu yakshanba — **are watching**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **are playing / watch** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **plays / are watching** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **play / watch** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. C) ✅ To'g'ri javob: is / using
is ... using.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **does / use** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **is / use** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **does / using** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. C) ✅ To'g'ri javob: used to be
used to be.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **is** — “is” — hozir sayyohlar ko'p
• **is being** — “is being” — mos emas
• **use to was** — “use to was” — used to + be

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. A) ✅ To'g'ri javob: chose / tried / bought
chose, tried on, bought.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **chose / wore / paid** — "wore it on ... paid it" — try on; pay FOR
• **saw / put / took** — "put it on ... took it" — tanlash va sotib olish izchilligi buziladi
• **looked / dressed / sold** — "sold it" — xaridor sotmaydi

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: "arrived the group" (arrive to'ldiruvchi olmaydi), "followed a goal", "looked the menu" (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. A) ✅ To'g'ri javob: was riding / fell
was riding / fell.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **rode / was falling** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **was riding / was falling** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **rides / fell** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. D) ✅ To'g'ri javob: It was better for carrying and using all day.
Yengil va batareyasi uzoq — har kuni olib yurish uchun.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **It had a better camera.** — kamerasi yaxshiroq — teskari, birinchisining kamerasi yaxshi
• **It was the most expensive.** — eng qimmat — matnda yo'q
• **His friend had the same one.** — do'stida xuddi shunday — matnda yo'q

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. A) ✅ To'g'ri javob: The team shared the preparation.
Tayyorgarlikni bo‘lishishdi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **The topics were easier.** — mavzular osonroq — matnda yo'q
• **The judges were friendlier.** — hakamlar do'stona — matnda yo'q
• **They had a new teacher only.** — faqat yangi o'qituvchi — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: 10-variant

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

A card that you show to get points every time you shop is a _____ card.

A) loyalty **B)** refund **C)** price **D)** receipt

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who keeps their promises?

A) careless **B)** moody **C)** trustworthy **D)** dishonest

3. B **Choose the best answer.** 2 ball

Thousands of _____ were cheering for their team in the stadium.

A) draws **B)** referees **C)** fans **D)** trophies

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a storm in which balls of ice fall from the sky?

A) wildfire **B)** drought **C)** landslide **D)** hailstorm

5. Q **Choose the correct sentence.** 2,8 ball

There isn't _____ traffic on the roads at night.

A) a few **B)** several **C)** much **D)** many

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) Can I have some waters, please? **B)** Can I have a water, please?

C) Can I have some water, please? **D)** Can I have many water, please?

7. Q **Which sentence is grammatically correct?** 2,8 ball

A) She keeps to forget her keys. **B)** She keeps forgetting her keys.

C) She keeps forgot her keys. **D)** She keeps forget her keys.

8. Q **Choose the best answer.** 2,8 ball

What _____ you _____ this evening? — I'm meeting Mia at the café.

A) did / do **B)** do / do **C)** have / done **D)** are / doing

9. Q **Choose the best answer.** 2,8 ball

Tom _____ coffee, but today he _____ green tea because he has a stomach ache.

A) drinks / is drinking **B)** drink / is drinking

C) drinks / drinks **D)** is drinking / drinks

10. Q **Choose the best answer.** 2,8 ball

Why _____ you _____ ? Is something wrong?

A) do / crying **B)** are / crying **C)** do / cry **D)** are / cry

11. Q **Choose the best answer.** 2,8 ball

Tom _____ video games all night, but now he goes to bed early.

A) is playing **B)** plays **C)** use to played **D)** used to play

12. Q **Choose the best answer.** 2,8 ball

When the storm started, the farmer _____ the animals, _____ the gates and _____ inside.

A) took / shut / came to **B)** fed / opened / went to

C) fed / closed / ran **D)** brought / locked / stayed to

13. Q **Choose the best answer.** 2,8 ball

We _____ TV when somebody _____ at the door.

A) watch / knocked **B)** were watching / was knocking

C) were watching / knocked **D)** watched / was knocking

14. M **Read the text and choose the correct answer to the question.** 3 ball

Supermarket Card

Lola's mum used a loyalty card every time she shopped. After a few months, she had enough points to get $ 20 off her shopping.

What did the loyalty card give Lola's mum?

A) A free holiday **B)** A refund for old shopping

C) A discount from collected points **D)** A new card

15. M **Read the text and answer question.** 3 ball

Birthday Surprise

Friends planned a surprise party for Lola. Kate made the cake, Ben decorated the room, and Sara invited the guests secretly. Lola was completely surprised and very happy.

Why did the surprise work well?

A) Everyone did their part and kept it secret.

B) Lola planned the party herself.

C) The party was cancelled.

D) Only Kate helped.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 10-variant

1 B A	2 B C	3 B C	4 Q D	5 Q C	6 Q C	7 Q B
8 Q D	9 Q A	10 Q B	11 Q D	12 Q C	13 Q C	14 M C
15 M A						

1. A) ✅ **To'g'ri javob: loyalty**
Bonus karta — **loyalty card**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **receipt** — chek
• **refund** — pul qaytarish
• **price** — narx

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. C) ✅ **To'g'ri javob: trustworthy**
Va'dasiga sodiq — **trustworthy**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **dishonest** — insofsiz — teskari
• **moody** — kayfiyati o'zgaruvchan
• **careless** — e'tiborsiz

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. C) ✅ To'g'ri javob: fans
Muxlislar — **fans**.

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** — durang; **break the rules** — qoidani buzmoq; **lift the trophy** — kubokni ko'tarmoq; **train** — mashq qilmoq; **referee** — hakam; **coach** — murabbiy; **fan** — muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **referees** — hakamlar — minglab bo'lmaydi va olqishlamaydi
• **trophies** — kuboklar
• **draws** — duranglar

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" — xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: hailstorm
Do‘l bo‘roni — **hailstorm**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** — aylanma kuchli shamol (quyun); **hurricane** — tropik dovul; **earthquake** — zilzila; **flood** — suv toshqini; **drought** — qurg'oqchilik; **avalanche** — qor ko'chkisi; **landslide** — tuproq/tosh ko'chkisi; **tsunami** — dengiz ostidagi zilziladan katta to'lqin; **wildfire** — o'rmon yong'ini; **volcanic eruption** — vulqon otilishi; **hailstorm** — do'lli bo'ron; **lightning** — chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **wildfire** — o'rmon yong'ini
• **drought** — qurg'oqchilik
• **landslide** — tuproq ko'chkisi

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) — ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** — to'lqin, **flood** — daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. C) ✅ To'g'ri javob: much
traffic — sanalmaydi: **much**.

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **many** — “many” — traffic sanalmaydi
• **a few** — “a few” — sanaladigan
• **several** — “several” — sanaladigan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. C) ✅ To'g'ri javob: Can I have some water, please?
water — sanalmaydi: **some water**.

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Can I have a water, please?** — “a water” — umumiy suv sanalmaydi (a glass of water)
• **Can I have some waters, please?** — “waters” — -s olmaydi
• **Can I have many water, please?** — “many water” — sanalmaydi

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. B) ✅ To'g'ri javob: She keeps forgetting her keys.
keep + -ing.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **She keeps to forget her keys.** — bu fe'ldan keyin “to + fe'l” emas, “-ing” shakl keladi
• **She keeps forget her keys.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **She keeps forgot her keys.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. D) ✅ To'g'ri javob: are / doing
Reja haqida savol — **are ... doing**.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / do** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **did / do** — Past Simple — harakat kelajakda, o'tmishda emas
• **have / done** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. A) ✅ To'g'ri javob: drinks / is drinking
Odat — **drinks**; bugun — **is drinking**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is drinking / drinks** — zamonlar almashgan: odat Continuousda, vaqtinchalik holat Simpleda
• **drink / is drinking** — ega bilan moslik xatosi (he/she → -s; we/they → -s yo'q)
• **drinks / drinks** — ikkinchi qism ham Present Simple — u vaqtinchalik holat (Continuous kerak)

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. B) ✅ To'g'ri javob: are / crying
are ... crying.

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / cry** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / cry** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / crying** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. D) ✅ To'g'ri javob: used to play
used to play.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **plays** — “plays” — hozir erta yotadi
• **is playing** — “is playing” — hozirgi zamon
• **use to played** — “use to played” — used to + asosiy shakl

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. C) ✅ To'g'ri javob: fed / closed / ran
fed, closed, ran inside.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **fed / opened / went to** — “opened the gates” — bo'ronda yopiladi; “went to inside” — to ortiqcha
• **took / shut / came to** — “came to inside” — to ortiqcha
• **brought / locked / stayed to** — “stayed to inside” — to ortiqcha

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: “arrived the group” (arrive to'ldiruvchi olmaydi), “followed a goal”, “looked the menu” (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. C) ✅ To'g'ri javob: were watching / knocked
were watching / knocked.

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **watched / was knocking** — zamonlar almashgan: uzoq harakat Simple, qisqa voqea Continuous bo'lib qolgan
• **were watching / was knocking** — ikkalasi ham Continuous — qisqa voqea (when-qism) Past Simple bo'ladi
• **watch / knocked** — hozirgi zamon — gap o'tmish haqida

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. C) ✅ To'g'ri javob: A discount from collected points
To‘plangan ballardan chegirma.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **A free holiday** — bepul sayohat — matnda yo'q
• **A new card** — yangi karta — matnda yo'q
• **A refund for old shopping** — eski xaridlar uchun pul qaytarish — matnda ballardan chegirma

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. A) ✅ To'g'ri javob: Everyone did their part and kept it secret.
Har kim o‘z ishini qildi va sir saqladi.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **Lola planned the party herself.** — Lola o'zi rejalashtirgan — teskari, bu syurpriz
• **The party was cancelled.** — ziyofat bekor qilingan — teskari
• **Only Kate helped.** — faqat Kate yordam bergan — teskari

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · INGLIZ TILI · CHSB-1

ChSB-1 tayyorgarlik: I chorak demo varianti

15 ta savol · 45 daqiqa · 40 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B **Choose the best answer.** 2 ball

If a shop gives you 10% off, you receive a _____.

A) receipt **B)** discount **C)** refund **D)** promotion

2. B **Choose the best answer.** 2 ball

Which adjective describes someone who can always be trusted?

A) jealous **B)** sensitive **C)** reliable **D)** anxious

3. B **Choose the best answer.** 2 ball

Our basketball team played confidently and finally _____ after extra time.

A) beat the other team **B)** reached a goalkeeper

C) scored the referee **D)** trained the stadium

4. Q **Choose the best answer.** 2,8 ball

Which natural event is a dangerous wind that moves in a circle?

A) flood **B)** earthquake **C)** lightning **D)** tornado

5. Q **Choose the correct sentence.** 2,8 ball

Although the website contained _____ useful information, we still needed to ask the teacher for help.

A) many **B)** a few **C)** a lot of **D)** several

6. Q **Choose the grammatically correct sentence.** 2,8 ball

A) There aren't many furnitures in our town.

B) We bought a few new furniture yesterday.

C) There isn't much information about the sale online.

D) She gave me many useful advice.

7. **Which sentence is grammatically correct?** 2,8 ball

A) My brother doesn't mind to help his classmates.

B) My brother doesn't mind helping his classmates.

C) My brother doesn't mind help his classmates.

D) My brother doesn't mind helped his classmates.

8. **Choose the best answer.** 2,8 ball

Emma has a presentation tomorrow. She _____ her classmates after school to prepare together.

A) meets **B)** met **C)** is meeting **D)** has met

9. **Choose the best answer.** 2,8 ball

My older brother _____ computer games every evening, but tonight he _____ for his exam.

A) is playing / studies **B)** played / studied

C) plays / is studying **D)** play / studies

10. **Choose the best answer.** 2,8 ball

Why _____ you _____ so much noise? I'm trying to concentrate.

A) do / make **B)** are / making **C)** are / make **D)** do / making

11. **Choose the best answer.** 2,8 ball

Last year, my grandfather _____ near the sea, but now he lives in the city.

A) is living **B)** lives **C)** used to live **D)** was living

12. **Choose the best answer.** 2,8 ball

During the hiking trip, the weather suddenly changed. The guide _____ everyone to stay together, _____ a safer route, and _____ the group back to the campsite.

A) warned / chose / led **B)** warned / selected / arrived

C) told / decided / returned **D)** called / looked / followed

13. Q **Choose the best answer.** 2,8 ball

I _____ home when I suddenly _____ my English teacher.

A) walked / was meeting
B) was walking / met
C) met / was walking
D) walk / met

14. M **Read the text and choose the correct answer to the question.** 3 ball

Shopping Smart

Nina wanted to buy a new backpack. The first shop sold a fashionable one for $ 60. She almost bought it immediately. Before paying, she checked two more shops. The second shop had exactly the same backpack for $ 45 and also offered a free water bottle. Nina bought it there and used the money she saved to buy a notebook for school.

Why was checking more than one shop a good decision?

A) She found a different colour to buy backpack.
B) She spent less money and received an extra gift.
C) She decided not to buy a backpack to save money.
D) She exchanged her old backpack to the new one.

15. M **Read the text and answer question.** 3 ball

Team Project

Four students had to prepare a presentation. Amir searched for information, Maya designed the slides, Leo practiced speaking, and Sara checked all the facts. They finished one day early because everyone completed their own task on time and helped each other whenever someone needed support.

Which statement best explains why the group succeeded?

A) One student did most of the work.
B) The project was very easy to explain and presenting.
C) Each member had a clear role and cooperated.
D) The teacher completed the difficult parts and helped for checking.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: I chorak demo varianti

1 B — B	2 B — C	3 B — A	4 Q — D	5 Q — C	6 Q — C	7 Q — B
8 Q — C	9 Q — C	10 Q — B	11 Q — C	12 Q — A	13 Q — B	14 M — B
15 M — C						

1. B) ✅ To'g'ri javob: discount
10% chegirma — **discount**.

📘 **Qoida.** Xarid so'zlari: **discount** — chegirma (10% off); **receipt** — chek; **refund** — pulni qaytarish; **bargain** — arzon, foydali xarid; **promotion** — aksiya (buy one, get one free); **instalments** — bo'lib to'lash; **checkout** — kassa; **customer** — xaridor; **cashier** — kassir; **supplier** — yetkazib beruvchi; **sale** — arzonlashtirilgan savdo; **loyalty card** — bonus (sodiqlik) kartasi; **queue** — navbat.

❌ **Nega boshqa variantlar noto'g'ri:**
• **receipt** — chek — chegirma emas, to'lov qog'ozi
• **refund** — pul qaytarish — mahsulot qaytarilganda
• **promotion** — aksiya — "bitta olsangiz, biri bepul" kabi maxsus taklif

⚠️ **Adashtiradigan joy.** Juda o'xshash so'zlar: **discount** (narx kamayishi) — **refund** (pul qaytishi) — **promotion** (maxsus taklif) — **sale** (arzonlashtirish davri). **customer** (sotib oluvchi) va **cashier** (pul oluvchi) — ikki xil odam.

💡 **Eslab qoling:** 10% off = discount; money back = refund; paper = receipt; low price, good thing = bargain.

2. C) ✅ To'g'ri javob: reliable
Doim ishonsa bo'ladigan — **reliable**.

📘 **Qoida.** Xarakter sifatlari: **reliable / trustworthy** — ishonchli; **sensitive** — ta'sirchan (tez xafa bo'ladi); **anxious** — xavotirli ↔ **relaxed** — xotirjam; **jealous** — hasadgo'y; **modest** — kamtar ↔ **arrogant** — takabbur; **hard-working** — mehnatkash ↔ **lazy**; **stubborn** — o'jar ↔ **flexible** — moslashuvchan; **outgoing** — dilkash, kirishimli ↔ **shy**; **creative** — ijodkor; **dishonest** — insofsiz.

❌ **Nega boshqa variantlar noto'g'ri:**
• **jealous** — hasadgo'y — ishonch haqida emas
• **sensitive** — ta'sirchan — tez xafa bo'lish
• **anxious** — xavotirli — ko'p tashvishlanish

⚠️ **Adashtiradigan joy. sensitive** — "sezgir, ta'sirchan" (sensible — "oqilona" bilan chalkashtirmang). **anxious** va **nervous** ma'nosi yaqin, ikkalasi ham relaxed ning teskarisi.

💡 **Eslab qoling:** Ta'rifdagi kalit fe'lni toping: trusted → reliable; feels hurt → sensitive; worries → anxious; wants what others have → jealous.

3. A) ✅ To'g'ri javob: beat the other team
Qo'shimcha vaqtdan keyin **beat the other team** (raqibni yengdi).

📘 **Qoida.** Sport iboralari (collocations): **beat** + raqib (beat the other team); **win** + o'yin/sovrin (win the match, win the cup); **score** + gol/ochko (score a goal); **draw** – durang; **break the rules** – qoidani buzmoq; **lift the trophy** – kubokni ko'tarmoq; **train** – mashq qilmoq; **referee** – hakam; **coach** – murabbiy; **fan** – muxlis.

❌ **Nega boshqa variantlar noto'g'ri:**
• **reached a goalkeeper** – "darvozabonga yetdi" – ma'nosiz
• **scored the referee** – "hakamni gol qildi" – ma'nosiz
• **trained the stadium** – "stadionni mashq qildirdi" – ma'nosiz

⚠️ **Adashtiradigan joy. win** va **beat** ni farqlang: **win** + match/game/cup (nima yutildi), **beat** + team/opponent (kim yengildi). Gol **score** qilinadi, "win/beat a goal" – xato.

💡 **Eslab qoling:** beat SOMEONE, win SOMETHING, score A GOAL.

4. D) ✅ To'g'ri javob: tornado
Aylanma harakatlanadigan xavfli shamol – **tornado**.

📘 **Qoida.** Tabiiy ofatlar: **tornado** – aylanma kuchli shamol (quyun); **hurricane** – tropik dovul; **earthquake** – zilzila; **flood** – suv toshqini; **drought** – qurg'oqchilik; **avalanche** – qor ko'chkisi; **landslide** – tuproq/tosh ko'chkisi; **tsunami** – dengiz ostidagi zilziladan katta to'lqin; **wildfire** – o'rmon yong'ini; **volcanic eruption** – vulqon otilishi; **hailstorm** – do'lli bo'ron; **lightning** – chaqmoq.

❌ **Nega boshqa variantlar noto'g'ri:**
• **flood** – toshqin – suv, aylanma shamol emas
• **earthquake** – zilzila – yer silkinishi
• **lightning** – chaqmoq – elektr razryadi

⚠️ **Adashtiradigan joy. avalanche** (qor) va **landslide** (tuproq, tosh) – ikkalasi ko'chki, lekin materiali farqli. **tornado** (aylanma) va **hurricane** (dengizda paydo bo'ladigan ulkan bo'ron). **tsunami** – to'lqin, **flood** – daryo toshishi.

💡 **Eslab qoling:** Har bir ofatni asosiy so'zi bilan eslang: circle → tornado, shakes → earthquake, snow → avalanche, wave → tsunami, lava → eruption.

5. C) ✅ To'g'ri javob: a lot of
information — sanalmaydigan ot: **a lot of** information (many, a few, several — sanaladiganlar uchun).

📘 **Qoida.** Miqdor so'zlari: **sanaladigan ko'plik** (friends, students, countries) — **many, a few, several, few**; **sanalmaydigan** (information, time, advice, milk, luggage, traffic) — **much, a little, little**; ikkalasi bilan ham — **a lot of, some, any, more**. **so many / so much ... that** — shunchalik ko'p ... ki.

❌ **Nega boshqa variantlar noto'g'ri:**
• **many** — "many" — information sanalmaydi
• **a few** — "a few" — sanaladigan otlar bilan
• **several** — "several" — sanaladigan ko'plik bilan

⚠️ **Adashtiradigan joy.** Avval otni aniqlang: sanaladimi? **information, advice, luggage, traffic, news, furniture** — o'zbek tilida sanalsa ham ingliz tilida SANALMAYDI (many informations — xato). **any** odatda inkor va so'roqda; tasdiq gapda "a lot of".

💡 **Eslab qoling:** many/few — sanaladigan; much/little — sanalmaydigan; a lot of/some — ikkalasi.

6. C) ✅ To'g'ri javob: There isn't much information about the sale online.
information — sanalmaydigan: There isn't **much** information. (furniture, advice ham sanalmaydi.)

📘 **Qoida.** Sanalmaydigan otlar: **advice, furniture, information, luggage, traffic, news, homework, money, progress, bread, water, cheese (umuman)** — ko'plik **-s** olmaydi, **a/an** olmaydi, **much / some / a lot of / a little** bilan keladi va birlik fe'l oladi (**is / was**). Dona qilish uchun: **a piece of advice, a loaf of bread, a glass of water**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **There aren't many furnitures in our town.** — "furnitures" — furniture sanalmaydi, -s olmaydi; many ham emas
• **We bought a few new furniture yesterday.** — "a few ... furniture" — a few sanaladigan otlar bilan
• **She gave me many useful advice.** — "many useful advice" — advice sanalmaydi: much/some advice

⚠️ **Adashtiradigan joy.** O'zbek tilidagi ma'noga aldanmang: "maslahatlar", "mebellar" deymiz, lekin inglizcha **advice, furniture** ko'plik bo'lmaydi. **news** -s bilan tugasa ham BIRLIK: The news **was** ...

💡 **Eslab qoling:** Sanalmaydigan ot + much / some / a lot of + is/was; -s yo'q, a/an yo'q.

7. B) ✅ To'g'ri javob: My brother doesn't mind helping his classmates.
mind + -ing: doesn't mind **helping**.

📘 **Qoida. Fe'l + -ing** (gerund): **enjoy, mind, finish, can't stand, avoid, give up, suggest, keep, would you mind → V-ing**: doesn't mind **helping**, enjoys **reading**. **Fe'l + to + V**: **decide, promise, want, hope, plan, need → to do**: decided **to go**, promised **to help**.

❌ **Nega boshqa variantlar noto'g'ri:**
• **My brother doesn't mind to help his classmates.** — bu fe'ldan keyin "to + fe'l" emas, "-ing" shakl keladi
• **My brother doesn't mind help his classmates.** — asosiy shakl — bu fe'ldan keyin -ing kerak
• **My brother doesn't mind helped his classmates.** — o'tgan zamon shakli — bu fe'ldan keyin -ing kerak

⚠️ **Adashtiradigan joy.** Fe'lni ro'yxatdan tekshiring: enjoy/mind/avoid/suggest — faqat -ing (enjoy to read — xato); decide/promise — faqat to (decided going — xato). Asosiy shakl (mind help) va o'tgan zamon (mind helped) bu yerda umuman ishlatilmaydi.

💡 **Eslab qoling:** Gerund guruhi: enjoy, mind, finish, avoid, suggest, keep, give up, can't stand. To-infinitive guruhi: decide, promise, want, hope, plan.

8. C) ✅ To'g'ri javob: is meeting
Ertangi kelishilgan reja — **is meeting**.

📘 **Qoida. Present Continuous kelasi zamon ma'nosida** — oldindan KELISHILGAN, rejalashtirilgan ish (chipta olingan, uchrashuv belgilangan): He **is flying** to Istanbul on Friday. Belgilari: tomorrow, on Friday, next week/month, tonight, this weekend + "has bought tickets / made an appointment / everything is arranged".

❌ **Nega boshqa variantlar noto'g'ri:**
• **meets** — Present Simple — kelishilgan shaxsiy reja uchun Continuous ishlatiladi
• **met** — Past Simple — harakat kelajakda, o'tmishda emas
• **has met** — Present Perfect — tugallangan harakat, kelajak reja emas

⚠️ **Adashtiradigan joy.** Present Simple kelasi zamonda faqat jadval (poyezd/kino vaqti) uchun; shaxsiy kelishuvlar — Continuous. Past Simple va Present Perfect — o'tgan harakatlar, ular "tomorrow / next week" bilan kelmaydi.

💡 **Eslab qoling:** Kelishilgan reja + vaqt (tomorrow, next ...) → am/is/are + -ing.

9. C) ✅ To'g'ri javob: plays / is studying
Har kuni — **plays**; bugun kechqurun — **is studying**.

📘 **Qoida. every evening / three times a week / usually** → odat → **Present Simple**; **tonight / this week / this term / today / at the moment** → vaqtinchalik → **Present Continuous**. Ega bilan moslik: he/she/it → **-s**, **is**; we/they → **are**, -s yo'q.

❌ **Nega boshqa variantlar noto'g'ri:**
• **is playing / studies** — “is playing / studies” — zamonlar almashgan
• **played / studied** — “played / studied” — o'tgan zamon; every evening va tonight hozirgi zamonni talab qiladi
• **play / studies** — “play / studies” — brother (he) → plays; tonight → is studying

⚠️ **Adashtiradigan joy.** Ikki bo'sh joyni alohida tekshiring: 1) zamonlar almashmaganmi; 2) ega bilan moslik (My friends **plays** — xato, Ali **have** — xato); 3) ikkinchi qism ham Simple bo'lib qolmaganmi.

💡 **Eslab qoling:** usually ... V(-s) / but today ... am/is/are + -ing.

10. B) ✅ To'g'ri javob: are / making
Hozir sodir bo'layotgan harakat: Why **are** you **making** so much noise?

📘 **Qoida. Present Continuous** hozir bo'layotgan (ko'pincha bezovta qiluvchi) harakat haqida so'roq: **Why are you making** so much noise? Tuzilma: **(Why/What) + am/is/are + ega + V-ing?**

❌ **Nega boshqa variantlar noto'g'ri:**
• **do / make** — “do/does + fe'l” — odat haqida so'roq; harakat hozir bo'layapti
• **are / make** — “am/is/are” dan keyin fe'l -ing shaklida bo'lishi kerak
• **do / making** — “do/does” bilan -ing shakl ishlatilmaydi

⚠️ **Adashtiradigan joy.** Gapdagi kontekst ("I'm trying to concentrate", "The baby is asleep") — harakat AYNI HOZIR bo'layapti; “do/does” bilan odat so'raladi. Aralash shakllar: “are ... make” (-ing yo'q), “do ... making” (do bilan -ing).

💡 **Eslab qoling:** Hozir bezovta qilayotgan harakat → Why are you ...-ing?

11. C) ✅ To'g'ri javob: used to live
O'tmishdagi, hozir yo'q holat: he **used to live** near the sea.

📘 **Qoida. used to + fe'l** — o'tmishdagi odat yoki holat, hozir esa yo'q: I **used to be** afraid of the dark (hozir qo'rqmayman). Inkor/so'roqda: didn't **use to**, Did you **use to**...? There **used to be** a cinema (avval bor edi).

❌ **Nega boshqa variantlar noto'g'ri:**
• **is living** — “is living” — hozirgi zamon; gap o'tgan yil haqida
• **lives** — “lives” — hozirgi zamon
• **was living** — “was living” — o'tmishdagi davomli jarayon, o'tmish odati/holati emas; "used to" aniqroq

⚠️ **Adashtiradigan joy.** "but now ..." — hozirgi holat o'tmishdagidan farq qiladi → used to. **used to + ASOSIY shakl** (used to live, used to play) — “use to played”, “use to was” xato. Hozirgi zamon (lives, is living) o'tmishni bildirmaydi. **be used to + -ing** — "ko'nikmoq", boshqa ma'no.

💡 **Eslab qoling:** Avval bor edi, hozir yo'q → USED TO + V1.

12. A) ✅ To'g'ri javob: warned / chose / led
Ketma-ket harakatlar: **warned** everyone, **chose** a route, **led** the group back.

📘 **Qoida.** Ketma-ket o'tgan harakatlar — Past Simple, va har bir fe'l o'zidan keyingi so'z bilan TO'G'RI birikma (collocation) hosil qilishi kerak: **warn** someone **to** do; **choose** a route; **lead** a group back; **check in**; **take** bags upstairs; **take off** shoes; **try on** a dress; **ask for** the bill; **unfasten** seat belts.

❌ **Nega boshqa variantlar noto'g'ri:**
• **warned / selected / arrived** — "selected ... arrived the group" — arrive to'ldiruvchi olmaydi (arrive at)
• **told / decided / returned** — "decided a safer route ... returned the group" — decide + route birikmaydi
• **called / looked / followed** — "called ... looked a route ... followed the group" — ma'no buziladi (look at, guide yetaklaydi)

⚠️ **Adashtiradigan joy.** Barcha variantlar o'tgan zamonda — farq MA'NO va BIRIKMADA. Har bir fe'lni o'z o'rniga qo'yib o'qing: "arrived the group" (arrive to'ldiruvchi olmaydi), "followed a goal", "looked the menu" (look AT) — xato.

💡 **Eslab qoling:** Fe'lni bo'sh joyga qo'yib, butun iborani ovoz chiqarib o'qing — ma'nosiz birikma darrov bilinadi.

13. B) ✅ To'g'ri javob: was walking / met
Davom etayotgan harakat (was walking) + to'satdan voqea (met).

📘 **Qoida. Past Continuous + Past Simple**: uzoq davom etgan fon harakat — **was/were + V-ing**, uni bo'lgan qisqa voqea — **Past Simple**, odatda **when** bilan: I **was walking** home when I suddenly **met** my teacher.

❌ **Nega boshqa variantlar noto'g'ri:**
• **walked / was meeting** — "walked / was meeting" — zamonlar almashgan
• **met / was walking** — "met / was walking" — ma'no teskari: uchrashish qisqa voqea, yurish — fon
• **walk / met** — "walk / met" — hozirgi zamon

⚠️ **Adashtiradigan joy.** Qaysi harakat UZOQ (fon) va qaysi QISQA (to'satdan) ekanini aniqlang: walking — uzoq, met — qisqa. Ikkalasini ham Continuous qilish, almashtirish yoki hozirgi zamon ishlatish — xato.

💡 **Eslab qoling:** was/were + -ing WHEN + V2.

14. B) ✅ To'g'ri javob: She spent less money and received an extra gift.
Ikkinchi do'konda 45 dollar — **arzonroq** va **bepul suv idishi** berildi.

📘 **Qoida.** Detallar va asoslangan qaror: savoldagi "Why was ... a good decision/choice?" — matndagi raqamlar va afzalliklarni solishtiring (narx, vaqt, kafolat, sovg'a) va BARCHASINI qamrab olgan javobni tanlang.

❌ **Nega boshqa variantlar noto'g'ri:**
• **She found a different colour to buy backpack.** — boshqa rang — matnda xuddi o'sha ryukzak
• **She decided not to buy a backpack to save money.** — ryukzak sotib olmaslik — u sotib olgan
• **She exchanged her old backpack to the new one.** — eskisini almashtirish — matnda yo'q

⚠️ **Adashtiradigan joy.** Qisman to'g'ri variantlar chalg'itadi ("only a free case" — faqat bir qismi). Matnda yo'q yoki teskari ma'lumot beruvchi variantlarni raqamlar bilan tekshiring ($ 45 < $ 60 → arzonroq).

💡 **Eslab qoling:** Raqamlarni yonma-yon yozing: narx, vaqt, masofa — farq javobni ko'rsatadi.

15. C) ✅ **To'g'ri javob: Each member had a clear role and cooperated.**
Har kimning **aniq vazifasi bo'lgan va ular hamkorlik qilgan**.

📘 **Qoida.** Asosiy fikr (main idea): guruh ishi haqidagi matnlarda kalit belgi — **har kimning aniq vazifasi (role/task)** va **hamkorlik (helped each other, together)**. To'g'ri javob ikkalasini qamrab oladi.

❌ **Nega boshqa variantlar noto'g'ri:**
• **One student did most of the work.** — bitta o'quvchi ko'p ish qilgan — teskari, har kim o'z vazifasini bajargan
• **The project was very easy to explain and presenting.** — loyiha juda oson — matnda yo'q
• **The teacher completed the difficult parts and helped for checking.** — o'qituvchi qiyin qismlarni qilgan — matnda yo'q

⚠️ **Adashtiradigan joy.** Chalg'ituvchilar teskari holatni (bitta odam hammasini qildi), matnda yo'q tashqi yordamni (o'qituvchi qildi, professional muhandislar) yoki tasodifiy omilni (oson edi) aytadi.

💡 **Eslab qoling:** "Everyone did their part + worked together" — guruh muvaffaqiyati haqidagi matnlarning odatiy asosiy fikri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 1-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $C_2H_5COONa + NaOH \xrightarrow{t^\circ} ...$

2) $CH_4 + Cl_2 \xrightarrow{h\nu} ...$

3) $CH_2{=}CH_2 + Br_2 \rightarrow ...$

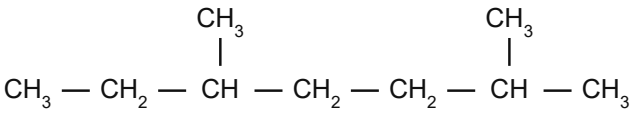
4) $Be_2C + H_2O \rightarrow ...$

5) $C_2H_5OH \xrightarrow{H_2SO_4,\ 170^\circ C} ...$

6) $CH_3COONa + H_2O \xrightarrow{elektroliz} ...$

A) 1, 2, 4 **B)** 1, 4, 6 **C)** 4, 6 **D)** 1, 4, 5, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 2-metilgeptan **B)** 5-metilgeptan

C) 2,5-dimetilgeptan **D)** 3,6-dimetilgeptan

3. B Sikloalkanlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) kumol; 2) 1,2-dimetilsiklopropan; 3) 3-metilbuten-1; 4) siklogeksan; 5) 1,3-dimetilsiklobutan; 6) atsetilen

A) 2, 4, 6 **B)** 2, 4, 5 **C)** 1, 2, 4, 5 **D)** 4, 5, 6

4. Q 11,6 gramm 2-metilpropan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 1-xlor-1,1-dimetiletan – 27,8 g **B)** 2-xlor-2-metilpropan – 18,5 g

C) 1-xlor-2-metilpropan – 18,5 g **D)** 2-xlor-2-metilpropan – 27,8 g

5. C_6H_{14} va C_8H_{18} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 8,6 mol CO_2 va 9,8 mol H_2O hosil bo'ldi. Dastlabki aralashma massasini (g) aniqlang. 1,5 ball

A) 126,1 **B)** 132 **C)** 122,8 **D)** 113

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 71,4 gramm aralashma mavjud. Aralashmadagi molekulalar soni $7{,}224 \cdot 10^{23}$ ga teng. Shu aralashma 1,2 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 138,35 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_7H_{14} **B)** C_6H_{12} **C)** C_5H_{10} **D)** C_8H_{16}

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 43,6 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 4% li bromli suvning 2400 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 2:3 butadiyen **B)** buten 3:1 butadiyen

C) buten 1:4 butadiyen **D)** buten 1:3 butadiyen

8. Kalsiy karbid va magniy nitrid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 26,88 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=23,75 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi magniy nitrid massasini (g) aniqlang. 1,5 ball

A) 57,6 **B)** 22,5 **C)** 30 **D)** 15

9. 148,4 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 138,8 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 329 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol siklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 0,6 **B)** 0,8 **C)** 1,6 **D)** 1

10. Q 0,4 mol organik modda tarkibida 38,4 gramm uglerod va 6,4 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_9H_{18} **B)** C_7H_{16} **C)** C_8H_{14} **D)** C_8H_{16}

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 4,8 mol kislorod sarflangani va 3,6 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda benzoy kislota va sirka kislota, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) 1-metil-2-etilbenzol **B)** 1-metil-4-etilbenzol

C) 1,4-dietilbenzol **D)** propilbenzol

12. Q Formulasi C_6H_{12} va asosiy halqasida 4 ta uglerod atomi bo'lgan (siklobutan hosilalari), o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 3 ta **B)** 5 ta **C)** 6 ta **D)** 4 ta

13. Q 3-metilgeptan molekulasida nechta uchlamchi uglerod atomi mavjud? 1,5 ball

A) 4 ta **B)** 2 ta **C)** 1 ta **D)** 3 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball

1) 2,2-dimetilpropan molekulasida 3 ta birlamchi uglerod atomi bor;
2) pentin-1 va izopren bir-biriga gomolog;
3) 2-metilpentan va 3-metilpentan molyar massalari bir xil;
4) siklopropan va propen xlor bilan birikkanda hosil bo'lgan mahsulotlarning tuzilishi bir xil bo'ladi;
5) atsetilen kumush oksidning ammiakdagi eritmasi bilan reaksiyaga kirishadi

A) 3, 4, 5 **B)** 1, 3, 5 **C)** 1, 5 **D)** 3, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_9H_{20} bo'lgan, asosiy zanjirida 6 ta yoki 6 tadan ko'proq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlarni ularning molekulasidagi birlamchi uglerod atomlari soniga ko'ra nechta guruhga bo'lish mumkin?

A) 6 ta **B)** 4 ta **C)** 5 ta **D)** 3 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 1-variant

1 B	B	2 B	C	3 B	B	4 Q B
5 Q	C	6 Q	B	7 Q	D	8 Q D
9 Q	A	10 Q	D	11 Q	D	12 Q D
13 Q	C	14 M	D	15 M	B	

1. B) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 1, 4, 6.

2. C) Eng uzun uglerod zanjiri 7 ta atomdan iborat (geptan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 2,5-dimetilgeptan.

3. B) Sikloalkanlar: 1,3-dimetilsiklobutan, metilsiklopentan, siklogeksan va h.k.. Javob: 2, 4, 5.

4. B) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2-metilpropan. n = 11,6/58 = 0,2 mol; m = 0,2·92,5 = 18,5 g.

5. C) m = m(C) + m(H) = 12·8,6 + 2·9,8 = 122,8 g.

6. B) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 1,2; mahsulotlar: 113a + (M + 34,5)b = 138,35; 42a + M·b = 71,4 → a = 0,7, b = 0,5, M = 84 → C_6H_{12}.

7. D) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2) = 2400·4/100/160 = 0,6$ mol = n(butadiyen). m(butadiyen) = 32,4 g; n(buten) = (43,6 − 32,4)/56 = 0,2 mol → buten 1:3 butadiyen.

8. D) Gazlar: C_2H_2 (26) va NH_3 (17). n = 1,2 mol; 26·x + 17·(1,2 − x) = 1,2·23,75 → $n(C_2H_2) = 0,9$ mol, $n(NH_3) = 0,3$ mol. $m(Mg_3N_2) = 15$ g.

9. A) $n(H_2) = (148,4 - 138,8)/2 = 4,8$ mol = 3·(a + b) → a + b = 1,6. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 329/329 = 1 mol = a → b = 0,6 mol.

10. D) n(C) = 38,4/12 = 3,2 mol, n(H) = 6,4 mol; bitta molekulada C: 3,2/0,4 = 8, H: 6,4/0,4 = 16 → C_8H_{16}.

11. D) $C_nH_{2n-6} + (3n - 3)/2\ O_2 \rightarrow nCO_2 + (n - 3)H_2O$. $n(O_2)/n(CO_2) = (3n - 3)/(2n) = 4,8/3,6$ → n = 9 (C_9H_{12}). Oksidlanish mahsulotlari bo'yicha — propilbenzol.

12. D) Halqaga 2 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 4 ta izomer.

13. C) 3-metilgeptan: birlamchi – 3, ikkilamchi – 4, uchlamchi – 1, to'rtlamchi – 0 ta uglerod atomi.

14. D) To'g'ri: 3, 5. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. B) Shartga mos 27 ta izomer bor; ulardagi birlamchi uglerod atomlari soni: 2, 3, 4, 5 → 4 ta guruh.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 2-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $CH_3COONa + H_2O \xrightarrow{\text{elektroliz}} ...$

2) $HC{\equiv}CH + 2H_2 \xrightarrow{Ni,\ t^\circ} ...$

3) $CH_3COONa + NaOH \xrightarrow{t^\circ} ...$

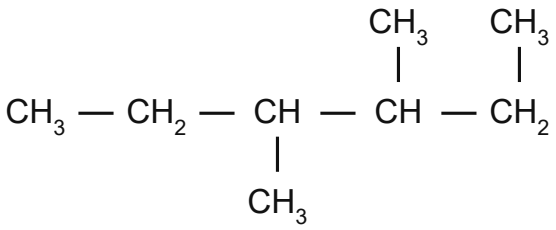
4) $Mg_2C_3 + H_2O \rightarrow ...$

5) $CH_4 + Cl_2 \xrightarrow{h\nu} ...$

6) $C_2H_5OH + Na \rightarrow ...$

A) 2, 3 **B)** 1, 2, 3 **C)** 1, 3 **D)** 1, 2, 3, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 3,4,5-trimetilpentan **B)** 3,4-dimetilgeksan

C) 3-metilgeksan **D)** 4-metilgeksan

3. B Alkanlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) naftalin; 2) 2,2-dimetilpropan; 3) 3-etilpentan; 4) izobutilen; 5) toluol; 6) butadiyen-1,3

A) 3 **B)** 2 **C)** 2, 3, 4 **D)** 2, 3

4. Q 34,8 gramm 2-metilpropan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 1-xlor-1,1-dimetiletan – 46,6 g **B)** 1-xlor-2-metilpropan – 55,5 g

C) 2-xlor-2-metilpropan – 46,6 g **D)** 2-xlor-2-metilpropan – 55,5 g

5. C_4H_{10} va C_8H_{18} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 5,2 mol CO_2 va 6 mol H_2O hosil bo'ldi. Dastlabki aralashma massasini (g) aniqlang. 1,5 ball

A) 68,4 **B)** 76,3 **C)** 74,4 **D)** 83,6

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 35 gramm aralashma mavjud. Aralashmadagi molekulalar soni $4{,}214 \cdot 10^{23}$ ga teng. Shu aralashma 0,7 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 77,4 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_8H_{16} **B)** C_5H_{10} **C)** C_7H_{14} **D)** C_4H_8

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 22 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 4% li bromli suvning 800 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 1:1 butadiyen **B)** buten 1:4 butadiyen

C) buten 1:2 butadiyen **D)** buten 2:1 butadiyen

8. Alyuminiy karbid va alyuminiy sulfid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 13,44 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=25 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi alyuminiy karbid massasini (g) aniqlang. 1,5 ball

A) 43,2 **B)** 14,4 **C)** 28,8 **D)** 15

9. 128,8 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 120,4 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 263,2 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol siklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 0,8 **B)** 1,4 **C)** 0,4 **D)** 0,6

10. Q 0,4 mol organik modda tarkibida 19,2 gramm uglerod va 3,2 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_5H_8 **B)** C_3H_8 **C)** C_4H_8 **D)** C_4H_6

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 3,375 mol kislorod sarflangani va 2,5 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda tereftal kislota va CO_2, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) butilbenzol **B)** 1-metil-4-propilbenzol

C) 1,4-dietilbenzol **D)** etilbenzol

12. Q Formulasi C_6H_{12} va asosiy halqasida 4 ta uglerod atomi bo'lgan (siklobutan hosilalari), halqaga aynan ikkita radikal birikkan, o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 2 ta **B)** 4 ta **C)** 3 ta **D)** 5 ta

13. Q 2,3,3-trimetilpentan molekulasida nechta ikkilamchi uglerod atomi mavjud? 1,5 ball

A) 3 ta **B)** 1 ta **C)** 5 ta **D)** 2 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball

1) pentin-1 va izopren bir-biriga izomer;
2) butan va izobutan bir-biriga gomolog;
3) butin-2 kumush oksidning ammiakdagi eritmasi bilan reaksiyaga kirishadi;
4) siklobutan va buten-1 bir-biriga izomer;
5) butadiyen-1,3 va butin-1 ning molekulyar formulasi bir xil

A) 1, 5 **B)** 1, 4, 5 **C)** 1, 3, 4 **D)** 4, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_7H_{16} bo'lgan, asosiy zanjirida 6 ta yoki 6 tadan ko'proq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlarni ularning molekulasidagi birlamchi uglerod atomlari soniga ko'ra nechta guruhga bo'lish mumkin?

A) 2 ta **B)** 4 ta **C)** 3 ta **D)** 1 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 2-variant

1 B	B	2 B	B	3 B	D	4 Q D
5 Q	C	6 Q	B	7 Q	A	8 Q B
9 Q	D	10 Q	C	11 Q	C	12 Q C
13 Q	B	14 M	B	15 M	A	

1. B) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 1, 2, 3.

2. B) Eng uzun uglerod zanjiri 6 ta atomdan iborat (geksan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 3,4-dimetilgeksan.

3. D) Alkanlar: izobutan, 2,2-dimetilpropan, neopentan va h.k.. Javob: 2, 3.

4. D) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2-metilpropan. n = 34,8/58 = 0,6 mol; m = 0,6·92,5 = 55,5 g.

5. C) m = m(C) + m(H) = 12·5,2 + 2·6 = 74,4 g.

6. B) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 0,7; mahsulotlar: 113a + (M + 34,5)b = 77,4; 42a + M·b = 35 → a = 0,5, b = 0,2, M = 70 → C_5H_{10}.

7. A) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2)$ = 800·4/100/160 = 0,2 mol = n(butadiyen). m(butadiyen) = 10,8 g; n(buten) = (22 − 10,8)/56 = 0,2 mol → buten 1:1 butadiyen.

8. B) Gazlar: CH_4 (16) va H_2S (34). n = 0,6 mol; 16·x + 34·(0,6 − x) = 0,6·25 → $n(CH_4)$ = 0,3 mol, $n(H_2S)$ = 0,3 mol. $m(Al_4C_3)$ = 14,4 g.

9. D) $n(H_2)$ = (128,8 − 120,4)/2 = 4,2 mol = 3·(a + b) → a + b = 1,4. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 263,2/329 = 0,8 mol = a → b = 0,6 mol.

10. C) n(C) = 19,2/12 = 1,6 mol, n(H) = 3,2 mol; bitta molekulada C: 1,6/0,4 = 4, H: 3,2/0,4 = 8 → C_4H_8.

11. C) $C_nH_{2n-6} + (3n - 3)/2\ O_2 \rightarrow nCO_2 + (n - 3)H_2O$. $n(O_2)/n(CO_2) = (3n - 3)/(2n) = 3{,}375/2{,}5$ → n = 10 ($C_{10}H_{14}$). Oksidlanish mahsulotlari bo'yicha — 1,4-dietilbenzol.

12. C) Halqaga 2 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 3 ta izomer.

13. B) 2,3,3-trimetilpentan: birlamchi – 5, ikkilamchi – 1, uchlamchi – 1, to'rtlamchi – 1 ta uglerod atomi.

14. B) To'g'ri: 1, 4, 5. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. A) Shartga mos 3 ta izomer bor; ulardagi birlamchi uglerod atomlari soni: 2, 3 → 2 ta guruh.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 3-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $CH_2{=}CH_2 + H_2 \xrightarrow{Ni,\ t^\circ} \ldots$

2) $HC{\equiv}CH + 2H_2 \xrightarrow{Ni,\ t^\circ} \ldots$

3) $CH_4 + Cl_2 \xrightarrow{h\nu} \ldots$

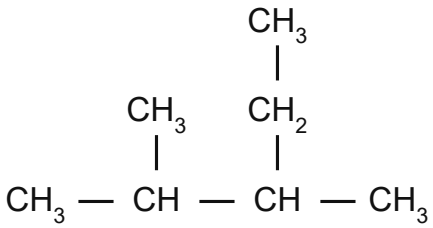
4) $C_2H_5OH \xrightarrow{H_2SO_4,\ 170^\circ C} \ldots$

5) $HC{\equiv}CH + H_2O \xrightarrow{Hg^{2+}} \ldots$

6) $CH_3Cl + Na \rightarrow \ldots$ (Vyurs reaksiyasi)

A) 1, 2, 3 **B)** 1, 2 **C)** 1, 2, 6 **D)** 1, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 2,3-dimetilpentan **B)** 2-metil-3-etilbutan

C) 3,4-dimetilpentan **D)** 3-metilpentan

3. B Alkinlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) propilatsetilen; 2) 2,2-dimetilpropan; 3) 2,3-dimetilbutadiyen-1,3; 4) naftalin; 5) penten-1; 6) atsetilen

A) 3, 6 **B)** 1, 6 **C)** 1, 4, 6 **D)** 1, 5, 6

4. Q 17,2 gramm 2,3-dimetilbutan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 1-xlor-2,3-dimetilbutan – 24,1 g **B)** 2-xlor-2,3-dimetilbutan – 24,1 g

C) 3-xlor-2,3-dimetilbutan – 24,1 g **D)** 2-xlor-2,3-dimetilbutan – 36,2 g

5. C_4H_{10} va C_5H_{12} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 2,3 mol CO_2 va 2,8 mol H_2O hosil bo'ldi. Dastlabki aralashma massasini (g) aniqlang. 1,5 ball

A) 33,7 **B)** 42,4 **C)** 33,2 **D)** 30,4

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 49 gramm aralashma mavjud. Aralashmadagi molekulalar soni $5{,}418 \cdot 10^{23}$ ga teng. Shu aralashma 0,9 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 105,6 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_7H_{14} **B)** C_6H_{12} **C)** C_4H_8 **D)** C_8H_{16}

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 32,8 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 8% li bromli suvning 800 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 1:1 butadiyen **B)** buten 1:2 butadiyen

C) buten 2:1 butadiyen **D)** buten 1:3 butadiyen

8. Berilliy karbid va alyuminiy sulfid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 40,32 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=19 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi alyuminiy sulfid massasini (g) aniqlang. 1,5 ball

A) 30 **B)** 22,5 **C)** 45 **D)** 15

9. 159,6 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 149,4 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 394,8 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol metilsiklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 1,2 **B)** 1,7 **C)** 0,5 **D)** 1,4

10. Q 0,5 mol organik modda tarkibida 54 gramm uglerod va 8 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) $C_{10}H_{16}$ **B)** C_8H_{16} **C)** C_9H_{18} **D)** C_9H_{16}

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 3 mol kislorod sarflangani va 2,25 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda tereftal kislota va CO_2, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) 1-metil-2-etilbenzol **B)** propilbenzol
C) etilbenzol **D)** 1-metil-4-etilbenzol

12. Q Formulasi C_8H_{16} va asosiy halqasida 6 ta uglerod atomi bo'lgan (siklogeksan hosilalari), o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 5 ta **B)** 4 ta **C)** 7 ta **D)** 6 ta

13. Q 2,5-dimetil-3-etilgeptan molekulasida nechta ikkilamchi uglerod atomi mavjud? 1,5 ball

A) 5 ta **B)** 3 ta **C)** 4 ta **D)** 2 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball
1) benzol va toluol bir-biriga gomolog;
2) metilsiklopentan va siklogeksan bir-biriga izomer;
3) butan va izobutan bir-biriga izomer;
4) propan va siklopropan bir-biriga izomer;
5) 2-metilpentan va 3-metilpentan molyar massalari har xil

A) 1, 2 **B)** 1, 2, 3, 4 **C)** 1, 2, 3 **D)** 1, 2, 3, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_7H_{16} bo'lgan, asosiy zanjirida 6 ta yoki 6 tadan kamroq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 9 ta **B)** 8 ta **C)** 7 ta **D)** 10 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 3-variant

1 B C	2 B A	3 B B	4 Q B	5 Q C	6 Q A	7 Q B
8 Q D	9 Q A	10 Q D	11 Q D	12 Q A	13 Q B	14 M C
15 M B						

1. C) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 1, 2, 6.

2. A) Eng uzun uglerod zanjiri 5 ta atomdan iborat (pentan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 2,3-dimetilpentan.

3. B) Alkinlar: 3-metilpentin-1, propilatsetilen, butin-2 va h.k.. Javob: 1, 6.

4. B) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2,3-dimetilbutan. n = 17,2/86 = 0,2 mol; m = 0,2·120,5 = 24,1 g.

5. C) m = m(C) + m(H) = 12·2,3 + 2·2,8 = 33,2 g.

6. A) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 0,9; mahsulotlar: 113a + (M + 34,5)b = 105,6; 42a + M·b = 49 → a = 0,7, b = 0,2, M = 98 → C_7H_{14}.

7. B) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2)$ = 800·8/100/160 = 0,4 mol = n(butadiyen). m(butadiyen) = 21,6 g; n(buten) = (32,8 − 21,6)/56 = 0,2 mol → buten 1:2 butadiyen.

8. D) Gazlar: CH_4 (16) va H_2S (34). n = 1,8 mol; 16·x + 34·(1,8 − x) = 1,8·19 → $n(CH_4)$ = 1,5 mol, $n(H_2S)$ = 0,3 mol. $m(Al_2S_3)$ = 15 g.

9. A) $n(H_2)$ = (159,6 − 149,4)/2 = 5,1 mol = 3·(a + b) → a + b = 1,7. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 394,8/329 = 1,2 mol = a → b = 0,5 mol.

10. D) n(C) = 54/12 = 4,5 mol, n(H) = 8 mol; bitta molekulada C: 4,5/0,5 = 9, H: 8/0,5 = 16 → C_9H_{16}.

11. D) $C_nH_{2n-6} + (3n - 3)/2\ O_2 \rightarrow nCO_2 + (n - 3)H_2O$. $n(O_2)/n(CO_2) = (3n - 3)/(2n) = 3/2{,}25$ → n = 9 (C_9H_{12}). Oksidlanish mahsulotlari bo'yicha — 1-metil-4-etilbenzol.

12. A) Halqaga 2 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 5 ta izomer.

13. B) 2,5-dimetil-3-etilgeptan: birlamchi – 5, ikkilamchi – 3, uchlamchi – 3, to'rtlamchi – 0 ta uglerod atomi.

14. C) To'g'ri: 1, 2, 3. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. B) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 4 C – 1 ta; 5 C – 5 ta; 6 C – 2 ta; 7 C – 1 ta. Shartga mos — 8 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 4-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $CH_2{=}CH_2 + H_2O \xrightarrow{H_2SO_4} \ldots$

2) $CH_2{=}CH_2 + Br_2 \rightarrow \ldots$

3) $C_2H_5COONa + NaOH \xrightarrow{t^\circ} \ldots$

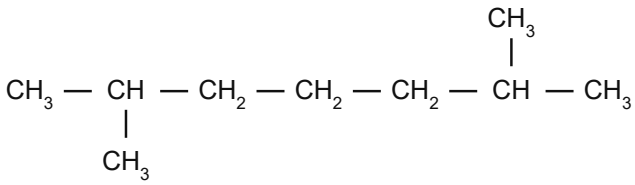
4) $C_6H_6 + HNO_3 \xrightarrow{H_2SO_4} \ldots$

5) $C_2H_5OH + Na \rightarrow \ldots$

6) $CH_2{=}CH_2 + H_2 \xrightarrow{Ni,\ t^\circ} \ldots$

A) 1, 3 **B)** 2, 3, 6 **C)** 3, 4, 6 **D)** 3, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 2,6-dimetilgeksan **B)** 6-metilgeptan

C) 2,6-dimetilgeptan **D)** 2-metilgeptan

3. B Alkadiyenlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) penten-1; 2) izopren; 3) etilbenzol; 4) pentadiyen-1,4; 5) propilatsetilen; 6) butadiyen-1,3

A) 2, 4, 5, 6 **B)** 2, 4, 6 **C)** 4, 6 **D)** 2, 4

4. Q 34,4 gramm 2-metilpentan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 2-xlor-2-metilpentan – 48,2 g **B)** 4-xlor-4-metilpentan – 48,2 g

C) 1-xlor-2-metilpentan – 48,2 g **D)** 2-xlor-2-metilpentan – 41,2 g

5. C_5H_{12} va C_6H_{14} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 5,5 mol CO_2 va 6,5 mol H_2O hosil bo'ldi. Dastlabki aralashma massasini (g) aniqlang. 1,5 ball

A) 88,2 **B)** 72,5 **C)** 79 **D)** 80,7

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 54,6 gramm aralashma mavjud. Aralashmadagi molekulalar soni $6{,}02\cdot10^{23}$ ga teng. Shu aralashma 1 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 114,65 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_5H_{10} **B)** C_6H_{12} **C)** C_8H_{16} **D)** C_4H_8

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 27,4 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 4% li bromli suvning 1200 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 2:1 butadiyen **B)** buten 2:3 butadiyen

C) buten 1:1 butadiyen **D)** buten 3:2 butadiyen

8. Magniy nitrid va kalsiy karbid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 40,32 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=18,5 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi kalsiy karbid massasini (g) aniqlang. 1,5 ball

A) 19,2 **B)** 75 **C)** 38,4 **D)** 28,8

9. 126 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 117,6 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 197,4 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmadagi siklogeksan massasini (g) aniqlang. 1,5 ball

A) 67,2 **B)** 62,4 **C)** 58,8 **D)** 50,4

10. 0,25 mol organik modda tarkibida 27 gramm uglerod va 3 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_9H_{10} **B)** C_9H_{12} **C)** $C_{10}H_{14}$ **D)** C_8H_{12}

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 2,625 mol kislorod sarflangani va 2 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda tereftal kislota, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) orto-ksilol **B)** etilbenzol **C)** butilbenzol **D)** para-ksilol

12. Q Formulasi C_6H_{12} va asosiy halqasida 3 ta uglerod atomi bo'lgan (siklopropan hosilalari), halqaga aynan ikkita radikal birikkan, o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 1 ta **B)** 2 ta **C)** 4 ta **D)** 3 ta

13. Q 2,2-dimetilgeksan molekulasida nechta birlamchi uglerod atomi mavjud? 1,5 ball

A) 5 ta **B)** 3 ta **C)** 4 ta **D)** 1 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball

1) butan va izobutan bir-biriga izomer;
2) benzol va siklogeksan bir-biriga izomer;
3) metilsiklopentan va siklogeksan bir-biriga izomer;
4) butadiyen-1,3 va buten-1 ning molekulyar formulasi bir xil;
5) siklobutan va butan bir-biriga izomer

A) 1, 3 **B)** 3 **C)** 1, 3, 4 **D)** 1, 2, 3

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_9H_{20} bo'lgan, asosiy zanjirida 7 ta yoki 7 tadan ko'proq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 15 ta **B)** 14 ta **C)** 16 ta **D)** 17 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 4-variant

1 B D	2 B C	3 B B	4 Q A	5 Q C	6 Q B	7 Q B
8 Q A	9 Q A	10 Q B	11 Q D	12 Q B	13 Q C	14 M A
15 M A						

1. D) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 3, 6.

2. C) Eng uzun uglerod zanjiri 7 ta atomdan iborat (geptan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 2,6-dimetilgeptan.

3. B) Alkadiyenlar: butadiyen-1,3, izopren, pentadiyen-1,4 va h.k.. Javob: 2, 4, 6.

4. A) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2-metilpentan. n = 34,4/86 = 0,4 mol; m = 0,4·120,5 = 48,2 g.

5. C) m = m(C) + m(H) = 12·5,5 + 2·6,5 = 79 g.

6. B) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 1; mahsulotlar: 113a + (M + 34,5)b = 114,65; 42a + M·b = 54,6 → a = 0,7, b = 0,3, M = 84 → C_6H_{12}.

7. B) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2) = 1200·4/100/160 = 0,3$ mol = n(butadiyen). m(butadiyen) = 16,2 g; n(buten) = (27,4 − 16,2)/56 = 0,2 mol → buten 2:3 butadiyen.

8. A) Gazlar: NH_3 (17) va C_2H_2 (26). n = 1,8 mol; 17·x + 26·(1,8 − x) = 1,8·18,5 → $n(NH_3) = 1,5$ mol, $n(C_2H_2) = 0,3$ mol. $m(CaC_2) = 19,2$ g.

9. A) $n(H_2) = (126 - 117,6)/2 = 4,2$ mol = 3·(a + b) → a + b = 1,4. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 197,4/329 = 0,6 mol = a → b = 0,8 mol.

10. B) n(C) = 27/12 = 2,25 mol, n(H) = 3 mol; bitta molekulada C: 2,25/0,25 = 9, H: 3/0,25 = 12 → C_9H_{12}.

11. D) $C_nH_{2n-6} + (3n - 3)/2\ O_2 \rightarrow nCO_2 + (n - 3)H_2O$. $n(O_2)/n(CO_2) = (3n - 3)/(2n) = 2,625/2$ → n = 8 (C_8H_{10}). Oksidlanish mahsulotlari bo'yicha — para-ksilol.

12. B) Halqaga 3 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 2 ta izomer.

13. C) 2,2-dimetilgeksan: birlamchi – 4, ikkilamchi – 3, uchlamchi – 0, to'rtlamchi – 1 ta uglerod atomi.

14. A) To'g'ri: 1, 3. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. A) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 5 C – 8 ta; 6 C – 12 ta; 7 C – 11 ta; 8 C – 3 ta; 9 C – 1 ta. Shartga mos — 15 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 5-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $CH_2{=}CH_2 + H_2 \xrightarrow{Ni,\ t^\circ} \ldots$

2) $C_2H_5OH + Na \rightarrow \ldots$

3) $HC{\equiv}CH + H_2O \xrightarrow{Hg^{2+}} \ldots$

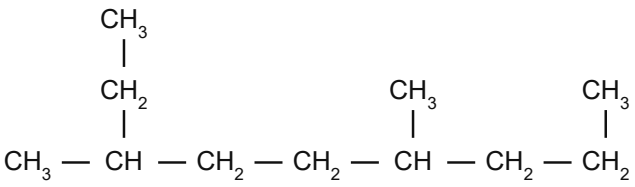
4) $CH_3COONa + NaOH \xrightarrow{t^\circ} \ldots$

5) $CH_2{=}CH_2 + H_2O \xrightarrow{H_2SO_4} \ldots$

6) $C + 2H_2 \xrightarrow{t^\circ,\ p} \ldots$

A) 1, 2, 4, 6 **B)** 4, 6 **C)** 4, 5, 6 **D)** 1, 4, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 3,6-dimetilnonan **B)** 4,7-dimetilnonan

C) 5,7-dimetil-2-etilgeptan **D)** 6-metilnonan

3. B Alkenlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball

1) siklogeksan; 2) 3-metilbuten-1; 3) 2-metilbuten-2; 4) etilen; 5) etilsiklopropan; 6) propilatsetilen

A) 3, 4 **B)** 1, 2, 3 **C)** 2, 3, 4, 6 **D)** 2, 3, 4

4. Q 20 gramm 2,4-dimetilpentan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 1-xlor-2,4-dimetilpentan – 26,9 g **B)** 2-xlor-2,4-dimetilpentan – 26,9 g

C) 2-xlor-2,4-dimetilpentan – 30 g **D)** 3-xlor-2,4-dimetilpentan – 26,9 g

5. Q C_3H_8 va C_4H_{10} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 3,8 mol CO_2 va 4,9 mol H_2O hosil bo'ldi. Aralashmadagi C_3H_8 miqdorini (mol) aniqlang. 1,5 ball

A) 0,7 **B)** 0,5 **C)** 1,1 **D)** 0,6

6. Q Siklopropan va noma'lum sikloalkandan tashkil topgan 70 gramm aralashma mavjud. Aralashmadagi molekulalar soni $6{,}02\cdot10^{23}$ ga teng. Shu aralashma 1 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 122,75 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_7H_{14} **B)** C_5H_{10} **C)** C_8H_{16} **D)** C_6H_{12}

7. Q Buten va butadiyen aralashmasi sopolimerlanganda massasi 66,4 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 4% li bromli suvning 1600 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 1:1 butadiyen **B)** buten 1:2 butadiyen

C) buten 3:1 butadiyen **D)** buten 2:1 butadiyen

8. Q Magniy nitrid va kalsiy karbid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 40,32 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=20 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi kalsiy karbid massasini (g) aniqlang. 1,5 ball

A) 76,8 **B)** 57,6 **C)** 60 **D)** 38,4

9. Q 198,8 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 185,6 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 329 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmadagi siklogeksan massasini (g) aniqlang. 1,5 ball

A) 84 **B)** 98 **C)** 93,6 **D)** 100,8

10. Q 0,25 mol organik modda tarkibida 30 gramm uglerod va 3,5 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) $C_{10}H_{22}$ **B)** $C_{10}H_{16}$ **C)** $C_{11}H_{16}$ **D)** $C_{10}H_{14}$

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 5,4 mol kislorod sarflangani va 4 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda benzoy kislota va propion kislota, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) 1,4-dietilbenzol **B)** 1-metil-4-propilbenzol
C) butilbenzol **D)** meta-ksilol

12. Q Formulasi C_7H_{14} va asosiy halqasida 3 ta uglerod atomi bo'lgan (siklopropan hosilalari), o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 14 ta **B)** 16 ta **C)** 15 ta **D)** 17 ta

13. Q 3-metiloktan molekulasida nechta ikkilamchi uglerod atomi mavjud? 1,5 ball

A) 3 ta **B)** 1 ta **C)** 6 ta **D)** 5 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball
1) butan va izobutan bir-biriga gomolog;
2) etilen kaliy permanganatning suvli eritmasi bilan reaksiyaga kirishganda ikki atomli spirt hosil bo'ladi;
3) siklopropan va propen xlor bilan birikkanda hosil bo'lgan mahsulotlarning tuzilishi bir xil bo'ladi;
4) atsetilen kumush oksidning ammiakdagi eritmasi bilan reaksiyaga kirishadi;
5) butadiyen-1,3 va buten-1 ning molekulyar formulasi bir xil

A) 2, 4, 5 **B)** 4 **C)** 4, 5 **D)** 2, 4

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_8H_{18} bo'lgan, asosiy zanjirida 7 ta yoki 7 tadan kamroq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 18 ta **B)** 16 ta **C)** 17 ta **D)** 19 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 5-variant

1 B D	2 B A	3 B D	4 Q B	5 Q D	6 Q A	7 Q D
8 Q D	9 Q D	10 Q D	11 Q C	12 Q C	13 Q D	14 M D
15 M C						

1. D) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 1, 4, 6.

2. A) Eng uzun uglerod zanjiri 9 ta atomdan iborat (nonan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 3,6-dimetilnonan.

3. D) Alkenlar: propilen, 2-metilbuten-2, penten-1 va h.k.. Javob: 2, 3, 4.

4. B) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2,4-dimetilpentan. n = 20/100 = 0,2 mol; m = 0,2·134,5 = 26,9 g.

5. D) n(alkanlar) = $n(H_2O)$ − $n(CO_2)$ = 1,1 mol. 3·x + 4·(1,1 − x) = 3,8 → x = 0,6 mol.

6. A) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 1; mahsulotlar: 113a + (M + 34,5)b = 122,75; 42a + M·b = 70 → a = 0,5, b = 0,5, M = 98 → C_7H_{14}.

7. D) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2)$ = 1600·4/100/160 = 0,4 mol = n(butadiyen). m(butadiyen) = 21,6 g; n(buten) = (66,4 − 21,6)/56 = 0,8 mol → buten 2:1 butadiyen.

8. D) Gazlar: NH_3 (17) va C_2H_2 (26). n = 1,8 mol; 17·x + 26·(1,8 − x) = 1,8·20 → $n(NH_3)$ = 1,2 mol, $n(C_2H_2)$ = 0,6 mol. $m(CaC_2)$ = 38,4 g.

9. D) $n(H_2)$ = (198,8 − 185,6)/2 = 6,6 mol = 3·(a + b) → a + b = 2,2. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 329/329 = 1 mol = a → b = 1,2 mol.

10. D) n(C) = 30/12 = 2,5 mol, n(H) = 3,5 mol; bitta molekulada C: 2,5/0,25 = 10, H: 3,5/0,25 = 14 → $C_{10}H_{14}$.

11. C) $C_nH_{2n-6} + (3n - 3)/2\ O_2 \rightarrow nCO_2 + (n - 3)H_2O$. $n(O_2)/n(CO_2)$ = (3n − 3)/(2n) = 5,4/4 → n = 10 ($C_{10}H_{14}$). Oksidlanish mahsulotlari bo'yicha — butilbenzol.

12. C) Halqaga 4 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 15 ta izomer.

13. D) 3-metiloktan: birlamchi – 3, ikkilamchi – 5, uchlamchi – 1, to'rtlamchi – 0 ta uglerod atomi.

14. D) To'g'ri: 2, 4. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. C) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 4 C – 1 ta; 5 C – 6 ta; 6 C – 7 ta; 7 C – 3 ta; 8 C – 1 ta. Shartga mos — 17 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 6-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $HC{\equiv}CH + H_2O \xrightarrow{Hg^{2+}} ...$

2) $C_2H_5Br + Na \rightarrow ...$ (Vyurs reaksiyasi)

3) $HC{\equiv}CH + 2H_2 \xrightarrow{Ni,\ t^\circ} ...$

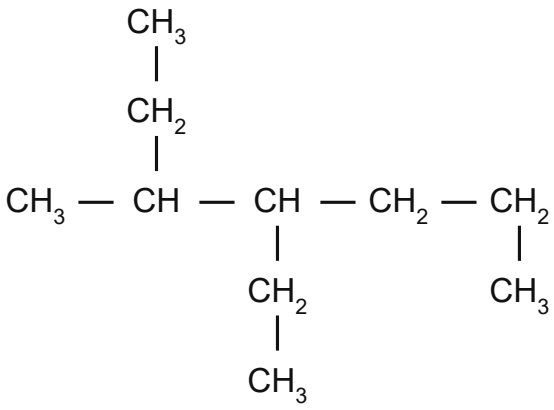
4) $C_2H_5OH + Na \rightarrow ...$

5) $Mg_2C_3 + H_2O \rightarrow ...$

6) $C_2H_5OH \xrightarrow{H_2SO_4,\ 170^\circ C} ...$

A) 3, 6 **B)** 2, 3, 5 **C)** 2, 3, 6 **D)** 2, 3

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 3-metil-4-etilgeptan **B)** 5-metil-2,3-dietilpentan

C) 5-metil-4-etilgeptan **D)** 4-etilgeptan

3. B Alkadiyenlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball

1) siklogeksan; 2) geksen-3; 3) izopren; 4) kumol; 5) butadiyen-1,3; 6) geksadiyen-1,5

A) 1, 3, 5, 6 **B)** 3, 5 **C)** 1, 5, 6 **D)** 3, 5, 6

4. Q 103,2 gramm 2-metilpentan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 4-xlor-4-metilpentan – 127,4 g **B)** 1-xlor-2-metilpentan – 144,6 g
C) 2-xlor-2-metilpentan – 127,4 g **D)** 2-xlor-2-metilpentan – 144,6 g

5. Q C_3H_8 va C_6H_{14} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 4,2 mol CO_2 va 5,3 mol H_2O hosil bo'ldi. Aralashmadagi C_3H_8 miqdorini (mol) aniqlang. 1,5 ball

A) 0,8 **B)** 1,1 **C)** 0,9 **D)** 0,3

6. Q Siklopropan va noma'lum sikloalkandan tashkil topgan 29,4 gramm aralashma mavjud. Aralashmadagi molekulalar soni $3{,}01\cdot10^{23}$ ga teng. Shu aralashma 0,5 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 53,95 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_5H_{10} **B)** C_8H_{16} **C)** C_6H_{12} **D)** C_4H_8

7. Q Buten va butadiyen aralashmasi sopolimerlanganda massasi 111 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 8% li bromli suvning 1000 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 3:2 butadiyen **B)** buten 3:1 butadiyen
C) buten 4:1 butadiyen **D)** buten 1:3 butadiyen

8. Q Alyuminiy karbid va alyuminiy sulfid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 33,6 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=26,8 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi alyuminiy karbid massasini (g) aniqlang. 1,5 ball

A) 86,4 **B)** 45 **C)** 28,8 **D)** 57,6

9. Q 100,8 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 94,2 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 197,4 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol siklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 0,7 **B)** 0,6 **C)** 0,5 **D)** 1,1

10. Q 0,4 mol organik modda tarkibida 14,4 gramm uglerod va 1,6 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_3H_8 **B)** C_3H_6 **C)** C_3H_2 **D)** C_3H_4

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 3,375 mol kislorod sarflangani va 2,5 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda tereftal kislota va CO_2, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) butilbenzol **B)** 1-metil-4-propilbenzol
C) para-ksilol **D)** 1,4-dietilbenzol

12. Q Formulasi C_8H_{16} va asosiy halqasida 4 ta uglerod atomi bo'lgan (siklobutan hosilalari), o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 23 ta **B)** 24 ta **C)** 25 ta **D)** 26 ta

13. Q 2,3,3-trimetilpentan molekulasida nechta birlamchi uglerod atomi mavjud? 1,5 ball

A) 6 ta **B)** 5 ta **C)** 7 ta **D)** 1 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball

1) atsetilen kumush oksidning ammiakdagi eritmasi bilan reaksiyaga kirishadi;
2) propan va siklopropan bir-biriga izomer;
3) butadiyen-1,3 va buten-1 ning molekulyar formulasi bir xil;
4) pentin-1 va izopren bir-biriga izomer;
5) etilen kaliy permanganatning suvli eritmasi bilan reaksiyaga kirishganda uch atomli spirt hosil bo'ladi

A) 2, 4 **B)** 1, 4 **C)** 3, 4 **D)** 1, 4, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_9H_{20} bo'lgan, asosiy zanjirida 6 ta yoki 6 tadan kamroq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 19 ta **B)** 20 ta **C)** 22 ta **D)** 21 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 6-variant

1 B D	2 B A	3 B D	4 Q D	5 Q A	6 Q A	7 Q B
8 Q C	9 Q C	10 Q D	11 Q D	12 Q B	13 Q B	14 M B
15 M B						

1. D) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 2, 3.

2. A) Eng uzun uglerod zanjiri 7 ta atomdan iborat (geptan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 3-metil-4-etilgeptan.

3. D) Alkadiyenlar: butadiyen-1,3, izopren, pentadiyen-1,4 va h.k.. Javob: 3, 5, 6.

4. D) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2-metilpentan. n = 103,2/86 = 1,2 mol; m = 1,2·120,5 = 144,6 g.

5. A) n(alkanlar) = $n(H_2O)$ − $n(CO_2)$ = 1,1 mol. 3·x + 6·(1,1 − x) = 4,2 → x = 0,8 mol.

6. A) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl$ + HCl). a + b = 0,5; mahsulotlar: 113a + (M + 34,5)b = 53,95; 42a + M·b = 29,4 → a = 0,2, b = 0,3, M = 70 → C_5H_{10}.

7. B) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2)$ = 1000·8/100/160 = 0,5 mol = n(butadiyen). m(butadiyen) = 27 g; n(buten) = (111 − 27)/56 = 1,5 mol → buten 3:1 butadiyen.

8. C) Gazlar: CH_4 (16) va H_2S (34). n = 1,5 mol; 16·x + 34·(1,5 − x) = 1,5·26,8 → $n(CH_4)$ = 0,6 mol, $n(H_2S)$ = 0,9 mol. $m(Al_4C_3)$ = 28,8 g.

9. C) $n(H_2)$ = (100,8 − 94,2)/2 = 3,3 mol = 3·(a + b) → a + b = 1,1. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 197,4/329 = 0,6 mol = a → b = 0,5 mol.

10. D) n(C) = 14,4/12 = 1,2 mol, n(H) = 1,6 mol; bitta molekulada C: 1,2/0,4 = 3, H: 1,6/0,4 = 4 → C_3H_4.

11. D) C_nH_{2n-6} + (3n − 3)/2 O_2 → nCO_2 + (n − 3)H_2O. $n(O_2)/n(CO_2)$ = (3n − 3)/(2n) = 3,375/2,5 → n = 10 ($C_{10}H_{14}$). Oksidlanish mahsulotlari bo'yicha — 1,4-dietilbenzol.

12. B) Halqaga 4 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 24 ta izomer.

13. B) 2,3,3-trimetilpentan: birlamchi – 5, ikkilamchi – 1, uchlamchi – 1, to'rtlamchi – 1 ta uglerod atomi.

14. B) To'g'ri: 1, 4. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. B) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 5 C – 8 ta; 6 C – 12 ta; 7 C – 11 ta; 8 C – 3 ta; 9 C – 1 ta. Shartga mos — 20 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 7-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $C_2H_5OH \xrightarrow{H_2SO_4,\ 170°C} ...$

2) $CH_3COONa + H_2O \xrightarrow{elektroliz} ...$

3) $C_6H_6 + HNO_3 \xrightarrow{H_2SO_4} ...$

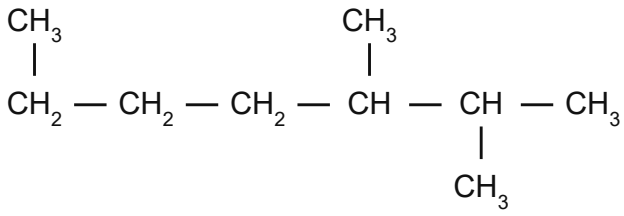
4) $CH_2{=}CH_2 + H_2 \xrightarrow{Ni,\ t°} ...$

5) $HC{\equiv}CH + 2H_2 \xrightarrow{Ni,\ t°} ...$

6) $CH_2{=}CH_2 + Br_2 \rightarrow ...$

A) 2, 4 **B)** 2, 4, 5 **C)** 2, 5 **D)** 2, 4, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 2,3-dimetilgeptan **B)** 3-metilgeptan

C) 5,6-dimetilgeptan **D)** 1,4,5-trimetilgeksan

3. B Sikloalkanlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) 3-metilbuten-1; 2) 1,3-dimetilsiklobutan; 3) 1,3,5-trimetilbenzol; 4) geksadiyen-1,5; 5) geksin-1; 6) metilsiklopentan

A) 3, 6 **B)** 2, 4, 6 **C)** 2, 6 **D)** 2, 4

4. Q 20 gramm 3-etilpentan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 1-xlor-3-etilpentan – 26,9 g **B)** 3-xlor-3-etilpentan – 30 g

C) 2-xlor-3-etilpentan – 26,9 g **D)** 3-xlor-3-etilpentan – 26,9 g

5. Q C_5H_{12} va C_6H_{14} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 3,2 mol CO_2 va 3,8 mol H_2O hosil bo'ldi. Aralashmadagi C_5H_{12} miqdorini (mol) aniqlang. 1,5 ball

A) 0,4 **B)** 0,5 **C)** 0,2 **D)** 0,6

6. Q Siklopropan va noma'lum sikloalkandan tashkil topgan 37,8 gramm aralashma mavjud. Aralashmadagi molekulalar soni $3{,}612 \cdot 10^{23}$ ga teng. Shu aralashma 0,6 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 69,45 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_6H_{12} **B)** C_7H_{14} **C)** C_5H_{10} **D)** C_4H_8

7. Q Buten va butadiyen aralashmasi sopolimerlanganda massasi 68 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 8% li bromli suvning 2000 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 1:1 butadiyen **B)** buten 1:5 butadiyen

C) buten 1:4 butadiyen **D)** buten 4:1 butadiyen

8. Q Alyuminiy sulfid va alyuminiy karbid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 33,6 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=30,4 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi alyuminiy sulfid massasini (g) aniqlang. 1,5 ball

A) 120 **B)** 14,4 **C)** 180 **D)** 60

9. Q 131,6 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 123,2 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 329 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol siklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 0,6 **B)** 0,4 **C)** 1,4 **D)** 1

10. Q 0,3 mol organik modda tarkibida 28,8 gramm uglerod va 4,8 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_8H_{18} **B)** C_8H_{16} **C)** C_9H_{16} **D)** C_8H_{14}

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 1,35 mol kislorod sarflangani va 1 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda tereftal kislota va CO_2, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) butilbenzol **B)** 1-metil-4-propilbenzol

C) 1,4-dietilbenzol **D)** propilbenzol

12. Q Formulasi C_6H_{12} va asosiy halqasida 3 ta uglerod atomi bo'lgan (siklopropan hosilalari), o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 8 ta **B)** 6 ta **C)** 5 ta **D)** 7 ta

13. Q 3-metil-5-etilgeptan molekulasida nechta ikkilamchi uglerod atomi mavjud? 1,5 ball

A) 5 ta **B)** 6 ta **C)** 2 ta **D)** 4 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball

1) siklobutan va buten-1 bir-biriga izomer;
2) butin-2 kumush oksidning ammiakdagi eritmasi bilan reaksiyaga kirishadi;
3) butan va izobutan bir-biriga gomolog;
4) etilen kaliy permanganatning suvli eritmasi bilan reaksiyaga kirishganda uch atomli spirt hosil bo'ladi;
5) butadiyen-1,3 va butin-1 ning molekulyar formulasi bir xil

A) 1, 5 **B)** 5 **C)** 1, 4 **D)** 1, 2, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_7H_{16} bo'lgan, asosiy zanjirida 5 ta yoki 5 tadan kamroq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 7 ta **B)** 8 ta **C)** 6 ta **D)** 5 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 7-variant

1 B	B	2 B	A	3 B	C	4 Q D
5 Q	A	6 Q	A	7 Q	C	8 Q D
9 Q	B	10 Q	B	11 Q	C	12 Q B
13 Q	D	14 M	A	15 M	C	

1. B) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 2, 4, 5.

2. A) Eng uzun uglerod zanjiri 7 ta atomdan iborat (geptan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 2,3-dimetilgeptan.

3. C) Sikloalkanlar: 1,3-dimetilsiklobutan, metilsiklopentan, siklogeksan va h.k.. Javob: 2, 6.

4. D) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 3-xlor-3-etilpentan. n = 20/100 = 0,2 mol; m = 0,2·134,5 = 26,9 g.

5. A) n(alkanlar) = $n(H_2O)$ − $n(CO_2)$ = 0,6 mol. 5·x + 6·(0,6 − x) = 3,2 → x = 0,4 mol.

6. A) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl$ + HCl). a + b = 0,6; mahsulotlar: 113a + (M + 34,5)b = 69,45; 42a + M·b = 37,8 → a = 0,3, b = 0,3, M = 84 → C_6H_{12}.

7. C) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2)$ = 2000·8/100/160 = 1 mol = n(butadiyen). m(butadiyen) = 54 g; n(buten) = (68 − 54)/56 = 0,25 mol → buten 1:4 butadiyen.

8. D) Gazlar: H_2S (34) va CH_4 (16). n = 1,5 mol; 34·x + 16·(1,5 − x) = 1,5·30,4 → $n(H_2S)$ = 1,2 mol, $n(CH_4)$ = 0,3 mol. $m(Al_2S_3)$ = 60 g.

9. B) $n(H_2)$ = (131,6 − 123,2)/2 = 4,2 mol = 3·(a + b) → a + b = 1,4. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 329/329 = 1 mol = a → b = 0,4 mol.

10. B) n(C) = 28,8/12 = 2,4 mol, n(H) = 4,8 mol; bitta molekulada C: 2,4/0,3 = 8, H: 4,8/0,3 = 16 → C_8H_{16}.

11. C) C_nH_{2n-6} + (3n − 3)/2 O_2 → nCO_2 + (n − 3)H_2O. $n(O_2)/n(CO_2)$ = (3n − 3)/(2n) = 1,35/1 → n = 10 ($C_{10}H_{14}$). Oksidlanish mahsulotlari bo'yicha — 1,4-dietilbenzol.

12. B) Halqaga 3 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 6 ta izomer.

13. D) 3-metil-5-etilgeptan: birlamchi – 4, ikkilamchi – 4, uchlamchi – 2, to'rtlamchi – 0 ta uglerod atomi.

14. A) To'g'ri: 1, 5. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. C) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 4 C – 1 ta; 5 C – 5 ta; 6 C – 2 ta; 7 C – 1 ta. Shartga mos — 6 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 8-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $Be_2C + H_2O \rightarrow ...$

2) $HC{\equiv}CH + 2H_2 \xrightarrow{Ni,\ t^\circ} ...$

3) $Mg_2C_3 + H_2O \rightarrow ...$

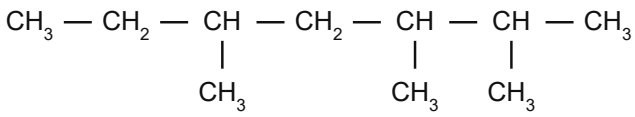
4) $CH_2{=}CH_2 + Br_2 \rightarrow ...$

5) $Al_4C_3 + H_2O \rightarrow ...$

6) $CH_4 + Cl_2 \xrightarrow{h\nu} ...$

A) 1, 2, 5 **B)** 1, 2, 6 **C)** 1, 2, 3, 5 **D)** 1, 5

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 2,3,5-trimetilgeptan **B)** 3,5-dimetilgeptan

C) 3,5,6-trimetilgeptan **D)** 2,3-dimetilgeptan

3. B Sikloalkanlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) 3-metilbuten-1; 2) siklooktan; 3) orto-ksilol; 4) metilsiklopentan; 5) etilsiklopropan; 6) 4-metilpentin-2

A) 2, 4, 5 **B)** 4, 5 **C)** 3, 4, 5 **D)** 2, 5

4. Q 120 gramm 2,4-dimetilpentan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 2-xlor-2,4-dimetilpentan – 126,1 g **B)** 2-xlor-2,4-dimetilpentan – 161,4 g

C) 3-xlor-2,4-dimetilpentan – 126,1 g **D)** 1-xlor-2,4-dimetilpentan – 161,4 g

5. C_6H_{14} va C_8H_{18} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 6,4 mol CO_2 va 7,4 mol H_2O hosil bo'ldi. Aralashmadagi C_6H_{14} miqdorini (mol) aniqlang. 1,5 ball

A) 0,2 **B)** 1 **C)** 0,8 **D)** 0,9

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 49 gramm aralashma mavjud. Aralashmadagi molekulalar soni $5{,}418 \cdot 10^{23}$ ga teng. Shu aralashma 0,9 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 105,6 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_4H_8 **B)** C_7H_{14} **C)** C_8H_{16} **D)** C_5H_{10}

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 111,2 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 8% li bromli suvning 800 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 5:1 butadiyen **B)** buten 4:1 butadiyen

C) buten 1:4 butadiyen **D)** buten 1:1 butadiyen

8. Alyuminiy karbid va kalsiy karbid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 13,44 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=21 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi kalsiy karbid massasini (g) aniqlang. 1,5 ball

A) 28,8 **B)** 38,4 **C)** 19,2 **D)** 14,4

9. 106,4 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 99,2 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 131,6 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol metilsiklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 0,8 **B)** 0,4 **C)** 1,2 **D)** 0,6

10. Q 0,3 mol organik modda tarkibida 21,6 gramm uglerod va 3,6 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_7H_{12} **B)** C_6H_{12} **C)** C_6H_{14} **D)** C_6H_{10}

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 1,35 mol kislorod sarflangani va 1 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda tereftal kislota va CO_2, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) 1-metil-4-propilbenzol **B)** 1,4-dietilbenzol
C) butilbenzol **D)** 1,3,5-trimetilbenzol

12. Q Formulasi C_7H_{14} va asosiy halqasida 3 ta uglerod atomi bo'lgan (siklopropan hosilalari), halqaga aynan ikkita radikal birikkan, o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 7 ta **B)** 6 ta **C)** 5 ta **D)** 8 ta

13. Q 2,3-dimetilpentan molekulasida nechta birlamchi uglerod atomi mavjud? 1,5 ball

A) 4 ta **B)** 1 ta **C)** 5 ta **D)** 2 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball
1) metilsiklopentan va siklogeksan bir-biriga izomer;
2) propan va siklopropan bir-biriga izomer;
3) butadiyen-1,3 va buten-1 ning molekulyar formulasi bir xil;
4) siklobutan va butan bir-biriga izomer;
5) butan va izobutan bir-biriga izomer

A) 1, 3, 5 **B)** 1, 5 **C)** 1, 2, 5 **D)** 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_9H_{20} bo'lgan, asosiy zanjirida 6 ta uglerod atomi bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 14 ta **B)** 13 ta **C)** 11 ta **D)** 12 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 8-variant

1 B A	2 B A	3 B A	4 Q B	5 Q C	6 Q B	7 Q B
8 Q C	9 Q B	10 Q B	11 Q B	12 Q B	13 Q A	14 M B
15 M D						

1. A) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 1, 2, 5.

2. A) Eng uzun uglerod zanjiri 7 ta atomdan iborat (geptan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 2,3,5-trimetilgeptan.

3. A) Sikloalkanlar: 1,3-dimetilsiklobutan, metilsiklopentan, siklogeksan va h.k.. Javob: 2, 4, 5.

4. B) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2,4-dimetilpentan. n = 120/100 = 1,2 mol; m = 1,2·134,5 = 161,4 g.

5. C) $n(alkanlar) = n(H_2O) - n(CO_2) = 1$ mol. $6·x + 8·(1 - x) = 6,4 → x = 0,8$ mol.

6. B) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 0,9; mahsulotlar: 113a + (M + 34,5)b = 105,6; 42a + M·b = 49 → a = 0,7, b = 0,2, M = 98 → C_7H_{14}.

7. B) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2) = 800·8/100/160 = 0,4$ mol = n(butadiyen). m(butadiyen) = 21,6 g; n(buten) = (111,2 - 21,6)/56 = 1,6 mol → buten 4:1 butadiyen.

8. C) Gazlar: CH_4 (16) va C_2H_2 (26). n = 0,6 mol; $16·x + 26·(0,6 - x) = 0,6·21 → n(CH_4) = 0,3$ mol, $n(C_2H_2) = 0,3$ mol. $m(CaC_2) = 19,2$ g.

9. B) $n(H_2) = (106,4 - 99,2)/2 = 3,6$ mol $= 3·(a + b) → a + b = 1,2$. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 131,6/329 = 0,4 mol = a → b = 0,8 mol.

10. B) n(C) = 21,6/12 = 1,8 mol, n(H) = 3,6 mol; bitta molekulada C: 1,8/0,3 = 6, H: 3,6/0,3 = 12 → C_6H_{12}.

11. B) $C_nH_{2n-6} + (3n - 3)/2\ O_2 → nCO_2 + (n - 3)H_2O$. $n(O_2)/n(CO_2) = (3n - 3)/(2n) = 1,35/1 → n = 10$ ($C_{10}H_{14}$). Oksidlanish mahsulotlari bo'yicha — 1,4-dietilbenzol.

12. B) Halqaga 4 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 6 ta izomer.

13. A) 2,3-dimetilpentan: birlamchi – 4, ikkilamchi – 1, uchlamchi – 2, to'rtlamchi – 0 ta uglerod atomi.

14. B) To'g'ri: 1, 5. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. D) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 5 C – 8 ta; 6 C – 12 ta; 7 C – 11 ta; 8 C – 3 ta; 9 C – 1 ta. Shartga mos — 12 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 9-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $HC{\equiv}CH + H_2O \xrightarrow{Hg^{2+}} ...$

2) $Al_4C_3 + H_2O \rightarrow ...$

3) $CH_2{=}CH_2 + H_2O \xrightarrow{H_2SO_4} ...$

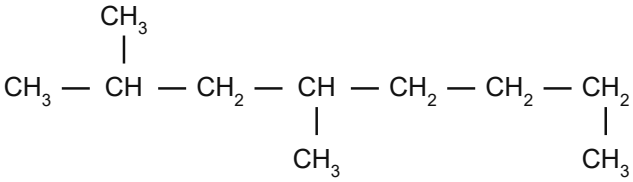
4) $HC{\equiv}CH + 2H_2 \xrightarrow{Ni,\ t^\circ} ...$

5) $CH_2{=}CH_2 + Br_2 \rightarrow ...$

6) $C_2H_5OH + Na \rightarrow ...$

A) 2, 4 **B)** 2, 3, 4 **C)** 3, 4 **D)** 2, 4, 5

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 5,7-dimetiloktan **B)** 2,4-dimetiloktan

C) 4-metiloktan **D)** 2,4,7-trimetilgeptan

3. B Arenlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) orto-ksilol; 2) 2-metilbuten-2; 3) geksadiyen-1,5; 4) metilsiklopentan; 5) neopentan; 6) 1,3,5-trimetilbenzol

A) 1, 5, 6 **B)** 1, 3, 6 **C)** 1 **D)** 1, 6

4. Q 34,4 gramm 2,3-dimetilbutan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 2-xlor-2,3-dimetilbutan – 51,6 g **B)** 2-xlor-2,3-dimetilbutan – 48,2 g

C) 1-xlor-2,3-dimetilbutan – 48,2 g **D)** 3-xlor-2,3-dimetilbutan – 51,6 g

5. C_4H_{10} va C_8H_{18} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 6 mol CO_2 va 7 mol H_2O hosil bo'ldi. Dastlabki aralashma massasini (g) aniqlang. 1,5 ball

A) 86 **B)** 88 **C)** 79 **D)** 95,2

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 63 gramm aralashma mavjud. Aralashmadagi molekulalar soni $6{,}622 \cdot 10^{23}$ ga teng. Shu aralashma 1,1 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 126,5 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_7H_{14} **B)** C_6H_{12} **C)** C_5H_{10} **D)** C_8H_{16}

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 27,5 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 2% li bromli suvning 2000 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 1:1 butadiyen **B)** buten 2:1 butadiyen

C) buten 1:2 butadiyen **D)** buten 1:4 butadiyen

8. Kalsiy karbid va alyuminiy karbid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 26,88 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=18,5 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi alyuminiy karbid massasini (g) aniqlang. 1,5 ball

A) 129,6 **B)** 19,2 **C)** 43,2 **D)** 86,4

9. 218,4 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 204 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 394,8 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol siklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 1,2 **B)** 1,4 **C)** 1 **D)** 2,4

10. 0,2 mol organik modda tarkibida 24 gramm uglerod va 3,6 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) $C_{10}H_{18}$ **B)** C_9H_{18} **C)** $C_{10}H_{20}$ **D)** $C_{10}H_{22}$

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 6 mol kislorod sarflangani va 4,5 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda ftal kislota va CO_2, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) 1-metil-4-etilbenzol **B)** 1-metil-2-etilbenzol
C) toluol **D)** propilbenzol

12. Q Formulasi C_6H_{12} va asosiy halqasida 4 ta uglerod atomi bo'lgan (siklobutan hosilalari), halqaga faqat metil radikallari birikkan, o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 4 ta **B)** 5 ta **C)** 3 ta **D)** 2 ta

13. Q 2,4,5-trimetilgeptan molekulasida nechta birlamchi uglerod atomi mavjud? 1,5 ball

A) 3 ta **B)** 6 ta **C)** 2 ta **D)** 5 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball
1) etilen kaliy permanganatning suvli eritmasi bilan reaksiyaga kirishganda ikki atomli spirt hosil bo'ladi;
2) benzol va toluol bir-biriga gomolog;
3) 2,2-dimetilpropan molekulasida 3 ta birlamchi uglerod atomi bor;
4) butadiyen-1,3 va buten-1 ning molekulyar formulasi bir xil;
5) propan va butan bir-biriga gomolog

A) 2, 5 **B)** 1, 2, 5 **C)** 1, 2, 4, 5 **D)** 1, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_7H_{16} bo'lgan, asosiy zanjirida 5 ta uglerod atomi bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 6 ta **B)** 7 ta **C)** 4 ta **D)** 5 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 9-variant

1 B A	2 B B	3 B D	4 Q B	5 Q A	6 Q B	7 Q A
8 Q C	9 Q A	10 Q A	11 Q B	12 Q C	13 Q D	14 M B
15 M D						

1. A) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 2, 4.

2. B) Eng uzun uglerod zanjiri 8 ta atomdan iborat (oktan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 2,4-dimetiloktan.

3. D) Arenlar: orto-ksilol, 1-metil-4-etilbenzol, toluol va h.k.. Javob: 1, 6.

4. B) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2,3-dimetilbutan. n = 34,4/86 = 0,4 mol; m = 0,4·120,5 = 48,2 g.

5. A) m = m(C) + m(H) = 12·6 + 2·7 = 86 g.

6. B) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 1,1; mahsulotlar: 113a + (M + 34,5)b = 126,5; 42a + M·b = 63 → a = 0,7, b = 0,4, M = 84 → C_6H_{12}.

7. A) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2) = 2000·2/100/160 = 0,25$ mol = n(butadiyen). m(butadiyen) = 13,5 g; n(buten) = (27,5 − 13,5)/56 = 0,25 mol → buten 1:1 butadiyen.

8. C) Gazlar: C_2H_2 (26) va CH_4 (16). n = 1,2 mol; 26·x + 16·(1,2 − x) = 1,2·18,5 → $n(C_2H_2) = 0,3$ mol, $n(CH_4) = 0,9$ mol. $m(Al_4C_3) = 43,2$ g.

9. A) $n(H_2) = (218,4 - 204)/2 = 7,2$ mol = 3·(a + b) → a + b = 2,4. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 394,8/329 = 1,2 mol = a → b = 1,2 mol.

10. A) n(C) = 24/12 = 2 mol, n(H) = 3,6 mol; bitta molekulada C: 2/0,2 = 10, H: 3,6/0,2 = 18 → $C_{10}H_{18}$.

11. B) $C_nH_{2n-6} + (3n - 3)/2\ O_2 \rightarrow nCO_2 + (n - 3)H_2O$. $n(O_2)/n(CO_2) = (3n - 3)/(2n) = 6/4,5$ → n = 9 (C_9H_{12}). Oksidlanish mahsulotlari bo'yicha — 1-metil-2-etilbenzol.

12. C) Halqaga 2 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 3 ta izomer.

13. D) 2,4,5-trimetilgeptan: birlamchi – 5, ikkilamchi – 2, uchlamchi – 3, to'rtlamchi – 0 ta uglerod atomi.

14. B) To'g'ri: 1, 2, 5. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. D) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 4 C – 1 ta; 5 C – 5 ta; 6 C – 2 ta; 7 C – 1 ta. Shartga mos — 5 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: 10-variant

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $C + 2H_2 \xrightarrow{t^\circ, p} \ldots$

2) $HC{\equiv}CH + H_2O \xrightarrow{Hg^{2+}} \ldots$

3) $CH_3COONa + H_2O \xrightarrow{\text{elektroliz}} \ldots$

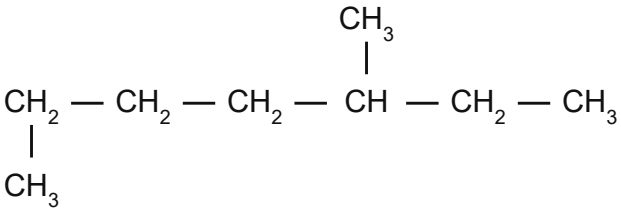
4) $CH_3COOH + C_2H_5OH \xrightarrow{H^+} \ldots$

5) $CH_4 + Cl_2 \xrightarrow{h\nu} \ldots$

6) $HC{\equiv}CH + 2H_2 \xrightarrow{Ni,\ t^\circ} \ldots$

A) 3, 4, 6 **B)** 1, 3, 6 **C)** 1, 3, 5, 6 **D)** 3, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 3-metilgeptan **B)** 5-metilgeptan

C) 1,4-dimetilgeksan **D)** 3-metilgeksan

3. B Arenlar sinfiga mansub bo'lgan moddalarni tanlang: 1,2 ball
1) penten-1; 2) orto-ksilol; 3) toluol; 4) siklogeksan; 5) etilsiklopropan; 6) 3-metilbuten-1

A) 2, 3, 4 **B)** 2, 3, 5 **C)** 1, 2, 3 **D)** 2, 3

4. Q 43 gramm 2,3-dimetilbutan yetarli miqdordagi xlor bilan yorug'likda reaksiyaga kirishganda alkan to'liq sarflandi va monoxloralkan hamda HCl hosil bo'ldi. Asosiy mahsulot bo'lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 1-xlor-2,3-dimetilbutan – 60,2 g **B)** 2-xlor-2,3-dimetilbutan – 60,2 g

C) 3-xlor-2,3-dimetilbutan – 50,2 g **D)** 2-xlor-2,3-dimetilbutan – 50,2 g

5. C_3H_8 va C_6H_{14} dan iborat alkanlar aralashmasi kislorodda to'liq yondirilganda 4,2 mol CO_2 va 5,2 mol H_2O hosil bo'ldi. Dastlabki aralashma massasini (g) aniqlang. 1,5 ball

A) 60,8 **B)** 70 **C)** 55,6 **D)** 61,6

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 60,2 gramm aralashma mavjud. Aralashmadagi molekulalar soni $5{,}418 \cdot 10^{23}$ ga teng. Shu aralashma 0,9 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 109,5 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_5H_{10} **B)** C_4H_8 **C)** C_8H_{16} **D)** C_7H_{14}

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 137 gramm bo'lgan sopolimer hosil bo'ldi. Ushbu sopolimer 3,2% li bromli suvning 7500 grammini rangsizlantira olishi ma'lum bo'lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo'lganligini aniqlang. 1,5 ball

A) buten 1:1 butadiyen **B)** buten 3:2 butadiyen

C) buten 2:3 butadiyen **D)** buten 2:1 butadiyen

8. Kalsiy karbid va alyuminiy karbid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 40,32 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=21 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi alyuminiy karbid massasini (g) aniqlang. 1,5 ball

A) 86,4 **B)** 43,2 **C)** 57,6 **D)** 129,6

9. 120,4 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 112,6 gramm bo'lgan arenlar aralashmasi hosil bo'ldi. Arenlar brom bilan reaksiyaga kirishganda 263,2 gramm 2,4,6-tribrom-1-metilbenzol hosil bo'ldi. Dastlabki aralashmada necha mol siklogeksan mavjud bo'lganligini aniqlang. 1,5 ball

A) 1,3 **B)** 0,5 **C)** 0,7 **D)** 0,8

10. Q 0,1 mol organik modda tarkibida 7,2 gramm uglerod va 1,2 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_7H_{12} **B)** C_7H_{14} **C)** C_6H_{10} **D)** C_6H_{12}

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi; reaksiyada 3 mol kislorod sarflangani va 2,25 mol CO_2 hosil bo'lgani ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda ftal kislota va CO_2, kaliy sulfat, marganes (II) sulfat va suv hosil bo'ladi. Aren nomini aniqlang. 1,5 ball

A) 1-metil-2-etilbenzol **B)** 1-metil-4-etilbenzol

C) para-ksilol **D)** propilbenzol

12. Q Formulasi C_8H_{16} va asosiy halqasida 5 ta uglerod atomi bo'lgan (siklopentan hosilalari), halqaga aynan ikkita radikal birikkan, o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 5 ta **B)** 3 ta **C)** 4 ta **D)** 2 ta

13. Q 2,3,3-trimetilgeksan molekulasida nechta birlamchi uglerod atomi mavjud? 1,5 ball

A) 2 ta **B)** 5 ta **C)** 6 ta **D)** 1 ta

14. M To'g'ri ma'lumotlarni tanlang: 2,5 ball

1) siklopropan va propen xlor bilan birikkanda hosil bo'lgan mahsulotlarning tuzilishi bir xil bo'ladi;
2) etilen kaliy permanganatning suvli eritmasi bilan reaksiyaga kirishganda uch atomli spirt hosil bo'ladi;
3) benzol va toluol bir-biriga gomolog;
4) siklobutan va buten-1 bir-biriga izomer;
5) 2,2-dimetilpropan molekulasida 4 ta birlamchi uglerod atomi bor

A) 3, 4, 5 **B)** 1, 3, 5 **C)** 3, 4 **D)** 1, 3, 4, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_8H_{18} bo'lgan, asosiy zanjirida 6 ta yoki 6 tadan ko'proq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlar nechta?

A) 12 ta **B)** 13 ta **C)** 11 ta **D)** 10 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 10-variant

1 B — B	2 B — A	3 B — D	4 Q — B	5 Q — A	6 Q — D	7 Q — C
8 Q — B	9 Q — B	10 Q — D	11 Q — A	12 Q — B	13 Q — B	14 M — A
15 M — C						

1. B) Alkanlar karbon kislota tuzlarini ishqor bilan suyuqlantirish, Al_4C_3 va Be_2C gidrolizi, to'yinmagan uglevodorodlarni gidrogenlash, Vyurs va Kolbe reaksiyalari orqali olinadi. Javob: 1, 3, 6.

2. A) Eng uzun uglerod zanjiri 7 ta atomdan iborat (geptan). O'rinbosarlar eng kichik raqamlarni oladigan tomondan raqamlanadi. Javob: 3-metilgeptan.

3. D) Arenlar: orto-ksilol, 1-metil-4-etilbenzol, toluol va h.k.. Javob: 2, 3.

4. B) Radikal almashinishda vodorod eng oson uchlamchi uglerod atomidan almashinadi → 2-xlor-2,3-dimetilbutan. n = 43/86 = 0,5 mol; m = 0,5·120,5 = 60,2 g.

5. A) m = m(C) + m(H) = 12·4,2 + 2·5,2 = 60,8 g.

6. D) Siklopropan xlorni biriktiradi ($C_3H_6Cl_2$), katta halqali sikloalkan esa almashinishga kirishadi ($C_nH_{2n-1}Cl + HCl$). a + b = 0,9; mahsulotlar: 113a + (M + 34,5)b = 109,5; 42a + M·b = 60,2 → a = 0,5, b = 0,4, M = 98 → C_7H_{14}.

7. C) Sopolimerda har bir butadiyen bo'g'inida bitta qo'sh bog' qoladi: $n(Br_2)$ = 7500·3,2/100/160 = 1,5 mol = n(butadiyen). m(butadiyen) = 81 g; n(buten) = (137 − 81)/56 = 1 mol → buten 2:3 butadiyen.

8. B) Gazlar: C_2H_2 (26) va CH_4 (16). n = 1,8 mol; 26·x + 16·(1,8 − x) = 1,8·21 → $n(C_2H_2)$ = 0,9 mol, $n(CH_4)$ = 0,9 mol. $m(Al_4C_3)$ = 43,2 g.

9. B) $n(H_2)$ = (120,4 − 112,6)/2 = 3,9 mol = 3·(a + b) → a + b = 1,3. Faqat toluoldan tribromtoluol hosil bo'ladi: n = 263,2/329 = 0,8 mol = a → b = 0,5 mol.

10. D) n(C) = 7,2/12 = 0,6 mol, n(H) = 1,2 mol; bitta molekulada C: 0,6/0,1 = 6, H: 1,2/0,1 = 12 → C_6H_{12}.

11. A) C_nH_{2n-6} + (3n − 3)/2 O_2 → nCO_2 + (n − 3)H_2O. $n(O_2)/n(CO_2)$ = (3n − 3)/(2n) = 3/2,25 → n = 9 (C_9H_{12}). Oksidlanish mahsulotlari bo'yicha — 1-metil-2-etilbenzol.

12. B) Halqaga 3 ta uglerod atomi radikallar ko'rinishida turli usullarda birikadi (bitta radikal, ikki radikal — 1,1-, 1,2-, 1,3-holatlar va h.k.). Jami 3 ta izomer.

13. B) 2,3,3-trimetilgeksan: birlamchi – 5, ikkilamchi – 2, uchlamchi – 1, to'rtlamchi – 1 ta uglerod atomi.

14. A) To'g'ri: 3, 4, 5. Izomerlar — molekulyar formulasi bir xil, tuzilishi har xil moddalar; gomologlar — tarkibi CH_2 guruhga farq qiluvchi bir sinf vakillari.

15. C) Izomerlarni asosiy zanjir uzunligi bo'yicha sanaymiz: 4 C – 1 ta; 5 C – 6 ta; 6 C – 7 ta; 7 C – 3 ta; 8 C – 1 ta. Shartga mos — 11 ta.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · KIMYO · CHSB-1

ChSB-1 tayyorgarlik: I chorak demo varianti

15 ta savol · 45 daqiqa · 23,6 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Qaysi reaksiyalar natijasida alkan hosil bo'ladi? 1,2 ball

1) $CH_3CH_2COONa + NaOH \xrightarrow{t^\circ} ...$

2) $C_{17}H_{35}COOH + CH_2(OH)CH(OH)CH_2OH \rightarrow ...$

3) $CaC_2 + H_2O \rightarrow ...$

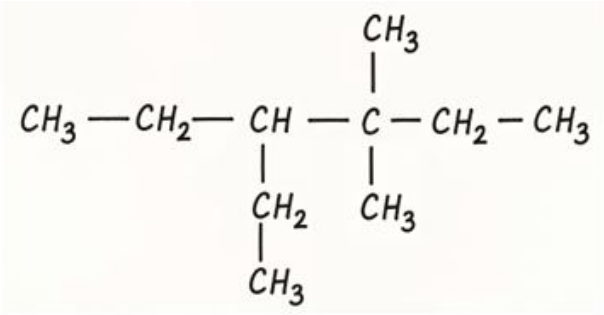
4) $Al_4C_3 + H_2O \rightarrow ...$

5) $CH_2{=}CH{-}CH_3 + H_2O \xrightarrow{H_2SO_4} ...$

6) $CH_2{=}CH{-}CH_3 + H_2 \xrightarrow{Ni,\ t^\circ} ...$

A) 2, 6 **B)** 3, 4 **C)** 1, 4, 6 **D)** 3, 5, 6

2. B Quyidagi moddaning nomini aniqlang. 1,2 ball



A) 3-etil-4,4-dimetilgeksan **B)** 1,1-dietil-2-metilbutan

C) 3,3-dimetil-4-etilgeksan **D)** 3-etil-4-metilgeksan

3. B Alkinlar sinfiga mansub bo‘lgan moddalarni tanlang: 1,2 ball

1) 1,3-dimetilsiklobutan; 2) 3-metilpentin-1; 3) orto-ksilol;
4) propilatsetilen; 5) izobutan; 6) 1-metil-4-etilbenzol

A) 3, 4 **B)** 3, 5 **C)** 2, 4 **D)** 1, 6

4. 43,2 gramm 2-metilbutan yetarli miqdordagi xlor bilan reaksiyaga kirishganda alkan reaksiyada to‘liq sarflandi va reaksiya natijasida monoxloralkan va HCl hosil bo‘ldi. Hosil bo‘lgan monoxloralkan nomini va massasini (g) aniqlang. 1,5 ball

A) 1-xlor-2-metilbutan – 74,2 g
B) 2-xlor-2-metilbutan – 63,9 g
C) 1-xlor-2-metilbutan – 106,5 g
D) 2-xlor-3-metilbutan – 106,5 g

5. C_5H_{12} va C_7H_{16} dan iborat alkanlar aralashmasi mavjud. Aralashmadagi alkanlar miqdorlari yig'indisi 1,2 mol. Alkanlar aralashmasi kislorodda to'liq yondirilganda 7,6 mol CO_2 va 8,8 mol H_2O hosil bo'ldi. Dastlabki aralashma massasini (g) aniqlang. 1,5 ball

A) 114,4 **B)** 99,0 **C)** 108,8 **D)** 92,0

6. Siklopropan va noma'lum sikloalkandan tashkil topgan 54,6 gramm aralashma mavjud. Sikloalkanlar aralashmasidagi molekulalar soni $6{,}02\cdot10^{23}$ ga teng. Shu aralashma 1 mol xlor bilan reaksiyaga kirishdi. Reaksiyada xlor va sikloalkanlar to'liq sarflandi va dixloralkan, HCl va X moddadan iborat bo'lgan aralashma hosil bo'ldi (X – molekulasida 1 ta xlor atomi bo'lgan organik modda). Dixloralkan va X massalarining yig'indisi 114,65 gramm. Noma'lum sikloalkan formulasini aniqlang. 1,5 ball

A) C_4H_8 **B)** C_7H_{14} **C)** C_6H_{12} **D)** C_5H_{10}

7. Buten va butadiyen aralashmasi sopolimerlanganda massasi 108,8 gramm bo‘lgan sopolimer hosil bo‘ldi. Ushbu sopolimer 4% li bromli suvning 6400 grammini rangsizlantira olishi ma’lum bo‘lsa, dastlabki aralashmada organik moddalar qanday miqdoriy (mol) nisbatda bo‘lganligini aniqlang. 1,5 ball

A) buten 6:1 butadiyen
B) buten 1:4 butadiyen
C) buten 1:3 butadiyen
D) buten 2:1 butadiyen

8. Kalsiy karbid va alyuminiy sulfid aralashmasi suv bilan reaksiyaga kirishdi. Reaksiya natijasida hajmi 44,8 litr (n.sh.) va o'rtacha molyar massasi $M_{o'rtacha}$=28,4 g/mol bo'lgan gazlar aralashmasi hosil bo'ldi. Dastlabki aralashmadagi alyuminiy sulfid massasini (g) aniqlang. 1,5 ball

A) 30 **B)** 60 **C)** 75 **D)** 45

9. Q 184,8 gramm metilsiklogeksan va siklogeksan aralashmasi degidrogenlanishi natijasida massasi 172,8 gramm bo‘lgan arenlar aralashmasi hosil bo‘ldi. Arenlar brom bilan reaksiyaga kirishganda 394,8 gramm 2,4,6-tribrom-1-metilbenzol hosil bo‘ldi. Dastlabki aralashmada necha mol siklogeksan mavjud bo‘lganligini aniqlang. 1,5 ball

A) 1,4 **B)** 1,0 **C)** 0,8 **D)** 1,2

10. Q 0,3 mol organik modda tarkibida 28,8 gramm uglerod va 3 gramm vodorod mavjud. Organik modda formulasini aniqlang. 1,5 ball

A) C_7H_{16} **B)** C_8H_{18} **C)** C_8H_{10} **D)** C_7H_{10}

11. Q Aren kislorodda yondirilganda CO_2 va H_2O hosil bo'ldi. Agar reaksiyada 4/3 mol kislorod sarflanganligi va 1 mol CO_2 hosil bo'lganligi ma'lum. Shu aren kaliy permanganat bilan sulfat kislota ishtirokida oksidlanganda benzoy kislota, CO_2, kaliy sulfat, marganes sulfat va suv hosil bo'lishi ma'lum. Aren nomini aniqlang. 1,5 ball

A) propilbenzol **B)** etilbenzol
C) 1,3-dimetilbenzol **D)** 2-metil-1-etilbenzol

12. Q Formulasi C_7H_{14} va asosiy halqasida 4 ta uglerod atomi bo'lgan o'zaro izomer hisoblanuvchi sikloalkanlar nechta? (geometrik va fazoviy izomerlar hisobga olinmasin) 1,5 ball

A) 10 ta **B)** 8 ta **C)** 9 ta **D)** 11 ta

13. Q 3-metil-4-etilgeptan molekulasida nechta ikkilamchi uglerod atomlari mavjud? 1,5 ball

A) 7 ta **B)** 4 ta **C)** 5 ta **D)** 6 ta

14. M To‘g‘ri ma’lumotlarni tanlang: 2,5 ball

1) 2-metilpentan va 3-metilpentan molyar massalari har xil;
2) 2-metilpentan va 3-metilpentan molyar massalari bir xil;
3) siklopropan xlor bilan birikish reaksiyasiga kirishganda, shuningdek propen xlor bilan birikish reaksiyasiga kirishganda hosil bo‘lgan mahsulotlar tuzilishi bir xil bo‘ladi;
4) siklopropan xlor bilan birikish reaksiyaga kirishganda, shuningdek propen xlor bilan birikish reaksiyasiga kirishganda hosil bo‘lgan mahsulotlarning tarkibi bir xil bo‘ladi;
5) etilen kaliy permanganatning suvli eritmasi bilan reaksiyaga kirishganda uch atomli spirt hosil bo‘ladi.

A) 1, 3 **B)** 2, 4, 5 **C)** 2, 4 **D)** 1, 3, 5

15. M Formulasi C_6H_{14} bo'lgan o'zaro izomer hisoblanuvchi 5 ta alkan mavjud. Ularni ulardagi birlamchi uglerod atomlari soniga ko'ra 3 guruhga bo'lish mumkin. 2,5 ball

Guruh tartib raqami	Molekuladagi birlamchi uglerod atomlari soni	Alkanlar
1	2 ta	n-geksan
2	3 ta	2-metilpentan; 3-metilpentan
3	4 ta	2,2-dimetilbutan; 2,3-dimetilbutan

Formulasi C_8H_{18} bo'lgan, asosiy zanjirida 6 ta yoki 6 tadan kamroq uglerod atomlari bo'lgan, o'zaro izomer hisoblanuvchi alkanlarni ularning molekulasidagi birlamchi uglerod atomlari soniga ko'ra nechta guruhga bo'lish mumkin?

A) 2 ta **B)** 5 ta **C)** 4 ta **D)** 3 ta

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: I chorak demo varianti

1 B C	2 B C	3 B C	4 Q B	5 Q C	6 Q C	7 Q B
8 Q A	9 Q C	10 Q C	11 Q A	12 Q B	13 Q B	14 M C
15 M C						

1. C) To'g'ri javob: **C) 1, 4, 6**

2. C) To'g'ri javob: **C) 3,3-dimetil-4-etilgeksan**

3. C) To'g'ri javob: **C) 2, 4**

4. B) To'g'ri javob: **B) 2-xlor-2-metilbutan – 63,9 g**

5. C) To'g'ri javob: **C) 108,8**

6. C) To'g'ri javob: **C) C_6H_{12}**

7. B) To'g'ri javob: **B) buten 1:4 butadiyen**

8. A) To'g'ri javob: **A) 30**

9. C) To'g'ri javob: **C) 0,8**

10. C) To'g'ri javob: **C) C_8H_{10}**

11. A) To'g'ri javob: **A) propilbenzol**

12. B) To'g'ri javob: **B) 8 ta**

13. B) To'g'ri javob: **B) 4 ta**

14. C) To'g'ri javob: **C) 2, 4**

15. C) To'g'ri javob: **C) 4 ta**

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 1-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning eng katta darajasi va bosh koeffitsiyenti yig‘indisini toping. 2 ball

$$P(x) = 9x^8 - 4x^2 - 5x^4 + 65 + 7x^9$$

A) 9 **B)** 16 **C)** 17 **D)** 74

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{14} + x^2 + 4$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) -89
II. $P(x) = x^3 + 6x + 6$ ko‘phadni $x^2 + 4$ ga bo‘lgandagi qoldiqni toping.	B) 4
III. $P(x) = -3x^{n-13} + 8x^{13-n} + x^6 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) $6x + 2$
IV. $P(x) = x^3 - 5x^2 + 3x - 8$ ko‘phadni $x + 3$ ga bo‘lgandagi qoldiqni toping.	D) 6
	E) 89
	F) $2x + 6$

A) I-B; II-C; III-F; IV-A **B)** I-C; II-F; III-D; IV-A

C) I-B; II-E; III-D; IV-C **D)** I-B; II-F; III-D; IV-A

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x+1)^7(x-6)^2(x+5)$$

A) -1 va -5 **B)** -5 va 6 **C)** hammasi **D)** -1 va 6

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 - x^2 - 21x - 4$ va $Q(x) = x^2 - 16$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x + 4$ **B)** $x + 5$ **C)** $x + 3$ **D)** $x - 4$

5. Q Quyida $P(x)$ ko‘phadni $x-1$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	2	6	0	4	5
1	2	8	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $2x^3+8x^2+8x+14$ va $r=-17$
B) $2x^3+8x^2+8x+12$ va $r=17$
C) $2x^3+8x^2-8x+12$ va $r=17$
D) $2x^3+8x^2+8x-12$ va $r=-17$

6. Q $P(x)=12x^3-29x^2+8x+4$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-\frac{1}{4};\frac{2}{3};2$
B) $-2;-\frac{2}{3};\frac{1}{4}$
C) $-\frac{2}{3};-\frac{1}{4};2$
D) $-\frac{1}{4};\frac{1}{2};\frac{2}{3}$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(8x-1)(6x-1)(24x-1)(12x-1)=57$$

A) $\frac{5}{12}$
B) $\frac{3}{8}$
C) $\frac{5}{48}$
D) $\frac{5}{24}$

8. Q Ushbu $x^2-x+a=0$ tenglamaning ildizlari $x^3+bx-4=0$ tenglamaning ham ildizlari bo‘lsa, ab ning qiymatini toping. 3,5 ball

A) 21
B) 18
C) -20
D) 20

9. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

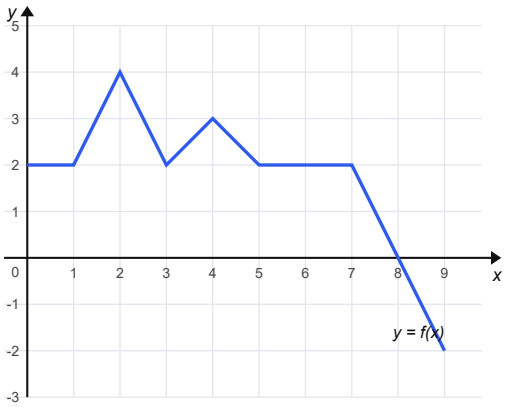
$$x^4+3x^3+2x^2+3x+1=0$$

A) -3
B) -6
C) 3
D) -2

10. B $y=\frac{2}{x-6}-4$ funksiyaning qiymatlar sohasini toping. 2 ball

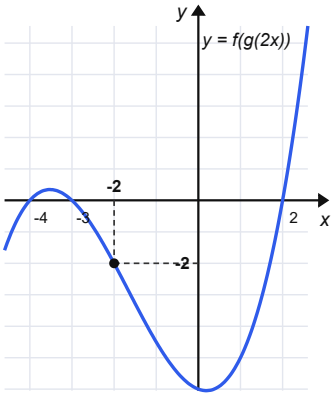
A) $(-\infty;6)\cup(6;\infty)$
B) $(-\infty;-4)\cup(-4;\infty)$
C) $(-4;0)$
D) $(-\infty;4)\cup(4;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 20 B) 22 C) 17 D) 19

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(-9)=-4$ bo‘lsa, $f^{-1}(-2)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) -9 B) 9 C) -4 D) -2

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Qirrasi a ga teng kubning yon sirti yuzi $6a^2$ ga teng.		
2.	Qirrasi a ga teng kubning diagonali $a\sqrt{3}$ ga teng.		
3.	To‘g‘ri prizmaning yon qirralari asos tekisligiga perpendikulyar.		
4.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = a^2 - h^2$ tenglik o‘rinli.		

A) 1-to‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

B) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

C) 1-noto‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

D) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

14. Q Muntazam ko‘pburchakning tomoni a, unga tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $\frac{a^2}{2(R^2-r^2)}$ ifodaning qiymatini toping. 5,2 ball

A) 1 **B)** 4 **C)** 3 **D)** 2

15. Q SABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning SG balandligi $SG = 2\sqrt{6}\ cm$ ga teng. Tetraedrning hajmini toping. 5,2 ball

A) $54\sqrt{2}\ cm^3$ **B)** $72\sqrt{2}\ cm^3$ **C)** $18\sqrt{2}\ cm^3$ **D)** $36\sqrt{2}\ cm^3$

16. B Berilgan ma’lumotlarga ko‘ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball
Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Silindrning to‘la sirt yuzi	A) $\pi r l$
II. Konusning to‘la sirt yuzi	B) $\frac{1}{3}\pi r^2 h$
III. Konusning yon sirt yuzi	C) $2\pi r(r+h)$
	D) $\pi r(r+l)$
	E) $\pi r^2 h$
	F) $4\pi r^2$

A) I–C; II–D; III–A **B)** I–F; II–D; III–A **C)** I–C; II–F; III–D **D)** I–C; II–D; III–F

17. Q To‘g‘ri doiraviy silindrning balandligi $5\ cm$, asos radiusi esa $3\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo‘lgan masofalari mos ravishda $1\ cm$ va $3\ cm$. Agar $AB = 6\ cm$ bo‘lsa, silindr o‘qidan AB kesmagacha bo‘lgan eng qisqa masofani toping. 5,2 ball

A) $1\ cm$ B) $\sqrt{3}\ cm$ C) $2\ cm$ D) $\sqrt{2}\ cm$

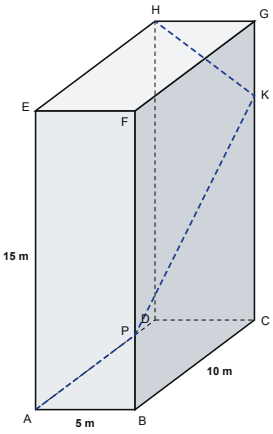
18. Q Yon tomonlari $15\ cm$ dan, asosi $18\ cm$ bo‘lgan teng yonli uchburchak o‘zining yon tomoni atrofida to‘liq aylantiriladi. Hosil bo‘lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $\frac{2376}{5}\pi\ cm^2$ B) $396\pi\ cm^2$ C) $216\pi\ cm^2$ D) $432\pi\ cm^2$

19. Q Shar shaklidagi tarvuz o‘zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $92\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 5 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $17\pi\ cm^2$ ga teng. Sharning sirt yuzini toping. 5,2 ball

A) $512\pi\ cm^2$ B) $468\pi\ cm^2$ C) $456\pi\ cm^2$ D) $368\pi\ cm^2$

20. M O‘lchamlari $5\ m,\ 10\ m$ va $15\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $30\ m$ B) $15\sqrt{2}\ m$ C) $26\ m$ D) $25\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 1-variant

1 B	B	2 Q	D	3 B	D	4 Q	A	5 Q	B	6 Q	A	7 Q	D
8 Q	D	9 Q	A	10 B	B	11 M	D	12 Q	A	13 B	C	14 Q	D
15 Q	C	16 B	A	17 Q	A	18 Q	A	19 Q	B	20 M	D		

1. B) Eng katta daraja 9, bosh koeffitsiyent 7: $9+7=16$.

2. D) I: ozod had 4. II: $x^3+6x+6=x(x^2+4)+2x+6$. III: ko‘phad bo‘lishi uchun $n=13$, daraja 6. IV: $P(-3)=-89$.

3. D) Karrali ildizlar: -1 (7 karrali), 6 (2 karrali). -5 – oddiy ildiz.

4. A) $Q=(x+4)(x-4)$. $P(-4)=0$, $P(4)=-40\neq 0\to$ EKUB $x+4$.

5. B) Gorner sxemasi: 2, 8, 8, 12, 17. Bo‘linma $2x^3+8x^2+8x+12$, qoldiq 17.

6. A) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p\mid 4$, $q\mid 12$). Tekshiramiz: $-\frac14;\frac23;2$.

7. D) $(8x-1)(12x-1)=96x^2-20x+1$, $(6x-1)(24x-1)=144x^2-30x+1$. $u=96x^2-20x$ desak, $(u+1)(\frac32u+1)=57\to u=-7$ yoki $u=\frac{16}{3}$. Faqat $u=\frac{16}{3}$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{5}{24}$.

8. D) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz -1. Ildizlar ko‘paytmasi: $a\cdot(-1)=4\to a=-4$; $b=a+(-1)(1)=-5$. $ab=20$.

9. A) x^2 ga bo‘lamiz va $t=x+\frac1x$ desak: $t^2+3t=0\to t=-3$ yoki $t=0$. Haqiqiy x faqat $|t|\geq 2$ da: $t=-3$. Har bir t uchun $x^2-tx+1=0$ (ko‘paytma 1, yig‘indi t). Javob: -3.

10. B) $\frac{2}{x-6}\neq 0\to y\neq -4$.

11. D) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=2$, $f(2)=4$, $f(3)=2$, $f(4)=3$, $f(5)=2$, $f(6)=2$, $f(7)=2$, $f(8)=0$, $f(9)=-2$. Eng katta qiymat $a=6, b=7, c=1, d=9, e=2$ da: $6+7+9-1-2=19$.

12. A) $x=-2$ da grafikdan $f(g(-4))=-2$. $g(-4)=-9\to f(-9)=-2\to f^{-1}(-2)=-9$.

13. C) 1) yon sirt $4a^2$ ✗. 2) $\sqrt{3a^2}$ ✓. 3) ta’rif ✓. 4) $f^2=h^2+\frac{a^2}{4}$ ✗.

14. D) Markazdan tomonga tushirilgan perpendikulyar: $\left(\frac{a}{2}\right)^2=R^2-r^2$, ya’ni $a^2=4(R^2-r^2)$. Qiymat 2.

15. C) $h=a\sqrt{\frac23}=\frac{a\sqrt6}{3}=2\sqrt6\to a=6$. Hajm $\frac{a^3\sqrt2}{12}=18\sqrt2\ cm^3$.

16. A) Silindrning to‘la sirt yuzi: $2\pi r(r+h)$. Konusning to‘la sirt yuzi: $\pi r(r+l)$. Konusning yon sirt yuzi: πrl.

17. A) Balandliklar farqi 2, asosdagi proyeksiya (vatar) $\sqrt{36-4}=4\sqrt2$. O‘qdan masofa $\sqrt{9-\frac{32}{4}}=1$ cm.

18. A) Uchburchak yuzi 108, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2\cdot108}{15}=\frac{72}{5}$. Ikki konus yon sirtlari: $S=\pi\cdot\frac{72}{5}(15+18)=\frac{2376}{5}\pi$.

19. B) $x^2+92=(x+5)^2+17\to x=5$, $R^2=117$. Javob 468π cm².

20. D) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(5+10+5)\times 15$ hosil bo‘ladi: $\sqrt{20^2+15^2}=25$ m.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 2-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning bosh koeffitsiyenti va ozod hadi ko‘paytmasini toping. 2 ball

$$P(x) = 5x^6 - 5x + 8x^9 - 8x^{10} + 47$$

A) 470 **B)** 39 **C)** -376 **D)** 235

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{10} + x^4 + 3$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) $3x + 4$
II. $P(x) = x^3 + 6x + 3$ ko‘phadni $x^2 + 2$ ga bo‘lgandagi qoldiqni toping.	B) 77
III. $P(x) = -3x^{n-10} + 8x^{10-n} + x^5 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) $4x + 3$
IV. $P(x) = x^3 - 2x^2 - 3x + 7$ ko‘phadni $x + 4$ ga bo‘lgandagi qoldiqni toping.	D) -77
	E) 5
	F) 3

A) I-B; II-C; III-E; IV-D **B)** I-F; II-C; III-E; IV-D

C) I-F; II-E; III-C; IV-D **D)** I-F; II-B; III-E; IV-D

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x-4)(x+5)^3(x-2)^5$$

A) -5 va 2 **B)** 4 va 2 **C)** -5 va 4 **D)** hammasi

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 + 7x^2 + 10x - 8$ va $Q(x) = x^2 + 3x - 4$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x - 1$ **B)** $x + 1$ **C)** $x + 4$ **D)** $x - 4$

5. Q Quyida $P(x)$ ko‘phadni $x-1$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	1	5	-1	-3	-4
1	1	6	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) x^3+6x^2+5x-2 va $r=2$
B) x^3+6x^2+5x+2 va $r=-2$
C) x^3+6x^2-5x+2 va $r=-2$
D) x^3+6x^2+5x+4 va $r=2$

6. Q $P(x)=4x^3+9x^2-x-6$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-2;-1;-\frac{3}{4}$
B) $-2;-1;\frac{3}{4}$
C) $-1;-\frac{1}{2};\frac{3}{4}$
D) $-\frac{3}{4};1;2$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(6x-1)(5x-1)(20x-1)(12x-1)=116$$

A) $\frac{1}{8}$
B) $\frac{1}{4}$
C) $\frac{1}{2}$
D) $\frac{5}{12}$

8. Q Ushbu $x^2+x+a=0$ tenglamaning ildizlari $x^3+bx-12=0$ tenglamaning ham ildizlari bo‘lsa, ab ning qiymatini toping. 3,5 ball

A) 132
B) 133
C) 130
D) -132

9. Q Tenglamaning haqiqiy ildizlari ko‘paytmasini toping. 3,5 ball

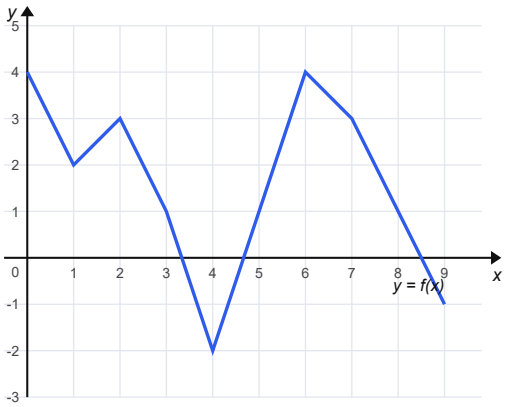
$$x^4+3x^3-16x^2+3x+1=0$$

A) -1
B) $-1{,}5$
C) $1{,}5$
D) 1

10. B $y=\frac{-6}{x-3}+2$ funksiyaning qiymatlar sohasini toping. 2 ball

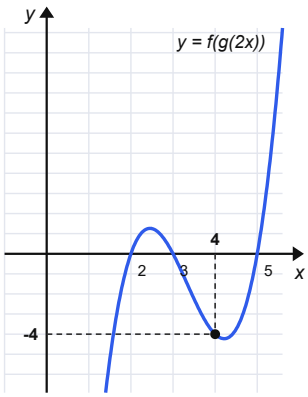
A) $(0;2)$
B) $(-\infty;2)\cup(2;\infty)$
C) $(-\infty;-2)\cup(-2;\infty)$
D) $(-\infty;3)\cup(3;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 16 **B)** 21 **C)** 18 **D)** 19

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(-6)=8$ bo‘lsa, $f^{-1}(-4)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) 8 **B)** 6 **C)** 4 **D)** -6

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Qirrasi a ga teng kubning diagonali $a\sqrt{3}$ ga teng.		
2.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = h^2 + \frac{a^2}{4}$ tenglik o‘rinli.		
3.	Qirrasi a ga teng kubning hajmi $3a^3$ ga teng.		
4.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = a^2 - h^2$ tenglik o‘rinli.		

A) 1-to‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

B) 1-to‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

C) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

D) 1-noto‘g‘ri, 2-noto‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

14. Q Tomoni 8 ga teng muntazam 9-burchakka tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $R^2 - r^2$ ning qiymatini toping. 5,2 ball

A) 4 **B)** 64 **C)** 16 **D)** 32

15. Q DABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning DG balandligi $DG = 3\sqrt{6}\ cm$ ga teng. Tetraedrning hajmini toping. 5,2 ball

A) $\frac{243}{4}\sqrt{2}\ cm^3$ **B)** $243\sqrt{2}\ cm^3$ **C)** $\frac{729}{4}\sqrt{2}\ cm^3$ **D)** $\frac{243}{2}\sqrt{2}\ cm^3$

16. B Berilgan ma'lumotlarga ko'ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball

Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Konusning yon sirt yuzi	A) $\pi r(r+l)$
II. Silindrning to'la sirt yuzi	B) $\frac{1}{3}\pi r^3$
III. Konusning to'la sirt yuzi	C) $2\pi r^2$
	D) $2\pi r(r+h)$
	E) $2\pi rh$
	F) πrl

A) I–F; II–D; III–C **B)** I–F; II–C; III–E **C)** I–F; II–D; III–A **D)** I–F; II–C; III–A

17. Q To'g'ri doiraviy silindrning balandligi $7\ cm$, asos radiusi esa $6\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo'lgan masofalari mos ravishda $1\ cm$ va $3\ cm$. Agar $AB = 7\ cm$ bo'lsa, silindr o'qidan AB kesmagacha bo'lgan eng qisqa masofani toping. 5,2 ball

A) $\sqrt{14}\ cm$ **B)** $\frac{\sqrt{95}}{2}\ cm$ **C)** $\frac{3\sqrt{11}}{2}\ cm$ **D)** $\frac{\sqrt{103}}{2}\ cm$

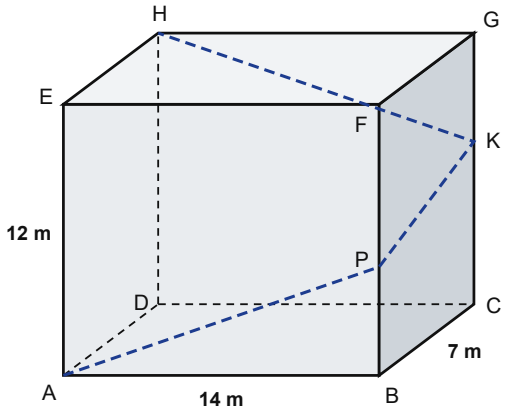
18. Q Yon tomonlari $13\ cm$ dan, asosi $24\ cm$ bo'lgan teng yonli uchburchak o'zining yon tomoni atrofida to'liq aylantiriladi. Hosil bo'lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $\frac{4440}{13}\pi\ cm^2$ **B)** $120\pi\ cm^2$ **C)** $185\pi\ cm^2$ **D)** $240\pi\ cm^2$

19. Q Metall shar o'zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $78\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 2 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $66\pi\ cm^2$ ga teng. Sharning sirt yuzini toping. 5,2 ball

A) $312\pi\ cm^2$ **B)** $348\pi\ cm^2$ **C)** $328\pi\ cm^2$ **D)** $316\pi\ cm^2$

20. M O‘lchamlari $14\ m,\ 7\ m$ va $12\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $38\ m$ **B)** $3\sqrt{65}\ m$ **C)** $33\ m$ **D)** $37\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 2-variant

1 B C	2 Q B	3 B A	4 Q C	5 Q B	6 Q B	7 Q B
8 Q A	9 Q D	10 B B	11 M C	12 Q D	13 B A	14 Q C
15 Q A	16 B C	17 Q C	18 Q A	19 Q C	20 M D	

1. C) Bosh koeffitsiyent -8, ozod had 47: $(-8)\cdot 47=-376$.

2. B) I: ozod had 3. II: $x^3+6x+3=x(x^2+2)+4x+3$. III: ko‘phad bo‘lishi uchun $n=10$, daraja 5. IV: $P(-4)=-77$.

3. A) Karrali ildizlar: -5 (3 karrali), 2 (5 karrali). 4 – oddiy ildiz.

4. C) $Q=(x+4)(x-1)$. $P(-4)=0$, $P(1)=10\neq 0$ → EKUB $x+4$.

5. B) Gorner sxemasi: 1, 6, 5, 2, -2. Bo‘linma x^3+6x^2+5x+2, qoldiq -2.

6. B) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p\mid 6$, $q\mid 4$). Tekshiramiz: $-2;-1;\frac{3}{4}$.

7. B) $(6x-1)(12x-1)=72x^2-18x+1$, $(5x-1)(20x-1)=100x^2-25x+1$. $u=72x^2-18x$ desak, $(u+1)(\frac{25}{18}u+1)=116\to u=-10$ yoki $u=\frac{207}{25}$. Faqat $u=\frac{207}{25}$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{1}{4}$.

8. A) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz 1. Ildizlar ko‘paytmasi: $a\cdot(1)=12\to a=12$; $b=a+(1)(-1)=11$. $ab=132$.

9. D) x^2 ga bo‘lamiz va $t=x+\frac{1}{x}$ desak: $t^2+3t-18=0\to t=3$ yoki $t=-6$. Haqiqiy x faqat $|t|\geq 2$ da: $t=3$ va $t=-6$. Har bir t uchun $x^2-tx+1=0$ (ko‘paytma 1, yig‘indi t). Javob: 1.

10. B) $\frac{-6}{x-3}\neq 0\to y\neq 2$.

11. C) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=2$, $f(2)=3$, $f(3)=1$, $f(4)=-2$, $f(5)=1$, $f(6)=4$, $f(7)=3$, $f(8)=1$, $f(9)=-1$. Eng katta qiymat $a=5, b=8, c=3, d=9, e=1$ da: $5+8+9-3-1=18$.

12. D) $x=4$ da grafikdan $f(g(8))=-4$. $g(8)=-6\to f(-6)=-4\to f^{-1}(-4)=-6$.

13. A) 1) $\sqrt{3a^2}$ ✓. 2) to‘g‘ri burchakli uchburchakdan ✓. 3) $V=a^3$ ✗. 4) $f^2=h^2+\frac{a^2}{4}$ ✗.

14. C) $\left(\frac{a}{2}\right)^2=R^2-r^2$ – burchaklar soniga bog‘liq emas: $R^2-r^2=16$.

15. A) $h=a\sqrt{\frac{2}{3}}=\frac{a\sqrt{6}}{3}=3\sqrt{6}\to a=9$. Hajm $\frac{a^3\sqrt{2}}{12}=\frac{243}{4}\sqrt{2}\ cm^3$.

16. C) Konusning yon sirt yuzi: πrl. Silindrning to‘la sirt yuzi: $2\pi r(r+h)$. Konusning to‘la sirt yuzi: $\pi r(r+l)$.

17. C) Balandliklar farqi 2, asosdagi proyeksiya (vatar) $\sqrt{49-4}=3\sqrt{5}$. O‘qdan masofa $\sqrt{36-\frac{45}{4}}=\frac{3\sqrt{11}}{2}$ cm.

18. A) Uchburchak yuzi 60, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2\cdot 60}{13}=\frac{120}{13}$. Ikki konus yon sirtlari: $S=\pi\cdot\frac{120}{13}(13+24)=\frac{4440}{13}\pi$.

19. C) $x^2+78=(x+2)^2+66\to x=2$, $R^2=82$. Javob 328π cm².

20. D) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(14+7+14)\times 12$ hosil bo‘ladi: $\sqrt{35^2+12^2}=37$ m.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 3-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning bosh koeffitsiyenti va ozod hadi ko‘paytmasini toping. 2 ball

$$P(x) = 4x^7 + 9x^{10} + 17 + 6x^9 + 5x^5$$

A) 153 **B)** 102 **C)** 26 **D)** 170

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{10} + x^5 + 2$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) $7x + 1$
II. $P(x) = x^3 + 3x + 7$ ko‘phadni $x^2 + 2$ ga bo‘lgandagi qoldiqni toping.	B) -14
III. $P(x) = -3x^{n-13} + 8x^{13-n} + x^6 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) 2
IV. $P(x) = x^3 - 4x^2 + 2x - 7$ ko‘phadni $x + 1$ ga bo‘lgandagi qoldiqni toping.	D) $x + 7$
	E) 14
	F) 6

A) I-C; II-D; III-F; IV-B **B)** I-E; II-D; III-F; IV-B

C) I-C; II-E; III-F; IV-B **D)** I-C; II-D; III-F; IV-E

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x+4)^2(x-3)^5(x-6)$$

A) -4 va 3 **B)** 6 va 3 **C)** hammasi **D)** -4 va 6

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 + 7x^2 + 14x + 8$ va $Q(x) = x^2 - 2x - 8$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x - 2$ **B)** $x + 2$ **C)** $x + 4$ **D)** $x - 4$

5. Q Quyida $P(x)$ ko‘phadni $x+2$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	3	-1	-6	-4	-3
-2	3	-7	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $3x^3-7x^2-8x-20$ va $r=37$
B) $3x^3-7x^2+8x-18$ va $r=-37$
C) $3x^3-7x^2+8x-20$ va $r=37$
D) $3x^3-7x^2+8x+20$ va $r=-37$

6. Q $P(x)=6x^3-11x^2-4x+4$ butun koeffitsiyentli ko‘phad berilgan. 3,5 ball
Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang.

A) $-\frac{2}{3};2;2$
B) $-2;-\frac{2}{3};\frac{1}{2}$
C) $-2;-\frac{1}{2};\frac{2}{3}$
D) $-\frac{2}{3};\frac{1}{2};2$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(12x-1)(10x-1)(30x-1)(20x-1)=30$$

A) $\frac{1}{15}$
B) $\frac{3}{10}$
C) $\frac{4}{15}$
D) $\frac{2}{15}$

8. Q Ushbu $x^2-x+a=0$ tenglamaning ildizlari $x^3+bx-6=0$ tenglamaning ham ildizlari bo‘lsa, $a-b$ ning qiymatini toping. 3,5 ball

A) -1
B) 1
C) 2
D) -11

9. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

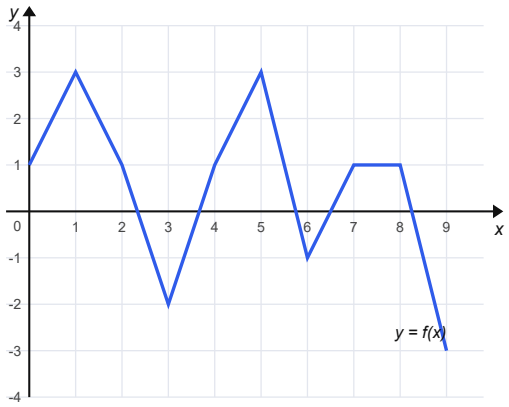
$$x^4-5x^3-4x^2-5x+1=0$$

A) 6
B) -5
C) 5
D) 12

10. B $y=\frac{5}{x-3}+5$ funksiyaning qiymatlar sohasini toping. 2 ball

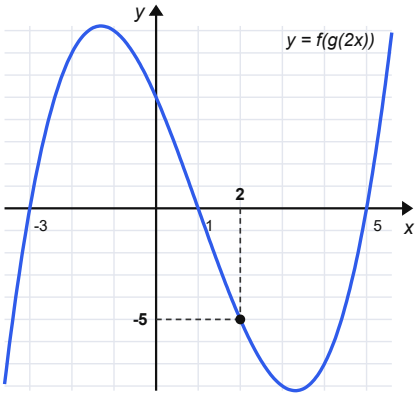
A) $(-\infty;3)\cup(3;\infty)$
B) $(-\infty;5)\cup(5;\infty)$
C) $(0;5)$
D) $(-\infty;-5)\cup(-5;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 19 B) 22 C) 21 D) 24

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(-1)=4$ bo‘lsa, $f^{-1}(-5)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) 1 B) 2 C) -1 D) 4

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Qirrasi a ga teng kubning hajmi $3a^3$ ga teng.		
2.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = h^2 + \frac{a^2}{4}$ tenglik o‘rinli.		
3.	Qirrasi a ga teng kubning to‘la sirti yuzi $6a^2$ ga teng.		
4.	Qirrasi a ga teng kubning yon sirti yuzi $6a^2$ ga teng.		

A) 1-to‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

B) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

C) 1-noto‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

D) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

14. Q Tomoni 3 ga teng muntazam 9-burchakka tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $R^2 - r^2$ ning qiymatini toping. 5,2 ball

A) $\frac{3}{2}$ **B)** 9 **C)** $\frac{9}{2}$ **D)** $\frac{9}{4}$

15. Q TABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning TG balandligi $TG = 4\sqrt{6}\ cm$ ga teng. Tetraedrning hajmini toping. 5,2 ball

A) $144\sqrt{2}\ cm^3$ **B)** $432\sqrt{2}\ cm^3$ **C)** $576\sqrt{2}\ cm^3$ **D)** $288\sqrt{2}\ cm^3$

16. B Berilgan ma’lumotlarga ko‘ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball
Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Silindrning to‘la sirt yuzi	A) $\pi r^2 h$
II. Shar sirt yuzi	B) $\pi r(r + l)$
III. Konusning to‘la sirt yuzi	C) $2\pi r(r + h)$
	D) $4\pi r^2$
	E) $2\pi r h$
	F) rh

A) I–C; II–D; III–B **B)** I–C; II–A; III–F **C)** I–C; II–D; III–E **D)** I–C; II–B; III–F

17. Q To‘g‘ri doiraviy silindrning balandligi $6\ cm$, asos radiusi esa $6\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo‘lgan masofalari mos ravishda $2\ cm$ va $5\ cm$. Agar $AB = 12\ cm$ bo‘lsa, silindr o‘qidan AB kesmagacha bo‘lgan eng qisqa masofani toping. 5,2 ball

A) $\frac{3}{2}\ cm$ B) $\frac{\sqrt{17}}{2}\ cm$ C) $\frac{\sqrt{5}}{2}\ cm$ D) $\frac{\sqrt{13}}{2}\ cm$

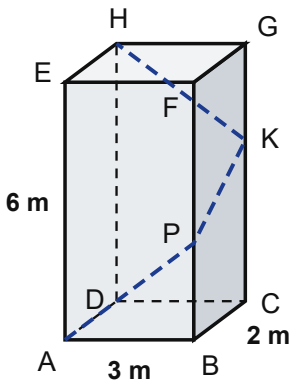
18. Q Yon tomonlari $10\ cm$ dan, asosi $12\ cm$ bo‘lgan teng yonli uchburchak o‘zining yon tomoni atrofida to‘liq aylantiriladi. Hosil bo‘lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $\frac{1056}{5}\pi\ cm^2$ B) $176\pi\ cm^2$ C) $96\pi\ cm^2$ D) $192\pi\ cm^2$

19. Q Shar shaklidagi pishloq o‘zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $49\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 5 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $14\pi\ cm^2$ ga teng. Sharning katta doira yuzini toping. 5,2 ball

A) $38\pi\ cm^2$ B) $50\pi\ cm^2$ C) $53\pi\ cm^2$ D) $49\pi\ cm^2$

20. M O‘lchamlari $3\ m,\ 2\ m$ va $6\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $\sqrt{61}\ m$ B) $11\ m$ C) $10\ m$ D) $14\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 3-variant

1 B A	2 Q A	3 B A	4 Q B	5 Q C	6 Q D	7 Q D
8 Q B	9 Q A	10 B B	11 M C	12 Q C	13 B C	14 Q D
15 Q A	16 B A	17 Q A	18 Q A	19 Q B	20 M C	

1. A) Bosh koeffitsiyent 9, ozod had 17: $9 \cdot 17 = 153$.

2. A) I: ozod had 2. II: $x^3 + 3x + 7 = x(x^2+2) + x + 7$. III: ko‘phad bo‘lishi uchun $n = 13$, daraja 6. IV: $P(-1) = -14$.

3. A) Karrali ildizlar: -4 (2 karrali), 3 (5 karrali). 6 – oddiy ildiz.

4. B) $Q = (x+2)(x-4)$. $P(-2) = 0$, $P(4) = 240 \neq 0$ → EKUB $x+2$.

5. C) Gorner sxemasi: 3, -7, 8, -20, 37. Bo‘linma $3x^3 - 7x^2 + 8x - 20$, qoldiq 37.

6. D) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p \mid 4$, $q \mid 6$). Tekshiramiz: $-\frac{2}{3}; \frac{1}{2}; 2$.

7. D) $(12x-1)(20x-1) = 240x^2 - 32x + 1$, $(10x-1)(30x-1) = 300x^2 - 40x + 1$. $u = 240x^2 - 32x$ desak, $(u+1)(\frac{5}{4}u + 1) = 30 \to u = 4$ yoki $u = -\frac{29}{5}$. Faqat $u = 4$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{2}{15}$.

8. B) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz -1. Ildizlar ko‘paytmasi: $a \cdot (-1) = 6 \to a = -6$; $b = a + (-1)(1) = -7$. $a - b = 1$.

9. A) x^2 ga bo‘lamiz va $t = x + \frac{1}{x}$ desak: $t^2 - 5t - 6 = 0 \to t = 6$ yoki $t = -1$. Haqiqiy x faqat $|t| \geq 2$ da: $t = 6$. Har bir t uchun $x^2 - tx + 1 = 0$ (ko‘paytma 1, yig‘indi t). Javob: 6.

10. B) $\frac{5}{x-3} \neq 0 \to y \neq 5$.

11. C) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=3, f(2)=1, f(3)=-2, f(4)=1, f(5)=3, f(6)=-1, f(7)=1, f(8)=1, f(9)=-3$. Eng katta qiymat $a=7, b=8, c=2, d=9, e=1$ da: $7+8+9-2-1 = 21$.

12. C) $x = 2$ da grafikdan $f(g(4)) = -5$. $g(4) = -1 \to f(-1) = -5 \to f^{-1}(-5) = -1$.

13. C) 1) $V = a^3$ ✗. 2) to‘g‘ri burchakli uchburchakdan ✓. 3) 6 ta yoq ✓. 4) yon sirt $4a^2$ ✗.

14. D) $\left(\frac{a}{2}\right)^2 = R^2 - r^2$ – burchaklar soniga bog‘liq emas: $R^2 - r^2 = \frac{9}{4}$.

15. A) $h = a\sqrt{\frac{2}{3}} = \frac{a\sqrt{6}}{3} = 4\sqrt{6} \to a = 12$. Hajm $\frac{a^3\sqrt{2}}{12} = 144\sqrt{2}\ cm^3$.

16. A) Silindrning to‘la sirt yuzi: $2\pi r(r+h)$. Shar sirt yuzi: $4\pi r^2$. Konusning to‘la sirt yuzi: $\pi r(r+l)$.

17. A) Balandliklar farqi 3, asosdagi proyeksiya (vatar) $\sqrt{144-9} = 3\sqrt{15}$. O‘qdan masofa $\sqrt{36 - \frac{135}{4}} = \frac{3}{2}$ cm.

18. A) Uchburchak yuzi 48, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2 \cdot 48}{10} = \frac{48}{5}$. Ikki konus yon sirtlari: $S = \pi \cdot \frac{48}{5}(10+12) = \frac{1056}{5}\pi$.

19. B) $x^2 + 49 = (x+5)^2 + 14 \to x = 1$, $R^2 = 50$. Javob 50π cm².

20. C) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(3+2+3) \times 6$ hosil bo‘ladi: $\sqrt{8^2 + 6^2} = 10$ m.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 4-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning eng katta darajasi va bosh koeffitsiyenti yig‘indisini toping. 2 ball

$$P(x) = 8x + 5x^5 - 2x^8 + 4x^2 + 41$$

A) 8 **B)** 6 **C)** 49 **D)** 7

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{13} + x^3 + 15$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) 19
II. $P(x) = x^3 + 4x + 8$ ko‘phadni $x^2 + 2$ ga bo‘lgandagi qoldiqni toping.	B) -19
III. $P(x) = -3x^{n-7} + 8x^{7-n} + x^3 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) $8x + 2$
IV. $P(x) = x^3 + 3x^2 + 5x - 13$ ko‘phadni $x + 2$ ga bo‘lgandagi qoldiqni toping.	D) 3
	E) 15
	F) $2x + 8$

A) I-E; II-F; III-D; IV-B **B)** I-E; II-A; III-D; IV-B

C) I-D; II-F; III-C; IV-B **D)** I-E; II-D; III-C; IV-B

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x-6)^5(x-1)(x+5)^5$$

A) hammasi **B)** 1 va -5 **C)** 6 va -5 **D)** 6 va 1

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 + 5x^2 + 5x + 1$ va $Q(x) = x^2 - 1$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x + 1$ **B)** $x - 1$ **C)** $x + 0$ **D)** $x + 2$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

5. Q Quyida $P(x)$ ko‘phadni $x-1$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	2	1	0	3	2
1	2	3	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $2x^3+3x^2+3x+8$ va $r=-8$
B) $2x^3+3x^2+3x-6$ va $r=-8$
C) $2x^3+3x^2-3x+6$ va $r=8$
D) $2x^3+3x^2+3x+6$ va $r=8$

6. Q $P(x)=2x^3-7x^2+9$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-3;-\frac{3}{2};1$
B) $1;\frac{3}{2};3$
C) $-1;\frac{3}{2};3$
D) $-1;\frac{1}{3};\frac{3}{2}$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(6x-1)(5x-1)(15x-1)(10x-1)=12$$

A) $\frac{8}{15}$
B) $\frac{4}{15}$
C) $\frac{13}{30}$
D) $\frac{2}{15}$

8. Q Ushbu $x^2-5x+a=0$ tenglamaning ildizlari $x^3+bx+25=0$ tenglamaning ham ildizlari bo‘lsa, ab ning qiymatini toping. 3,5 ball

A) -102
B) -99
C) 100
D) -100

9. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

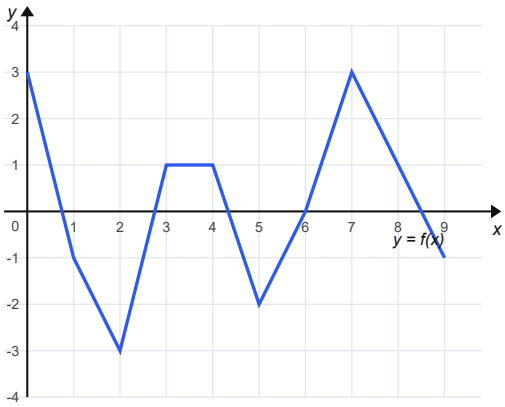
$$x^4+8x^3+17x^2+8x+1=0$$

A) -8
B) -7
C) 8
D) -10

10. B $y=\frac{8}{x+6}-1$ funksiyaning qiymatlar sohasini toping. 2 ball

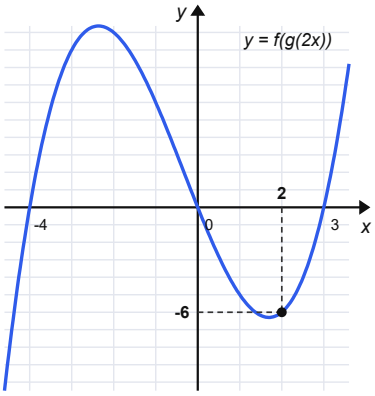
A) $(-\infty;1)\cup(1;\infty)$
B) $(-\infty;-1)\cup(-1;\infty)$
C) $(-1;0)$
D) $(-\infty;-6)\cup(-6;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 9 B) 14 C) 12 D) 11

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(-3)=4$ bo‘lsa, $f^{-1}(-6)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) 2 B) -3 C) 3 D) 4

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	To‘g‘ri to‘rtburchakli parallelepipedning qirralari $a,\ b$ va c bo‘lsa, uning diagonali $\sqrt{a^2+b^2+c^2}$ ga teng.		
2.	To‘g‘ri uchburchakli prizmaning asosi tomoni $a\ cm$ bo‘lgan muntazam uchburchak, yon qirrasi $h\ cm$. Prizmaning yon sirt yuzi $3ah\ cm^2$ ga teng.		
3.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2=a^2-h^2$ tenglik o‘rinli.		
4.	To‘g‘ri prizmaning yon qirralari asos tekisligiga perpendikulyar.		

A) 1-to‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

B) 1-to‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

C) 1-to‘g‘ri, 2-noto‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

D) 1-noto‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

14. Q Muntazam ko‘pburchakning tomoni a, unga tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $\frac{R^2-r^2}{a^2}$ ifodaning qiymatini toping. 5,2 ball

A) $\frac{1}{2}$ **B)** $\frac{1}{8}$ **C)** $\frac{5}{4}$ **D)** $\frac{1}{4}$

15. Q DABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning DG balandligi $DG=2\sqrt{6}\ cm$ ga teng. Tetraedrning qirrasi uzunligini toping. 5,2 ball

A) $7\ cm$ **B)** $4\ cm$ **C)** $8\ cm$ **D)** $6\ cm$

16. B Berilgan ma'lumotlarga ko'ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball

Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Konusning to'la sirt yuzi	A) $\pi r(r+l)$
II. Silindrning yon sirt yuzi	B) πrl
III. Silindrning o'q kesimining yuzi	C) rl
	D) $2\pi rh$
	E) $\frac{1}{3}\pi r^3$
	F) $2rh$

A) I–A; II–C; III–F **B)** I–B; II–C; III–F **C)** I–A; II–D; III–F **D)** I–B; II–D; III–F

17. Q To'g'ri doiraviy silindrning balandligi $11\ cm$, asos radiusi esa $4\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo'lgan masofalari mos ravishda $1\ cm$ va $2\ cm$. Agar $AB = 3\ cm$ bo'lsa, silindr o'qidan AB kesmagacha bo'lgan eng qisqa masofani toping. 5,2 ball

A) $2\sqrt{3}\ cm$ **B)** $\sqrt{13}\ cm$ **C)** $\sqrt{15}\ cm$ **D)** $\sqrt{14}\ cm$

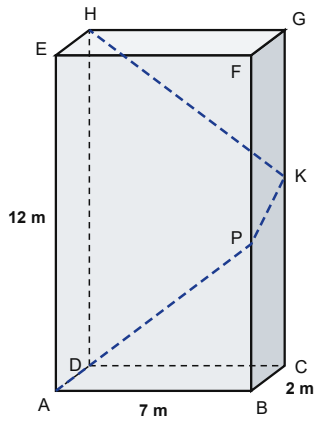
18. Q Yon tomonlari $13\ cm$ dan, asosi $10\ cm$ bo'lgan teng yonli uchburchak o'zining yon tomoni atrofida to'liq aylantiriladi. Hosil bo'lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $120\pi\ cm^2$ **B)** $240\pi\ cm^2$ **C)** $\frac{2760}{13}\pi\ cm^2$ **D)** $276\pi\ cm^2$

19. Q Shar shaklidagi tarvuz o'zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $77\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 1 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $70\pi\ cm^2$ ga teng. Sharning sirt yuzini toping. 5,2 ball

A) $308\pi\ cm^2$ **B)** $372\pi\ cm^2$ **C)** $332\pi\ cm^2$ **D)** $344\pi\ cm^2$

20. M O‘lchamlari $7\ m,\ 2\ m$ va $12\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $28\ m$ **B)** $15\ m$ **C)** $21\ m$ **D)** $20\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 4-variant

1 B	B	2 Q	A	3 B	C	4 Q	A	5 Q	D	6 Q	C	7 Q	B
8 Q	D	9 Q	A	10 B	B	11 M	D	12 Q	B	13 B	A	14 Q	D
15 Q	D	16 B	C	17 Q	D	18 Q	C	19 Q	D	20 M	D		

1. B) Eng katta daraja 8, bosh koeffitsiyent -2: $8+(-2)=6$.

2. A) I: ozod had 15. II: $x^3+4x+8=x(x^2+2)+2x+8$. III: ko‘phad bo‘lishi uchun $n=7$, daraja 3. IV: $P(-2)=-19$.

3. C) Karrali ildizlar: 6 (5 karrali), -5 (5 karrali). 1 – oddiy ildiz.

4. A) $Q=(x+1)(x-1)$. $P(-1)=0$, $P(1)=12\neq 0$ → EKUB $x+1$.

5. D) Gorner sxemasi: 2, 3, 3, 6, 8. Bo‘linma $2x^3+3x^2+3x+6$, qoldiq 8.

6. C) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p\mid 9$, $q\mid 2$). Tekshiramiz: $-1;\ \frac{3}{2};\ 3$.

7. B) $(6x-1)(10x-1)=60x^2-16x+1$, $(5x-1)(15x-1)=75x^2-20x+1$. $u=60x^2-16x$ desak, $(u+1)(\frac{5}{4}u+1)=12\to u=-4$ yoki $u=\frac{11}{5}$. Faqat $u=\frac{11}{5}$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{4}{15}$.

8. D) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz -5. Ildizlar ko‘paytmasi: $a\cdot(-5)=-25\to a=5$; $b=a+(-5)(5)=-20$. $ab=-100$.

9. A) x^2 ga bo‘lamiz va $t=x+\frac{1}{x}$ desak: $t^2+8t+15=0\to t=-5$ yoki $t=-3$. Haqiqiy x faqat $|t|\geq 2$ da: $t=-5$ va $t=-3$. Har bir t uchun $x^2-tx+1=0$ (ko‘paytma 1, yig‘indi t). Javob: -8.

10. B) $\frac{8}{x+6}\neq 0\to y\neq -1$.

11. D) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=-1$, $f(2)=-3$, $f(3)=1$, $f(4)=1$, $f(5)=-2$, $f(6)=0$, $f(7)=3$, $f(8)=1$, $f(9)=-1$. Eng katta qiymat $a=4, b=8, c=3, d=9, e=7$ da: $4+8+9-3-7=11$.

12. B) $x=2$ da grafikdan $f(g(4))=-6$. $g(4)=-3\to f(-3)=-6\to f^{-1}(-6)=-3$.

13. A) 1) fazoviy Pifagor ✓. 2) 3 ta yon yoq ✓. 3) $f^2=h^2+\frac{a^2}{4}$ ✗. 4) ta’rif ✓.

14. D) Markazdan tomonga tushirilgan perpendikulyar: $\left(\frac{a}{2}\right)^2=R^2-r^2$, ya’ni $a^2=4(R^2-r^2)$. Qiymat $\frac{1}{4}$.

15. D) $h=a\sqrt{\frac{2}{3}}=\frac{a\sqrt{6}}{3}=2\sqrt{6}\to a=6$. Qirra 6 cm.

16. C) Konusning to‘la sirt yuzi: $\pi r(r+l)$. Silindrning yon sirt yuzi: $2\pi rh$. Silindrning o‘q kesimining yuzi: $2rh$.

17. D) Balandliklar farqi 1, asosdagi proyeksiya (vatar) $\sqrt{9-1}=2\sqrt{2}$. O‘qdan masofa $\sqrt{16-\frac{8}{4}}=\sqrt{14}$ cm.

18. C) Uchburchak yuzi 60, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2\cdot 60}{13}=\frac{120}{13}$. Ikki konus yon sirtlari: $S=\pi\cdot\frac{120}{13}(13+10)=\frac{2760}{13}\pi$.

19. D) $x^2+77=(x+1)^2+70\to x=3$, $R^2=86$. Javob 344π cm².

20. D) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(7+2+7)\times 12$ hosil bo‘ladi: $\sqrt{16^2+12^2}=20$ m.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 5-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning darajasi va ozod hadi yig‘indisini toping. 2 ball

$$P(x) = 6x^{10} + 62 + 8x + 6x^{11} + 8x^5$$

A) 17 **B)** 73 **C)** 62 **D)** 74

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{14} + x^4 + 9$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) -66
II. $P(x) = x^3 + 4x + 6$ ko‘phadni $x^2 + 3$ ga bo‘lgandagi qoldiqni toping.	B) $6x + 1$
III. $P(x) = -3x^{n-13} + 8x^{13-n} + x^5 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) 5
IV. $P(x) = x^3 - 4x^2 - x - 6$ ko‘phadni $x + 3$ ga bo‘lgandagi qoldiqni toping.	D) 66
	E) $x + 6$
	F) 9

A) I-F; II-E; III-B; IV-A **B)** I-C; II-E; III-D; IV-A

C) I-D; II-E; III-C; IV-A **D)** I-F; II-E; III-C; IV-A

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x+2)^5(x-5)^6(x+3)$$

A) hammasi **B)** -2 va 5 **C)** -3 va 5 **D)** -2 va -3

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 + 4x^2 + 8x + 15$ va $Q(x) = x^2 + 4x + 3$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x + 1$ **B)** $x + 3$ **C)** $x - 1$ **D)** $x - 3$

5. Q Quyida $P(x)$ ko‘phadni $x+3$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	1	3	5	2	-2
-3	1	0	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $x^3+5x-13$ va $r=37$
B) $x^3+5x+13$ va $r=-37$
C) $x^3-5x-13$ va $r=37$
D) $x^3+5x-11$ va $r=-37$

6. Q $P(x)=2x^3+7x^2-9$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-1;\frac{3}{2};3$
B) $-3;1;\frac{3}{2}$
C) $-3;-\frac{3}{2};-1$
D) $-3;-\frac{3}{2};1$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(4x-1)(12x-1)(3x-1)(6x-1)=51$$

A) $\frac{7}{12}$
B) $\frac{5}{12}$
C) $\frac{5}{6}$
D) $\frac{5}{24}$

8. Q Ushbu $x^2+x+a=0$ tenglamaning ildizlari $x^3+bx+2=0$ tenglamaning ham ildizlari bo‘lsa, $a-b$ ning qiymatini toping. 3,5 ball

A) 1
B) -3
C) 2
D) -1

9. Q Tenglamaning haqiqiy ildizlari ko‘paytmasini toping. 3,5 ball

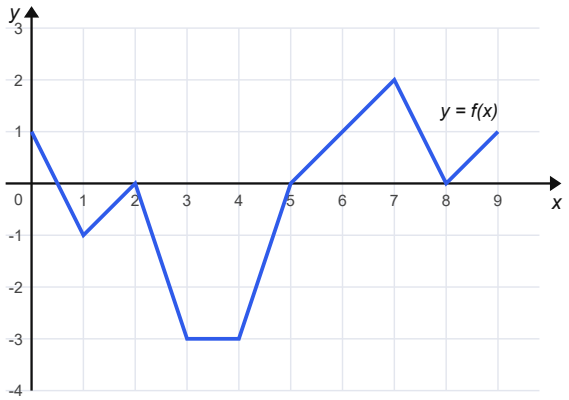
$$x^4-10x^3+26x^2-10x+1=0$$

A) -1
B) -5
C) 1
D) 5

10. B $y=\frac{2}{x-3}+1$ funksiyaning qiymatlar sohasini toping. 2 ball

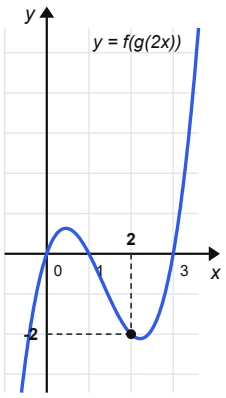
A) $(-\infty;3)\cup(3;\infty)$
B) $(-\infty;-1)\cup(-1;\infty)$
C) $(0;1)$
D) $(-\infty;1)\cup(1;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 7 B) 12 C) 9 D) 10

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(6)=4$ bo‘lsa, $f^{-1}(-2)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) 2 B) -6 C) 6 D) 4

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	To‘g‘ri to‘rtburchakli parallelepipedning qirralari $a,\ b$ va c bo‘lsa, uning diagonali $\sqrt{a^2+b^2+c^2}$ ga teng.		
2.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2=a^2-h^2$ tenglik o‘rinli.		
3.	Qirrasi a ga teng kubning to‘la sirti yuzi $6a^2$ ga teng.		
4.	To‘g‘ri uchburchakli prizmaning asosi tomoni $a\ cm$ bo‘lgan muntazam uchburchak, yon qirrasi $h\ cm$. Prizmaning yon sirt yuzi $3ah\ cm^2$ ga teng.		

A) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

B) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

C) 1-noto‘g‘ri, 2-noto‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

D) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

14. Q Tomoni 4 ga teng muntazam 10-burchakka tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, R^2-r^2 ning qiymatini toping. 5,2 ball

A) 16 **B)** 8 **C)** 2 **D)** 4

15. Q DABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning DG balandligi $DG=5\sqrt{6}\ cm$ ga teng. Tetraedrning qirrasi uzunligini toping. 5,2 ball

A) $20\ cm$ **B)** $16\ cm$ **C)** $10\ cm$ **D)** $15\ cm$

16. B Berilgan ma'lumotlarga ko'ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball

Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Shar hajmi	A) $\pi r l$
II. Silindrning yon sirt yuzi	B) $\frac{4}{3}\pi r^3$
III. Konusning yon sirt yuzi	C) $\pi r^2 h$
	D) $\pi r(r+l)$
	E) rl
	F) $2\pi r h$

A) I–B; II–F; III–A **B)** I–B; II–D; III–E **C)** I–B; II–C; III–A **D)** I–B; II–F; III–E

17. Q To'g'ri doiraviy silindrning balandligi $4\ cm$, asos radiusi esa $3\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo'lgan masofalari mos ravishda $1\ cm$ va $3\ cm$. Agar $AB = 3\ cm$ bo'lsa, silindr o'qidan AB kesmagacha bo'lgan eng qisqa masofani toping. 5,2 ball

A) $\sqrt{7}\ cm$ **B)** $\frac{3\sqrt{3}}{2}\ cm$ **C)** $\frac{\sqrt{35}}{2}\ cm$ **D)** $\frac{\sqrt{31}}{2}\ cm$

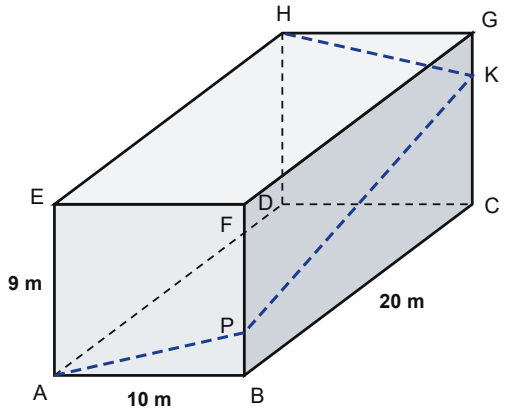
18. Q Yon tomonlari $5\ cm$ dan, asosi $8\ cm$ bo'lgan teng yonli uchburchak o'zining yon tomoni atrofida to'liq aylantiriladi. Hosil bo'lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $48\pi\ cm^2$ **B)** $\frac{312}{5}\pi\ cm^2$ **C)** $39\pi\ cm^2$ **D)** $24\pi\ cm^2$

19. Q Shar shaklidagi mum o'zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $63\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 4 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $39\pi\ cm^2$ ga teng. Sharning sirt yuzini toping. 5,2 ball

A) $268\pi\ cm^2$ **B)** $252\pi\ cm^2$ **C)** $256\pi\ cm^2$ **D)** $244\pi\ cm^2$

20. M O‘lchamlari $10\ m,\ 20\ m$ va $9\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $42\ m$ **B)** $41\ m$ **C)** $3\sqrt{109}\ m$ **D)** $39\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 5-variant

1 B	B	2 Q	D	3 B	B	4 Q	B	5 Q	A	6 Q	D	7 Q	B
8 Q	A	9 Q	C	10 B	D	11 M	C	12 Q	C	13 B	A	14 Q	D
15 Q	D	16 B	A	17 Q	D	18 Q	B	19 Q	C	20 M	B		

1. B) Daraja 11, ozod had 62: $11 + 62 = 73$.

2. D) I: ozod had 9. II: $x^3 + 4x + 6 = x(x^2 + 3) + x + 6$. III: ko‘phad bo‘lishi uchun $n = 13$, daraja 5. IV: $P(-3) = -66$.

3. B) Karrali ildizlar: -2 (5 karrali), 5 (6 karrali). -3 – oddiy ildiz.

4. B) $Q = (x+3)(x+1)$. $P(-3) = 0$, $P(-1) = 10 \neq 0 \to$ EKUB $x + 3$.

5. A) Gorner sxemasi: 1, 0, 5, -13, 37. Bo‘linma $x^3 + 5x - 13$, qoldiq 37.

6. D) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p \mid 9, q \mid 2$). Tekshiramiz: $-3; -\frac{3}{2}; 1$.

7. B) $(4x-1)(6x-1) = 24x^2 - 10x + 1$, $(12x-1)(3x-1) = 36x^2 - 15x + 1$. $u = 24x^2 - 10x$ desak, $(u+1)(\frac{3}{2}u + 1) = 51 \to u = 5$ yoki $u = -\frac{20}{3}$. Faqat $u = 5$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{5}{12}$.

8. A) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz 1. Ildizlar ko‘paytmasi: $a \cdot (1) = -2 \to a = -2$; $b = a + (1)(-1) = -3$. $a - b = 1$.

9. C) x^2 ga bo‘lamiz va $t = x + \frac{1}{x}$ desak: $t^2 - 10t + 24 = 0 \to t = 4$ yoki $t = 6$. Haqiqiy x faqat $|t| \geq 2$ da: $t = 4$ va $t = 6$. Har bir t uchun $x^2 - tx + 1 = 0$ (ko‘paytma 1, yig‘indi t). Javob: 1.

10. D) $\frac{2}{x-3} \neq 0 \to y \neq 1$.

11. C) Grafikdan butun nuqtalardagi qiymatlar: $f(1) = -1$, $f(2) = 0$, $f(3) = -3$, $f(4) = -3$, $f(5) = 0$, $f(6) = 1$, $f(7) = 2$, $f(8) = 0$, $f(9) = 1$. Eng katta qiymat $a = 5, b = 8, c = 2, d = 4, e = 6$ da: $5 + 8 + 4 - 2 - 6 = 9$.

12. C) $x = 2$ da grafikdan $f(g(4)) = -2$. $g(4) = 6 \to f(6) = -2 \to f^{-1}(-2) = 6$.

13. A) 1) fazoviy Pifagor ✓. 2) $f^2 = h^2 + \frac{a^2}{4}$ ✗. 3) 6 ta yoq ✓. 4) 3 ta yon yoq ✓.

14. D) $\left(\frac{a}{2}\right)^2 = R^2 - r^2$ – burchaklar soniga bog‘liq emas: $R^2 - r^2 = 4$.

15. D) $h = a\sqrt{\frac{2}{3}} = \frac{a\sqrt{6}}{3} = 5\sqrt{6} \to a = 15$. Qirra 15 cm.

16. A) Shar hajmi: $\frac{4}{3}\pi r^3$. Silindrning yon sirt yuzi: $2\pi rh$. Konusning yon sirt yuzi: πrl.

17. D) Balandliklar farqi 2, asosdagi proyeksiya (vatar) $\sqrt{9-4} = \sqrt{5}$. O‘qdan masofa $\sqrt{9 - \frac{5}{4}} = \frac{\sqrt{31}}{2}$ cm.

18. B) Uchburchak yuzi 12, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2 \cdot 12}{5} = \frac{24}{5}$. Ikki konus yon sirtlari: $S = \pi \cdot \frac{24}{5}(5 + 8) = \frac{312}{5}\pi$.

19. C) $x^2 + 63 = (x+4)^2 + 39 \to x = 1$, $R^2 = 64$. Javob 256π cm².

20. B) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(10 + 20 + 10) \times 9$ hosil bo‘ladi: $\sqrt{40^2 + 9^2} = 41$ m.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 6-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning bosh koeffitsiyenti va ozod hadi ko‘paytmasini toping. 2 ball

$$P(x) = 6x^4 + 6x^8 + 8x^9 + 5 - 5x^3$$

A) 13 **B)** 30 **C)** 45 **D)** 40

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{10} + x^4 + 6$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) 6
II. $P(x) = x^3 + 1x + 1$ ko‘phadni $x^2 + 5$ ga bo‘lgandagi qoldiqni toping.	B) 4
III. $P(x) = -3x^{n-5} + 8x^{5-n} + x^4 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) $x - 4$
IV. $P(x) = x^3 - x^2 + x + 7$ ko‘phadni $x + 2$ ga bo‘lgandagi qoldiqni toping.	D) -7
	E) 7
	F) $-4x + 1$

A) I-A; II-E; III-B; IV-D **B)** I-A; II-F; III-B; IV-C

C) I-A; II-E; III-F; IV-D **D)** I-A; II-F; III-B; IV-D

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x - 6)(x + 4)^4(x + 3)^3$$

A) hammasi **B)** -4 va 6 **C)** -4 va -3 **D)** 6 va -3

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 + x^2 - 8x + 6$ va $Q(x) = x^2 - 3x + 2$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x + 2$ **B)** $x - 2$ **C)** $x + 1$ **D)** $x - 1$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

5. Q Quyida $P(x)$ ko‘phadni $x+3$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	2	4	-3	-2	-4
-3	2	-2	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $2x^3-2x^2+3x-11$ va $r=29$
B) $2x^3-2x^2+3x+11$ va $r=-29$
C) $2x^3-2x^2+3x-9$ va $r=-29$
D) $2x^3-2x^2-3x-11$ va $r=29$

6. Q $P(x)=6x^3+5x^2-3x-2$ butun koeffitsiyentli ko‘phad berilgan. 3,5 ball
Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang.

A) $-2;-1;\frac{2}{3}$
B) $-\frac{2}{3};\frac{1}{2};1$
C) $-\frac{1}{2};\frac{2}{3};1$
D) $-1;-\frac{1}{2};\frac{2}{3}$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(4x-1)(24x-1)(3x-1)(8x-1)=154$$

A) $\frac{3}{8}$
B) $\frac{3}{4}$
C) $\frac{13}{24}$
D) $\frac{3}{16}$

8. Q Ushbu $x^2-5x+a=0$ tenglamaning ildizlari $x^3+bx+10=0$ tenglamaning ham ildizlari bo‘lsa, $a+b$ ning qiymatini toping. 3,5 ball

A) -23
B) 21
C) -20
D) -21

9. Q Tenglamaning haqiqiy ildizlari ko‘paytmasini toping. 3,5 ball

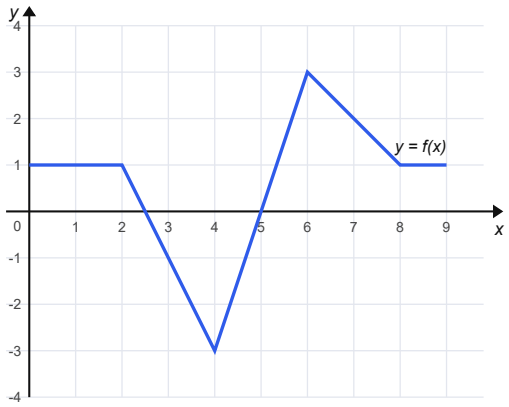
$$x^4+5x^3+6x^2+5x+1=0$$

A) $2{,}5$
B) -1
C) 1
D) $-2{,}5$

10. B $y=\frac{-8}{x+1}+2$ funksiyaning qiymatlar sohasini toping. 2 ball

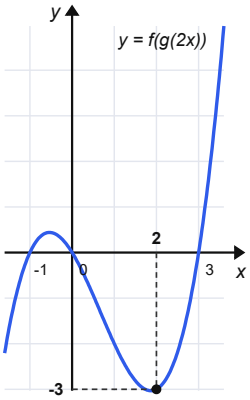
A) $(-\infty;2)\cup(2;\infty)$
B) $(-\infty;-1)\cup(-1;\infty)$
C) $(0;2)$
D) $(-\infty;-2)\cup(-2;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 13 B) 18 C) 15 D) 16

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(7)=4$ bo‘lsa, $f^{-1}(-3)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) 4 B) 7 C) 2 D) -7

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = h^2 + \frac{a^2}{4}$ tenglik o‘rinli.		
2.	Qirrasi a ga teng kubning yon sirti yuzi $6a^2$ ga teng.		
3.	Qirrasi a ga teng kubning to‘la sirti yuzi $6a^2$ ga teng.		
4.	Qirrasi a ga teng kubning hajmi $3a^3$ ga teng.		

A) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

B) 1-to‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

C) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

D) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

14. Q Tomoni 12 ga teng muntazam 10-burchakka tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $R^2 - r^2$ ning qiymatini toping. 5,2 ball

A) 144 **B)** 36 **C)** 72 **D)** 6

15. Q TABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning TG balandligi $TG = 2\sqrt{6}\ cm$ ga teng. Tetraedrning yon sirt yuzini toping. 5,2 ball

A) $9\sqrt{3}\ cm^2$ **B)** $36\sqrt{3}\ cm^2$ **C)** $54\sqrt{3}\ cm^2$ **D)** $27\sqrt{3}\ cm^2$

16. B Berilgan ma’lumotlarga ko‘ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball
Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Shar hajmi	A) $2rh$
II. Shar sirt yuzi	B) $\pi r^2 h$
III. Silindr hajmi	C) $4\pi r^2$
	D) $\frac{4}{3}\pi r^3$
	E) $\pi r(2r + h)$
	F) $2\pi rh$

A) I–D; II–F; III–B **B)** I–D; II–C; III–E **C)** I–E; II–C; III–A **D)** I–D; II–C; III–B

17. Q To‘g‘ri doiraviy silindrning balandligi $11\ cm$, asos radiusi esa $3\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo‘lgan masofalari mos ravishda $5\ cm$ va $10\ cm$. Agar $AB = 6\ cm$ bo‘lsa, silindr o‘qidan AB kesmagacha bo‘lgan eng qisqa masofani toping. 5,2 ball

A) $\frac{\sqrt{21}}{2}\ cm$ B) $\frac{5}{2}\ cm$ C) $2\ cm$ D) $\frac{\sqrt{29}}{2}\ cm$

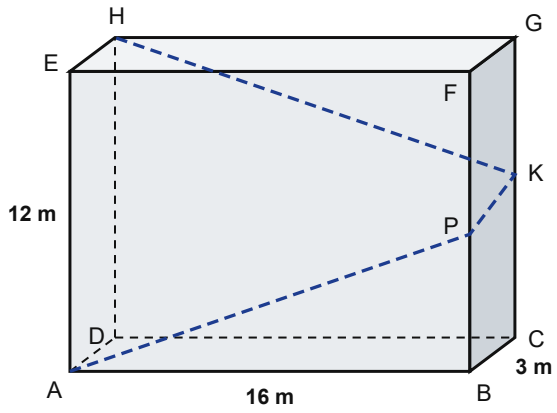
18. Q Yon tomonlari $17\ cm$ dan, asosi $16\ cm$ bo‘lgan teng yonli uchburchak o‘zining yon tomoni atrofida to‘liq aylantiriladi. Hosil bo‘lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $\frac{7920}{17}\pi\ cm^2$ B) $495\pi\ cm^2$ C) $240\pi\ cm^2$ D) $480\pi\ cm^2$

19. Q Shar shaklidagi tarvuz o‘zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $52\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 3 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $13\pi\ cm^2$ ga teng. Sharning katta doira yuzini toping. 5,2 ball

A) $77\pi\ cm^2$ B) $65\pi\ cm^2$ C) $88\pi\ cm^2$ D) $52\pi\ cm^2$

20. M O‘lchamlari $16\ m,\ 3\ m$ va $12\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $31\ m$ B) $\sqrt{505}\ m$ C) $38\ m$ D) $37\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 6-variant

1 B D	2 Q D	3 B C	4 Q D	5 Q A	6 Q D	7 Q A
8 Q D	9 Q C	10 B A	11 M C	12 Q B	13 B D	14 Q B
15 Q D	16 B D	17 Q B	18 Q A	19 Q A	20 M D	

1. D) Bosh koeffitsiyent 8, ozod had 5: $8 \cdot 5 = 40$.

2. D) I: ozod had 6. II: $x^3 + 1x + 1 = x(x^2 + 5) + -4x + 1$. III: ko‘phad bo‘lishi uchun $n = 5$, daraja 4. IV: $P(-2) = -7$.

3. C) Karrali ildizlar: -4 (4 karrali), -3 (3 karrali). 6 – oddiy ildiz.

4. D) $Q = (x-1)(x-2)$. $P(1) = 0$, $P(2) = 2 \neq 0 \to$ EKUB $x - 1$.

5. A) Gorner sxemasi: 2, -2, 3, -11, 29. Bo‘linma $2x^3 - 2x^2 + 3x - 11$, qoldiq 29.

6. D) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p \mid 2, q \mid 6$). Tekshiramiz: $-1; -\frac{1}{2}; \frac{2}{3}$.

7. A) $(4x-1)(8x-1) = 32x^2 - 12x + 1$, $(24x-1)(3x-1) = 72x^2 - 27x + 1$. $u = 32x^2 - 12x$ desak, $(u+1)(\frac{9}{4}u + 1) = 154 \to u = -9$ yoki $u = \frac{68}{9}$. Faqat $u = \frac{68}{9}$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{3}{8}$.

8. D) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz -5. Ildizlar ko‘paytmasi: $a \cdot (-5) = -10 \to a = 2; b = a + (-5)(5) = -23$. $a + b = -21$.

9. C) x^2 ga bo‘lamiz va $t = x + \frac{1}{x}$ desak: $t^2 + 5t + 4 = 0 \to t = -4$ yoki $t = -1$. Haqiqiy x faqat $|t| \geq 2$ da: $t = -4$. Har bir t uchun $x^2 - tx + 1 = 0$ (ko‘paytma 1, yig‘indi t). Javob: 1.

10. A) $\frac{-8}{x+1} \neq 0 \to y \neq 2$.

11. C) Grafikdan butun nuqtalardagi qiymatlar: $f(1) = 1, f(2) = 1, f(3) = -1, f(4) = -3, f(5) = 0, f(6) = 3, f(7) = 2, f(8) = 1, f(9) = 1$. Eng katta qiymat $a = 8, b = 9, c = 1, d = 5, e = 6$ da: $8 + 9 + 5 - 1 - 6 = 15$.

12. B) $x = 2$ da grafikdan $f(g(4)) = -3$. $g(4) = 7 \to f(7) = -3 \to f^{-1}(-3) = 7$.

13. D) 1) to‘g‘ri burchakli uchburchakdan ✓. 2) yon sirt $4a^2$ ✗. 3) 6 ta yoq ✓. 4) $V = a^3$ ✗.

14. B) $\left(\frac{a}{2}\right)^2 = R^2 - r^2$ – burchaklar soniga bog‘liq emas: $R^2 - r^2 = 36$.

15. D) $h = a\sqrt{\frac{2}{3}} = \frac{a\sqrt{6}}{3} = 2\sqrt{6} \to a = 6$. Yon sirt $3 \cdot \frac{\sqrt{3}}{4}a^2 = 27\sqrt{3}\ cm^2$.

16. D) Shar hajmi: $\frac{4}{3}\pi r^3$. Shar sirt yuzi: $4\pi r^2$. Silindr hajmi: $\pi r^2 h$.

17. B) Balandliklar farqi 5, asosdagi proyeksiya (vatar) $\sqrt{36 - 25} = \sqrt{11}$. O‘qdan masofa $\sqrt{9 - \frac{11}{4}} = \frac{5}{2}$ cm.

18. A) Uchburchak yuzi 120, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2 \cdot 120}{17} = \frac{240}{17}$. Ikki konus yon sirtlari: $S = \pi \cdot \frac{240}{17}(17 + 16) = \frac{7920}{17}\pi$.

19. A) $x^2 + 52 = (x+3)^2 + 13 \to x = 5$, $R^2 = 77$. Javob 77π cm².

20. D) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(16 + 3 + 16) \times 12$ hosil bo‘ladi: $\sqrt{35^2 + 12^2} = 37$ m.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 7-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning darajasi va ozod hadi yig‘indisini toping. 2 ball

$$P(x) = 9x^3 + 70 + 7x^5 + 9x^8 - 9x^9$$

A) 0 **B)** 70 **C)** 80 **D)** 79

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{13} + x^3 + 16$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) 16
II. $P(x) = x^3 + 4x + 7$ ko‘phadni $x^2 + 3$ ga bo‘lgandagi qoldiqni toping.	B) -48
III. $P(x) = -3x^{n-11} + 8x^{11-n} + x^6 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) $x + 7$
IV. $P(x) = x^3 + 5x^2 - 5x + 15$ ko‘phadni $x + 3$ ga bo‘lgandagi qoldiqni toping.	D) 48
	E) $7x + 1$
	F) 6

A) I-B; II-C; III-F; IV-D **B)** I-A; II-C; III-F; IV-D

C) I-A; II-C; III-E; IV-D **D)** I-A; II-C; III-F; IV-E

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x-1)^4(x+1)^3(x+5)$$

A) 1 va -5 **B)** -5 va -1 **C)** hammasi **D)** 1 va -1

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 + 7x^2 + 13x + 3$ va $Q(x) = x^2 + 5x + 6$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x - 2$ **B)** $x + 2$ **C)** $x - 3$ **D)** $x + 3$

5. Q Quyida $P(x)$ ko‘phadni $x+1$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	2	-2	6	-5	-1
-1	2	-4	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $2x^3-4x^2+10x-15$ va $r=14$
B) $2x^3-4x^2+10x-13$ va $r=-14$
C) $2x^3-4x^2+10x+15$ va $r=-14$
D) $2x^3-4x^2-10x-15$ va $r=14$

6. Q $P(x)=9x^3-7x-2$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-\frac{2}{3};-\frac{1}{3};1$
B) $-3;-\frac{2}{3};1$
C) $-1;\frac{1}{3};\frac{2}{3}$
D) $-1;-\frac{2}{3};-\frac{1}{3}$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(5x-1)(4x-1)(30x-1)(12x-1)=45$$

A) $\frac{17}{120}$
B) $\frac{9}{20}$
C) $\frac{17}{60}$
D) $\frac{17}{30}$

8. Q Ushbu $x^2-3x+a=0$ tenglamaning ildizlari $x^3+bx+3=0$ tenglamaning ham ildizlari bo‘lsa, $a+b$ ning qiymatini toping. 3,5 ball

A) -6
B) 7
C) -9
D) -7

9. Q Tenglamaning haqiqiy ildizlari ko‘paytmasini toping. 3,5 ball

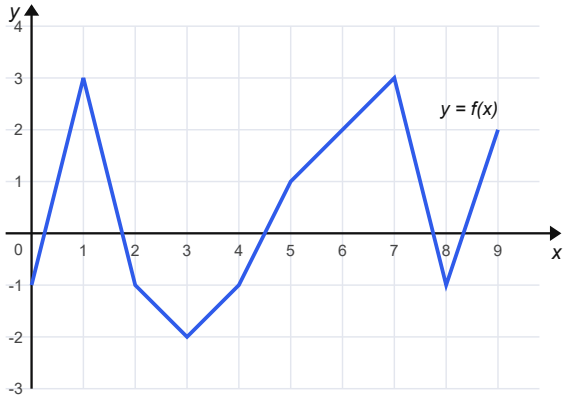
$$x^4-7x^3+8x^2-7x+1=0$$

A) $-3{,}5$
B) -1
C) $3{,}5$
D) 1

10. B $y=\frac{9}{x-4}-1$ funksiyaning qiymatlar sohasini toping. 2 ball

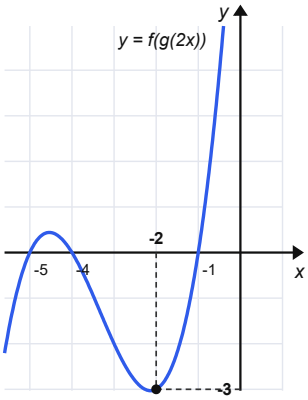
A) $(-\infty;-1)\cup(-1;\infty)$
B) $(-\infty;1)\cup(1;\infty)$
C) $(-1;0)$
D) $(-\infty;4)\cup(4;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 10 B) 12 C) 15 D) 13

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(-8)=-4$ bo‘lsa, $f^{-1}(-3)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) 8 B) -8 C) -4 D) -2

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = a^2 - h^2$ tenglik o‘rinli.		
2.	To‘g‘ri to‘rtburchakli parallelepipedning qirralari $a,\ b$ va c bo‘lsa, uning diagonali $\sqrt{a^2+b^2+c^2}$ ga teng.		
3.	To‘g‘ri to‘rtburchakli parallelepipedning qirralari $a,\ b$ va c bo‘lsa, uning diagonali $a+b+c$ ga teng.		
4.	Qirrasi a ga teng kubning to‘la sirti yuzi $6a^2$ ga teng.		

A) 1-to‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

B) 1-noto‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

C) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

D) 1-noto‘g‘ri, 2-noto‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

14. Q Tomoni 5 ga teng muntazam 12-burchakka tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $R^2 - r^2$ ning qiymatini toping. 5,2 ball

A) $\frac{25}{2}$ **B)** $\frac{25}{4}$ **C)** $\frac{5}{2}$ **D)** 25

15. Q TABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning TG balandligi $TG = 2\sqrt{6}\ cm$ ga teng. Tetraedrning to‘la sirt yuzini toping. 5,2 ball

A) $72\sqrt{3}\ cm^2$ **B)** $27\sqrt{3}\ cm^2$ **C)** $36\sqrt{3}\ cm^2$ **D)** $32\sqrt{3}\ cm^2$

16. B Berilgan ma’lumotlarga ko‘ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball

Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Konusning to‘la sirt yuzi	A) $4\pi r^2$
II. Konusning yon sirt yuzi	B) $\pi r l$
III. Silindrning o‘q kesimining yuzi	C) $\pi r(r+l)$
	D) $2rh$
	E) $2\pi r(r+h)$
	F) $2\pi rh$

A) I–C; II–B; III–F **B)** I–A; II–C; III–D **C)** I–C; II–B; III–D **D)** I–C; II–F; III–D

17. Q To‘g‘ri doiraviy silindrning balandligi $6\ cm$, asos radiusi esa $4\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo‘lgan masofalari mos ravishda $4\ cm$ va $5\ cm$. Agar $AB = 5\ cm$ bo‘lsa, silindr o‘qidan AB kesmagacha bo‘lgan eng qisqa masofani toping. 5,2 ball

A) $\sqrt{11}\ cm$ **B)** $2\ cm$ **C)** $3\ cm$ **D)** $\sqrt{10}\ cm$

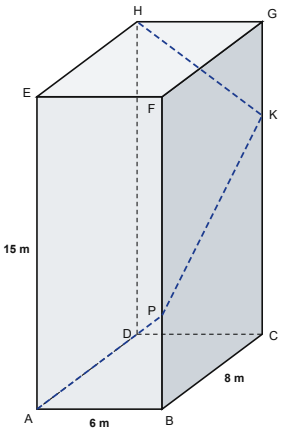
18. Q Yon tomonlari $25\ cm$ dan, asosi $14\ cm$ bo‘lgan teng yonli uchburchak o‘zining yon tomoni atrofida to‘liq aylantiriladi. Hosil bo‘lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $\frac{13104}{25}\pi\ cm^2$ **B)** $672\pi\ cm^2$ **C)** $336\pi\ cm^2$ **D)** $936\pi\ cm^2$

19. Q Shar shaklidagi mum o‘zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $69\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 1 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $62\pi\ cm^2$ ga teng. Sharning katta doira yuzini toping. 5,2 ball

A) $66\pi\ cm^2$ **B)** $69\pi\ cm^2$ **C)** $85\pi\ cm^2$ **D)** $78\pi\ cm^2$

20. M O‘lchamlari $6\ m,\ 8\ m$ va $15\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $\sqrt{421}\ m$ B) $25\ m$ C) $29\ m$ D) $26\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 7-variant

1 B D	2 Q B	3 B D	4 Q D	5 Q A	6 Q A	7 Q C
8 Q D	9 Q D	10 B A	11 M B	12 Q B	13 B C	14 Q B
15 Q C	16 B C	17 Q D	18 Q A	19 Q D	20 M B	

1. D) Daraja 9, ozod had 70: $9+70=79$.

2. B) I: ozod had 16. II: $x^3+4x+7=x(x^2+3)+x+7$. III: ko‘phad bo‘lishi uchun $n=11$, daraja 6. IV: $P(-3)=48$.

3. D) Karrali ildizlar: 1 (4 karrali), -1 (3 karrali). -5 – oddiy ildiz.

4. D) $Q=(x+3)(x+2)$. $P(-3)=0$, $P(-2)=-3\neq 0\to$ EKUB $x+3$.

5. A) Gorner sxemasi: 2, -4, 10, -15, 14. Bo‘linma $2x^3-4x^2+10x-15$, qoldiq 14.

6. A) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p\mid 2, q\mid 9$). Tekshiramiz: $-\frac{2}{3};-\frac{1}{3};1$.

7. C) $(5x-1)(12x-1)=60x^2-17x+1$, $(4x-1)(30x-1)=120x^2-34x+1$. $u=60x^2-17x$ desak, $(u+1)(2u+1)=45\to u=4$ yoki $u=-\frac{11}{2}$. Faqat $u=4$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{17}{60}$.

8. D) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz -3. Ildizlar ko‘paytmasi: $a\cdot(-3)=-3\to a=1; b=a+(-3)(3)=-8$. $a+b=-7$.

9. D) x^2 ga bo‘lamiz va $t=x+\frac{1}{x}$ desak: $t^2-7t+6=0\to t=6$ yoki $t=1$. Haqiqiy x faqat $|t|\geq 2$ da: $t=6$. Har bir t uchun $x^2-tx+1=0$ (ko‘paytma 1, yig‘indi t). Javob: 1.

10. A) $\frac{9}{x-4}\neq 0\to y\neq -1$.

11. B) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=3, f(2)=-1, f(3)=-2, f(4)=-1, f(5)=1, f(6)=2, f(7)=3, f(8)=-1, f(9)=2$. Eng katta qiymat $a=4, b=8, c=2, d=3, e=1$ da: $4+8+3-2-1=12$.

12. B) $x=-2$ da grafikdan $f(g(-4))=-3$. $g(-4)=-8\to f(-8)=-3\to f^{-1}(-3)=-8$.

13. C) 1) $f^2=h^2+\frac{a^2}{4}$ ✗. 2) fazoviy Pifagor ✓. 3) diagonal $\sqrt{a^2+b^2+c^2}$ ✗. 4) 6 ta yoq ✓.

14. B) $\left(\frac{a}{2}\right)^2=R^2-r^2$ – burchaklar soniga bog‘liq emas: $R^2-r^2=\frac{25}{4}$.

15. C) $h=a\sqrt{\frac{2}{3}}=\frac{a\sqrt{6}}{3}=2\sqrt{6}\to a=6$. To‘la sirt $\sqrt{3}a^2=36\sqrt{3}\ cm^2$.

16. C) Konusning to‘la sirt yuzi: $\pi r(r+l)$. Konusning yon sirt yuzi: πrl. Silindrning o‘q kesimining yuzi: $2rh$.

17. D) Balandliklar farqi 1, asosdagi proyeksiya (vatar) $\sqrt{25-1}=2\sqrt{6}$. O‘qdan masofa $\sqrt{16-\frac{24}{4}}=\sqrt{10}$ cm.

18. A) Uchburchak yuzi 168, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2\cdot 168}{25}=\frac{336}{25}$. Ikki konus yon sirtlari: $S=\pi\cdot\frac{336}{25}(25+14)=\frac{13104}{25}\pi$.

19. D) $x^2+69=(x+1)^2+62\to x=3, R^2=78$. Javob 78π cm².

20. B) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(6+8+6)\times 15$ hosil bo‘ladi: $\sqrt{20^2+15^2}=25$ m.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 8-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning darajasi va ozod hadi yig‘indisini toping. 2 ball

$$P(x) = 14 + 9x^5 - 4x^{11} + 8x^3 + 5x^9$$

A) 7 **B)** 25 **C)** 26 **D)** 14

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{13} + x^5 + 14$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) $5x - 4$
II. $P(x) = x^3 + 1x + 5$ ko‘phadni $x^2 + 5$ ga bo‘lgandagi qoldiqni toping.	B) 14
III. $P(x) = -3x^{n-14} + 8x^{14-n} + x^3 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) $-4x + 5$
IV. $P(x) = x^3 - 2x^2 - 4x - 13$ ko‘phadni $x + 1$ ga bo‘lgandagi qoldiqni toping.	D) -12
	E) 3
	F) 12

A) I-A; II-C; III-E; IV-B **B)** I-B; II-C; III-E; IV-D

C) I-B; II-C; III-A; IV-D **D)** I-B; II-A; III-E; IV-C

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x+1)^3(x-5)(x+2)^5$$

A) hammasi **B)** -1 va -2 **C)** 5 va -2 **D)** -1 va 5

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 + x^2 - 21x + 4$ va $Q(x) = x^2 - 7x + 12$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x + 4$ **B)** $x + 3$ **C)** $x - 3$ **D)** $x - 4$

5. Q Quyida $P(x)$ ko‘phadni $x-2$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	2	-2	0	2	-6
2	2	2	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $2x^3+2x^2+4x-10$ va $r=-14$
B) $2x^3+2x^2-4x+10$ va $r=14$
C) $2x^3+2x^2+4x+10$ va $r=14$
D) $2x^3+2x^2+4x+12$ va $r=-14$

6. Q $P(x)=6x^3+11x^2-3x-2$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-2;-\frac{1}{3};\frac{1}{2}$
B) $-\frac{1}{3};\frac{1}{2};2$
C) $-\frac{1}{2};\frac{1}{3};2$
D) $-2;-\frac{1}{3};2$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(7x-1)(6x-1)(21x-1)(14x-1)=65$$

A) $\frac{3}{28}$
B) $\frac{3}{14}$
C) $\frac{3}{7}$
D) $\frac{8}{21}$

8. Q Ushbu $x^2-2x+a=0$ tenglamaning ildizlari $x^3+bx-20=0$ tenglamaning ham ildizlari bo‘lsa, $a+b$ ning qiymatini toping. 3,5 ball

A) 24
B) -24
C) -26
D) -23

9. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

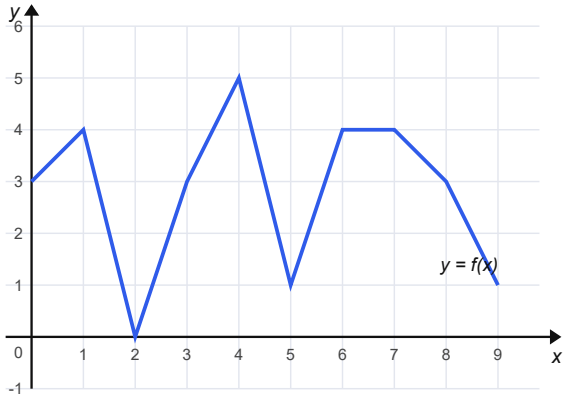
$$x^4+11x^3+30x^2+11x+1=0$$

A) 11
B) -11
C) -14
D) -10

10. B $y=\frac{-2}{x+4}+1$ funksiyaning qiymatlar sohasini toping. 2 ball

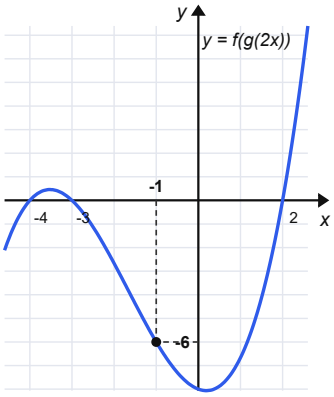
A) $(-\infty;-1)\cup(-1;\infty)$
B) $(-\infty;1)\cup(1;\infty)$
C) $(-\infty;-4)\cup(-4;\infty)$
D) $(0;1)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 20 B) 18 C) 15 D) 17

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(2)=-2$ bo‘lsa, $f^{-1}(-6)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) -2 B) -1 C) 2 D) 4

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Qirrasi a ga teng kubning yon sirti yuzi $6a^2$ ga teng.		
2.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = h^2 + \frac{a^2}{4}$ tenglik o‘rinli.		
3.	To‘g‘ri to‘rtburchakli parallelepipedning qirralari $a,\ b$ va c bo‘lsa, uning diagonali $a + b + c$ ga teng.		
4.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = a^2 - h^2$ tenglik o‘rinli.		

A) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

B) 1-noto‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

C) 1-noto‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

D) 1-noto‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

14. Q Muntazam ko‘pburchakning tomoni a, unga tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $\frac{3a^2}{R^2 - r^2}$ ifodaning qiymatini toping. 5,2 ball

A) 13 **B)** 12 **C)** 6 **D)** 24

15. Q DABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning DG balandligi $DG = \sqrt{6}\ cm$ ga teng. Tetraedrning qirrasi uzunligini toping. 5,2 ball

A) $3\ cm$ **B)** $4\ cm$ **C)** $2\ cm$ **D)** $6\ cm$

16. B Berilgan ma'lumotlarga ko'ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball
Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Konusning yon sirt yuzi	A) $2\pi r(r+h)$
II. Konus hajmi	B) $\pi r^2 l$
III. Shar sirt yuzi	C) $4\pi r^2$
	D) $\pi r l$
	E) $\frac{1}{3}\pi r^2 h$
	F) rl

A) I–D; II–F; III–E **B)** I–D; II–E; III–C **C)** I–D; II–E; III–B **D)** I–F; II–E; III–C

17. Q To'g'ri doiraviy silindrning balandligi $11\ cm$, asos radiusi esa $3\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo'lgan masofalari mos ravishda $7\ cm$ va $9\ cm$. Agar $AB = 4\ cm$ bo'lsa, silindr o'qidan AB kesmagacha bo'lgan eng qisqa masofani toping. 5,2 ball

A) $\sqrt{6}\ cm$ **B)** $\sqrt{3}\ cm$ **C)** $\sqrt{5}\ cm$ **D)** $\sqrt{7}\ cm$

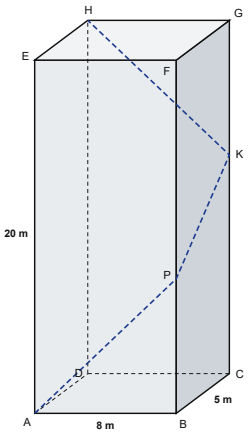
18. Q Yon tomonlari $25\ cm$ dan, asosi $48\ cm$ bo'lgan teng yonli uchburchak o'zining yon tomoni atrofida to'liq aylantiriladi. Hosil bo'lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $672\pi\ cm^2$ **B)** $511\pi\ cm^2$ **C)** $\frac{24528}{25}\pi\ cm^2$ **D)** $336\pi\ cm^2$

19. Q Shar shaklidagi mum o'zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $80\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 4 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $16\pi\ cm^2$ ga teng. Sharning sirt yuzini toping. 5,2 ball

A) $320\pi\ cm^2$ **B)** $464\pi\ cm^2$ **C)** $452\pi\ cm^2$ **D)** $516\pi\ cm^2$

20. M O‘lchamlari $8\ m,\ 5\ m$ va $20\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $33\ m$ B) $29\ m$ C) $\sqrt{569}\ m$ D) $30\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 8-variant

1 B	B	2 Q	B	3 B	B	4 Q D	5 Q C	6 Q A	7 Q B
8 Q	B	9 Q	B	10 B	B	11 M D	12 Q C	13 B A	14 Q B
15 Q	A	16 B	B	17 Q	A	18 Q C	19 Q B	20 M B	

1. B) Daraja 11, ozod had 14: $11+14=25$.

2. B) I: ozod had 14. II: $x^3+1x+5=x(x^2+5)+-4x+5$. III: ko‘phad bo‘lishi uchun $n=14$, daraja 3. IV: $P(-1)=-12$.

3. B) Karrali ildizlar: -1 (3 karrali), -2 (5 karrali). 5 – oddiy ildiz.

4. D) $Q=(x-4)(x-3)$. $P(4)=0$, $P(3)=-23\neq 0\to$ EKUB $x-4$.

5. C) Gorner sxemasi: 2, 2, 4, 10, 14. Bo‘linma $2x^3+2x^2+4x+10$, qoldiq 14.

6. A) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p\mid 2, q\mid 6$). Tekshiramiz: $-2;-\frac{1}{3};\frac{1}{2}$.

7. B) $(7x-1)(14x-1)=98x^2-21x+1$, $(6x-1)(21x-1)=126x^2-27x+1$. $u=98x^2-21x$ desak, $(u+1)(\frac{9}{7}u+1)=65\to u=-8$ yoki $u=\frac{56}{9}$. Faqat $u=\frac{56}{9}$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{3}{14}$.

8. B) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz -2. Ildizlar ko‘paytmasi: $a\cdot(-2)=20\to a=-10; b=a+(-2)(2)=-14$. $a+b=-24$.

9. B) x^2 ga bo‘lamiz va $t=x+\frac{1}{x}$ desak: $t^2+11t+28=0\to t=-7$ yoki $t=-4$. Haqiqiy x faqat $|t|\geq 2$ da: $t=-7$ va $t=-4$. Har bir t uchun $x^2-tx+1=0$ (ko‘paytma 1, yig‘indi t). Javob: -11.

10. B) $\frac{-2}{x+4}\neq 0\to y\neq 1$.

11. D) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=4$, $f(2)=0$, $f(3)=3$, $f(4)=5$, $f(5)=1$, $f(6)=4$, $f(7)=4$, $f(8)=3$, $f(9)=1$. Eng katta qiymat $a=6, b=7, c=1, d=9, e=4$ da: $6+7+9-1-4=17$.

12. C) $x=-1$ da grafikdan $f(g(-2))=-6$. $g(-2)=2\to f(2)=-6\to f^{-1}(-6)=2$.

13. A) 1) yon sirt $4a^2$ ✗. 2) to‘g‘ri burchakli uchburchakdan ✓. 3) diagonal $\sqrt{a^2+b^2+c^2}$ ✗. 4) $f^2=h^2+\frac{a^2}{4}$ ✗.

14. B) Markazdan tomonga tushirilgan perpendikulyar: $\left(\frac{a}{2}\right)^2=R^2-r^2$, ya’ni $a^2=4(R^2-r^2)$. Qiymat 12.

15. A) $h=a\sqrt{\frac{2}{3}}=\frac{a\sqrt{6}}{3}=\sqrt{6}\to a=3$. Qirra 3 cm.

16. B) Konusning yon sirt yuzi: πrl. Konus hajmi: $\frac{1}{3}\pi r^2 h$. Shar sirt yuzi: $4\pi r^2$.

17. A) Balandliklar farqi 2, asosdagi proyeksiya (vatar) $\sqrt{16-4}=2\sqrt{3}$. O‘qdan masofa $\sqrt{9-\frac{12}{4}}=\sqrt{6}$ cm.

18. C) Uchburchak yuzi 168, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2\cdot 168}{25}=\frac{336}{25}$. Ikki konus yon sirtlari: $S=\pi\cdot\frac{336}{25}(25+48)=\frac{24528}{25}\pi$.

19. B) $x^2+80=(x+4)^2+16\to x=6$, $R^2=116$. Javob 464π cm².

20. B) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(8+5+8)\times 20$ hosil bo‘ladi: $\sqrt{21^2+20^2}=29$ m.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 9-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning darajasi va ozod hadi yig‘indisini toping. 2 ball

$$P(x) = 9x^2 + 33 + 5x^3 + 9x^9 - 2x^5$$

A) 18 **B)** 33 **C)** 43 **D)** 42

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{15} + x^5 + 15$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) 7
II. $P(x) = x^3 + 2x + 8$ ko‘phadni $x^2 + 4$ ga bo‘lgandagi qoldiqni toping.	B) $-2x + 8$
III. $P(x) = -3x^{n-13} + 8x^{13-n} + x^7 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) -6
IV. $P(x) = x^3 - 5x^2 - 4x + 8$ ko‘phadni $x + 1$ ga bo‘lgandagi qoldiqni toping.	D) 15
	E) 6
	F) $8x - 2$

A) I-D; II-B; III-F; IV-E **B)** I-D; II-B; III-A; IV-E

C) I-D; II-B; III-F; IV-A **D)** I-D; II-B; III-A; IV-F

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x-2)^7(x-6)(x+5)^3$$

A) hammasi **B)** 2 va 6 **C)** 2 va -5 **D)** 6 va -5

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 - 4x^2 - x + 4$ va $Q(x) = x^2 - 16$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x - 5$ **B)** $x - 4$ **C)** $x - 3$ **D)** $x + 4$

5. Q Quyida $P(x)$ ko‘phadni $x+2$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	1	0	1	-6	3
-2	1	-2	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $x^3-2x^2+5x-16$ va $r=35$
B) $x^3-2x^2-5x-16$ va $r=35$
C) $x^3-2x^2+5x+16$ va $r=-35$
D) $x^3-2x^2+5x-14$ va $r=-35$

6. Q $P(x)=4x^3-16x^2+9x+9$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-\frac{1}{2};\frac{1}{3};\frac{3}{2}$
B) $-\frac{3}{2};-\frac{1}{2};3$
C) $-3;-\frac{3}{2};\frac{1}{2}$
D) $-\frac{1}{2};\frac{3}{2};3$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(20x-1)(12x-1)(15x-1)(10x-1)=33$$

A) $\frac{19}{60}$
B) $\frac{3}{10}$
C) $\frac{3}{20}$
D) $\frac{3}{40}$

8. Q Ushbu $x^2+x+a=0$ tenglamaning ildizlari $x^3+bx+3=0$ tenglamaning ham ildizlari bo‘lsa, ab ning qiymatini toping. 3,5 ball

A) 13
B) 12
C) -12
D) 10

9. Q Tenglamaning haqiqiy ildizlari ko‘paytmasini toping. 3,5 ball

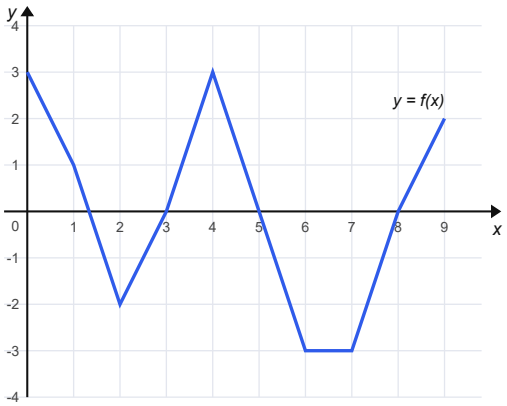
$$x^4+x^3-10x^2+x+1=0$$

A) 1
B) $-0{,}5$
C) $0{,}5$
D) -1

10. B $y=\frac{6}{x+2}-4$ funksiyaning qiymatlar sohasini toping. 2 ball

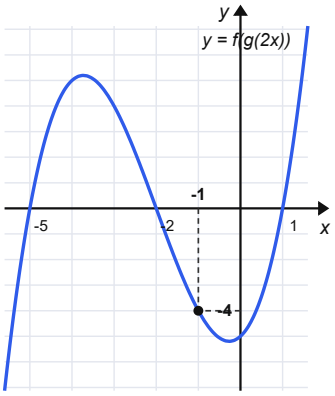
A) $(-\infty;-2)\cup(-2;\infty)$
B) $(-\infty;-4)\cup(-4;\infty)$
C) $(-4;0)$
D) $(-\infty;4)\cup(4;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 14 B) 19 C) 17 D) 16

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(-9)=-2$ bo‘lsa, $f^{-1}(-4)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) -1 B) -9 C) -2 D) 9

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = a^2 - h^2$ tenglik o‘rinli.		
2.	To‘g‘ri prizmaning yon qirralari asos tekisligiga perpendikulyar.		
3.	Qirrasi a ga teng kubning to‘la sirti yuzi $6a^2$ ga teng.		
4.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = h^2 + \frac{a^2}{4}$ tenglik o‘rinli.		

A) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

B) 1-noto‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

C) 1-to‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

D) 1-noto‘g‘ri, 2-noto‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

14. Q Tomoni 6 ga teng muntazam 9-burchakka tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $R^2 - r^2$ ning qiymatini toping. 5,2 ball

A) 18 **B)** 36 **C)** 9 **D)** 3

15. Q TABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning TG balandligi $TG = 3\sqrt{6}\ cm$ ga teng. Tetraedrning to‘la sirt yuzini toping. 5,2 ball

A) $81\sqrt{3}\ cm^2$ **B)** $77\sqrt{3}\ cm^2$ **C)** $83\sqrt{3}\ cm^2$ **D)** $162\sqrt{3}\ cm^2$

16. B Berilgan ma’lumotlarga ko‘ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball

Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Shar sirt yuzi	A) $2\pi r^2$
II. Konusning o‘q kesimining yuzi	B) $\pi r(2r+h)$
III. Silindrning o‘q kesimining yuzi	C) $2rh$
	D) $4\pi r^2$
	E) rh
	F) $\frac{1}{3}\pi r^3$

A) I–D; II–B; III–C **B)** I–B; II–D; III–C **C)** I–D; II–E; III–C **D)** I–D; II–E; III–A

17. Q To‘g‘ri doiraviy silindrning balandligi $4\ cm$, asos radiusi esa $3\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo‘lgan masofalari mos ravishda $1\ cm$ va $2\ cm$. Agar $AB = 6\ cm$ bo‘lsa, silindr o‘qidan AB kesmagacha bo‘lgan eng qisqa masofani toping. 5,2 ball

A) $\frac{1}{2}\ cm$ **B)** $\frac{3}{2}\ cm$ **C)** $\frac{\sqrt{5}}{2}\ cm$ **D)** $\frac{\sqrt{13}}{2}\ cm$

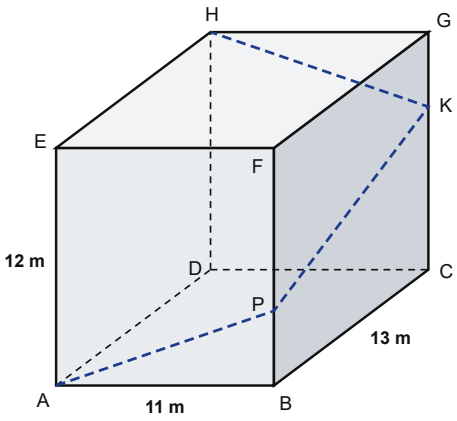
18. Q Yon tomonlari $10\ cm$ dan, asosi $16\ cm$ bo‘lgan teng yonli uchburchak o‘zining yon tomoni atrofida to‘liq aylantiriladi. Hosil bo‘lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $\frac{1248}{5}\pi\ cm^2$ **B)** $156\pi\ cm^2$ **C)** $192\pi\ cm^2$ **D)** $96\pi\ cm^2$

19. Q Shar shaklidagi mum o‘zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $53\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 2 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $29\pi\ cm^2$ ga teng. Sharning katta doira yuzini toping. 5,2 ball

A) $66\pi\ cm^2$ **B)** $53\pi\ cm^2$ **C)** $78\pi\ cm^2$ **D)** $89\pi\ cm^2$

20. M O‘lchamlari $11\ m,\ 13\ m$ va $12\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $36\ m$ **B)** $12\sqrt{5}\ m$ **C)** $37\ m$ **D)** $38\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 9-variant

1 B D	2 Q B	3 B C	4 Q B	5 Q A	6 Q D	7 Q C
8 Q B	9 Q A	10 B B	11 M D	12 Q B	13 B B	14 Q C
15 Q A	16 B C	17 Q A	18 Q A	19 Q C	20 M C	

1. D) Daraja 9, ozod had 33: $9+33=42$.

2. B) I: ozod had 15. II: $x^3+2x+8=x(x^2+4)+-2x+8$. III: ko‘phad bo‘lishi uchun $n=13$, daraja 7. IV: $P(-1)=6$.

3. C) Karrali ildizlar: 2 (7 karrali), -5 (3 karrali). 6 – oddiy ildiz.

4. B) $Q=(x-4)(x+4)$. $P(4)=0$, $P(-4)=-120\neq 0\to$ EKUB $x-4$.

5. A) Gorner sxemasi: 1, -2, 5, -16, 35. Bo‘linma $x^3-2x^2+5x-16$, qoldiq 35.

6. D) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p\mid 9, q\mid 4$). Tekshiramiz: $-\frac{1}{2};\frac{3}{2};3$.

7. C) $(20x-1)(10x-1)=200x^2-30x+1$, $(12x-1)(15x-1)=180x^2-27x+1$. $u=200x^2-30x$ desak, $(u+1)(\frac{9}{10}u+1)=33\to u=5$ yoki $u=-\frac{64}{9}$. Faqat $u=5$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{3}{20}$.

8. B) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz 1. Ildizlar ko‘paytmasi: $a\cdot(1)=-3\to a=-3$; $b=a+(1)(-1)=-4$. $ab=12$.

9. A) x^2 ga bo‘lamiz va $t=x+\frac{1}{x}$ desak: $t^2+t-12=0\to t=-4$ yoki $t=3$. Haqiqiy x faqat $|t|\geq 2$ da: $t=-4$ va $t=3$. Har bir t uchun $x^2-tx+1=0$ (ko‘paytma 1, yig‘indi t). Javob: 1.

10. B) $\frac{6}{x+2}\neq 0\to y\neq -4$.

11. D) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=1$, $f(2)=-2$, $f(3)=0$, $f(4)=3$, $f(5)=0$, $f(6)=-3$, $f(7)=-3$, $f(8)=0$, $f(9)=2$. Eng katta qiymat $a=5, b=8, c=3, d=7, e=1$ da: $5+8+7-3-1=16$.

12. B) $x=-1$ da grafikdan $f(g(-2))=-4$. $g(-2)=-9\to f(-9)=-4\to f^{-1}(-4)=-9$.

13. B) 1) $f^2=h^2+\frac{a^2}{4}$ ✗. 2) ta’rif ✓. 3) 6 ta yoq ✓. 4) to‘g‘ri burchakli uchburchakdan ✓.

14. C) $\left(\frac{a}{2}\right)^2=R^2-r^2$ – burchaklar soniga bog‘liq emas: $R^2-r^2=9$.

15. A) $h=a\sqrt{\frac{2}{3}}=\frac{a\sqrt{6}}{3}=3\sqrt{6}\to a=9$. To‘la sirt $\sqrt{3}a^2=81\sqrt{3}\ cm^2$.

16. C) Shar sirt yuzi: $4\pi r^2$. Konusning o‘q kesimining yuzi: rh. Silindrning o‘q kesimining yuzi: $2rh$.

17. A) Balandliklar farqi 1, asosdagi proyeksiya (vatar) $\sqrt{36-1}=\sqrt{35}$. O‘qdan masofa $\sqrt{9-\frac{35}{4}}=\frac{1}{2}$ cm.

18. A) Uchburchak yuzi 48, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2\cdot 48}{10}=\frac{48}{5}$. Ikki konus yon sirtlari: $S=\pi\cdot\frac{48}{5}(10+16)=\frac{1248}{5}\pi$.

19. C) $x^2+53=(x+2)^2+29\to x=5$, $R^2=78$. Javob 78π cm².

20. C) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(11+13+11)\times 12$ hosil bo‘ladi: $\sqrt{35^2+12^2}=37$ m.

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: 10-variant

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning darajasi va ozod hadi yig‘indisini toping. 2 ball

$$P(x) = 6x^{10} + 7x^9 + 6x^8 + 5 - 3x^4$$

A) 15 **B)** 14 **C)** 5 **D)** 16

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^8 + x^5 + 15$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) $x + 5$
II. $P(x) = x^3 + 2x + 5$ ko‘phadni $x^2 + 1$ ga bo‘lgandagi qoldiqni toping.	B) 15
III. $P(x) = -3x^{n-10} + 8x^{10-n} + x^6 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang.	C) $5x + 1$
IV. $P(x) = x^3 + x^2 + 2x + 13$ ko‘phadni $x + 2$ ga bo‘lgandagi qoldiqni toping.	D) -5
	E) 5
	F) 6

A) I-B; II-A; III-F; IV-D **B)** I-D; II-B; III-F; IV-E

C) I-B; II-A; III-F; IV-E **D)** I-B; II-E; III-F; IV-D

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x-3)^2(x+5)(x+1)^3$$

A) hammasi **B)** 3 va -5 **C)** 3 va -1 **D)** -5 va -1

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 - 2x^2 - 5x + 6$ va $Q(x) = x^2 - 1$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x + 1$ **B)** $x - 1$ **C)** $x + 0$ **D)** $x - 2$

5. Q Quyida $P(x)$ ko‘phadni $x-1$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	1	3	-2	5	-6
1	1	4	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) x^3+4x^2+2x+7 va $r=1$
B) x^3+4x^2+2x+9 va $r=-1$
C) x^3+4x^2+2x-7 va $r=-1$
D) x^3+4x^2-2x+7 va $r=1$

6. Q $P(x)=12x^3-25x^2+x+2$ butun koeffitsiyentli ko‘phad berilgan. Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang. 3,5 ball

A) $-\frac{1}{4};\frac{1}{3};2$
B) $-2;-\frac{1}{3};\frac{1}{4}$
C) $-4;\frac{1}{3};2$
D) $-2;-\frac{1}{4};\frac{1}{3}$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(30x-1)(6x-1)(15x-1)(5x-1)=56$$

A) $\frac{7}{30}$
B) $\frac{7}{15}$
C) $\frac{7}{60}$
D) $\frac{2}{5}$

8. Q Ushbu $x^2-x+a=0$ tenglamaning ildizlari $x^3+bx-4=0$ tenglamaning ham ildizlari bo‘lsa, $a-b$ ning qiymatini toping. 3,5 ball

A) 1
B) 2
C) -1
D) -7

9. Q Tenglamaning haqiqiy ildizlari ko‘paytmasini toping. 3,5 ball

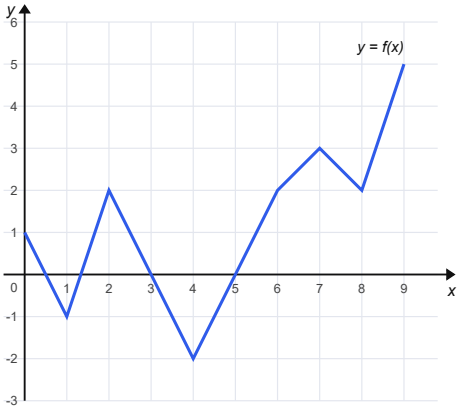
$$x^4+4x^3+5x^2+4x+1=0$$

A) -2
B) 1
C) 2
D) -1

10. B $y=\frac{-2}{x+4}-5$ funksiyaning qiymatlar sohasini toping. 2 ball

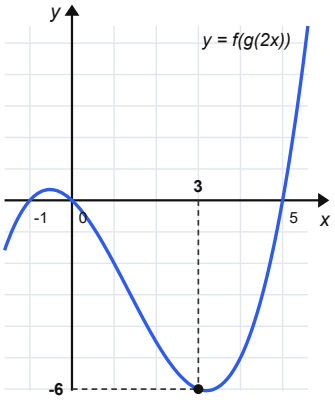
A) $(-5;0)$
B) $(-\infty;5)\cup(5;\infty)$
C) $(-\infty;-4)\cup(-4;\infty)$
D) $(-\infty;-5)\cup(-5;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan (grafik butun koordinatali nuqtalarni kesmalar bilan tutashtiradi). Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 11 B) 10 C) 13 D) 8

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(-1)=6$ bo‘lsa, $f^{-1}(-6)$ ning qiymatini toping. Bu yerda $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) -1 B) 6 C) 3 D) 1

13. B Quyidagi tasdiqlarning to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Qirrasi a ga teng kubning hajmi $3a^3$ ga teng.		
2.	To‘g‘ri prizmaning yon qirralari asos tekisligiga perpendikulyar.		
3.	Qirrasi a ga teng kubning yon sirti yuzi $6a^2$ ga teng.		
4.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2 = a^2 - h^2$ tenglik o‘rinli.		

A) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

B) 1-noto‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

C) 1-to‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

D) 1-to‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

14. Q Tomoni 10 ga teng muntazam 6-burchakka tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $R^2 - r^2$ ning qiymatini toping. 5,2 ball

A) 100 **B)** 5 **C)** 25 **D)** 50

15. Q SABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning SG balandligi $SG = 3\sqrt{6}\ cm$ ga teng. Tetraedrning qirrasi uzunligini toping. 5,2 ball

A) $6\ cm$ **B)** $10\ cm$ **C)** $9\ cm$ **D)** $12\ cm$

16. B Berilgan ma’lumotlarga ko‘ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball
Aylanish jismlari. Bu yerda r — radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Konusning yon sirt yuzi	A) $\frac{4}{3}\pi r^3$
II. Shar sirt yuzi	B) $\pi r l$
III. Silindr hajmi	C) $\pi r^2 h$
	D) $2\pi r^2$
	E) $4\pi r^2$
	F) $\frac{1}{3}\pi r^3$

A) I–B; II–C; III–F **B)** I–B; II–F; III–A **C)** I–B; II–E; III–C **D)** I–A; II–E; III–C

17. Q To‘g‘ri doiraviy silindrning balandligi $7\ cm$, asos radiusi esa $4\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo‘lgan masofalari mos ravishda $1\ cm$ va $6\ cm$. Agar $AB = 6\ cm$ bo‘lsa, silindr o‘qidan AB kesmagacha bo‘lgan eng qisqa masofani toping. 5,2 ball

A) $\frac{\sqrt{53}}{2}\ cm$ B) $\frac{\sqrt{57}}{2}\ cm$ C) $\sqrt{11}\ cm$ D) $\frac{7}{2}\ cm$

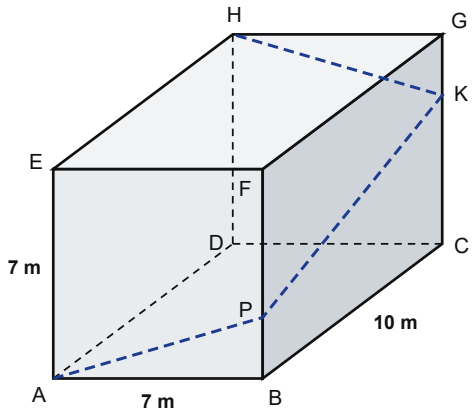
18. Q Yon tomonlari $15\ cm$ dan, asosi $24\ cm$ bo‘lgan teng yonli uchburchak o‘zining yon tomoni atrofida to‘liq aylantiriladi. Hosil bo‘lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $432\pi\ cm^2$ B) $\frac{2808}{5}\pi\ cm^2$ C) $351\pi\ cm^2$ D) $216\pi\ cm^2$

19. Q Metall shar o‘zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan. Birinchi tekislik hosil qilgan doiraviy kesimning yuzi $74\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchisidan shar markazidan 5 cm uzoqroqda joylashgan va u hosil qilgan doiraviy kesimning yuzi $9\pi\ cm^2$ ga teng. Sharning katta doira yuzini toping. 5,2 ball

A) $74\pi\ cm^2$ B) $78\pi\ cm^2$ C) $99\pi\ cm^2$ D) $90\pi\ cm^2$

20. M O‘lchamlari $7\ m,\ 10\ m$ va $7\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak. Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $26\ m$ B) $13\sqrt{2}\ m$ C) $25\ m$ D) $24\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 10-variant

1 B A	2 Q C	3 B C	4 Q B	5 Q A	6 Q A	7 Q A
8 Q A	9 Q B	10 B D	11 M B	12 Q A	13 B A	14 Q C
15 Q C	16 B C	17 Q A	18 Q B	19 Q D	20 M C	

1. A) Daraja 10, ozod had 5: $10+5=15$.

2. C) I: ozod had 15. II: $x^3+2x+5=x(x^2+1)+x+5$. III: ko‘phad bo‘lishi uchun $n=10$, daraja 6. IV: $P(-2)=5$.

3. C) Karrali ildizlar: 3 (2 karrali), -1 (3 karrali). -5 – oddiy ildiz.

4. B) $Q=(x-1)(x+1)$. $P(1)=0$, $P(-1)=8\neq 0\to$ EKUB $x-1$.

5. A) Gorner sxemasi: 1, 4, 2, 7, 1. Bo‘linma x^3+4x^2+2x+7, qoldiq 1.

6. A) Ratsional ildizlar $\frac{p}{q}$ ko‘rinishida ($p\mid 2, q\mid 12$). Tekshiramiz: $-\frac14;\frac13;2$.

7. A) $(30x-1)(5x-1)=150x^2-35x+1$, $(6x-1)(15x-1)=90x^2-21x+1$. $u=150x^2-35x$ desak, $(u+1)(\frac35u+1)=56\to u=-11$ yoki $u=\frac{25}{3}$. Faqat $u=\frac{25}{3}$ da haqiqiy ildizlar bor (ikkinchisida diskriminant manfiy), yig‘indisi $\frac{7}{30}$.

8. A) Kubik tenglamada x^2 qatnashmaydi, ildizlar yig‘indisi 0: uchinchi ildiz -1. Ildizlar ko‘paytmasi: $a\cdot(-1)=4\to a=-4$; $b=a+(-1)(1)=-5$. $a-b=1$.

9. B) x^2 ga bo‘lamiz va $t=x+\frac1x$ desak: $t^2+4t+3=0\to t=-3$ yoki $t=-1$. Haqiqiy x faqat $|t|\geq 2$ da: $t=-3$. Har bir t uchun $x^2-tx+1=0$ (ko‘paytma 1, yig‘indi t). Javob: 1.

10. D) $\frac{-2}{x+4}\neq 0\to y\neq -5$.

11. B) Grafikdan butun nuqtalardagi qiymatlar: $f(1)=-1$, $f(2)=2$, $f(3)=0$, $f(4)=-2$, $f(5)=0$, $f(6)=2$, $f(7)=3$, $f(8)=2$, $f(9)=5$. Eng katta qiymat $a=6, b=8, c=2, d=5, e=7$ da: $6+8+5-2-7=10$.

12. A) $x=3$ da grafikdan $f(g(6))=-6$. $g(6)=-1\to f(-1)=-6\to f^{-1}(-6)=-1$.

13. A) 1) $V=a^3$ ✗. 2) ta’rif ✓. 3) yon sirt $4a^2$ ✗. 4) $f^2=h^2+\frac{a^2}{4}$ ✗.

14. C) $\left(\frac{a}{2}\right)^2=R^2-r^2$ – burchaklar soniga bog‘liq emas: $R^2-r^2=25$.

15. C) $h=a\sqrt{\frac23}=\frac{a\sqrt6}{3}=3\sqrt6\to a=9$. Qirra 9 cm.

16. C) Konusning yon sirt yuzi: πrl. Shar sirt yuzi: $4\pi r^2$. Silindr hajmi: πr^2h.

17. A) Balandliklar farqi 5, asosdagi proyeksiya (vatar) $\sqrt{36-25}=\sqrt{11}$. O‘qdan masofa $\sqrt{16-\frac{11}{4}}=\frac{\sqrt{53}}{2}$ cm.

18. B) Uchburchak yuzi 108, aylanish o‘qiga (yon tomonga) qarama-qarshi uchdan tushirilgan balandlik $\frac{2\cdot108}{15}=\frac{72}{5}$. Ikki konus yon sirtlari: $S=\pi\cdot\frac{72}{5}(15+24)=\frac{2808}{5}\pi$.

19. D) $x^2+74=(x+5)^2+9\to x=4$, $R^2=90$. Javob 90π cm².

20. C) Uchta yoqni bir tekislikka yoysak, to‘g‘ri to‘rtburchak $(7+10+7)\times 7$ hosil bo‘ladi: $\sqrt{24^2+7^2}=25$ m.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · MATEMATIKA · CHSB-1

ChSB-1 tayyorgarlik: I chorak demo varianti

20 ta savol · 45 daqiqa · 80 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. B Ko‘phadning eng katta darajasi va bosh koeffitsiyenti yig‘indisini toping. 2 ball

$$P(x) = 56 + 3x + 6x^7 + 2x^9 + 8x^4$$

A) 11 **B)** 12 **C)** 13 **D)** 4

2. Q Berilgan ma’lumotlarga ko‘ra, I–IV bandlarni A–F javoblar bilan moslashtiring. 3,5 ball

Bandlar	Javoblar
I. $P(x) = x^{13} + x^3 + 9$ ko‘phad uchun $P(x) = Q(x)$ bo‘lsa, $Q(x)$ ko‘phadning ozod hadini toping.	A) 2
II. $P(x) = x^3 + 5x + 2$ ko‘phadni $x^2 + 3$ ga bo‘lgandagi qoldiqni toping.	B) $2 + 8x$
III. $P(x) = -3x^{n-11} + 8x^{11-n} + x^6 + 12$ ifoda ko‘phad bo‘lsa, uning darajasini aniqlang	C) 6
IV. $P(x) = x^3 - 4x^2 + 2x + 14$ ko‘phadni $x + 2$ ga bo‘lgandagi qoldiqni toping.	D) 9
	E) $2x + 2$
	F) -14

A) I-D; II-E; III-C; IV-F **B)** I-F; II-B; III-C; IV-A

C) I-D; II-E; III-A; IV-F **D)** I-E; II-C; III-A; IV-F

3. B Ko‘phadning karrali ildizlarini aniqlang. 2 ball

$$P(x) = (x + 2)^7(x - 4)^3(x + 3)$$

A) -2 va 4 **B)** -2 va -3 **C)** -3 va 4 **D)** hammasi

4. Q Yevklid algoritmidan foydalanib, $P(x) = x^3 - 4x^2 + 5x + 10$ va $Q(x) = x^2 - x - 2$ ko‘phadlarning eng katta umumiy bo‘luvchisini (EKUB) toping. 3,5 ball

A) $x + 1$ **B)** $x - 3$ **C)** $x - 2$ **D)** $x - 1$

5. Q Quyida $P(x)$ ko‘phadni $x-2$ ga bo‘lish uchun tuzilgan Gorner sxemasi berilgan. 3,5 ball

	2	−4	5	1	−6		2	2	0	...	...	...

Jadvaldagi yetishmayotgan qiymatlarni aniqlab, bo‘linma va qoldiqni toping.

A) $2x^3+5x+11$ va $r=16$
B) $2x^3+5x-11$ va $r=-16$
C) $2x^3-5x+11$ va $r=16$
D) $2x^3+5x+9$ va $r=-16$

6. Q $P(x)=6x^3-17x^2-5x+6$ butun koeffitsiyentli ko‘phad berilgan. 3,5 ball
Ratsional ildizlar haqidagi teoremadan foydalanib, ko‘phadning barcha ratsional ildizlari berilgan javobni aniqlang.

A) $-\frac{2}{3};\frac{1}{2};3$
B) $-\frac{1}{2};\frac{2}{3};3$
C) $-2;-\frac{2}{3};\frac{1}{3}$
D) $-2;\frac{1}{2};\frac{2}{3}$

7. Q Tenglamaning haqiqiy ildizlari yig‘indisini toping. 3,5 ball

$$(12x-1)(6x-1)(4x-1)(3x-1)=5$$

A) $\frac{5}{12}$
B) $\frac{7}{12}$
C) $\frac{1}{6}$
D) $\frac{11}{12}$

8. Q Ushbu $x^2-3x+a=0$ tenglamaning ildizlari $x^3+bx+18=0$ tenglamaning ham ildizlari bo‘lsa, $a+b$ ning qiymatini toping. 3,5 ball

A) 3
B) 4
C) 5
D) 2

9. Q Tenglamaning haqiqiy ildizlari ko‘paytmasini toping. 3,5 ball

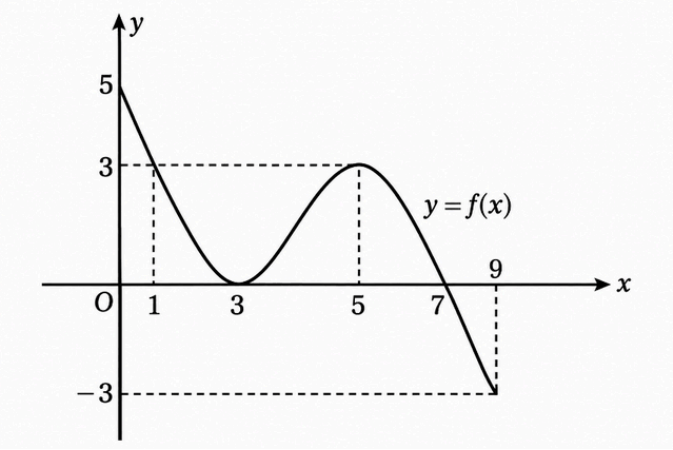
$$x^4+5x^3+2x^2+5x+1=0$$

A) 1
B) -1
C) $-2,5$
D) $2,5$

10. B $y=\frac{7}{x-1}+2$ funksiyaning qiymatlar sohasini toping. 2 ball

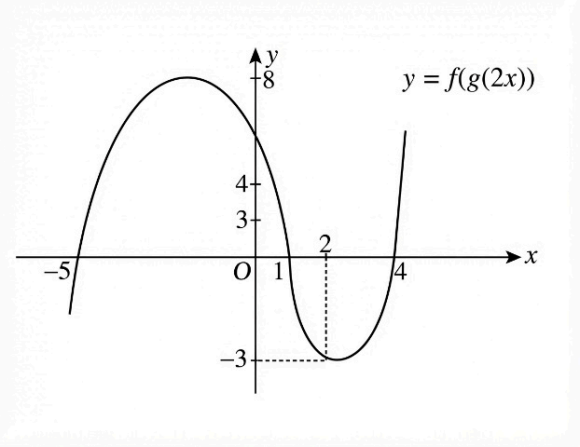
A $(-\infty;2)\cup(2;\infty)$
B) $(0;2)$
C) $(1;2)$
D $(-\infty;1)\cup(1;\infty)$

11. M Dekart koordinatalar tekisligida $[0;9]$ oraliqda aniqlangan $y=f(x)$ funksiyaning grafigi berilgan. Aniqlanish sohasiga tegishli, o‘zaro turli (a,b,c,d,e) musbat butun sonlar uchun $f(a)=f(b)=f(c)$ va $f(d)<f(c)<f(e)$ shartlar bajariladi.
Grafikdan foydalanib, $a+b+d-c-e$ ifodaning eng katta qiymatini toping. 6 ball



A) 16 **B)** 19 **C)** 10 **D)** 17

12. Q Quyida $y=f(g(2x))$ funksiyaning grafigi berilgan.
Agar $g^{-1}(6)=4$ bo‘lsa, $f^{-1}(-3)$ ning qiymatini toping. Bu yerda funksiya uchun $y=f^{-1}(x)$ va $y=g^{-1}(x)$ mos ravishda $y=f(x)$ va $y=g(x)$ funksiyalarning teskari funksiyalaridir. 3,5 ball



A) 6 **B)** 8 **C)** 7 **D)** 4

13. B Quyidagi tasdiqlarni to‘g‘ri yoki noto‘g‘ri ekanligini aniqlang. 4 ball

	Tasdiqlar	to‘g‘ri	noto‘g‘ri
1.	Qirrasi a ga teng kubning yon sirti yuzi $6a^2$ ga teng.		
2.	To‘g‘ri to‘rtburchakli parallelepipedning qirralari $a,\ b$ va c bo‘lsa, uning diagonali $\sqrt{a^2+b^2+c^2}$ ga teng.		
3.	To‘g‘ri uchburchakli prizmaning asosi tomoni $a\ cm$ bo‘lgan muntazam uchburchak, yon qirrasi esa $h\ cm$ ga teng. Prizmaning yon sirt yuzi $3ah\ cm^2$ ga teng.		
4.	Muntazam to‘rtburchakli piramida asosining tomoni a, balandligi h, apofemasi f bo‘lsa, $f^2=a^2-h^2$ tenglik o‘rinli.		

A) 1-noto‘g‘ri, 2-to‘g‘ri, 3-to‘g‘ri, 4-noto‘g‘ri

B) 1-noto‘g‘ri, 2-to‘g‘ri, 3-noto‘g‘ri, 4-noto‘g‘ri

C) 1-to‘g‘ri, 2-noto‘g‘ri, 3-to‘g‘ri, 4-to‘g‘ri

D) 1-to‘g‘ri, 2-noto‘g‘ri, 3-noto‘g‘ri, 4-to‘g‘ri

14. Q Muntazam ko‘pburchakning tomoni a, unga tashqi va ichki chizilgan aylanalarning radiuslari mos ravishda R va r bo‘lsa, $\frac{a^2}{R^2-r^2}$ ifodaning qiymatini toping. 5,2 ball

A) 4 **B)** 5 **C)** 2 **D)** $\sqrt{2}$

15. Q TABC muntazam tetraedrda G nuqta ABC asos medianalarining kesishish nuqtasi. Tetraedrning TG balandligi $TG=2\sqrt{6}\ cm$ ga teng. Tetraedrning to‘la sirt yuzini toping. 5,2 ball

A) $36\sqrt{3}\ cm^2$ **B)** $24\sqrt{3}\ cm^2$ **C)** $48\sqrt{3}\ cm^2$ **D)** $56\sqrt{3}\ cm^2$

16. B Berilgan ma'lumotlarga ko'ra, I–III bandlarni A–F javoblar bilan moslashtiring. 4 ball

Aylanish jismlari Bu yerda r— radius, h — balandlik, l — konusning yasovchisi.

Bandlar	Javoblar
I. Silindrning to'la sirt yuzi	A) $2\pi r^2$
II. Konusning o'q kesimining yuzi	B) $2\pi r(r+h)$
III. Shar sirt yuzi	C) rl
	D) $4\pi r^2$
	E) rh
	F) $\pi r(2r+h)$

A) I–B; II–E; III–D;
B) I–B; II–E; III–A;
C) I–F; II–C; III–D;
D) I–F; II–C; III–A;

17. Q To'g'ri doiraviy silindrning balandligi $5\ cm$, asos radiusi esa $4\ cm$ ga teng. Silindrning yon sirtida joylashgan A va B nuqtalarning asos tekisligigacha bo'lgan masofalari mos ravishda $2\ cm$ va $3\ cm$. Agar $AB = 5\ cm$ bo'lsa, silindr o'qidan AB kesmagacha bo'lgan eng qisqa masofani toping. 5,2 ball

A) $\sqrt{10}\ cm$ **B)** $\sqrt{11}\ cm$ **C)** $\frac{\sqrt{39}}{2}\ cm$ **D)** $\frac{\sqrt{43}}{2}\ cm$

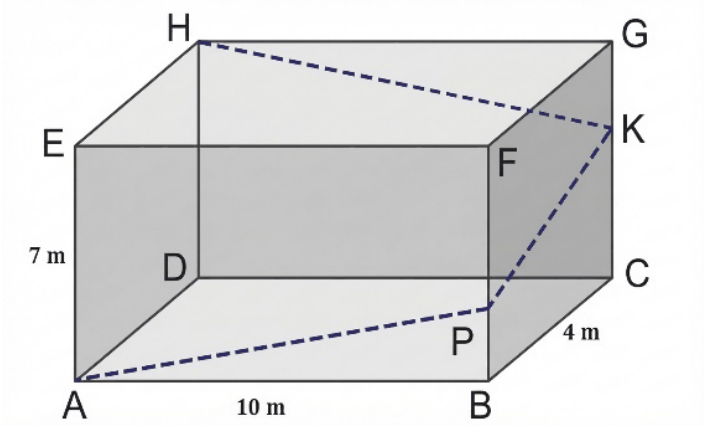
18. Q Yon tomonlari $5\ cm$ dan, asosi $6\ cm$ bo'lgan teng yonli uchburchak o'zining yon tomoni atrofida to'liq aylantiriladi. Hosil bo'lgan aylanma jismning sirt yuzini toping. 5,2 ball

A) $\frac{264\pi}{5}\ cm^2$ **B)** $\frac{256\pi}{25}\ cm^2$ **C)** $\frac{192\pi}{5}\ cm^2$ **D)** $\frac{243\pi}{25}\ cm^2$

19. Q Shar shaklidagi mum o'zaro parallel ikkita tekislik bilan kesilgan. Ikkala tekislik ham shar markazining bir tomonida joylashgan.Birinchi tekislik shar markazidan $x\ cm$ masofada joylashgan bo'lib, hosil bo'lgan doiraviy kesimning yuzi $45\pi\ cm^2$ ga teng. Ikkinchi tekislik birinchi tekislikka nisbatan shar markazidan $3\ cm$ uzoqroqda joylashgan va bu tekislik hosil qilgan doiraviy kesimning yuzi $24\pi\ cm^2$ ga teng. Sharning sirt yuzini toping. 5,2 ball

A) $196\pi\ cm^2$ **B)** $144\pi\ cm^2$ **C)** $169\pi\ cm^2$ **D)** $225\pi\ cm^2$

20. M O‘lchamlari $10\ m,\ 4\ m$ va $7\ m$ bo‘lgan to‘g‘ri to‘rtburchakli parallelepiped shaklidagi binoning tashqi sirtiga signal kabeli yotqizilmoqda. Kabel A nuqtadan H nuqtaga binoning old, o‘ng yon va orqa sirtlari bo‘ylab ketma-ket o‘tishi kerak.Kabelning A nuqtadan H nuqtaga borishdagi mumkin bo‘lgan eng qisqa uzunligini toping. 6 ball



A) $25\ m$ **B)** $17\ m$ **C)** $30\ m$ **D)** $20\ m$

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: I chorak demo varianti

1 B A	2 Q A	3 B A	4 Q A	5 Q A	6 Q A	7 Q A
8 Q A	9 Q A	10 B A	11 M A	12 Q A	13 B A	14 Q A
15 Q A	16 B A	17 Q A	18 Q A	19 Q A	20 M A	

1. A) Eng katta daraja 9, bosh koeffitsiyent 2: $9+2=11$.

2. A) I: ozod had 9 (D). II: $x^3+5x+2=x(x^2+3)+2x+2$ (E). III: $n=11$, daraja 6 (C). IV: $P(-2)=-14$ (F).

3. A) Karrali ildizlar: -2 (7 karrali) va 4 (3 karrali).

4. A) $Q=(x-2)(x+1)$. $P(-1)=0$, $P(2)=12\neq 0$ → EKUB $x+1$.

5. A) 2 | 2, 0, 5, 11, 16: bo'linma $2x^3+5x+11$, qoldiq 16.

6. A) $P(3)=0$; $6x^2+x-2=(3x+2)(2x-1)$ → $-\frac{2}{3};\frac{1}{2};3$.

7. A) $(36x^2-15x+1)(24x^2-10x+1)=5$, $t=12x^2-5x$: $(3t+1)(2t+1)=5$ → $t=\frac{1}{2}$ (ikkinchisida haqiqiy ildiz yo'q). Yig'indi $\frac{5}{12}$.

8. A) Kubik tenglamada uchinchi ildiz -3 (yig'indi 0). $a\cdot(-3)=-18\to a=6$; $b=6-9=-3$. $a+b=3$.

9. A) $t=x+\frac{1}{x}$: $t^2+5t=0\to t=-5$: $x^2+5x+1=0$, ko'paytma 1.

10. A) $\frac{7}{x-1}\neq 0\to y\neq 2$.

11. A) Teng qiymatli uchta butun nuqta $f(2)=f(4)=f(6)$, c eng kichigi 2. $d=9$, $e=1$: $4+6+9-2-1=16$.

12. A) $x=2$ da $f(g(4))=-3$, $g(4)=6\to f(6)=-3\to f^{-1}(-3)=6$.

13. A) 1) Yon sirt $4a^2$ ✗ 2) ✓ 3) ✓ 4) $f^2=h^2+\frac{a^2}{4}$ ✗.

14. A) $\left(\frac{a}{2}\right)^2=R^2-r^2\to\frac{a^2}{R^2-r^2}=4$.

15. A) $h=a\sqrt{\frac{2}{3}}=2\sqrt{6}\to a=6$. To'la sirt $\sqrt{3}a^2=36\sqrt{3}$ cm².

16. A) Silindr $2\pi r(r+h)$ (B), konus o'q kesimi rh (E), shar $4\pi r^2$ (D).

17. A) Balandliklar farqi 1, vatar $\sqrt{25-1}=2\sqrt{6}$. O'qdan masofa $\sqrt{16-6}=\sqrt{10}$.

18. A) Uchidan yon tomonga balandlik $\frac{24}{5}$. Ikki konus: $S=\pi\cdot\frac{24}{5}(5+6)=\frac{264\pi}{5}$.

19. A) $x^2+45=(x+3)^2+24\to x=2$, $R^2=49$. Sirt 196π.

20. A) Yoqlarni yoysak: $\sqrt{(10+4+10)^2+7^2}=25$ m.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 1-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Tadqiqotchi turli materiklardagi tuyaqushsimon qushlarning skelet tuzilishini o‘zaro solishtirib, o‘xshash va farqli belgilarini aniqladi. Qaysi metoddan foydalanilgan? 2 ball

A) Eksperimental metod
B) Modellashtirish metodi
C) Taqqoslash metodi
D) Kuzatish metodi

2. B Ma’lum hududda yashovchi bir turga mansub bo‘rilar guruhining soni, yosh tarkibi va ko‘payishi o‘rganildi. Qaysi tuzilish darajasi? 1 ball

A) Biosfera darajasi
B) Organizm darajasi
C) Biogeosenoz darajasi
D) Populyatsiya–tur darajasi

3. Q Suvning biologik xususiyatlariga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Suv molekulasi qutbli bo‘lgani uchun tuzlar va qandlarni yaxshi eritadi.
2. Suvning issiqlik sig‘imi pastligi tana haroratini barqaror saqlaydi.
3. Bug‘lanishda ko‘p issiqlik yutilgani uchun terlash tanani sovitadi.
4. Lipidlar suvda yaxshi eriydi.
5. Fotosintezda suv boshlang‘ich modda sifatida qatnashadi.

A) 1, 2, 3
B) 1, 4, 5
C) 1, 3, 5
D) 2, 3, 4

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	20	10
Y	10	30

A) X – 758,6 kJ; Y – 1360,6 kJ
B) X – 702,1 kJ; Y – 1304,1 kJ
C) X – 1343 kJ; Y – 741 kJ
D) X – 741 kJ; Y – 1343 kJ

5. Q Qon plazmasidagi X oqsil kamaygach, jarohatda qon laxtasi kech hosil bo‘ldi; kislorod va lipidlar tashilishi me’yorda. X va uning funksiyasi? 2 ball

A) Fibrinogen : himoya
B) Gemoglobin : transport
C) Albumin : transport
D) Insulin : gormonal

6. M Joydagi qon izi (D) va gumondorlar (G1–G4) DNKsining bir xil sohasi ko‘paytirilib, gel-elektroforezda fragment uzunliklari (bp) aniqlandi. 3 ball

Namuna	Fragmentlar, bp
D	250, 400, 700
G1	250, 500, 700
G2	250, 400, 700
G3	300, 400, 600
G4	150, 400, 900

Mulohazalarni baholang (T/N).
1. Iz qoldirgan G2 bo‘lishi mumkin, chunki uning uchala fragmenti D bilan mos.
2. G1 ham iz qoldirgan bo‘lishi mumkin, chunki ikki fragmentning mosligi yetarli.
3. Qisqa fragmentlar gelda uzunlariga qaraganda uzoqroq masofaga siljiydi.

A) TTT **B)** TNT **C)** TNN **D)** NNT

7. Q Ferment eritmasidan uch namuna 10 °C, 37 °C va 80 °C da saqlanib, so‘ng 37 °C ga qaytarildi. 2 ball

Namuna	Saqlash haroratida faollik, %	37 °C ga qaytgandan so‘ng, %
1 (10 °C)	20	95
2 (37 °C)	100	100
3 (80 °C)	0	0

To‘g‘ri xulosani aniqlang.

A) 2-namuna optimal harorat 80 °C ekanini ko‘rsatadi.
B) 3-namunada fermentning tuzilishi to‘liq tiklangan.
C) 1-namunada ferment qaytmas denaturatsiyaga uchragan.
D) 3-namunada faollikning tiklanmasligi qaytmas denaturatsiyani ko‘rsatadi.

8. B Hujayra nazariyasining qoidasiga mos fikrni aniqlang. 1 ball

A) Barcha tirik organizmlar hujayralardan tuzilgan.

B) Hujayralar o‘lik moddadan o‘z-o‘zidan paydo bo‘ladi.

C) Viruslar hujayralardan tuzilgan.

D) Barcha hujayralar bir xil shaklga ega.

9. Q 2 ball

Belgi	X	Y	Z
Yadro	yo‘q	bor	bor
Hujayra devori	bor (murein)	bor (sellyuloza)	yo‘q
Xloroplast	yo‘q	bor	yo‘q

To‘g‘ri fikrlarni aniqlang.

1. X hujayrada ribosomalar mavjud.
2. Y hujayrada mitoxondriyalar bo‘lmaydi.
3. Z hujayra hayvon organizmiga tegishli.
4. Y da irsiy axborot faqat yadroda.
5. Uchala hujayrada ham plazmatik membrana mavjud.

A) 1, 3, 5 **B)** 3, 4, 5 **C)** 2, 4, 5 **D)** 1, 3, 4

10. Q Energetik almashinuvda hosil bo‘lgan 1000 kJ energiyaning 40 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 200 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang. 2 ball

A) a – 10 mol; b – 5 mol **B)** a – 16 mol; b – 5 mol

C) a – 25 mol; b – 5 mol **D)** a – 10 mol; b – 2,5 mol

11. Q Hujayrada 6 mol glyukoza glikolizga uchradi, shundan 4 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang. 2 ball

A) 156 mol **B)** 228 mol **C)** 144 mol **D)** 12 mol

12. Q O‘simlik fotosintezda 18 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 6 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol). 2 ball

A) 720 g **B)** 180 g **C)** 540 g **D)** 360 g

13. Q 2n = 4 bo‘lgan hayvon hujayrasining mitoz bosqichlari aralash berilgan: 2 ball

1. Xromosomalar ekvator tekisligida bir qator bo‘lib joylashgan.
2. Xromosomalar spirallashib, yadro qobig‘i erimoqda.
3. Qutblarda yangi yadro qobiqlari hosil bo‘lmoqda.
4. Xromatidalar qutblarga tortilmoqda.

To‘g‘ri ketma-ketlikni aniqlang.

A) 2 → 1 → 3 → 4 **B)** 1 → 2 → 4 → 3 **C)** 2 → 4 → 1 → 3 **D)** 2 → 1 → 4 → 3

14. M Lavlagi bo‘laklari yuvilib, turli haroratli suvga solindi; 5 daqiqadan so‘ng pigment miqdori o‘lchandi. 3 ball

Harorat, °C	20	40	60	80
Pigment (nisbiy)	2	4	30	60

Mulohazalarni baholang (T/N).

1. Yuqori haroratda pigmentning ko‘p chiqishi membrana o‘tkazuvchanligi ortganini ko‘rsatadi.
2. Harorat oshishi bilan pigment chiqishi kamaygan.
3. Bo‘laklar yuvilmasa, kesishda chiqqan pigment natijaga xalaqit beradi.

A) NNT **B)** TTT **C)** TNT **D)** TNN

15. B Nerv va gumoral boshqaruvga xos umumiy xususiyatni aniqlang. 1 ball

A) Organlar faoliyatini muvofiqlashtirib, ichki muhit barqarorligini saqlaydi.

B) Ta’sirni nerv tolalari orqali yetkazadi.

C) Javobni bir xil tezlikda yuzaga keltiradi.

D) Biologik faol moddalarni qon orqali tarqatadi.

16. Q Organizmlar xususiyatlari: a) yorug‘lik energiyasidan foydalanib organik modda hosil qiladi; b) anorganik moddalar oksidlanishidan energiya olib organik modda sintezlaydi; c) o‘lik organik qoldiqlar bilan oziqlanadi. 2 ball

Misollar: 1. Nitrifikator bakteriya; 2. Saprofit zamburug‘; 3. Sianobakteriya; 4. Parazit zamburug‘.

Moslashtiring.

A) a–1; b–3; c–2 **B)** a–3; b–2; c–1 **C)** a–3; b–1; c–2 **D)** a–3; b–1; c–4

17. B Qaysi juftlikdagi organizmlar fragmentatsiya orqali ko‘payadi? 1 ball

A) Spirogira va oq planariya **B)** Spirogira va achitqi

C) Planariya va xlamidomonada **D)** Achitqi va tufelka

18. Q Hayvon hujayrasida 2n = 8. 3 ta birlamchi spermatotsit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan spermatozoidlar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 3 ta spermatozoid; har birida 4 ta xromosoma.

B) 3 ta spermatozoid; har birida 8 ta xromosoma.

C) 12 ta spermatozoid; har birida 8 ta xromosoma.

D) 12 ta spermatozoid; har birida 4 ta xromosoma.

19. Q Rivojlanish xususiyatlari: a) tuxumdan chiqqan lichinka kattasidan tuzilishi va yashash muhiti bilan farq qiladi; b) embrion tuxumdagi sariqlik hisobiga rivojlanadi; c) embrion ona organizmida yo‘ldosh orqali oziqlanadi.
Tiplar: 1. Tuxumda rivojlanish; 2. Bachadonda rivojlanish; 3. Lichinkali rivojlanish.
Moslashtiring. 2 ball

A) a–3; b–1; c–2 **B)** a–3; b–2; c–1 **C)** a–1; b–3; c–2 **D)** a–2; b–1; c–3

20. M O‘simlik urug‘lari 30 tadan ikki guruhga ajratilib, 20 °C va 30 °C da saqlandi (boshqa sharoitlar bir xil). Jami ungan urug‘lar soni: 3 ball

Kun	2	4	6	8
20 °C	2	8	18	27
30 °C	6	16	25	27

Unib chiqqandan 5 kun o‘tgach nihollar o‘rtacha bo‘yi: 20 °C — 3 cm, 30 °C — 5 cm.
Mulohazalarni baholang (T/N).
1. 4-kuni 30 °C da ungan urug‘lar 20 °C dagidan ikki marta ko‘p.
2. 8-kuni sonlar tengligi harorat unish tezligiga ta’sir qilmaganini ko‘rsatadi.
3. Bir xil yoshdagi nihollar bo‘yi farqi harorat o‘sishga ta’sir qilganini ko‘rsatadi.

A) TTT **B)** TNT **C)** TNN **D)** NNT

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 1-variant

1 Q C	2 B D	3 Q C	4 Q D	5 Q A	6 M B	7 Q D
8 B A	9 Q A	10 Q A	11 Q A	12 Q D	13 Q D	14 M C
15 B A	16 Q C	17 B A	18 Q D	19 Q A	20 M B	

1. C) To‘g‘ri javob: **Taqqoslash metodi**
Solishtirish — taqqoslash.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. D) To‘g‘ri javob: **Populyatsiya–tur darajasi**
Bir tur guruhi — populyatsiya.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. C) To‘g‘ri javob: **1, 3, 5**
Issiqlik sig‘imi yuqori; lipidlar gidrofob.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. D) To‘g‘ri javob: **X – 741 kJ; Y – 1343 kJ**
X: 20·17,6 + 10·38,9 = 741 kJ; Y: 10·17,6 + 30·38,9 = 1343 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. A) To‘g‘ri javob: **Fibrinogen : himoya**
Ivish — fibrinogen.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. B) To‘g‘ri javob: **TNT**
To‘liq moslik kerak; kichik fragmentlar tezroq siljiydi.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. D) To‘g‘ri javob: **3-namunada faollikning tiklanmasligi qaytmas denaturatsiyani ko‘rsatadi.**
Yuqori haroratda qaytmas o‘zgarish.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. A) To‘g‘ri javob: **Barcha tirik organizmlar hujayralardan tuzilgan.**
Asosiy qoida.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. A) To‘g‘ri javob: **1, 3, 5**
Y o‘simlik — mitoxondriya bor; DNK xloroplastda ham.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. A) To‘g‘ri javob: **a – 10 mol; b – 5 mol**
a = 1000·40 % : 40 = 10 mol; b = 200 : 40 = 5 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. A) To‘g‘ri javob: **156 mol**
2·6 + 36·4 = 156 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. D) To‘g‘ri javob: **360 g**
Hosil bo‘lgan: 18 : 6 = 3 mol; sarflangan: 6 : 6 = 1 mol; qolgan: 2 · 180 = 360 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. D) To‘g‘ri javob: **2 → 1 → 4 → 3**
Profaza → metafaza → anafaza → telofaza.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. C) To‘g‘ri javob: **TNT**
Pigment ortgan.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. A) To‘g‘ri javob: **Organlar faoliyatini muvofiqlashtirib, ichki muhit barqarorligini saqlaydi.**
Umumiy — muvofiqlashtirish.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. C) To‘g‘ri javob: **a–3; b–1; c–2**
Fotoavtotrof, xemoavtotrof, saprotrof.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. A) To‘g‘ri javob: **Spirogira va oq planariya**
Bo‘laklanish.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. D) To‘g‘ri javob: **12 ta spermatozoid; har birida 4 ta xromosoma.**
Har bir spermatotsitdan 4 ta spermatozoid hosil bo‘ladi; gametada n = 4.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. A) To‘g‘ri javob: **a–3; b–1; c–2**
Lichinkali, tuxumda, bachadonda.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. B) To‘g‘ri javob: **TNT**
Tezlik farqi oldingi kunlarda ko‘rinadi.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 2-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Olim chumolilar uyasi yonida bir oy davomida ularning xatti-harakatini aralashmasdan yozib bordi. Qaysi metod qo‘llanilgan? 2 ball

A) Taqqoslash metodi **B)** Eksperimental metod
C) Modellashtirish metodi **D)** Kuzatish metodi

2. B Ko‘l suvi, tubidagi tuproq, undagi o‘simliklar, baliqlar va bakteriyalar orasidagi modda almashinuvi o‘rganildi. Qaysi daraja? 1 ball

A) Populyatsiya–tur darajasi **B)** Hujayra darajasi
C) Biogeosenoz darajasi **D)** Molekula darajasi

3. Q Mineral tuzlarga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Kalsiy ionlari qon ivishida qatnashadi.
2. Temir gemoglobin tarkibiga kiradi.
3. Yod insulin tarkibiga kiradi.
4. Magniy xlorofill tarkibiga kiradi.
5. Natriy ionlari faqat suyak tarkibida bo‘ladi.

A) 2, 3, 4 **B)** 1, 2, 4 **C)** 2, 3, 5 **D)** 1, 3, 5

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	30	10
Y	15	20

A) X – 878,1 kJ; Y – 1003,1 kJ **B)** X – 934,6 kJ; Y – 1059,6 kJ
C) X – 1042 kJ; Y – 917 kJ **D)** X – 917 kJ; Y – 1042 kJ

5. Q Bemor eritrotsitlarida X oqsil kam bo‘lgani uchun to‘qimalarga kislorod yetishmadi, qon ivishi me’yorda. X va funksiyasi? 2 ball

A) Fibrinogen : himoya
B) Gemoglobin : transport
C) Aktin : harakat
D) Keratin : tayanch

6. M 3 ball

Namuna	Fragmentlar, bp
D	300, 550, 800
G1	300, 550, 800
G2	300, 600, 800
G3	200, 550, 900

Mulohazalarni baholang (T/N).
1. G1 iz qoldirgan bo‘lishi mumkin.
2. G3 iz qoldirgan bo‘lishi mumkin, chunki bitta fragment mos.
3. Fragmentlar gelda massasi (uzunligi)ga ko‘ra ajraladi.

A) TTT **B)** TNT **C)** TNN **D)** NNT

7. Q 2 ball

Namuna	Saqlash, faollik %	37 °C da, %
1 (5 °C)	10	98
2 (40 °C)	90	90
3 (75 °C)	0	5

To‘g‘ri xulosa?

A) 3-namuna to‘liq renaturatsiyaga uchragan.
B) Harorat ferment faolligiga ta’sir qilmaydi.
C) Past harorat fermentni qaytmas denaturatsiyaga uchratadi.
D) Past harorat fermentni vaqtincha sekinlashtiradi, tuzilishini buzmaydi.

8. B Hujayra nazariyasiga ko‘ra yangi hujayralar qanday hosil bo‘ladi? 1 ball

A) Faqat urug‘lanish natijasida
B) Mavjud hujayralarning bo‘linishi natijasida
C) Anorganik moddalardan o‘z-o‘zidan
D) Hujayralararo moddadan

9. Q 2 ball

Belgi	X	Y	Z
Yadro	bor	yo‘q	bor
Hujayra devori	xitin	murein	yo‘q
Xloroplast	yo‘q	yo‘q	yo‘q

To‘g‘ri fikrlarni aniqlang.
1. X zamburug‘ hujayrasi.
2. Y prokariot hujayra.
3. Z da sentriolalar bo‘lishi mumkin.
4. X fotosintez qiladi.
5. Y da mitoxondriya bor.

A) 2, 3, 4 **B)** 1, 4, 5 **C)** 1, 2, 4 **D)** 1, 2, 3

10. Q Energetik almashinuvda hosil bo‘lgan 1500 kJ energiyaning 40 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 400 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang. 2 ball

A) a – 15 mol; b – 5 mol **B)** a – 37,5 mol; b – 10 mol
C) a – 15 mol; b – 10 mol **D)** a – 21 mol; b – 10 mol

11. Q Hujayrada 8 mol glyukoza glikolizga uchradi, shundan 5 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang. 2 ball

A) 16 mol **B)** 304 mol **C)** 180 mol **D)** 196 mol

12. Q O‘simlik fotosintezda 30 mol CO2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 6 mol O2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol). 2 ball

A) 180 g **B)** 1080 g **C)** 720 g **D)** 900 g

13. Q 1. Xromatidalar ajralib qutblarga harakatlanmoqda. 2 ball
2. Yadro qobiqlari qayta hosil bo‘lib, sitoplazma bo‘linmoqda.
3. Xromosomalar ekvatorda joylashgan.
4. Xromatin spirallashib, ko‘rinadigan xromosomalar hosil bo‘lmoqda.
To‘g‘ri ketma-ketlik?

A) 3 → 4 → 1 → 2 **B)** 4 → 3 → 2 → 1 **C)** 4 → 1 → 3 → 2 **D)** 4 → 3 → 1 → 2

14. M 3 ball

Harorat, °C	10	30	50	70
Pigment	1	3	20	45

Mulohazalarni baholang (T/N).
1. 50 °C dan yuqorida membrana oqsillari denaturatsiyasi pigment chiqishini kuchaytirgan.
2. 70 °C da pigment 30 °C dagidan 15 marta ko‘p.
3. 10 °C da membranalar butunlay buzilgan.

A) TTN **B)** NTN **C)** TNN **D)** TTT

15. B Faqat gumoral boshqaruvga xos xususiyatni aniqlang. 1 ball

A) Ta’sir nerv impulsi orqali tez uzatiladi.
B) Refleks yoyi ishtirok etadi.
C) Faqat bitta muskulga ta’sir qiladi.
D) Ta’sir qon orqali tarqalib, sekinroq va uzoqroq davom etadi.

16. Q a) tirik organizm hisobiga yashab, unga zarar yetkazadi; b) fotosintez qiladi; c) chirindi bilan oziqlanadi. 2 ball
Misollar: 1. Mog‘or zamburug‘i (mukor); 2. Qo‘ng‘irbosh zamburug‘i; 3. Xlorella.
Moslashtiring.

A) a–2; b–1; c–3 **B)** a–1; b–3; c–2 **C)** a–3; b–2; c–1 **D)** a–2; b–3; c–1

17. B Qaysi juftlikdagi organizmlar kurtaklanish orqali ko‘payadi? 1 ball

A) Amyoba va spirogira
B) Xlamidomonada va amyoba
C) Planariya va tufelka
D) Achitqi va gidra

18. Q Hayvon hujayrasida 2n = 20. 5 ta birlamchi spermatotsit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan spermatozoidlar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 20 ta spermatozoid; har birida 10 ta xromosoma.

B) 5 ta spermatozoid; har birida 20 ta xromosoma.

C) 5 ta spermatozoid; har birida 10 ta xromosoma.

D) 20 ta spermatozoid; har birida 20 ta xromosoma.

19. Q a) qurbaqa; b) toshbaqa; c) quyon. 2 ball
Tiplar: 1. Tuxumda rivojlanish; 2. Bachadonda rivojlanish; 3. Lichinkali rivojlanish.
Moslashtiring.

A) a–1; b–3; c–2 **B)** a–3; b–1; c–2 **C)** a–3; b–2; c–1 **D)** a–2; b–1; c–3

20. M 3 ball

Kun	2	4	6	8
15 °C	1	5	12	22
25 °C	4	15	24	28

Nihollar bo‘yi (5 kun): 15 °C — 2 cm, 25 °C — 4 cm.
Mulohazalarni baholang (T/N).
1. 4-kuni 25 °C da ungan urug‘lar 15 °C dagidan 3 marta ko‘p.
2. 8-kuni 15 °C da barcha urug‘lar ungan.
3. 25 °C da nihollar tezroq o‘sgan.

A) NNT **B)** TTT **C)** TNN **D)** TNT

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 2-variant

1 Q D	2 B C	3 Q B	4 Q D	5 Q B	6 M B	7 Q D
8 B B	9 Q D	10 Q C	11 Q D	12 Q C	13 Q D	14 M A
15 B D	16 Q D	17 B D	18 Q A	19 Q B	20 M D	

1. D) To‘g‘ri javob: **Kuzatish metodi**
Aralashmasdan qayd etish — kuzatish.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. C) To‘g‘ri javob: **Biogeosenoz darajasi**
Jonli va jonsiz tarkib — biogeosenoz.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. B) To‘g‘ri javob: **1, 2, 4**
Yod — tiroksinda; natriy — nerv impulsida.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. D) To‘g‘ri javob: **X – 917 kJ; Y – 1042 kJ**
X: 30·17,6 + 10·38,9 = 917 kJ; Y: 15·17,6 + 20·38,9 = 1042 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. B) To‘g‘ri javob: **Gemoglobin : transport**
Kislorod tashish.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. B) To‘g‘ri javob: **TNT**
Bitta moslik yetarli emas.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. D) To‘g‘ri javob: **Past harorat fermentni vaqtincha sekinlashtiradi, tuzilishini buzmaydi.**
Sovuq — qaytar.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. B) To‘g‘ri javob: **Mavjud hujayralarning bo‘linishi natijasida**
Hujayra — hujayradan.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. D) To‘g‘ri javob: **1, 2, 3**
Zamburug‘ — geterotrof; prokariotda mitoxondriya yo‘q.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. C) To‘g‘ri javob: **a – 15 mol; b – 10 mol**
a = 1500·40 % : 40 = 15 mol; b = 400 : 40 = 10 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. D) To‘g‘ri javob: **196 mol**
2·8 + 36·5 = 196 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. C) To‘g‘ri javob: **720 g**
Hosil bo‘lgan: 30 : 6 = 5 mol; sarflangan: 6 : 6 = 1 mol; qolgan: 4 · 180 = 720 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. D) To‘g‘ri javob: **4 → 3 → 1 → 2**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. A) To‘g‘ri javob: **TTN**
45 = 15×3.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. D) To‘g‘ri javob: **Ta’sir qon orqali tarqalib, sekinroq va uzoqroq davom etadi.**
Gumoral — qon orqali.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. D) To‘g‘ri javob: **a–2; b–3; c–1**
Parazit — qo‘ng‘irbosh.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. D) To‘g‘ri javob: **Achitqi va gidra**
Kurtaklanish.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. A) To‘g‘ri javob: **20 ta spermatozoid; har birida 10 ta xromosoma.**
Har bir spermatotsitdan 4 ta spermatozoid hosil bo‘ladi; gametada n = 10.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. B) To‘g‘ri javob: **a–3; b–1; c–2**
Itbaliq — lichinka.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. D) To‘g‘ri javob: **TNT**
22 < 30.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 3-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Tadqiqotchi o‘simliklarni turli yorug‘lik sharoitida o‘stirib, boshqa sharoitlarni bir xil saqlab, o‘sishini o‘lchadi. Qaysi metod? 2 ball

A) Eksperimental metod **B)** Taqqoslash metodi

C) Tarixiy metod **D)** Kuzatish metodi

2. B Yer sayyorasidagi uglerodning global aylanishi o‘rganildi. Qaysi daraja? 1 ball

A) Hujayra darajasi **B)** Organizm darajasi

C) Biogeosenoz darajasi **D)** Biosfera darajasi

3. Q Suvga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Suv molekulalari o‘zaro vodorod bog‘lar hosil qiladi.
2. Muz suvdan zichroq bo‘lgani uchun cho‘kadi.
3. Suv hujayradagi ko‘plab reaksiyalar uchun muhit hisoblanadi.
4. Kondensatsiya reaksiyasida suv ajraladi.
5. Suv qutbsiz erituvchi.

A) 2, 4, 5 **B)** 1, 3, 4 **C)** 1, 2, 3 **D)** 1, 2, 5

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	10	30
Y	25	10

A) X – 1360,6 kJ; Y – 846,6 kJ **B)** X – 829 kJ; Y – 1343 kJ

C) X – 1304,1 kJ; Y – 790,1 kJ **D)** X – 1343 kJ; Y – 829 kJ

5. Q Organizmda X oqsillar yetishmagani uchun infeksiyaga qarshi kurashish susaydi. X va funksiyasi? 2 ball

A) Miozin : harakat
B) Kollagen : tayanch
C) Antitanachalar : himoya
D) Gemoglobin : transport

6. M 3 ball

Namuna	Fragmentlar, bp
D	200, 450, 650
G1	200, 450, 700
G2	250, 450, 650
G3	200, 400, 650

Mulohazalarni baholang (T/N).
1. Gumondorlardan hech biri D bilan to‘liq mos emas.
2. G1 iz qoldirgan, chunki ikki fragment mos.
3. DNK manfiy zaryadli bo‘lgani uchun gelda musbat qutb tomon siljiydi.

A) NNT **B)** TNN **C)** TTT **D)** TNT

7. Q Pepsin fermenti pH 2 da faol. Uni pH 8 eritmaga o‘tkazgandan so‘ng faolligi yo‘qoldi va pH 2 ga qaytarilganda ham tiklanmadi. To‘g‘ri xulosa? 2 ball

A) pH fermentga ta’sir qilmaydi.
B) Ferment renaturatsiyalangan.
C) pH o‘zgarishi qaytmas denaturatsiyaga olib kelgan.
D) Ferment substratga aylangan.

8. B Hujayra nazariyasiga mos fikrni aniqlang. 1 ball

A) Hujayralar faqat hayvonlarda bo‘ladi.
B) Hujayralar tarkibi organizmlarda butunlay farqli.
C) Hujayra — tiriklikning tuzilish va funksional birligi.
D) Organizmning xossalari hujayralarga bog‘liq emas.

9. 2 ball

Belgi	X	Y	Z
Yadro	bor	bor	yo‘q
Hujayra devori	sellyuloza	yo‘q	murein
Vakuola	yirik	mayda	yo‘q

To‘g‘ri fikrlarni aniqlang.
1. X o‘simlik hujayrasi bo‘lishi mumkin.
2. Y da lizosomalar bo‘lishi mumkin.
3. Z da membranali organoidlar bor.
4. Z da halqasimon DNK bor.
5. X da ribosomalar yo‘q.

A) 1, 2, 5 **B)** 1, 2, 4 **C)** 1, 2, 3 **D)** 1, 3, 4

10. 2 ball

Energetik almashinuvda hosil bo‘lgan 2000 kJ energiyaning 40 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 480 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang.

A) a – 20 mol; b – 12 mol **B)** a – 20 mol; b – 6 mol
C) a – 50 mol; b – 12 mol **D)** a – 26 mol; b – 12 mol

11. 2 ball

Hujayrada 4 mol glyukoza glikolizga uchradi, shundan 2 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang.

A) 8 mol **B)** 80 mol **C)** 152 mol **D)** 72 mol

12. 2 ball

O‘simlik fotosintezda 18 mol CO2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 12 mol O2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol).

A) 360 g **B)** 180 g **C)** 900 g **D)** 540 g

13. Q 1. Yadro qobig‘i yo‘qolgan, bo‘linish duki shakllanmoqda. 2 ball
2. Har bir qutbda 4 tadan xromosoma to‘plangan, yangi yadrolar hosil bo‘lmoqda.
3. Xromatidalar qutblarga tortilmoqda.
4. Xromosomalar ekvatorda bir qator.
To‘g‘ri ketma-ketlik?

A) 1 → 3 → 4 → 2 **B)** 1 → 4 → 3 → 2 **C)** 1 → 4 → 2 → 3 **D)** 4 → 1 → 3 → 2

14. M 3 ball

Etanol, %	0	10	20	40
Pigment	2	5	18	50

Mulohazalarni baholang (T/N).
1. Etanol konsentratsiyasi oshishi membrana lipidlarini eritib, o‘tkazuvchanlikni oshirgan.
2. Bu tajribada erkli o‘zgaruvchi — harorat.
3. Bo‘laklar o‘lchami bir xil bo‘lishi natijani ishonchli qiladi.

A) TTT **B)** TNN **C)** TNT **D)** NNT

15. B Faqat nerv boshqaruviga xos xususiyatni aniqlang. 1 ball

A) Gormonlar qonga chiqariladi. **B)** Butun organizmga bir vaqtda tarqaladi.
C) Javob juda tez va aniq manzilga yetadi. **D)** Ta’sir uzoq vaqt saqlanadi.

16. Q a) xemosintez; b) fotosintez; c) tayyor organik moddani yutadi. 2 ball
Misollar: 1. Temirbakteriya; 2. Amyoba; 3. Yashil o‘simlik.
Moslashtiring.

A) a–1; b–2; c–3 **B)** a–1; b–3; c–2 **C)** a–3; b–1; c–2 **D)** a–2; b–3; c–1

17. B Qaysi juftlikdagi organizmlar ikkiga bo‘linish orqali ko‘payadi? 1 ball

A) Spirogira va planariya **B)** Amyoba va infuzoriya tufelka
C) Qulupnay va kartoshka **D)** Achitqi va gidra

18. Q Hayvon hujayrasida 2n = 14. 4 ta birlamchi oosit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan tuxum hujayralar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 4 ta tuxum hujayra; har birida 7 ta xromosoma.

B) 4 ta tuxum hujayra; har birida 14 ta xromosoma.

C) 16 ta tuxum hujayra; har birida 14 ta xromosoma.

D) 16 ta tuxum hujayra; har birida 7 ta xromosoma.

19. Q a) kapalak (qurt bosqichi bilan); b) tovuq; c) it. 2 ball
Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.
Moslashtiring.

A) a–3; b–1; c–2 **B)** a–2; b–1; c–3 **C)** a–1; b–3; c–2 **D)** a–3; b–2; c–1

20. M 3 ball

Kun	3	6	9
Yorug‘da	5	15	25
Qorong‘ida	5	14	25

Mulohazalarni baholang (T/N).
1. Bu o‘simlik urug‘larining unishiga yorug‘lik deyarli ta’sir qilmagan.
2. Qorong‘ida urug‘lar umuman unmagan.
3. Bu tajribada erkli o‘zgaruvchi — yorug‘lik.

A) TNT **B)** TTT **C)** TNN **D)** NNT

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 3-variant

1 Q A	2 B D	3 Q B	4 Q D	5 Q C	6 M D	7 Q C
8 B C	9 Q B	10 Q A	11 Q B	12 Q B	13 Q B	14 M C
15 B C	16 Q B	17 B B	18 Q A	19 Q A	20 M A	

1. A) To‘g‘ri javob: **Eksperimental metod**
Sharoitni o‘zgartirish — eksperiment.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. D) To‘g‘ri javob: **Biosfera darajasi**
Global — biosfera.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. B) To‘g‘ri javob: **1, 3, 4**
Muz yengil; suv qutbli.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. D) To‘g‘ri javob: **X – 1343 kJ; Y – 829 kJ**
X: 10·17,6 + 30·38,9 = 1343 kJ; Y: 25·17,6 + 10·38,9 = 829 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. C) To‘g‘ri javob: **Antitanachalar : himoya**
Immunoglobulinlar.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. D) To‘g‘ri javob: **TNT**
Fosfat guruhlari — manfiy.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. C) To‘g‘ri javob: **pH o‘zgarishi qaytmas denaturatsiyaga olib kelgan.**
Qaytmas.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. C) To‘g‘ri javob: **Hujayra — tiriklikning tuzilish va funksional birligi.**
Birlik.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. B) To‘g‘ri javob: **1, 2, 4**
Prokariot — membranali organoid yo‘q.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. A) To‘g‘ri javob: **a – 20 mol; b – 12 mol**
a = 2000·40 % : 40 = 20 mol; b = 480 : 40 = 12 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. B) To‘g‘ri javob: **80 mol**
2·4 + 36·2 = 80 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. B) To‘g‘ri javob: **180 g**
Hosil bo‘lgan: 18 : 6 = 3 mol; sarflangan: 12 : 6 = 2 mol; qolgan: 1 · 180 = 180 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. B) To‘g‘ri javob: **1 → 4 → 3 → 2**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. C) To‘g‘ri javob: **TNT**
Erkli o‘zgaruvchi — etanol.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. C) To‘g‘ri javob: **Javob juda tez va aniq manzilga yetadi.**
Nerv — tez, aniq.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. B) To‘g‘ri javob: **a–1; b–3; c–2**
Temirbakteriya — xemosintez.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. B) To‘g‘ri javob: **Amyoba va infuzoriya tufelka**
Oddiy bo‘linish.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. A) To‘g‘ri javob: **4 ta tuxum hujayra; har birida 7 ta xromosoma.**
Har bir oositdan 1 ta tuxum hujayra va qutb tanachalari hosil bo‘ladi; gametada n = 7.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. A) To‘g‘ri javob: **a–3; b–1; c–2**
Kapalak — lichinkali.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. A) To‘g‘ri javob: **TNT**
Unish deyarli bir xil.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 4-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Olim kompyuterda populyatsiya sonining o‘zgarishini matematik tenglamalar orqali bashorat qildi. Qaysi metod? 2 ball

A) Eksperimental metod **B)** Kuzatish metodi

C) Modellashtirish metodi **D)** Taqqoslash metodi

2. B Gemoglobin oqsilining tuzilishi va kislorod biriktirishi o‘rganildi. Qaysi daraja? 1 ball

A) Populyatsiya darajasi **B)** Molekula darajasi

C) Hujayra darajasi **D)** Organizm darajasi

3. Q Hujayradagi anorganik moddalarga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Fosfat ionlari ATF va nuklein kislotalar tarkibiga kiradi.
2. Kaliy va natriy ionlari membrana potensialini hosil qilishda qatnashadi.
3. Glyukoza anorganik modda hisoblanadi.
4. Suv hujayra massasining eng katta qismini tashkil qiladi.
5. Kalsiy fosfat oqsillarning asosiy monomeri.

A) 1, 4, 5 **B)** 2, 4, 5 **C)** 3, 4, 5 **D)** 1, 2, 4

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	40	10
Y	20	20

A) X – 1110,6 kJ; Y – 1147,6 kJ **B)** X – 1093 kJ; Y – 1130 kJ

C) X – 1054,1 kJ; Y – 1091,1 kJ **D)** X – 1130 kJ; Y – 1093 kJ

5. Q Qondagi glyukoza miqdori doimiy yuqori, X oqsil-gormon yetishmaydi. X va funksiyasi? 2 ball

A) Albumin : transport
B) Fibrinogen : himoya
C) Keratin : tayanch
D) Insulin : boshqaruv (gormonal)

6. M 3 ball

Namuna	Fragmentlar, bp
D	150, 350, 600
G1	150, 350, 600
G2	150, 350, 600
G3	100, 350, 600

Mulohazalarni baholang (T/N).
1. G1 va G2 ning ikkalasi ham D ga mos, shu sohalar bo‘yicha ularni ajratib bo‘lmaydi.
2. G3 aniq iz qoldirgan.
3. Aniqroq xulosa uchun ko‘proq DNK sohalarini tahlil qilish kerak.

A) TNT **B)** TNN **C)** TTT **D)** NNT

7. Q Oqsil kuchsiz denaturatsiyalovchi modda ta’siridan so‘ng modda yuvib tashlangach, faolligi tiklandi. Bu qanday ataladi? 2 ball

A) Renaturatsiya **B)** Denaturatsiya **C)** Kondensatsiya **D)** Gidroliz

8. B Hujayra nazariyasining qaysi qoidasi organizmlarning kelib chiqishi umumiyligini ko‘rsatadi? 1 ball

A) Hujayralar shakli turlicha.
B) Hujayralar yashash davomiyligi turlicha.
C) Hujayralar o‘lchami turlicha.
D) Barcha hujayralarning tuzilishi va kimyoviy tarkibi o‘xshash.

9. 2 ball

Belgi	X	Y	Z
Yadro	yo‘q	bor	bor
Xloroplast	yo‘q	yo‘q	bor
Zaxira modda	–	glikogen	kraxmal

To‘g‘ri fikrlarni aniqlang.
1. Y hayvon yoki zamburug‘ hujayrasi bo‘lishi mumkin.
2. Z avtotrof oziqlanadi.
3. X eukariot hujayra.
4. Uchala hujayrada ribosoma bor.
5. Z da hujayra devori bo‘lmaydi.

A) 1, 3, 5 **B)** 1, 2, 4 **C)** 1, 3, 4 **D)** 1, 4, 5

10. 2 ball

Energetik almashinuvda hosil bo‘lgan 1200 kJ energiyaning 50 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 360 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang.

A) a – 15 mol; b – 9 mol **B)** a – 21 mol; b – 9 mol
C) a – 30 mol; b – 9 mol **D)** a – 15 mol; b – 4,5 mol

11. 2 ball

Hujayrada 7 mol glyukoza glikolizga uchradi, shundan 3 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang.

A) 108 mol **B)** 14 mol **C)** 266 mol **D)** 122 mol

12. 2 ball

O‘simlik fotosintezda 30 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 12 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol).

A) 360 g **B)** 1260 g **C)** 900 g **D)** 540 g

13. Q 1. Hujayra markazida xromosomalar metafaza plastinkasini hosil qilgan. 2 ball
2. Despirallashish va ikki qiz hujayra hosil bo‘lishi.
3. Sentromeralar bo‘linib, xromatidalar ajralmoqda.
4. Yadrocha yo‘qolib, xromosomalar quyuqlashmoqda.
To‘g‘ri ketma-ketlik?

A) 1 → 4 → 3 → 2 **B)** 4 → 1 → 3 → 2 **C)** 4 → 3 → 1 → 2 **D)** 4 → 1 → 2 → 3

14. M 3 ball

Harorat, °C	20	40	60	80
Pigment	3	5	28	55

Mulohazalarni baholang (T/N).
1. 40 °C dan 60 °C gacha pigment chiqishi keskin ortgan.
2. Pigment hujayra vakuolasida joylashgan, u membranalar buzilganda chiqadi.
3. Natija haroratning membranaga ta’sir qilmasligini ko‘rsatadi.

A) TTN **B)** NTN **C)** TNN **D)** TTT

15. B Nerv va gumoral boshqaruvning o‘zaro bog‘liqligini ko‘rsatuvchi tuzilma qaysi? 1 ball

A) Gipotalamus–gipofiz tizimi **B)** Oshqozon
C) Teri **D)** Taloq

16. Q a) yorug‘likda fotosintez, qorong‘ida tayyor organik modda bilan oziqlanadi; b) faqat tayyor organik modda bilan; c) faqat fotosintez. 2 ball
Misollar: 1. Yashil evglena; 2. Infuzoriya tufelka; 3. Xlamidomonada.
Moslashtiring.

A) a–1; b–3; c–2 **B)** a–3; b–2; c–1 **C)** a–2; b–1; c–3 **D)** a–1; b–2; c–3

17. B Qaysi juftlikdagi organizmlar spora orqali ko‘payadi? 1 ball

A) Gidra va achitqi **B)** Planariya va spirogira
C) Amyoba va tufelka **D)** Mog‘or zamburug‘i va qirqbo‘g‘im

18. Q Hayvon hujayrasida 2n = 16. 4 ta birlamchi spermatotsit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan spermatozoidlar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 4 ta spermatozoid; har birida 8 ta xromosoma.

B) 4 ta spermatozoid; har birida 16 ta xromosoma.

C) 16 ta spermatozoid; har birida 8 ta xromosoma.

D) 16 ta spermatozoid; har birida 16 ta xromosoma.

19. Q a) embrion bachadon devoriga birikib, yo‘ldosh hosil qiladi; b) lichinka metamorfoz orqali voyaga yetadi; c) tuxum qattiq po‘choq bilan qoplangan, quruqlikda qo‘yiladi. 2 ball
Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.
Moslashtiring.

A) a–1; b–3; c–2 **B)** a–3; b–2; c–1 **C)** a–2; b–3; c–1 **D)** a–2; b–1; c–3

20. M 3 ball

Kun	2	4	6	8
10 °C	0	2	6	10
20 °C	3	12	22	28

Mulohazalarni baholang (T/N).
1. Past haroratda unish sekinroq kechgan.
2. 4-kuni 20 °C da ungan urug‘lar 10 °C dagidan 6 marta ko‘p.
3. 8-kuni 10 °C da ungan urug‘lar 20 °C dagidan ko‘p.

A) TTN **B)** NTN **C)** TNN **D)** TTT

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 4-variant

1 Q C	2 B B	3 Q D	4 Q B	5 Q D	6 M A	7 Q A
8 B D	9 Q B	10 Q A	11 Q D	12 Q D	13 Q B	14 M A
15 B A	16 Q D	17 B D	18 Q C	19 Q C	20 M A	

1. C) To‘g‘ri javob: **Modellashtirish metodi**
Model.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. B) To‘g‘ri javob: **Molekula darajasi**
Molekula.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. D) To‘g‘ri javob: **1, 2, 4**
Glyukoza organik; oqsil monomeri — aminokislota.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. B) To‘g‘ri javob: **X – 1093 kJ; Y – 1130 kJ**
X: 40·17,6 + 10·38,9 = 1093 kJ; Y: 20·17,6 + 20·38,9 = 1130 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. D) To‘g‘ri javob: **Insulin : boshqaruv (gormonal)**
Insulin.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. A) To‘g‘ri javob: **TNT**
G3 mos emas.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. A) To‘g‘ri javob: **Renaturatsiya**
Tiklanish.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. D) To‘g‘ri javob: **Barcha hujayralarning tuzilishi va kimyoviy tarkibi o‘xshash.**
O‘xshashlik — umumiy kelib chiqish.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. B) To‘g‘ri javob: **1, 2, 4**
X — prokariot; Z o‘simlik — devori bor.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. A) To‘g‘ri javob: **a – 15 mol; b – 9 mol**
a = 1200·50 % : 40 = 15 mol; b = 360 : 40 = 9 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. D) To‘g‘ri javob: **122 mol**
2·7 + 36·3 = 122 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. D) To‘g‘ri javob: **540 g**
Hosil bo‘lgan: 30 : 6 = 5 mol; sarflangan: 12 : 6 = 2 mol; qolgan: 3 · 180 = 540 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. B) To‘g‘ri javob: **4 → 1 → 3 → 2**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. A) To‘g‘ri javob: **TTN**
Ta’sir qiladi.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. A) To‘g‘ri javob: **Gipotalamus–gipofiz tizimi**
Neyroendokrin bog‘lanish.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. D) To‘g‘ri javob: **a–1; b–2; c–3**
Evglena — miksotrof.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. D) To‘g‘ri javob: **Mog‘or zamburug‘i va qirqbo‘g‘im**
Sporalar.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. C) To‘g‘ri javob: **16 ta spermatozoid; har birida 8 ta xromosoma.**
Har bir spermatotsitdan 4 ta spermatozoid hosil bo‘ladi; gametada n = 8.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. C) To‘g‘ri javob: **a–2; b–3; c–1**
Moslash.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. A) To‘g‘ri javob: **TTN**
10 < 28.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 5-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Paleontolog qatlamlardagi qazilma qoldiqlar asosida otlarning evolyutsion o‘zgarishini tikladi. Qaysi metod? 2 ball

A) Kuzatish metodi **B)** Eksperimental metod

C) Modellashtirish metodi **D)** Tarixiy metod

2. B Mitoxondriya va xloroplastlarning ishi o‘rganildi. Qaysi daraja? 1 ball

A) Hujayra darajasi **B)** Biosfera darajasi

C) Molekula darajasi **D)** Organizm darajasi

3. Q Suvning xossalariga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Kapillyarlik suvning o‘simlik tomirlari bo‘ylab ko‘tarilishiga yordam beradi.
2. Suv yaxshi issiqlik o‘tkazadi va organizmda issiqlikni taqsimlaydi.
3. Suv molekulasi elektroneytral emas, zaryadlangan ion hisoblanadi.
4. Gidroliz reaksiyalarida suv sarflanadi.
5. Suv havzalari pastdan yuqoriga qarab muzlaydi.

A) 1, 2, 4 **B)** 1, 2, 5 **C)** 1, 2, 3 **D)** 1, 3, 4

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	15	30
Y	35	10

A) X – 1431 kJ; Y – 1005 kJ **B)** X – 1448,6 kJ; Y – 1022,6 kJ

C) X – 1392,1 kJ; Y – 966,1 kJ **D)** X – 1005 kJ; Y – 1431 kJ

5. Q Muskul tolalarida X oqsillar shikastlangani uchun muskul qisqarishi buzildi. X va funksiyasi? 2 ball

A) Insulin : gormonal
B) Gemoglobin : transport
C) Aktin va miozin : harakat (qisqarish)
D) Fibrinogen : himoya

6. M 3 ball

Namuna	Fragmentlar, bp
D	400, 500, 900
G1	400, 900
G2	400, 500, 900
G3	500, 900, 1000

Mulohazalarni baholang (T/N).
1. G2 iz qoldirgan bo‘lishi mumkin.
2. G1 ham mos, chunki uning barcha fragmentlari D da bor.
3. 900 bp fragment 400 bp fragmentga qaraganda startga yaqinroq joylashadi.

A) TNN **B)** NNT **C)** TTT **D)** TNT

7. Q 2 ball

Namuna	Saqlash, faollik %	37 °C da, %
1 (15 °C)	30	100
2 (37 °C)	100	100
3 (60 °C)	10	40

To‘g‘ri xulosa?

A) 3-namunada denaturatsiya qisman qaytar bo‘lgan.
B) 3-namunada ferment to‘liq tiklangan.
C) Optimal harorat 15 °C.
D) 1-namunada ferment buzilgan.

8. B Ko‘p hujayrali organizmga oid hujayra nazariyasi qoidasini aniqlang. 1 ball

A) Ko‘p hujayrali organizm hujayralari ixtisoslashib, to‘qima va organlarni hosil qiladi.
B) Ko‘p hujayralilar hujayrasiz to‘qimalardan iborat.
C) Ko‘p hujayralilarda barcha hujayralar bir xil vazifani bajaradi.
D) Ko‘p hujayrali organizm hujayralari bir-biridan mustaqil yashaydi.

9. 2 ball

Belgi	X	Y	Z
Yadro	bor	bor	yo‘q
Mitoxondriya	bor	bor	yo‘q
Hujayra devori	yo‘q	sellyuloza	murein

To‘g‘ri fikrlarni aniqlang.
1. X hayvon hujayrasi bo‘lishi mumkin.
2. Y da yirik markaziy vakuola bo‘lishi mumkin.
3. Z da ribosomalar yo‘q.
4. Z prokariot.
5. X da sellyulozali devor bor.

A) 1, 2, 3 **B)** 1, 2, 4 **C)** 2, 3, 4 **D)** 1, 3, 5

10. 2 ball

Energetik almashinuvda hosil bo‘lgan 1600 kJ energiyaning 50 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 400 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang.

A) a – 20 mol; b – 10 mol **B)** a – 20 mol; b – 5 mol
C) a – 26 mol; b – 10 mol **D)** a – 40 mol; b – 10 mol

11. 2 ball

Hujayrada 10 mol glyukoza glikolizga uchradi, shundan 6 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang.

A) 236 mol **B)** 380 mol **C)** 216 mol **D)** 20 mol

12. 2 ball

O‘simlik fotosintezda 36 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 12 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol).

A) 1080 g **B)** 720 g **C)** 360 g **D)** 1440 g

13. Q 1. Xromosomalar qutblarga yetib, despirallashmoqda. 2 ball
2. Xromosomalar spirallashmoqda.
3. Xromatidalar ajralmoqda.
4. Xromosomalar ekvatorda.
To‘g‘ri ketma-ketlik?

A) 2 → 3 → 4 → 1 **B)** 4 → 2 → 3 → 1 **C)** 2 → 4 → 1 → 3 **D)** 2 → 4 → 3 → 1

14. M Bir guruh bo‘laklar yuvilmasdan, ikkinchi guruh yuvilib 30 °C suvga solindi. 3 ball
Yuvilmaganlarida pigment ko‘p bo‘ldi. Mulohazalarni baholang (T/N).
1. Yuvilmagan bo‘laklarda kesilgan hujayralardan chiqqan pigment natijani oshirgan.
2. Yuvish membranalarni buzgan.
3. Bu natija tajribada bo‘laklarni yuvish zarurligini ko‘rsatadi.

A) TTT **B)** NNT **C)** TNN **D)** TNT

15. B Qaysi misol ikkala boshqaruvning birgalikdagi ishini ko‘rsatadi? 1 ball

A) Xavf paytida nerv signali buyrakusti bezidan adrenalin ajralishini kuchaytiradi.

B) Ko‘z qisish refleksi.

C) Qo‘lni issiqdan tortib olish.

D) Tizza refleksi.

16. Q a) oltingugurt birikmalarini oksidlaydi; b) chirigan qoldiqlarni parchalaydi; c) yashil pigment yordamida fotosintez qiladi. 2 ball
Misollar: 1. Chirituvchi bakteriya; 2. Oltingugurtbakteriya; 3. Spirogira.
Moslashtiring.

A) a–2; b–3; c–1 **B)** a–1; b–2; c–3 **C)** a–2; b–1; c–3 **D)** a–3; b–1; c–2

17. B Qaysi juftlikdagi o‘simliklar vegetativ (ildizpoya) bilan ko‘payadi? 1 ball

A) Mog‘or va paporotnik **B)** Achitqi va gidra

C) Ajriq va qamish **D)** Amyoba va xlorella

18. Q Hayvon hujayrasida 2n = 10. 6 ta birlamchi oosit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan tuxum hujayralar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 6 ta tuxum hujayra; har birida 5 ta xromosoma.

B) 6 ta tuxum hujayra; har birida 10 ta xromosoma.

C) 24 ta tuxum hujayra; har birida 5 ta xromosoma.

D) 24 ta tuxum hujayra; har birida 10 ta xromosoma.

19. Q a) ilon; b) chivin; c) ot. 2 ball
Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.
Moslashtiring.

A) a–1; b–2; c–3 **B)** a–3; b–1; c–2 **C)** a–2; b–3; c–1 **D)** a–1; b–3; c–2

20. M Urug‘lar turli namlikda undirildi (30 tadan, 25 °C): 3 ball

Kun	3	6	9
Quruq	0	0	0
Nam	8	20	28
Suvga botirilgan	2	5	7

Mulohazalarni baholang (T/N).
1. Unish uchun suv zarur.
2. Suvga to‘liq botirilgan urug‘larning kam unishi kislorod yetishmasligi bilan izohlanishi mumkin.
3. Quruq urug‘lar tirik emasligi isbotlangan.

A) TNN **B)** TTT **C)** TTN **D)** NTN

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 5-variant

1 Q D	2 B A	3 Q A	4 Q A	5 Q C	6 M D	7 Q A
8 B A	9 Q B	10 Q A	11 Q A	12 Q B	13 Q D	14 M D
15 B A	16 Q C	17 B C	18 Q A	19 Q D	20 M C	

1. D) To‘g‘ri javob: **Tarixiy metod**
O‘tmish rivojlanishini tiklash — tarixiy.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. A) To‘g‘ri javob: **Hujayra darajasi**
Organoidlar — hujayra.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. A) To‘g‘ri javob: **1, 2, 4**
Suv qutbli, ion emas; yuzadan muzlaydi.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. A) To‘g‘ri javob: **X – 1431 kJ; Y – 1005 kJ**
X: 15·17,6 + 30·38,9 = 1431 kJ; Y: 35·17,6 + 10·38,9 = 1005 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. C) To‘g‘ri javob: **Aktin va miozin : harakat (qisqarish)**
Qisqaruvchi oqsillar.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. D) To‘g‘ri javob: **TNT**
G1 da 500 yo‘q.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. A) To‘g‘ri javob: **3-namunada denaturatsiya qisman qaytar bo‘lgan.**
40% — qisman tiklanish.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. A) To‘g‘ri javob: **Ko‘p hujayrali organizm hujayralari ixtisoslashib, to‘qima va organlarni hosil qiladi.**
Ixtisoslashuv.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. B) To‘g‘ri javob: **1, 2, 4**
Prokariotda ham ribosoma bor.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. A) To‘g‘ri javob: **a – 20 mol; b – 10 mol**
a = 1600·50 % : 40 = 20 mol; b = 400 : 40 = 10 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. A) To‘g‘ri javob: **236 mol**
2·10 + 36·6 = 236 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. B) To‘g‘ri javob: **720 g**
Hosil bo‘lgan: 36 : 6 = 6 mol; sarflangan: 12 : 6 = 2 mol; qolgan: 4 · 180 = 720 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. D) To‘g‘ri javob: **2 → 4 → 3 → 1**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. D) To‘g‘ri javob: **TNT**
Yuvish faqat sirtdagi pigmentni ketkazadi.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. A) To‘g‘ri javob: **Xavf paytida nerv signali buyrakusti bezidan adrenalin ajralishini kuchaytiradi.**
Stress javobi.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. C) To‘g‘ri javob: **a–2; b–1; c–3**
Xemotrof, saprotrof, fototrof.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. C) To‘g‘ri javob: **Ajriq va qamish**
Ildizpoya.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. A) To‘g‘ri javob: **6 ta tuxum hujayra; har birida 5 ta xromosoma.**
Har bir oositdan 1 ta tuxum hujayra va qutb tanachalari hosil bo‘ladi; gametada n = 5.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. D) To‘g‘ri javob: **a–1; b–3; c–2**
Chivin — lichinka.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. C) To‘g‘ri javob: **TTN**
Quruq urug‘ tinim holatida.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 6-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Tadqiqotchi sut emizuvchilar va qushlar yurak tuzilishini solishtirib xulosa chiqardi. Qaysi metod? 2 ball

A) Modellashtirish metodi **B)** Taqqoslash metodi

C) Eksperimental metod **D)** Tarixiy metod

2. B Bitta quyonning ovqat hazm qilish, nafas olish va harakat tizimlari o‘zaro bog‘liqlikda o‘rganildi. Qaysi daraja? 1 ball

A) Populyatsiya–tur darajasi **B)** Organizm darajasi

C) Biogeosenoz darajasi **D)** Hujayra darajasi

3. Q Mineral elementlar yetishmovchiligiga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Temir yetishmasa kamqonlik kuzatilishi mumkin.
2. Yod yetishmasa qalqonsimon bez kattalashishi mumkin.
3. Kalsiy yetishmasa suyaklar mo‘rtlashishi mumkin.
4. Magniy yetishmasa o‘simlik barglari to‘q yashil bo‘ladi.
5. Azot yetishmasa o‘simlik tez o‘sadi.

A) 1, 3, 4 **B)** 2, 4, 5 **C)** 1, 2, 3 **D)** 2, 3, 4

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	25	20
Y	45	10

A) X – 1181 kJ; Y – 1218 kJ **B)** X – 1235,6 kJ; Y – 1198,6 kJ

C) X – 1179,1 kJ; Y – 1142,1 kJ **D)** X – 1218 kJ; Y – 1181 kJ

5. Q Teri va paylarda X oqsil yetishmagani uchun ularning mustahkamligi pasaydi. X va funksiyasi? 2 ball

A) Kollagen : tuzilish (tayanch)
B) Amilaza : fermentativ
C) Gemoglobin : transport
D) Insulin : gormonal

6. M 3 ball

Namuna	Fragmentlar, bp
D	250, 600, 750
G1	300, 600, 750
G2	250, 650, 750
G3	250, 600, 750

Mulohazalarni baholang (T/N).
1. G3 iz qoldirgan bo‘lishi mumkin.
2. G1 va G2 da ikki fragment mos, shuning uchun ular ham iz qoldirgan bo‘lishi mumkin.
3. PZR DNK ning kerakli sohasini ko‘paytirish uchun ishlatiladi.

A) TNN **B)** TNT **C)** TTT **D)** NNT

7. Q Tuxum oqi qaynatilgandan keyin sovitilganda qayta suyuq holatga qaytmaydi. Sababi nima? 2 ball

A) Oqsil glyukozaga aylangan.
B) Oqsil renaturatsiyalangan.
C) Oqsil qaytmas denaturatsiyaga uchragan.
D) Oqsil gidrolizlangan.

8. B Hujayra nazariyasining irsiyatga oid qoidasiga mos fikrni aniqlang. 1 ball

A) Hujayralarda modda almashinuvi bo‘lmaydi.
B) Hujayralar irsiy axborotga ega emas.
C) Hujayrada irsiy axborot saqlanadi va avlodga beriladi.
D) Hujayralar faqat yadrodan iborat.

9. Q 2 ball

Belgi	X	Y	Z
Yadro	yo‘q	bor	bor
Xivchin	bor	bor	yo‘q
Xloroplast	yo‘q	bor	yo‘q

To‘g‘ri fikrlarni aniqlang.
1. Y xlamidomonada kabi bir hujayrali suvo‘t bo‘lishi mumkin.
2. X bakteriya bo‘lishi mumkin.
3. Z ning harakatlanishi xivchin orqali.
4. Y fotosintez qila oladi.
5. X da yadro qobig‘i bor.

A) 1, 3, 4 **B)** 1, 2, 4 **C)** 1, 4, 5 **D)** 1, 2, 3

10. Q Energetik almashinuvda hosil bo‘lgan 800 kJ energiyaning 60 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 240 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang. 2 ball

A) a – 20 mol; b – 6 mol **B)** a – 12 mol; b – 6 mol
C) a – 12 mol; b – 3 mol **D)** a – 18 mol; b – 6 mol

11. Q Hujayrada 9 mol glyukoza glikolizga uchradi, shundan 4 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang. 2 ball

A) 18 mol **B)** 162 mol **C)** 144 mol **D)** 342 mol

12. Q O‘simlik fotosintezda 42 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 12 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol). 2 ball

A) 900 g **B)** 1260 g **C)** 1620 g **D)** 360 g

13. Q 1. Xromatidalar qarama-qarshi qutblarga yo‘nalgan. 2 ball
2. Xromosomalar ekvatorda, dukka birikkan.
3. Ikki qiz yadro shakllangan.
4. Yadro qobig‘i erib, sentriolalar qutblarga ajralmoqda.
To‘g‘ri ketma-ketlik?

A) 4 → 2 → 1 → 3 **B)** 2 → 4 → 1 → 3 **C)** 4 → 1 → 2 → 3 **D)** 4 → 2 → 3 → 1

14. M 3 ball

Harorat, °C	0	20	40	60
Pigment	6	2	4	25

Mulohazalarni baholang (T/N).
1. 0 °C da pigment ko‘payishi muzlash membranalarni shikastlagani bilan izohlanishi mumkin.
2. 60 °C da pigment chiqishi eng kam.
3. 20–40 °C da membranalar nisbatan butun saqlangan.

A) TNN **B)** NNT **C)** TTT **D)** TNT

15. B Nerv va gumoral boshqaruvga xos umumiy xususiyat qaysi? 1 ball

A) Ikkalasi ham faqat qon orqali ishlaydi.

B) Ikkalasi ham faqat nerv tolalari orqali ishlaydi.

C) Ikkalasi ham faqat o‘simliklarda bo‘ladi.

D) Ikkalasi ham axborotni signal orqali uzatib, nishon organlarning javobini keltirib chiqaradi.

16. Q a) avtotrof; b) geterotrof parazit; c) geterotrof saprotrof. 2 ball
Misollar: 1. Tasmasimon chuvalchang; 2. Qayin; 3. Yomg‘ir chuvalchangi.
Moslashtiring.

A) a–2; b–3; c–1 **B)** a–3; b–1; c–2 **C)** a–2; b–1; c–3 **D)** a–1; b–2; c–3

17. B Qaysi organizm bo‘laklarga ajralganda har bir bo‘lakdan butun organizm tiklanadi (fragmentatsiya)? 1 ball

A) Amyoba **B)** Gidra (kurtaklanish)

C) Achitqi **D)** Dengiz yulduzi

18. Q Hayvon hujayrasida 2n = 24. 6 ta birlamchi spermatotsit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan spermatozoidlar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 6 ta spermatozoid; har birida 24 ta xromosoma.

B) 24 ta spermatozoid; har birida 12 ta xromosoma.

C) 24 ta spermatozoid; har birida 24 ta xromosoma.

D) 6 ta spermatozoid; har birida 12 ta xromosoma.

19. Q a) ninachi; b) kaltakesak; c) sigir. 2 ball
Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.
Moslashtiring.

A) a–1; b–3; c–2 **B)** a–3; b–2; c–1 **C)** a–2; b–1; c–3 **D)** a–3; b–1; c–2

20. M 3 ball

Kun	2	4	6	8
25 °C	4	14	24	29
40 °C	2	6	9	10

Mulohazalarni baholang (T/N).
1. Juda yuqori harorat unishni susaytirgan.
2. 8-kuni 25 °C da ungan urug‘lar 40 °C dagidan taxminan 3 marta ko‘p.
3. 40 °C bu o‘simlik uchun optimal harorat.

A) NTN **B)** TTN **C)** TTT **D)** TNN

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 6-variant

1 Q B	2 B B	3 Q C	4 Q D	5 Q A	6 M B	7 Q C
8 B C	9 Q B	10 Q B	11 Q B	12 Q A	13 Q A	14 M D
15 B D	16 Q C	17 B D	18 Q B	19 Q D	20 M B	

1. B) To‘g‘ri javob: **Taqqoslash metodi**
Taqqoslash.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. B) To‘g‘ri javob: **Organizm darajasi**
Bitta individ.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. C) To‘g‘ri javob: **1, 2, 3**
Mg va N yetishmasa barglar sarg‘ayadi, o‘sish sekinlashadi.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. D) To‘g‘ri javob: **X – 1218 kJ; Y – 1181 kJ**
X: 25·17,6 + 20·38,9 = 1218 kJ; Y: 45·17,6 + 10·38,9 = 1181 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. A) To‘g‘ri javob: **Kollagen : tuzilish (tayanch)**
Kollagen.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. B) To‘g‘ri javob: **TNT**
To‘liq moslik zarur.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. C) To‘g‘ri javob: **Oqsil qaytmas denaturatsiyaga uchragan.**
Qaytmas.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. C) To‘g‘ri javob: **Hujayrada irsiy axborot saqlanadi va avlodga beriladi.**
Irsiy axborot.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. B) To‘g‘ri javob: **1, 2, 4**
Z da xivchin yo‘q; X yadrosiz.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. B) To‘g‘ri javob: **a – 12 mol; b – 6 mol**
a = 800·60 % : 40 = 12 mol; b = 240 : 40 = 6 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. B) To‘g‘ri javob: **162 mol**
2·9 + 36·4 = 162 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. A) To‘g‘ri javob: **900 g**
Hosil bo‘lgan: 42 : 6 = 7 mol; sarflangan: 12 : 6 = 2 mol; qolgan: 5 · 180 = 900 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. A) To‘g‘ri javob: **4 → 2 → 1 → 3**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. D) To‘g‘ri javob: **TNT**
60 ˚C da eng ko‘p.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. D) To‘g‘ri javob: **Ikkalasi ham axborotni signal orqali uzatib, nishon organlarning javobini keltirib chiqaradi.**
Signal–javob.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. C) To‘g‘ri javob: **a–2; b–1; c–3**
Yomg‘ir chuvalchangi — chirindi bilan.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. D) To‘g‘ri javob: **Dengiz yulduzi**
Regeneratsiya bilan fragmentatsiya.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. B) To‘g‘ri javob: **24 ta spermatozoid; har birida 12 ta xromosoma.**
Har bir spermatotsitdan 4 ta spermatozoid hosil bo‘ladi; gametada n = 12.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. D) To‘g‘ri javob: **a–3; b–1; c–2**
Ninachi — suvda lichinka.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. B) To‘g‘ri javob: **TTN**
Optimal emas.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 7-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q O‘quvchi sichqonlarga turli ozuqa berib, boshqa sharoitlarni bir xil saqlab, vazni o‘zgarishini o‘lchadi. Qaysi metod? 2 ball

A) Tarixiy metod
B) Taqqoslash metodi
C) Kuzatish metodi
D) Eksperimental metod

2. B Ikki xil o‘rmondagi qayin daraxtlari guruhlari orasidagi genetik farqlar o‘rganildi. Qaysi daraja? 1 ball

A) Molekula darajasi
B) Organizm darajasi
C) Populyatsiya–tur darajasi
D) Biosfera darajasi

3. Q Hujayradagi suv almashinuviga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Suv hujayraga osmos yo‘li bilan kiradi.
2. Turgor bosimi hujayradagi suv bilan bog‘liq.
3. Suv yog‘larni eritadigan asosiy erituvchi.
4. Suvning yuqori issiqlik sig‘imi suv havzalarida haroratni barqaror saqlaydi.
5. Suv molekulalari orasida peptid bog‘lar bor.

A) 1, 3, 5
B) 2, 4, 5
C) 1, 2, 4
D) 1, 4, 5

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	50	10
Y	5	30

A) X – 1255 kJ; Y – 1269 kJ
B) X – 1286,6 kJ; Y – 1272,6 kJ
C) X – 1230,1 kJ; Y – 1216,1 kJ
D) X – 1269 kJ; Y – 1255 kJ

5. Q So‘lakda X oqsil yo‘qligi sababli og‘izda kraxmal parchalanmadi. X va funksiyasi? 2 ball

A) Amilaza : fermentativ (katalitik)
B) Albumin : transport
C) Kollagen : tayanch
D) Fibrinogen : himoya

6. M 3 ball

Namuna	Fragmentlar, bp
D	180, 420, 760
G1	180, 420, 760
G2	180, 420, 800

Mulohazalarni baholang (T/N).
1. G1 iz qoldirgan bo‘lishi mumkin.
2. G2 ham iz qoldirgan bo‘lishi mumkin, chunki 800 va 760 bir-biriga yaqin.
3. Har bir odamda bu fragmentlar uzunligi irsiy farqlar tufayli turlicha bo‘ladi.

A) TNN
B) NNT
C) TTT
D) TNT

7. Q Denaturatsiyada oqsilning qaysi tuzilmasi odatda saqlanib qoladi? 2 ball

A) Birlamchi struktura (aminokislotalar ketma-ketligi)
B) Ikkilamchi struktura
C) Uchlamchi struktura
D) To‘rtlamchi struktura

8. B Hujayra nazariyasiga zid fikrni aniqlang. 1 ball

A) Barcha hujayralar o‘xshash kimyoviy tarkibga ega.
B) Hujayralar hujayrasiz moddadan paydo bo‘ladi.
C) Hujayralar bo‘linish orqali ko‘payadi.
D) Hujayra — tiriklikning rivojlanish birligi.

9. 2 ball

Belgi	X	Y	Z
DNK joylashuvi	sitoplazma	yadro	yadro
Devor moddasi	murein	xitin	sellyuloza
Fotosintez	bor	yo‘q	bor

To‘g‘ri fikrlarni aniqlang.
1. X sianobakteriya bo‘lishi mumkin.
2. Y zamburug‘ hujayrasi.
3. Z hayvon hujayrasi.
4. X da xloroplastlar bor.
5. Y geterotrof.

A) 1, 2, 5 **B)** 2, 3, 4 **C)** 3, 4, 5 **D)** 1, 3, 5

10. 2 ball

Energetik almashinuvda hosil bo‘lgan 2400 kJ energiyaning 40 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 560 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang.

A) a – 24 mol; b – 14 mol **B)** a – 60 mol; b – 14 mol
C) a – 30 mol; b – 14 mol **D)** a – 24 mol; b – 7 mol

11. 2 ball

Hujayrada 5 mol glyukoza glikolizga uchradi, shundan 2 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang.

A) 82 mol **B)** 72 mol **C)** 10 mol **D)** 190 mol

12. 2 ball

O‘simlik fotosintezda 24 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 6 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol).

A) 540 g **B)** 900 g **C)** 720 g **D)** 180 g

13. Q 1. Xromosomalar spirallashgan, yadro qobig‘i yo‘qolmoqda. 2 ball
2. Qiz xromosomalar qutblarga harakatlanmoqda.
3. Xromosomalar ekvator tekisligida.
4. Sitokinez yakunlanmoqda.
To‘g‘ri ketma-ketlik?

A) 1 → 3 → 4 → 2 **B)** 1 → 2 → 3 → 4 **C)** 1 → 3 → 2 → 4 **D)** 3 → 1 → 2 → 4

14. M 3 ball

Vaqt, daqiqa (70 °C)	1	3	5	10
Pigment	10	25	40	60

Mulohazalarni baholang (T/N).
1. Yuqori haroratda saqlash vaqti oshgan sari pigment chiqishi ortgan.
2. Bu tajribada erkli o‘zgaruvchi — vaqt.
3. 10 daqiqada pigment 1 daqiqadagidan 10 marta ko‘p.

A) TTN **B)** NTN **C)** TNN **D)** TTT

15. B Gumoral boshqaruvning signal molekulalari qanday ataladi? 1 ball

A) Refleks yoylari
B) Nerv impulslari
C) Retseptorlar
D) Gormonlar va boshqa biologik faol moddalar

16. Q a) azotni o‘zlashtirib, o‘simlik bilan simbioz hosil qiladi; b) o‘simlik hujayrasiga kirib, kasallik keltiradi; c) o‘lik barglarni parchalaydi. 2 ball
Misollar: 1. Tuganak bakteriyalar; 2. Fitoftora; 3. Penitsill.
Moslashtiring.

A) a–3; b–2; c–1 **B)** a–1; b–2; c–3 **C)** a–2; b–1; c–3 **D)** a–1; b–3; c–2

17. B Qaysi juftlikdagi o‘simliklar tugunak bilan ko‘payadi? 1 ball

A) Qulupnay va ajriq **B)** Piyoz va lola
C) Bug‘doy va makkajo‘xori **D)** Kartoshka va yer noki (topinambur)

18. Q Hayvon hujayrasida 2n = 18. 3 ta birlamchi oosit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan tuxum hujayralar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 3 ta tuxum hujayra; har birida 9 ta xromosoma.

B) 3 ta tuxum hujayra; har birida 18 ta xromosoma.

C) 12 ta tuxum hujayra; har birida 18 ta xromosoma.

D) 12 ta tuxum hujayra; har birida 9 ta xromosoma.

19. Q a) tuxum sariqligi ko‘p, embrion tashqi muhitda rivojlanadi; b) embrion ona organizmida oziqlanadi; c) lichinka mustaqil oziqlanib, keyin metamorfozga uchraydi. 2 ball
Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.
Moslashtiring.

A) a–1; b–3; c–2 **B)** a–3; b–2; c–1 **C)** a–1; b–2; c–3 **D)** a–2; b–1; c–3

20. M 3 ball

Kun	2	4	6	8
20 °C	3	9	18	26
30 °C	9	18	26	26

Nihollar bo‘yi (5 kun): 20 °C — 4 cm, 30 °C — 4 cm.
Mulohazalarni baholang (T/N).
1. 2-kuni 30 °C da ungan urug‘lar 20 °C dagidan 3 marta ko‘p.
2. Nihollar bo‘yi bir xilligi harorat bu o‘simlik nihollari o‘sishiga sezilarli ta’sir qilmaganini ko‘rsatadi.
3. 30 °C da 6-kundan keyin ham unish davom etgan.

A) TNN **B)** TTN **C)** TTT **D)** NTN

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 7-variant

1 Q D	2 B C	3 Q C	4 Q D	5 Q A	6 M D	7 Q A
8 B B	9 Q A	10 Q A	11 Q A	12 Q A	13 Q C	14 M A
15 B D	16 Q B	17 B D	18 Q A	19 Q C	20 M B	

1. D) To‘g‘ri javob: **Eksperimental metod**
Eksperiment.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. C) To‘g‘ri javob: **Populyatsiya–tur darajasi**
Tur ichidagi guruhlar.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. C) To‘g‘ri javob: **1, 2, 4**
Yog‘ — gidrofob; peptid bog‘ — oqsilda.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. D) To‘g‘ri javob: **X – 1269 kJ; Y – 1255 kJ**
X: 50·17,6 + 10·38,9 = 1269 kJ; Y: 5·17,6 + 30·38,9 = 1255 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. A) To‘g‘ri javob: **Amilaza : fermentativ (katalitik)**
Ferment.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. D) To‘g‘ri javob: **TNT**
Uzunlik aniq mos kelishi kerak.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. A) To‘g‘ri javob: **Birlamchi struktura (aminokislotalar ketma-ketligi)**
Peptid bog‘lar uzilmaydi.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. B) To‘g‘ri javob: **Hujayralar hujayrasiz moddadan paydo bo‘ladi.**
Zid fikr.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. A) To‘g‘ri javob: **1, 2, 5**
Sianobakteriyada xloroplast yo‘q, tilakoidlar bor.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. A) To‘g‘ri javob: **a – 24 mol; b – 14 mol**
a = 2400·40 % : 40 = 24 mol; b = 560 : 40 = 14 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. A) To‘g‘ri javob: **82 mol**
2·5 + 36·2 = 82 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. A) To‘g‘ri javob: **540 g**
Hosil bo‘lgan: 24 : 6 = 4 mol; sarflangan: 6 : 6 = 1 mol; qolgan: 3 · 180 = 540 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. C) To‘g‘ri javob: **1 → 3 → 2 → 4**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. A) To‘g‘ri javob: **TTN**
60 = 6×10.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. D) To‘g‘ri javob: **Gormonlar va boshqa biologik faol moddalar**
Gormonlar.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. B) To‘g‘ri javob: **a–1; b–2; c–3**
Simbiont, parazit, saprotrof.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. D) To‘g‘ri javob: **Kartoshka va yer noki (topinambur)**
Tugunak.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. A) To‘g‘ri javob: **3 ta tuxum hujayra; har birida 9 ta xromosoma.**
Har bir oositdan 1 ta tuxum hujayra va qutb tanachalari hosil bo‘ladi; gametada n = 9.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. C) To‘g‘ri javob: **a–1; b–2; c–3**
Moslash.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. B) To‘g‘ri javob: **TTN**
26 → 26.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 8-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Olim DNK molekulasining fazoviy tuzilishini sim va sharchalardan yasalgan maket orqali tushuntirdi. Qaysi metod? 2 ball

A) Kuzatish metodi **B)** Modellashtirish metodi

C) Tarixiy metod **D)** Eksperimental metod

2. B Cho‘l ekotizimida yorug‘lik, harorat, o‘simliklar va hayvonlar o‘rtasidagi oziq zanjirlari o‘rganildi. Qaysi daraja? 1 ball

A) Organizm darajasi **B)** Hujayra darajasi

C) Biogeosenoz darajasi **D)** Molekula darajasi

3. Q Hujayradagi ionlarga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Bikarbonat ionlari qon pH ini barqarorlashtirishda qatnashadi.
2. Kalsiy ionlari muskul qisqarishida qatnashadi.
3. Xlor ionlari hujayrada umuman uchramaydi.
4. Ruh (sink) ba’zi fermentlar tarkibiga kiradi.
5. Kaliy faqat o‘simliklarda bo‘ladi.

A) 1, 2, 4 **B)** 2, 4, 5 **C)** 2, 3, 5 **D)** 1, 2, 5

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	35	20
Y	10	40

A) X – 1394 kJ; Y – 1732 kJ **B)** X – 1411,6 kJ; Y – 1749,6 kJ

C) X – 1355,1 kJ; Y – 1693,1 kJ **D)** X – 1732 kJ; Y – 1394 kJ

5. Q Soch va tirnoqlarning mo‘rtlashishi X oqsil tuzilishi buzilishi bilan bog‘liq. X va funksiyasi? 2 ball

A) Gemoglobin : transport
B) Insulin : gormonal
C) Keratin : tuzilish (himoya-tayanch)
D) Pepsin : fermentativ

6. M Ota-onalikni aniqlash: bolaning har bir fragmenti ota yoki onadan birida bo‘lishi kerak. 3 ball

Namuna	Fragmentlar, bp
Ona	200, 500
Bola	200, 350
Erkak A	350, 600
Erkak B	400, 600

Mulohazalarni baholang (T/N).
1. Bola 200 bp fragmentni onadan olgan bo‘lishi mumkin.
2. Erkak B bolaning otasi bo‘lishi mumkin.
3. Erkak A bolaning otasi bo‘lishi mumkin.

A) NNT
B) TNN
C) TTT
D) TNT

7. Q 2 ball

Namuna	Saqlash, faollik %	37 °C da, %
1 (0 °C)	5	97
2 (45 °C)	70	85
3 (90 °C)	0	0

To‘g‘ri xulosa?

A) 45 °C fermentning optimal harorati.
B) 90 °C da faollik ortgan.
C) 0 °C da ferment buzilgan.
D) 90 °C da ferment faol markazining shakli qaytmas buzilgan.

8. B Hujayralarning hayotiy jarayonlari haqidagi hujayra nazariyasi qoidasini aniqlang. 1 ball

A) Hujayralarda organoidlar soni bir xil.
B) Hujayralarda moddalar almashinuvi o‘xshash tarzda kechadi.
C) Hujayralar bir xil vazifani bajaradi.
D) Hujayralar bir xil tezlikda o‘sadi.

9. 2 ball

Belgi	X	Y	Z
Yadro	bor	yo‘q	bor
Sentriola	bor	yo‘q	yo‘q
Plastidalar	yo‘q	yo‘q	bor

To‘g‘ri fikrlarni aniqlang.
1. X hayvon hujayrasi bo‘lishi mumkin.
2. Y prokariot.
3. Z o‘simlik hujayrasi bo‘lishi mumkin.
4. Y da mitoxondriya bor.
5. Z da yadro yo‘q.

A) 1, 2, 3 **B)** 2, 3, 4 **C)** 1, 3, 5 **D)** 1, 2, 5

10. Energetik almashinuvda hosil bo‘lgan 1800 kJ energiyaning 60 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 640 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang. 2 ball

A) a – 27 mol; b – 16 mol **B)** a – 33 mol; b – 16 mol
C) a – 45 mol; b – 16 mol **D)** a – 27 mol; b – 8 mol

11. Hujayrada 12 mol glyukoza glikolizga uchradi, shundan 8 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang. 2 ball

A) 456 mol **B)** 24 mol **C)** 312 mol **D)** 288 mol

12. O‘simlik fotosintezda 48 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 30 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol). 2 ball

A) 2340 g **B)** 1440 g **C)** 900 g **D)** 540 g

13. Q 1. Ekvatorda 4 ta ikki xromatidali xromosoma. 2 ball
2. Har bir qutbga 4 tadan bir xromatidali xromosoma tortilmoqda.
3. Xromatin spirallashib, xromosomalar ko‘rinmoqda.
4. Ikki qiz hujayra hosil bo‘lmoqda.
To‘g‘ri ketma-ketlik?

A) 1 → 3 → 2 → 4 **B)** 3 → 1 → 2 → 4 **C)** 3 → 1 → 4 → 2 **D)** 3 → 2 → 1 → 4

14. M 3 ball

Harorat, °C	25	45	65	85
Pigment	2	6	35	70

Mulohazalarni baholang (T/N).
1. 85 °C da pigment 65 °C dagidan ikki baravar ko‘p.
2. Pigment chiqishining ortishi membrana oqsillarining denaturatsiyasi bilan bog‘liq.
3. Turli harorat uchun suv hajmi har xil bo‘lishi natijani ishonchli qiladi.

A) TTT **B)** TNN **C)** TTN **D)** NTN

15. B Nerv boshqaruvida signal neyronlar orasida qanday uzatiladi? 1 ball

A) Gormonlar orqali
B) Sinapslarda neyromediatorlar yordamida
C) Faqat qon orqali
D) Limfa orqali

16. Q a) ammiakni nitritlarga oksidlaydi; b) o‘lik organizmlar qoldiqlari bilan oziqlanadi; c) quyosh nuri energiyasidan foydalanadi. 2 ball
Misollar: 1. Achitqi; 2. Nitrifikator; 3. Sianobakteriya; 4. Bezgak plazmodiysi.
Moslashtiring.

A) a–1; b–2; c–3 **B)** a–2; b–1; c–3 **C)** a–3; b–1; c–2 **D)** a–2; b–4; c–3

17. B Qaysi juftlikdagi o‘simliklar piyozbosh bilan ko‘payadi? 1 ball

A) Ajriq va qamish
B) Piyoz va lola
C) Kartoshka va qulupnay
D) Paporotnik va mox

18. Q Hayvon hujayrasida 2n = 10. 7 ta birlamchi spermatotsit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan spermatozoidlar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 28 ta spermatozoid; har birida 10 ta xromosoma.

B) 28 ta spermatozoid; har birida 5 ta xromosoma.

C) 7 ta spermatozoid; har birida 5 ta xromosoma.

D) 7 ta spermatozoid; har birida 10 ta xromosoma.

19. Q a) mushuk; b) baqa; c) o‘rdak. 2 ball
Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.
Moslashtiring.

A) a–3; b–2; c–1 **B)** a–2; b–1; c–3 **C)** a–2; b–3; c–1 **D)** a–1; b–3; c–2

20. M Tuz konsentratsiyasining unishga ta’siri (30 tadan urug‘, 9-kun): 3 ball

NaCl, %	0	0,5	1	2
Ungan urug‘lar	28	22	12	2

Mulohazalarni baholang (T/N).
1. Tuz konsentratsiyasi oshgan sari unish kamaygan.
2. 1 % li eritmada ungan urug‘lar 0 % dagidan ikki marta ko‘p.
3. Sho‘r muhitda urug‘ga suv kirishi qiyinlashadi.

A) TNN **B)** NNT **C)** TNT **D)** TTT

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 8-variant

1 Q B	2 B C	3 Q A	4 Q A	5 Q C	6 M D	7 Q D
8 B B	9 Q A	10 Q A	11 Q C	12 Q D	13 Q B	14 M C
15 B B	16 Q B	17 B B	18 Q B	19 Q C	20 M C	

1. B) To‘g‘ri javob: **Modellashtirish metodi**
Maket — model.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. C) To‘g‘ri javob: **Biogeosenoz darajasi**
Ekotizim.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. A) To‘g‘ri javob: **1, 2, 4**
Cl va K hamma hujayralarda bor.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. A) To‘g‘ri javob: **X – 1394 kJ; Y – 1732 kJ**
X: 35·17,6 + 20·38,9 = 1394 kJ; Y: 10·17,6 + 40·38,9 = 1732 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. C) To‘g‘ri javob: **Keratin : tuzilish (himoya-tayanch)**
Keratin.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. D) To‘g‘ri javob: **TNT**
350 bp faqat A da.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. D) To‘g‘ri javob: **90 °C da ferment faol markazining shakli qaytmas buzilgan.**
Qaytmas buzilish.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. B) To‘g‘ri javob: **Hujayralarda moddalar almashinuvi o‘xshash tarzda kechadi.**
Asosiy jarayonlar o‘xshash.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. A) To‘g‘ri javob: **1, 2, 3**
Prokariotda mitoxondriya yo‘q; Z da yadro bor.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. A) To‘g‘ri javob: **a – 27 mol; b – 16 mol**
a = 1800·60 % : 40 = 27 mol; b = 640 : 40 = 16 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. C) To‘g‘ri javob: **312 mol**
2·12 + 36·8 = 312 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. D) To‘g‘ri javob: **540 g**
Hosil bo‘lgan: 48 : 6 = 8 mol; sarflangan: 30 : 6 = 5 mol; qolgan: 3 · 180 = 540 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. B) To‘g‘ri javob: **3 → 1 → 2 → 4**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. C) To‘g‘ri javob: **TTN**
Suv hajmi bir xil bo‘lishi kerak.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. B) To‘g‘ri javob: **Sinapslarda neyromediatorlar yordamida**
Sinaps.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. B) To‘g‘ri javob: **a–2; b–1; c–3**
Achitqi — saprotrof.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. B) To‘g‘ri javob: **Piyoz va lola**
Piyozbosh.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. B) To‘g‘ri javob: **28 ta spermatozoid; har birida 5 ta xromosoma.**
Har bir spermatotsitdan 4 ta spermatozoid hosil bo‘ladi; gametada n = 5.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. C) To‘g‘ri javob: **a–2; b–3; c–1**
Moslash.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. C) To‘g‘ri javob: **TNT**
12 < 28.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 9-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Ornitolog qushlarning bahorgi uchib kelish vaqtlarini har yili tabiatda qayd etib bordi. Qaysi metod? 2 ball

A) Kuzatish metodi **B)** Taqqoslash metodi

C) Eksperimental metod **D)** Modellashtirish metodi

2. B DNK replikatsiyasi fermentlari o‘rganildi. Qaysi daraja? 1 ball

A) Biosfera darajasi **B)** Organizm darajasi

C) Molekula darajasi **D)** Hujayra darajasi

3. Q Suvning biologik ahamiyatiga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Suv qon plazmasida moddalarni tashiydi.
2. Suv bo‘g‘imlarda moylovchi suyuqlik tarkibida bo‘ladi.
3. Suv energiya manbai sifatida parchalanib ATF hosil qiladi.
4. Suv ko‘z, miya kabi organlarni zarbdan himoya qiluvchi suyuqliklar tarkibida bo‘ladi.
5. Suv hujayraga faqat faol transport bilan kiradi.

A) 1, 2, 5 **B)** 1, 2, 4 **C)** 1, 4, 5 **D)** 1, 3, 5

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	20	30
Y	60	10

A) X – 1480,1 kJ; Y – 1406,1 kJ **B)** X – 1536,6 kJ; Y – 1462,6 kJ

C) X – 1445 kJ; Y – 1519 kJ **D)** X – 1519 kJ; Y – 1445 kJ

5. Q Muskulda kislorod zaxirasini saqlovchi X oqsil kam bo‘lgani uchun chidamlilik pasaydi. X va funksiyasi? 2 ball

A) Mioglobin : kislorod zaxiralash
B) Fibrinogen : himoya
C) Keratin : tayanch
D) Insulin : gormonal

6. M 3 ball

Namuna	Fragmentlar, bp
D	320, 480, 880
G1	320, 480, 880
G2	320, 500, 880
G3	300, 480, 860

Mulohazalarni baholang (T/N).
1. G1 iz qoldirgan bo‘lishi mumkin.
2. G3 iz qoldirgan bo‘lishi mumkin, chunki farqlar 20 bp dan oshmaydi.
3. Fragmentlar uzunligini aniqlash uchun ma’lum uzunlikdagi marker (M) bilan taqqoslanadi.

A) TTT
B) TNT
C) TNN
D) NNT

7. Q Ferment faolligi harorat 10 °C dan 40 °C gacha ortib, keyin keskin kamaydi. Kamayish sababi nima? 2 ball

A) Substrat ko‘payishi
B) Yuqori haroratda ferment denaturatsiyasi
C) Fermentning renaturatsiyasi
D) Harorat ta’sir qilmaydi

8. B Hujayra nazariyasi bo‘yicha organizm rivojlanishi nimadan boshlanadi? 1 ball

A) To‘qimadan
B) Hujayralararo moddadan
C) Bitta hujayradan (masalan, zigotadan)
D) Organdan

9. 2 ball

Belgi	X	Y	Z
Yadro	bor	bor	yo‘q
Hujayra devori	yo‘q	xitin	murein
Zaxira modda	glikogen	glikogen	–

To‘g‘ri fikrlarni aniqlang.
1. X va Y da glikogen zaxiralanadi.
2. Y zamburug‘ hujayrasi bo‘lishi mumkin.
3. Z eukariot.
4. X o‘simlik hujayrasi.
5. Uchala hujayrada plazmatik membrana bor.

A) 2, 3, 4 **B)** 1, 3, 4 **C)** 2, 3, 5 **D)** 1, 2, 5

10. 2 ball

Energetik almashinuvda hosil bo‘lgan 1400 kJ energiyaning 40 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 120 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang.

A) a – 35 mol; b – 3 mol **B)** a – 14 mol; b – 1,5 mol
C) a – 14 mol; b – 3 mol **D)** a – 20 mol; b – 3 mol

11. 2 ball

Hujayrada 6 mol glyukoza glikolizga uchradi, shundan 1 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang.

A) 12 mol **B)** 228 mol **C)** 48 mol **D)** 36 mol

12. 2 ball

O‘simlik fotosintezda 54 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 24 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol).

A) 900 g **B)** 2340 g **C)** 720 g **D)** 1620 g

13. Q 1. Duk iplari qisqarib, xromatidalarni ajratmoqda. 2 ball
2. Xromosomalar ko‘rina boshladi, yadrocha yo‘qolmoqda.
3. Yangi yadro qobig‘i shakllanmoqda.
4. Xromosomalar ekvatorda saf tortgan.
To‘g‘ri ketma-ketlik?

A) 2 → 4 → 1 → 3 **B)** 4 → 2 → 1 → 3 **C)** 2 → 4 → 3 → 1 **D)** 2 → 1 → 4 → 3

14. M 3 ball

Harorat, °C	20	40	60	80
Pigment	1	2	16	32

Mulohazalarni baholang (T/N).
1. 60 °C da pigment 40 °C dagidan 8 marta ko‘p.
2. 80 °C da membranalar 20 °C dagidan butunroq.
3. Har bir haroratda tajribani takrorlash natija ishonchliligini oshiradi.

A) NNT **B)** TNN **C)** TTT **D)** TNT

15. B Nerv va gumoral boshqaruvning umumiy maqsadi nima? 1 ball

A) Gomeostazni saqlash va organizmni muhit o‘zgarishiga moslashtirish
B) Faqat o‘sishni to‘xtatish
C) Faqat ko‘payish
D) Faqat ovqat hazm qilish

16. Q a) hasharotlarni tutib, qo‘shimcha oziqlanadi, fotosintez ham qiladi; b) faqat fotosintez; c) faqat tayyor organik modda. 2 ball
Misollar: 1. Drozera (yirtqich o‘simlik); 2. Bug‘doy; 3. Shampinyon.
Moslashtiring.

A) a–1; b–3; c–2 **B)** a–1; b–2; c–3 **C)** a–2; b–1; c–3 **D)** a–3; b–2; c–1

17. B Qaysi juftlikdagi o‘simliklar gajak (uzun yer ustki novda) bilan ko‘payadi? 1 ball

A) Lola va sarimsoq **B)** Bug‘doy va arpa
C) Kartoshka va piyoz **D)** Qulupnay va g‘ozpanja

18. Q Hayvon hujayrasida 2n = 22. 5 ta birlamchi oosit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan tuxum hujayralar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 20 ta tuxum hujayra; har birida 22 ta xromosoma.

B) 20 ta tuxum hujayra; har birida 11 ta xromosoma.

C) 5 ta tuxum hujayra; har birida 11 ta xromosoma.

D) 5 ta tuxum hujayra; har birida 22 ta xromosoma.

19. Q a) asalari; b) timsoh; c) odam. 2 ball
Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.
Moslashtiring.

A) a–2; b–3; c–1 **B)** a–1; b–3; c–2 **C)** a–3; b–1; c–2 **D)** a–3; b–2; c–1

20. M 3 ball

Kun	2	4	6	8
18 °C	2	7	15	24
28 °C	7	21	27	28

Nihollar bo‘yi (5 kun): 18 °C — 2,5 cm, 28 °C — 5 cm.
Mulohazalarni baholang (T/N).
1. 4-kuni 28 °C da ungan urug‘lar 18 °C dagidan 3 marta ko‘p.
2. 28 °C da nihollar 18 °C dagidan ikki marta baland.
3. 8-kuni ikkala guruhda ungan urug‘lar soni teng.

A) NTN **B)** TTT **C)** TTN **D)** TNN

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 9-variant

1 Q A	2 B C	3 Q B	4 Q D	5 Q A	6 M B	7 Q B
8 B C	9 Q D	10 Q C	11 Q C	12 Q A	13 Q A	14 M D
15 B A	16 Q B	17 B D	18 Q C	19 Q C	20 M C	

1. A) To‘g‘ri javob: **Kuzatish metodi**
Kuzatish.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. C) To‘g‘ri javob: **Molekula darajasi**
Molekula.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. B) To‘g‘ri javob: **1, 2, 4**
Suv energiya manbai emas; osmos — passiv.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. D) To‘g‘ri javob: **X – 1519 kJ; Y – 1445 kJ**
X: 20·17,6 + 30·38,9 = 1519 kJ; Y: 60·17,6 + 10·38,9 = 1445 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. A) To‘g‘ri javob: **Mioglobin : kislorod zaxiralash**
Mioglobin.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. B) To‘g‘ri javob: **TNT**
Fragmentlar aniq mos bo‘lishi kerak.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. B) To‘g‘ri javob: **Yuqori haroratda ferment denaturatsiyasi**
Denaturatsiya.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. C) To‘g‘ri javob: **Bitta hujayradan (masalan, zigotadan)**
Rivojlanish birligi.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. D) To‘g‘ri javob: **1, 2, 5**
Z prokariot; X — hayvon.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. C) To‘g‘ri javob: **a – 14 mol; b – 3 mol**
a = 1400·40 % : 40 = 14 mol; b = 120 : 40 = 3 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. C) To‘g‘ri javob: **48 mol**
2·6 + 36·1 = 48 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. A) To‘g‘ri javob: **900 g**
Hosil bo‘lgan: 54 : 6 = 9 mol; sarflangan: 24 : 6 = 4 mol; qolgan: 5 · 180 = 900 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. A) To‘g‘ri javob: **2 → 4 → 1 → 3**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. D) To‘g‘ri javob: **TNT**
80 ˚C da ko‘proq buzilgan.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. A) To‘g‘ri javob: **Gomeostazni saqlash va organizmni muhit o‘zgarishiga moslashtirish**
Gomeostaz.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. B) To‘g‘ri javob: **a–1; b–2; c–3**
Miksotrof o‘simlik.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. D) To‘g‘ri javob: **Qulupnay va g‘ozpanja**
Gajak.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. C) To‘g‘ri javob: **5 ta tuxum hujayra; har birida 11 ta xromosoma.**
Har bir oositdan 1 ta tuxum hujayra va qutb tanachalari hosil bo‘ladi; gametada n = 11.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. C) To‘g‘ri javob: **a–3; b–1; c–2**
Asalari — lichinkali.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. C) To‘g‘ri javob: **TTN**
24 ≠ 28.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: 10-variant

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

Sinf: _____ Sana: _____ Ball: _____

1. Q Tadqiqotchi baliq, qurbaqa va kaltakesak embrionlarini bir xil bosqichda solishtirdi. Qaysi metod? 2 ball

A) Taqqoslash metodi
B) Eksperimental metod
C) Modellashtirish metodi
D) Kuzatish metodi

2. B Okeanlar, atmosfera va quruqlikdagi tirik organizmlarning kislorod balansiga qo‘shgan hissasi o‘rganildi. Qaysi daraja? 1 ball

A) Organizm darajasi
B) Populyatsiya–tur darajasi
C) Biosfera darajasi
D) Biogeosenoz darajasi

3. Q Anorganik birikmalarga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Suv hujayradagi eng ko‘p uchraydigan birikma.
2. Mineral tuzlar hujayrada ionlar holida bo‘ladi.
3. Karbonat angidrid organik modda hisoblanadi.
4. Fosfor suyak to‘qimasi tarkibiga kiradi.
5. Suv molekulasida ikkita kislorod atomi bor.

A) 3, 4, 5
B) 1, 2, 4
C) 1, 4, 5
D) 2, 4, 5

4. Q Faqat uglevod va lipiddan iborat ikki ozuqa namunasining tarkibi jadvalda berilgan. 1 g uglevod to‘liq oksidlanganda 17,6 kJ, 1 g lipid esa 38,9 kJ energiya ajratadi. Namunalar to‘liq oksidlanganda ajraladigan energiyani aniqlang. 2 ball

Namuna	Uglevod, g	Lipid, g
X	45	20
Y	30	30

A) X – 1695 kJ; Y – 1570 kJ
B) X – 1531,1 kJ; Y – 1656,1 kJ
C) X – 1587,6 kJ; Y – 1712,6 kJ
D) X – 1570 kJ; Y – 1695 kJ

5. Q Qon plazmasida X oqsil kamaygani uchun yog‘ kislotalari va ba’zi dorilarning tashilishi buzildi, ivish me’yorda. X va funksiyasi? 2 ball

A) Gemoglobin : transport
B) Insulin : gormonal
C) Albumin : transport
D) Fibrinogen : himoya

6. M 3 ball

Namuna	Fragmentlar, bp
D	100, 300, 500
G1	100, 300
G2	100, 300, 500
G3	100, 300, 500, 700

Mulohazalarni baholang (T/N).
1. G2 iz qoldirgan bo‘lishi mumkin.
2. G3 ham mos, chunki D ning barcha fragmentlari unda bor.
3. Fragmentlar to‘liq mos kelishi uchun ortiqcha yoki yetishmaydigan fragment bo‘lmasligi kerak.

A) NNT **B)** TNT **C)** TTT **D)** TNN

7. Q Ferment 50 °C da 5 daqiqa saqlanib 37 °C ga qaytarilganda faolligi 90 % ga tiklandi, 50 °C da 60 daqiqa saqlanganda esa 10 % ga. To‘g‘ri xulosa? 2 ball

A) Vaqt denaturatsiyaga ta’sir qilmaydi.
B) Yuqori haroratda uzoq saqlash qaytmas buzilishni kuchaytiradi.
C) Ikkala holatda ham ferment to‘liq buzilgan.
D) Uzoq saqlash fermentni faollashtiradi.

8. B Hujayra nima uchun tiriklikning elementar birligi deb ataladi? Nazariyaga mos fikrni aniqlang. 1 ball

A) Molekulalar mustaqil tirik birlik hisoblanadi.
B) Organoidlar hujayradan tashqarida mustaqil yashaydi.
C) Hujayralar mustaqil yashay olmaydi.
D) Hujayra — tiriklikning eng kichik mustaqil yashay oladigan birligi.

9. 2 ball

Belgi	X	Y	Z
Yadro	yo‘q	bor	bor
Lizosoma	yo‘q	bor	bor
Xloroplast	yo‘q	yo‘q	bor

To‘g‘ri fikrlarni aniqlang.
1. X da membranali organoidlar yo‘q.
2. Y avtotrof.
3. Z da mitoxondriya bor.
4. Y da ribosoma bor.
5. X ning DNKsi yadroda.

A) 1, 2, 4 **B)** 2, 3, 5 **C)** 1, 3, 4 **D)** 1, 2, 3

10. 2 ball

Energetik almashinuvda hosil bo‘lgan 3000 kJ energiyaning 40 % i ATF da to‘plandi. Plastik almashinuvga shu ATF hisobidan 720 kJ sarflandi. 1 mol ATF 40 kJ energiya saqlaydi. Hosil bo‘lgan (a) va sarflangan (b) ATF miqdorini (mol) aniqlang.

A) a – 36 mol; b – 18 mol **B)** a – 30 mol; b – 18 mol
C) a – 75 mol; b – 18 mol **D)** a – 30 mol; b – 9 mol

11. 2 ball

Hujayrada 3 mol glyukoza glikolizga uchradi, shundan 2 mol glyukoza kislorodli bosqichda to‘liq parchalandi. Glikolizda 1 mol glyukozadan 2 mol, kislorodli bosqichda esa 36 mol ATF hosil bo‘ladi. Jami hosil bo‘lgan ATF miqdorini aniqlang.

A) 114 mol **B)** 72 mol **C)** 6 mol **D)** 78 mol

12. 2 ball

O‘simlik fotosintezda 60 mol CO_2 yutdi va hosil bo‘lgan glyukozaning bir qismini nafas olishda sarfladi; bunda 36 mol O_2 sarflandi. Hosil bo‘lgan glyukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang (glyukoza – 180 g/mol).

A) 1080 g **B)** 1800 g **C)** 720 g **D)** 2880 g

13. Q 1. Xromatidalar qutblarga tortilmoqda. 2 ball
2. Xromosomalar ekvatorda.
3. Xromosomalar spirallashmoqda.
4. Xromosomalar despirallashib, ikki yadro hosil bo‘lmoqda.
To‘g‘ri ketma-ketlik?

A) 3 → 2 → 4 → 1 **B)** 2 → 3 → 1 → 4 **C)** 3 → 2 → 1 → 4 **D)** 3 → 1 → 2 → 4

14. M 3 ball

Harorat, °C	30	50	70
Pigment	3	12	40

Mulohazalarni baholang (T/N).
1. Harorat oshishi bilan pigment chiqishi ortgan.
2. 50 °C da pigment 30 °C dagidan 4 marta ko‘p.
3. Pigment chiqishi membrana fosfolipidlari qattiqlashgani bilan izohlanadi.

A) TNN **B)** TTN **C)** TTT **D)** NTN

15. B Qondagi glyukoza oshganda insulin ajralishi qaysi boshqaruvga misol? 1 ball

A) Gumoral boshqaruv **B)** Faqat nerv boshqaruvi
C) Refleks yoyi **D)** Mexanik boshqaruv

16. Q a) vodorodni oksidlab energiya oladi; b) tirik baliq terisida yashab, uning hisobiga oziqlanadi; c) fotosintez qiladi. 2 ball
Misollar: 1. Vodorod bakteriya; 2. Saproleginiya; 3. Ulotriks.
Moslashtiring.

A) a–1; b–2; c–3 **B)** a–2; b–1; c–3 **C)** a–3; b–2; c–1 **D)** a–1; b–3; c–2

17. B Ipsimon suvo‘t ipi bir necha bo‘lakka uzilib, har biri yangi ipga aylandi. Bu qanday ko‘payish? 1 ball

A) Kurtaklanish **B)** Sporalar bilan **C)** Fragmentatsiya **D)** Jinsiy ko‘payish

18. Q Hayvon hujayrasida 2n = 46. 2 ta birlamchi spermatotsit meyoz yo‘li bilan bo‘linib, gametalar hosil qildi. Hosil bo‘lgan spermatozoidlar soni va har biridagi xromosomalar miqdorini aniqlang. 2 ball

A) 2 ta spermatozoid; har birida 23 ta xromosoma.

B) 8 ta spermatozoid; har birida 23 ta xromosoma.

C) 2 ta spermatozoid; har birida 46 ta xromosoma.

D) 8 ta spermatozoid; har birida 46 ta xromosoma.

19. Q a) tuxumdan chiqqan ijtimoiy hasharot qurtlari ishchilar tomonidan boqiladi; b) jo‘ja tuxumdan to‘liq shakllanib chiqadi; c) bola tug‘ilgach ona suti bilan oziqlanadi. 2 ball

Tiplar: 1. Tuxumda; 2. Bachadonda; 3. Lichinkali.

Moslashtiring.

A) a–1; b–3; c–2 **B)** a–2; b–1; c–3 **C)** a–3; b–2; c–1 **D)** a–3; b–1; c–2

20. M Urug‘lar oldindan 0, 6 va 12 soat ivitilib, 25 °C da undirildi (30 tadan): 3 ball

Ivitish, soat	0	6	12
3-kuni ungan	4	10	16

Mulohazalarni baholang (T/N).

1. Ivitish vaqti oshgan sari 3-kuni ungan urug‘lar ko‘paygan.
2. 12 soat ivitilgan urug‘lar ivitilmaganlaridan 4 marta ko‘p ungan.
3. Bu tajribada erkli o‘zgaruvchi — harorat.

A) TTT **B)** TTN **C)** NTN **D)** TNN

CHSB_PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: 10-variant

1 Q A	2 B C	3 Q B	4 Q D	5 Q C	6 M B	7 Q B
8 B D	9 Q C	10 Q B	11 Q D	12 Q C	13 Q C	14 M B
15 B A	16 Q A	17 B C	18 Q B	19 Q D	20 M B	

1. A) To‘g‘ri javob: **Taqqoslash metodi**
Taqqoslash.
Tekshiriladigan ko‘nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. C) To‘g‘ri javob: **Biosfera darajasi**
Biosfera.
Tekshiriladigan ko‘nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. B) To‘g‘ri javob: **1, 2, 4**
CO_2 anorganik; H_2O da bitta O.
Tekshiriladigan ko‘nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. D) To‘g‘ri javob: **X – 1570 kJ; Y – 1695 kJ**
X: 45·17,6 + 20·38,9 = 1570 kJ; Y: 30·17,6 + 30·38,9 = 1695 kJ.
Tekshiriladigan ko‘nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. C) To‘g‘ri javob: **Albumin : transport**
Albumin.
Tekshiriladigan ko‘nikma: Oqsillarning tuzilishi va funksiyalari.

6. B) To‘g‘ri javob: **TNT**
G3 da ortiqcha 700 bp bor.
Tekshiriladigan ko‘nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. B) To‘g‘ri javob: **Yuqori haroratda uzoq saqlash qaytmas buzilishni kuchaytiradi.**
Vaqt omili.
Tekshiriladigan ko‘nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. D) To‘g‘ri javob: **Hujayra — tiriklikning eng kichik mustaqil yashay oladigan birligi.**
Elementar birlik.
Tekshiriladigan ko‘nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. C) To‘g‘ri javob: **1, 3, 4**
Y — xloroplastsiz, geterotrof.
Tekshiriladigan ko‘nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. B) To‘g‘ri javob: **a – 30 mol; b – 18 mol**
a = 3000·40 % : 40 = 30 mol; b = 720 : 40 = 18 mol.
Tekshiriladigan ko‘nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. D) To‘g‘ri javob: **78 mol**
2·3 + 36·2 = 78 mol.
Tekshiriladigan ko‘nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. C) To‘g‘ri javob: **720 g**
Hosil bo‘lgan: 60 : 6 = 10 mol; sarflangan: 36 : 6 = 6 mol; qolgan: 4 · 180 = 720 g.
Tekshiriladigan ko‘nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. C) To‘g‘ri javob: **3 → 2 → 1 → 4**
P → M → A → T.
Tekshiriladigan ko‘nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. B) To‘g‘ri javob: **TTN**
Yuqori haroratda lipidlar suyuqlashadi.
Tekshiriladigan ko‘nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. A) To‘g‘ri javob: **Gumoral boshqaruv**
Gormon — gumoral.
Tekshiriladigan ko‘nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. A) To‘g‘ri javob: **a–1; b–2; c–3**
Xemotrof, parazit, fototrof.
Tekshiriladigan ko‘nikma: Tirik organizmlarning oziqlanishiga ko‘ra turlari.

17. C) To‘g‘ri javob: **Fragmentatsiya**
Bo‘laklanish.
Tekshiriladigan ko‘nikma: Organizmlarning ko‘payishi. Jinssiz ko‘payish. Jinssiz ko‘payishning turlari.

18. B) To‘g‘ri javob: **8 ta spermatozoid; har birida 23 ta xromosoma.**
Har bir spermatotsitdan 4 ta spermatozoid hosil bo‘ladi; gametada n = 23.
Tekshiriladigan ko‘nikma: Hayvonlarda jinsiy ko‘payish shakllari. Hayvonlarda gametogenez jarayoni. Urug‘lanish. Partenogenez.

19. D) To‘g‘ri javob: **a–3; b–1; c–2**
Moslash.
Tekshiriladigan ko‘nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. B) To‘g‘ri javob: **TTN**
Erkli — ivitish vaqti.
Tekshiriladigan ko‘nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

10-SINF · TABIIY FAN · CHSB-1

ChSB-1 tayyorgarlik: I chorak demo varianti

20 ta savol · 45 daqiqa · 39 ball

F.I.Sh.: _____

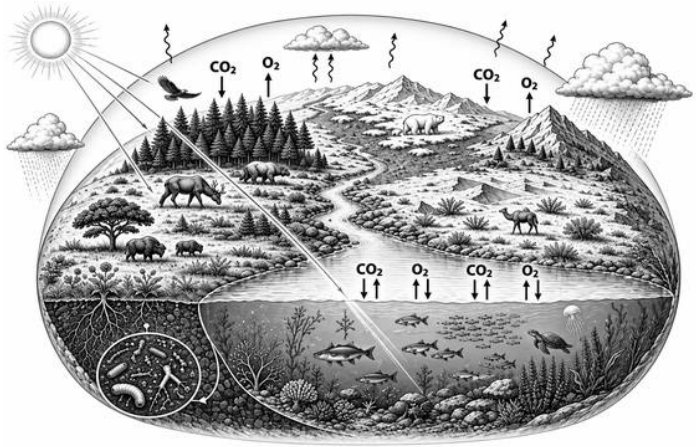
Sinf: _____ Sana: _____ Ball: _____

1. Q Tadqiqotchi tovuq va odam embrionlarining bir xil rivojlanish bosqichlaridagi tuzilish belgilarini o‘zaro o‘rganib, ularning o‘xshash va farqli jihatlarini aniqladi. Tadqiqotchi qaysi biologik metoddan foydalangan? 2 ball

A) taqqoslash metodi **B)** kuzatish metodi

C) eksperimental metodi **D)** modellashtirish metodi

2. B Rasm ma’lumotlari asosida hayotning qaysi tuzilish darajasi ifodalanganini aniqlang. 1 ball



A) Biosfera darajasi **B)** Biogeosenoz darajasi

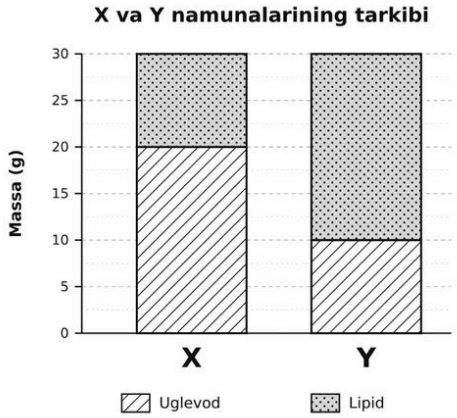
C) Populatsiya–tur darajasi **D)** Molekula darajasi

3. Q Suvning biologik xususiyatlariga oid to‘g‘ri fikrlarni aniqlang. 2 ball

1. Suv molekulasining qutbliligi qutbli moddalarni eritishga yordam beradi.
2. Organizm haroratining barqarorligi suvning issiqlik sig‘imi pastligi bilan izohlanadi.
3. Muzning zichligi suyuq suvnikidan kichik bo‘lgani uchun suv havzasi yuzasidan muzlaydi.
4. Suv molekulasining qutbliligi lipidlarning sitoplazmada erkin erishini ta’minlaydi.
5. Gidroliz reaksiyalarida suv boshlang‘ich moddalardan biri sifatida qatnashadi.

A) 1, 3, 5 **B)** 1, 2, 4 **C)** 2, 3, 5 **D)** 1, 4, 5

4. Q Grafikda faqat uglevod va lipiddan iborat X va Y ozuqa namunalarining tarkibi berilgan. Ular to‘liq oksidlanganda ajraladigan energiya miqdorini (kJ) aniqlang. 2 ball



A) X - 741 kJ; Y - 954 kJ

B) X - 954 kJ; Y - 741 kJ

C) X - 704 kJ; Y - 917 kJ

D) X - 778 kJ; Y - 991 kJ

5. Q Qon plazmasidagi X oqsil kamaygach, jarohatlangan tomirda qon laxtasi kech hosil bo‘ldi. Lipidlar va kislorod tashilishi esa me’yorida saqlandi. X oqsilni va uning funksiyasini aniqlang. 2 ball

A) Fibrinogen : himoya

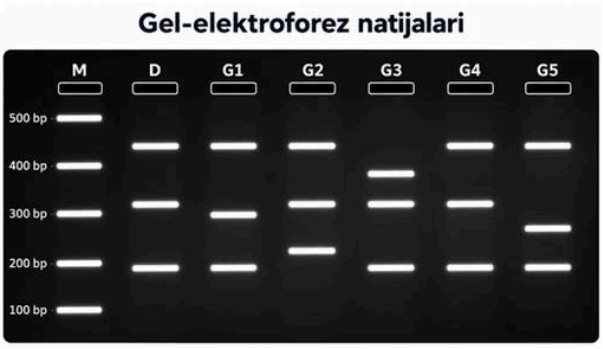
B) Albumin : transport

C) Gemoglobin : transport

D) Insulin : gormonal

6. M Biotexnologiya markazida o‘g‘rilik sodir bo‘ldi. Kameraga o‘g‘rining yuzi tushmagan, ammo uning ehtiyotsizlik bilan qo‘lini jarohatlab, eshik tutqichida qon izi qoldirgani qayd etilgan. Eshikdagi qon izi (D) va besh gumondorning og‘iz epiteliyasidan (G1–G5) olingan DNKning bir xil sohasi PZR yordamida ko‘paytirildi. Hosil bo‘lgan fragmentlar gel-elektroforezda uzunligiga ko‘ra ajratilib, profillar o‘zaro taqqoslandi. Rasm asosida mulohazalarni mulohazalarning **to‘g‘ri (T)** yoki **noto‘g‘ri (N)** ekanligini aniqlang. 3 ball

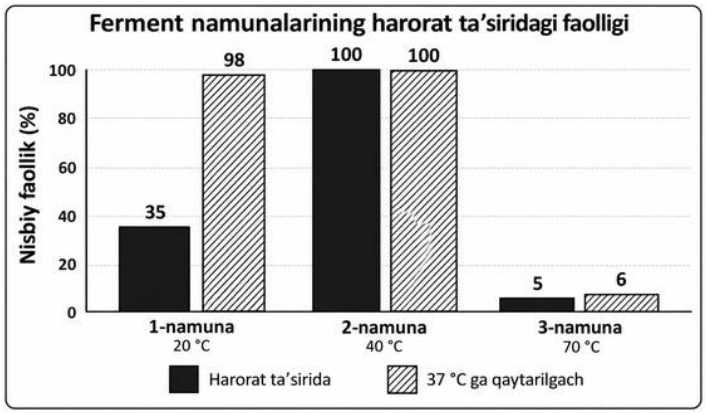
1. O‘g‘ri to‘rtinchi gumondor bo‘lishi mumkin, chunki uning uchala fragmenti eshikdagi qon izi bilan mos keladi.
2. O‘g‘ri ikkinchi gumondor bo‘lishi mumkin, chunki uch fragmentdan ikkitasining mosligi shaxsni aniqlash uchun yetarli.
3. O‘g‘ri birinchi gumondor bo‘lishi mumkin, chunki uning fragmentlari M markeridagi aniq bp chiziqlariga tushgan.



M: marker – fragment uzunligini aniqlash uchun o‘lchov namunasi
bp: DNK fragmentidagi azotli asoslar jufti soni
D: eshik tutqichidagi qon izi
G1 – G5: gumondorlarning og‘iz epiteliyasi

A) TNN **B)** TTN **C)** NNT **D)** NTT

7. Q Bir xil ferment eritmasidan teng hajmda olingan uchta namuna mos ravishda 20 °C, 40 °C va 70 °C da bir xil vaqt davomida saqlandi. Ferment faolligi ushbu haroratlarda va namunalar 37 °C ga qaytarilgandan so‘ng qayta o‘lchandi. Grafik ma’lumotlarini tahlil qilib, to‘g‘ri xulosani aniqlang. 2 ball



A) Uchinchi namunadagi faollikning qayta tiklanmasligi, bu oqsilning renaturatsiya qila olmasligini ifodalaydi.

B) Birinchi namunadagi faollikning qayta tiklanishi denaturatsiya yuz bermaganini isbotlaydi.

C) Ikkinchi namunadagi barqaror faollik 40 °C fermentning optimal harorati emasligini ko‘rsatadi.

D) Uchinchi namunada faollikning ortishi fermentning dastlabki tuzilishi to‘liq tiklanganini ko‘rsatadi.

8. B Hujayra nazariyasining asosiy qoidalariga mos to‘g‘ri fikrni aniqlang. 1 ball

A) Hujayralar tuzilishi va asosiy hayotiy jarayonlari jihatidan o‘xshash.

B) Hujayralar shakli va bajaradigan vazifalari jihatidan bir xil.

C) Hujayralar organoidlarining soni va tarkibi jihatidan bir xil.

D) Hujayralar o‘sish tezligi va yashash davomiyligi jihatidan o‘xshash.

9. Q Jadval ma’lumotlarini tahlil qilib, to‘g‘ri fikrlar berilgan javobni aniqlang. 2 ball

1. X hujayrada ribosomalar mavjud.
2. Y hujayrada mitoxondriyalar bo‘lmaydi.
3. Z hujayra hayvon organizmiga tegishli.
4. Y hujayrada irsiy axborot faqat yadroda joylashgan.
5. Uchala hujayrada ham plazmatik membrana mavjud.

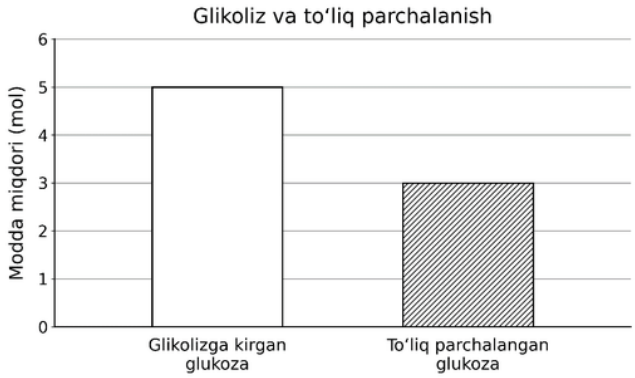
Belgi	X hujayra	Y hujayra	Z hujayra
Yadro	yo‘q	bor	bor
Hujayra devori	mureinli	sellulozali	yo‘q
Xloroplast	yo‘q	bor	yo‘q

A) 1, 3, 5 **B)** 1, 2, 3 **C)** 2, 4, 5 **D)** 3, 4, 5

10. Q Energetik almashinuvda hosil bo‘lgan 1200 kJ energiyaning 40% i ATFda to‘plandi. Plastik almashinuvga shu ATF hisobidan 320 kJ sarflandi. Hosil bo‘lgan (a) va plastik almashinuvda sarflangan (b) ATF miqdorini (mol) aniqlang. 2 ball

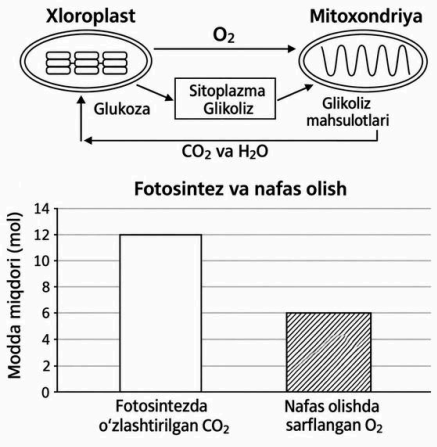
A) a–12 mol; b–8 mol **B)** a–30 mol; b–8 mol
C) a–12 mol; b–4 mol **D)** a–18 mol; b–8 mol

11. Q Grafikda glikolizga kirgan glukozaning umumiy miqdori va uning to‘liq parchalangan qismi ko‘rsatilgan. Grafik ma’lumotlari asosida jami hosil bo‘lgan ATF miqdorini aniqlang. 2 ball



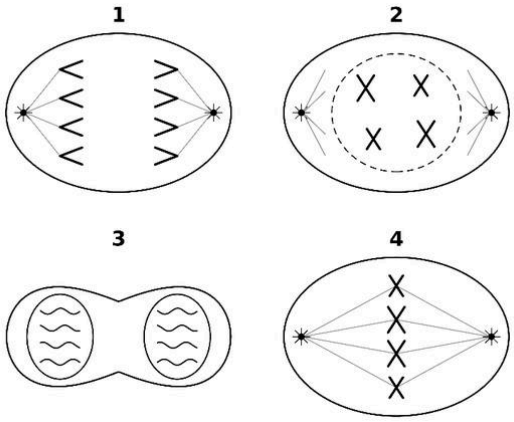
A) 118 mol **B)** 108 mol **C)** 190 mol **D)** 10 mol

12. Q Fotosintezda hosil bo‘lgan glukozaning nafas olishdan keyin saqlanib qolgan massasini aniqlang. 2 ball



A) 180 gr **B)** 360 gr **C)** 540 gr **D)** 1080 gr

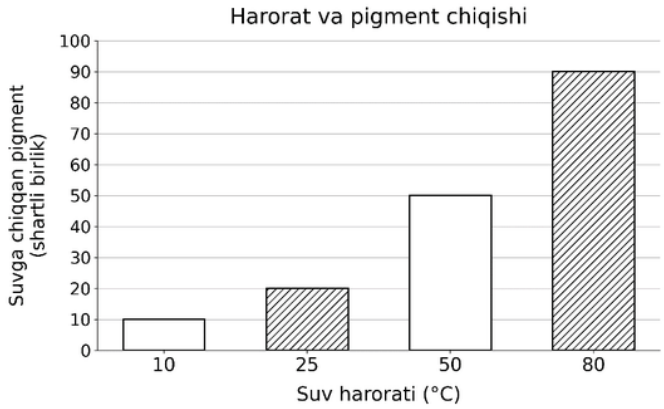
13. Q Quyidagi rasmda 2n = 4 bo‘lgan hayvon hujayrasining mitoz bosqichlari aralash tartibda berilgan. Xromosomalarning holatiga asoslanib, bosqichlarning to‘g‘ri ketma-ketligini aniqlang. 2 ball



A) 2 → 4 → 1 → 3 **B)** 2 → 1 → 4 → 3 **C)** 4 → 2 → 1 → 3 **D)** 2 → 4 → 3 → 1

14. M Lavlagi ildizmevasidan bir xil o‘lchamda kesilgan bo‘laklar yuvilib, teng hajmdagi turli haroratli suvga solindi. Besh daqiqadan keyin suvga chiqqan pigment miqdori aniqlandi. Grafik asosida quyidagi mulohazalarning **to‘g‘ri (T)** yoki **noto‘g‘ri (N)** ekanligini aniqlang. 3 ball

1. Yuqori haroratda pigmentning ko‘proq chiqishi hujayra membranalari o‘tkazuvchanligining ortishi bilan izohlanadi.
2. Harorat oshishi bilan pigment chiqishining kamayishi membrana oqsillari denaturatsiyasi bilan izohlanadi.
3. Bo‘laklar oldindan yuvilmasa, kesishda ajralgan pigment harorat ta’sirini baholashga xalaqit beradi.



A) TNT **B)** TTN **C)** NTT **D)** NNT

15. B Nerv va gumoral boshqaruvga xos umumiy xususiyatni aniqlang. 1 ball

A) Organlar faoliyatini muvofiqlashtirib, ichki muhit barqarorligini saqlaydi.

B) Boshqaruvchi ta’sirni nerv tolalari orqali tegishli organga yetkazadi.

C) Biologik faol moddalarni organizm suyuqliklari orqali tarqatadi.

D) Organlarning javob reaksiyasini bir xil tezlikda yuzaga keltiradi.

16. Q Uchta organizmning oziqlanish xususiyatlari o‘rganildi. Jadvaldagi a, b va c organizmlarni tegishli misollar bilan moslashtiring. 2 ball
1. Nitrifikator bakteriya; 2. Saprofit zamburug‘; 3. Sianobakteriya; 4. Parazit zamburug‘.

Organizm	Organik modda sintezi uchun uglerod manbai	Energiya manbai
a	CO_2	Yorug‘lik
b	CO_2	Ammiakning oksidlanishi
c	Nobud bo‘lgan organizmlar qoldig‘idagi organik moddalar	Organik moddalarning parchalanishi

A) a–3; b–1; c–2 **B)** a–3; b–2; c–1 **C)** a–1; b–3; c–2 **D)** a–3; b–1; c–4

17. B Qaysi juftlikdagi organizmlar fragmentatsiya orqali ko‘payadi? 1 ball

A) Spirogira va oq planariya **B)** Spirogira va achitqi zamburug‘i
C) Oq planariya va xlamidomonada **D)** Achitqi zamburug‘i va infuzoriya tufelka

18. Q Hayvonda hujayrasida 2n = 12. Ikkita birlamchi spermatotsit meyoz yo‘li bilan bo‘linib, spermatidalar hosil qildi. Hosil bo‘lgan barcha spermatidalar spermatozoidlarga aylansa, ularning soni va har biridagi xromosomalar miqdori to‘g‘ri berilgan javobni aniqlang. 2 ball

A) 8 ta spermatozoid; har birida 6 ta xromosoma.
B) 4 ta spermatozoid; har birida 6 ta xromosoma.
C) 8 ta spermatozoid; har birida 12 ta xromosoma.
D) 4 ta spermatozoid; har birida 12 ta xromosoma.

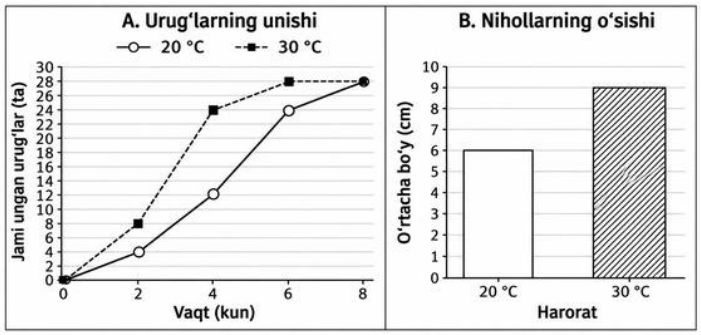
19. Q Jadvalda berilgan rivojlanish xususiyatlarini ontogenez tiplari bilan moslashtiring. 2 ball
1. Tuxumda rivojlanish; 2. Bachadonda rivojlanish; 3. Lichinkali rivojlanish.

Belgi	Kuzatilgan rivojlanish xususiyati
a	Tuxumdan voyaga yetgan organizmdan tuzilishi bilan farqlanadigan, mustaqil oziqlanuvchi yosh organizm chiqadi.
b	Embrion sariqlik zaxirasi hisobiga uzoq vaqt tuxum ichida rivojlanadi.
c	Embrionning oziqlanishi va gaz almashinuvi yo‘ldosh orqali ona organizmi hisobiga ta’minlanadi.

A) a–3; b–1; c–2 **B)** a–1; b–3; c–2 **C)** a–3; b–2; c–1 **D)** a–2; b–1; c–3

20. M O‘simlik urug‘lari 30 tadan ikki guruhga ajratilib, 20 ˚C va 30 ˚C da saqlandi. Suv, yorug‘lik va boshqa sharoitlar bir xil holatda saqlandi. A grafikda jami ungan urug‘lar soni, B grafikda esa har bir urug‘ unib chiqqanidan 5 kun o‘tgach o‘lchangan nihollar bo‘yining o‘rtacha qiymati berilgan. Grafik asosida quyidagi mulohazalarning **to‘g‘ri (T)** yoki **noto‘g‘ri (N)** ekanligini aniqlang. 3 ball

1. To‘rtinchi kuni 30 ˚C da ungan urug‘lar soni 20 ˚C dagidan ikki marta ko‘p bo‘lgan.
2. Sakkizinchi kuni ungan urug‘lar sonining teng bo‘lishi harorat unish tezligiga ta’sir qilmaganini ko‘rsatadi.
3. Bir xil yoshdagi nihollar bo‘yining farq qilishi harorat ularning o‘sishiga ta’sir qilganini ko‘rsatadi.



A) TNT **B)** TTN **C)** NTT **D)** NNT

CHSB _PRO Tayyorladi: Shokirboyev Asadbek

Javoblar kaliti — ChSB-1 tayyorgarlik: I chorak demo varianti

1 Q A	2 B A	3 Q A	4 Q A	5 Q A	6 M A	7 Q A
8 B A	9 Q A	10 Q A	11 Q A	12 Q A	13 Q A	14 M A
15 B A	16 Q A	17 B A	18 Q A	19 Q A	20 M A	

1. A) To'g'ri javob: **A) taqqoslash metodi**
Tekshiriladigan ko'nikma: Biologiya hayot haqidagi fan. Biologiyaning istiqbolli yo‘nalishlari. Tirik organizmlarni o‘rganish metodlari.

2. A) To'g'ri javob: **A) Biosfera darajasi**
Tekshiriladigan ko'nikma: Tirik organizmlarga xos xususiyatlar. Tiriklikning tuzilish darajalari.

3. A) To'g'ri javob: **A) 1, 3, 5**
Tekshiriladigan ko'nikma: Tirik organizmlarning kimyoviy tarkibi va uning doimiyligi. Hujayraning anorganik birikmalari.

4. A) To'g'ri javob: **A) X - 741 kJ; Y - 954 kJ**
Tekshiriladigan ko'nikma: Uglevod va lipidlar. Ularlarning tuzilishi, xilma-xilligi, biologik ahamiyati.

5. A) To'g'ri javob: **A) Fibrinogen : himoya**
Tekshiriladigan ko'nikma: Oqsillarning tuzilishi va funksiyalari.

6. A) To'g'ri javob: **A) TNN**
Tekshiriladigan ko'nikma: Nuklein kislotalar. DNK va RNKning tuzilishi, funksiyalari. ATF.

7. A) To'g'ri javob: **A) Uchinchi namunadagi faollikning qayta tiklanmasligi, bu oqsilning renaturatsiya qila olmasligini ifodalaydi.**
Tekshiriladigan ko'nikma: Hayotning molekula darajasi va uning o‘ziga xos jihatlari.

8. A) To'g'ri javob: **A) Hujayralar tuzilishi va asosiy hayotiy jarayonlari jihatidan o‘xshash.**
Tekshiriladigan ko'nikma: Hujayra – tiriklikning tuzilish, funksional, rivojlanish birligi. Hujayra nazariyasining asosiy qoidalari.

9. A) To'g'ri javob: **A) 1, 3, 5**
Tekshiriladigan ko'nikma: Tirik organizmlar hujayralarining qiyosiy xarakteristikasi.

10. A) To'g'ri javob: **A) a–12 mol; b–8 mol**
Tekshiriladigan ko'nikma: Moddalar almashinuvi – hujayra hayotiy faoliyatining asosi.

11. A) To'g'ri javob: **A) 118 mol**
Tekshiriladigan ko'nikma: Energetik almashinuv – dissimilyatsiya. Energetik almashinuv bosqichlari.

12. A) To'g'ri javob: **A) 180 gr**
Tekshiriladigan ko'nikma: Plastik almashinuv. Fotosintez. Xemosintez.

13. A) To'g'ri javob: **A) 2 → 4 → 1 → 3**
Tekshiriladigan ko'nikma: Hujayraning hayot sikli. Mitoz. Mitozning biologik ahamiyati. Meyoz. Meyozning biologik ahamiyati.

14. A) To'g'ri javob: **A) TNT**
Tekshiriladigan ko'nikma: Hujayra membranasiga haroratning ta’sirini o‘rganish.

15. A) To'g'ri javob: **A) Organlar faoliyatini muvofiqlashtirib, ichki muhit barqarorligini saqlaydi.**
Tekshiriladigan ko'nikma: Hayotning organizm darajasi va uning o‘ziga xos jihatlari. Organizm – o‘zini o‘zi boshqaruvchi biologik sistema. Nerv va gumoral boshqarilishning o‘ziga xos xususiyatlari.

16. A) To'g'ri javob: **A) a–3; b–1; c–2**
Tekshiriladigan ko'nikma: Tirik organizmlarning oziqlanishiga koʻra turlari.

17. A) To'g'ri javob: **A) Spirogira va oq planariya**
Tekshiriladigan ko'nikma: Organizmlarning koʻpayishi. Jinssiz koʻpayish. Jinssiz koʻpayishning turlari.

18. A) To'g'ri javob: **A) 8 ta spermatozoid; har birida 6 ta xromosoma.**
Tekshiriladigan ko'nikma: Hayvonlarda jinsiy koʻpayish shakllari. Hayvonlarda gametogenez jarayoni. Urugʻlanish. Partenogenez.

19. A) To'g'ri javob: **A) a–3; b–1; c–2**
Tekshiriladigan ko'nikma: Ontogenez – tirik organizmlarning individual rivojlanishi. Ontogenez tiplari. Embrional va postembrional rivojlanish davri.

20. A) To'g'ri javob: **A) TNT**
Tekshiriladigan ko'nikma: Tashqi muhit omillarining organizmning rivojlanishiga ta’siri.